DevOrSysAdminScripts

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Abstract

Some "small" useful scripts for enhancing development or system-administrators tasks

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Current number of commits: 735
Current git SHA1: 31f5ff427c42bf68a00d5f65572e37bd55f65a43
Code lines: 11714 total lines, 10662 not empty lines, 1052 empty lines.

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1 Files tree

```
.
├── build_and_checks_dependencies
│   ├── build_and_checks.exec.sh.tpl
│   ├── build_dependencies_notes.exec.php
│   ├── build_md_from_printable_md.exec.sh
│   ├── check_shell_scripts_beginnings.libr.sh
│   ├── check_shell_scripts_indentation.libr.sh
│   ├── check_URLs.libr.sh
│   ├── common_build_and_checks.exec.sh
│   ├── common_build_and_checks.exec.sh.tpl
│   ├── common_options.libr.sh
│   ├── comparisons.libr.sh
│   ├── create_PDF.exec.sh
│   ├── generate_from_template.libr.sh
│   ├── get_common_text_glob_patterns.libr.sh
│   ├── grammar_and_spelling_check.libr.sh
│   ├── licenses_templates
│   │   ├── build_licenses_templates.exec.sh
│   │   ├── license_file_header_AGPLv3+.tpl
│   │   ├── license_file_header_GPLv3+.tpl
│   │   └── license_file_header_LGPLv3+.tpl
│   ├── lines_counts.libr.sh
│   ├── lines_filters.libr.sh
│   ├── lines_maps.libr.sh
│   ├── overwrite_if_not_equal.libr.sh
│   ├── python_black_complement.libr.sh
│   ├── python_isort_complement.libr.sh
│   ├── repository_data.libr.sh
│   ├── shell_checks_complement.libr.sh
│   ├── split_score.exec.php
│   ├── split_score.libr.php
│   ├── strings_functions.libr.sh
│   ├── too_long_code_lines.libr.sh
│   ├── update_common_build_and_checks.exec.sh
│   └── update_or_check_files_names_listing.exec.sh
├── build_and_checks.exec.sh
├── build_and_checks_variables
│   ├── bandit_baseline.json
│   ├── bandit.ini
│   ├── black.toml
│   ├── dependencies_data.json
│   ├── dependencies_notes.txt
│   ├── DevOrSysAdminScripts.html.tpl
│   ├── DevOrSysAdminScripts.tex
```

```
│   ├── DevOrSysAdminScripts.tex.tpl
│   ├── files_listing.tex.tpl
│   ├── files_names_listing.txt
│   ├── .gitignore
│   ├── licenses_notes.txt
│   ├── phpmd_baseline.xml
│   ├── pylintrc.toml
│   ├── repository_data.txt
│   ├── repository_data.txt.tpl
│   ├── ruff.toml
│   ├── shellcheck.ini
│   └── shellcheck_notes.txt
├── COPYING
├── COPYING.LESSER
├── DevOrSysAdminScripts.html
├── DevOrSysAdminScripts.pdf
├── git_helpers
│   ├── git_checks.libr.sh
│   ├── post-commit
│   ├── post_commit_renaming_comment.libr.sh
│   └── pre-commit
├── IMAP_transfert.py
├── LiensUtiles.txt
├── miscellaneous
│   ├── 1234567890123456789012345678901234567890123456789012345678901234567890123456789
│   │   0123\
│   ├── 1234567890123456789012345678901234567890123456789012345678901234567890123456789
│   │   01234\
│   ├── 1234567890123456789012345678901234567890123456789012345678901234567890123456789
│   │   012345
│   ├── 1234567890123456789012345678901234567890123456789012345678901234567890123456789
│   │   012345\
│   ├── 1234567890123456789012345678901234567890123456789012345678901234567890123456789
│   │   0123456
│   ├── 1234567890123456789012345678901234567890123456789012345678901234567890123456789
│   │   0123456\
│   ├── a_long_long_long_long_long_long_long_long_long_filename_so_that_you_c
│   │   an_see_that_it_works.txt
│   ├── funny.sh
```

```
│   ├── spampty_data.txt
│   ├── string_escaping.html
│   ├── string_escaping.xhtml
│   └──
│     yet_another_long_long_long_long_long_long_long_long_long_filename_so_
│     that_you_can_see_that_it_works.txt
├── README.html
├── README.md
├── README.md.tpl
├── string_escaping.libr.php
├── unicode.libr.php
├── upgrade_build_and_checks.exec.sh
└── wget_sha512.libr.sh

6 directories, 81 files

[ 12K Mar 23 18:03]  ./
├── [4.0K Mar 23 18:03]  build_and_checks_dependencies/
│   ├── [3.9K Mar 22 13:27]  build_and_checks.exec.sh.tpl*
│   ├── [4.4K Mar 23 17:50]  build_dependencies_notes.exec.php*
│   ├── [4.2K Mar 23 08:55]  build_md_from_printable_md.exec.sh*
│   ├── [3.2K Mar 23 09:48]  check_shell_scripts_beginnings.libr.sh
│   ├── [5.3K Mar 23 09:47]  check_shell_scripts_indentation.libr.sh
│   ├── [3.6K Mar 23 10:11]  check_URLs.libr.sh
│   ├── [ 30K Mar 23 18:00]  common_build_and_checks.exec.sh*
│   ├── [ 25K Mar 23 17:30]  common_build_and_checks.exec.sh.tpl
│   ├── [ 16K Mar 23 15:17]  common_options.libr.sh
│   ├── [6.1K Mar 22 16:03]  comparisons.libr.sh
│   ├── [ 20K Mar 23 18:00]  create_PDF.exec.sh*
│   ├── [3.7K Mar 22 16:36]  generate_from_template.libr.sh
│   ├── [2.0K Mar  5 21:22]  get_common_text_glob_patterns.libr.sh
│   ├── [3.3K Mar 23 10:18]  grammar_and_spelling_check.libr.sh
│   ├── [4.0K Mar 23 10:24]  licenses_templates/
│   │   ├── [ 14K Mar 23 10:24]  build_licenses_templates.exec.sh*
│   │   ├── [ 728 Feb 28 20:38]  license_file_header_AGPLv3+.tpl
│   │   ├── [ 707 Feb 28 20:38]  license_file_header_GPLv3+.tpl
│   │   └── [ 728 Feb 28 20:38]  license_file_header_LGPLv3+.tpl
│   ├── [6.2K Mar 23 15:40]  lines_counts.libr.sh
│   ├── [ 14K Mar 22 18:05]  lines_filters.libr.sh
│   ├── [2.7K Feb 19 16:16]  lines_maps.libr.sh
│   ├── [2.7K Mar 22 18:33]  overwrite_if_not_equal.libr.sh
│   ├── [2.5K Mar 23 15:29]  python_black_complement.libr.sh
│   ├── [3.0K Mar 23 15:30]  python_isort_complement.libr.sh
│   ├── [2.1K Mar 23 10:14]  repository_data.libr.sh
│   ├── [3.4K Mar 23 15:30]  shell_checks_complement.libr.sh
│   ├── [1.6K Feb  3 17:57]  split_score.exec.php
```

```
│   ├── [1.9K Mar 22 19:01]  split_score.libr.php
│   ├── [ 48K Mar 23 10:16]  strings_functions.libr.sh
│   ├── [4.5K Mar 23 10:12]  too_long_code_lines.libr.sh
│   ├── [4.2K Mar 23 17:15]  update_common_build_and_checks.exec.sh*
│   └── [6.1K Mar 23 17:56]  update_or_check_files_names_listing.exec.sh*
├── [4.1K Mar 23 18:00]  build_and_checks.exec.sh*
├── [4.0K Mar 23 18:03]  build_and_checks_variables/
│   ├── [  15 Jan  5 20:00]  bandit_baseline.json
│   ├── [  25 Jan  5 18:49]  bandit.ini
│   ├── [  30 Feb 28 20:30]  black.toml
│   ├── [4.6K Mar 23 16:44]  dependencies_data.json
│   ├── [3.8K Mar 23 16:44]  dependencies_notes.txt
│   ├── [5.7K Mar 23 17:43]  DevOrSysAdminScripts.html.tpl
│   ├── [ 35K Mar 23 18:03]  DevOrSysAdminScripts.tex
│   ├── [4.2K Mar 23 17:40]  DevOrSysAdminScripts.tex.tpl
│   ├── [ 16K Mar 23 17:18]  files_listing.tex.tpl
│   ├── [5.4K Mar 23 17:13]  files_names_listing.txt
│   ├── [  24 Jan 15 22:26]  .gitignore
│   ├── [1.4K Mar 23 16:11]  licenses_notes.txt
│   ├── [2.2K Mar 23 17:31]  phpmd_baseline.xml
│   ├── [ 678 Feb 24 22:38]  pylintrc.toml
│   ├── [4.5K Mar 22 15:51]  repository_data.txt
│   ├── [ 786 Mar 22 15:39]  repository_data.txt.tpl
│   ├── [1.4K Feb 24 23:25]  ruff.toml
│   ├── [ 167 Feb 19 18:39]  shellcheck.ini
│   └── [3.5K Mar  9 14:46]  shellcheck_notes.txt
├── [ 34K May  6  2024]  COPYING
├── [7.5K Dec 27 21:48]  COPYING.LESSER
├── [451K Mar 23 18:03]  DevOrSysAdminScripts.html
├── [396K Mar 23 18:03]  DevOrSysAdminScripts.pdf
├── [4.0K Mar 23 10:34]  git_helpers/
│   ├── [3.8K Mar 22 21:06]  git_checks.libr.sh
│   ├── [2.1K Mar 22 21:09]  post-commit*
│   ├── [ 15K Mar 23 10:34]  post_commit_renaming_comment.libr.sh
│   └── [1.8K Mar 22 21:11]  pre-commit*
├── [8.3K Feb 28 20:31]  IMAP_transfert.py
├── [ 830 Dec 29 19:10]  LiensUtiles.txt
├── [4.0K Mar 18 17:27]  miscellaneous/
│   ├── [   0 Mar  4 10:37]
│   │ 1234567890123456789012345678901234567890123456789012345678901234567890123456789
│   │ 0123\
│   ├── [   0 Mar  4 10:03]
│   │ 1234567890123456789012345678901234567890123456789012345678901234567890123456789
│   │ 01234\
│   ├── [   0 Mar  4 10:01]
│   │ 1234567890123456789012345678901234567890123456789012345678901234567890123456789
```

```
│   │ 012345
│   ├── [   0 Mar  4 10:37]
│   │ 1234567890123456789012345678901234567890123456789012345678901234567890123456789
│   │ 012345\
│   ├── [   0 Mar  4 10:37]
│   │ 1234567890123456789012345678901234567890123456789012345678901234567890123456789
│   │ 0123456
│   ├── [   0 Mar  4 10:37]
│   │ 1234567890123456789012345678901234567890123456789012345678901234567890123456789
│   │ 0123456\
│   ├── [   0 Mar  4 09:13]
│   │ a_long_long_long_long_long_long_long_long_long_filename_so_that_you_c
│   │ an_see_that_it_works.txt
│   ├── [1.3K Mar  9 15:15]  funny.sh
│   ├── [  94 Mar  3 19:56]  spampty_data.txt
│   ├── [ 442 Mar 18 16:50]  string_escaping.html
│   ├── [ 654 Mar 18 17:27]  string_escaping.xhtml
│   └── [   0 Mar  4 09:13]
│     yet_another_long_long_long_long_long_long_long_long_long_filename_so_
│     that_you_can_see_that_it_works.txt
├── [8.1K Mar 22 21:29]  README.html
├── [4.3K Mar 22 21:29]  README.md
├── [4.3K Mar 22 21:29]  README.md.tpl
├── [3.2K Mar 22 21:36]  string_escaping.libr.php
├── [ 13K Mar 22 21:59]  unicode.libr.php
├── [1.4K Mar 10 16:37]  upgrade_build_and_checks.exec.sh*
└── [1.5K Mar 22 22:04]  wget_sha512.libr.sh

6 directories, 81 files
```

2 Notes on licenses and dependencies

This repository is dual licensed. All the Bash scripts source code is covered by GPLv3+. All the PHP scripts source code is covered by LGPLv3+. All the Python scripts source code is covered by LGPLv3+.

The text of the licenses is available at: `https://www.gnu.org/licenses/`. The git repository of this source code is also available at: `https://github.com/LLyaudet/DevOrSysAdminScripts/`. Clone it with this URL: `git@github.com:LLyaudet/DevOrSysAdminScripts.git`.

Here is a listing of the dependencies of the code in this repository:

- The following dependencies are GPLv3+ licensed:
  basename, bash (and its builtins), cat, chmod, cp, cut, date, diff, dirname, env, find, grep, head, mkdir, mv, realpath, rm, sed, sha512sum, shellcheck, sort, tail, touch, wc, wget, xargs;

- The following dependencies are GPLv2+ licensed:
  more, pandoc, pdftex, pdflatex, tree;
- The following dependencies are GPLv2 licensed:
  git, pylint;
- The following dependencies are LGPLv2.1+ licensed:
  notify-send;
- The following dependencies are MIT licensed:
  black, isort, mypy, pip, ruff;
- The following dependencies are Apache-2.0 licensed:
  bandit;
- The following dependencies are BSD-3-Clause licensed:
  phpmd;
- php is licensed under PHP License v3.01;
- tex is licensed under Knuth license;
- pcregrep, pcre2grep has a custom license from Cambridge University, but PCRE2 has BSD-3-Clause WITH PCRE2-exception,
- python3 is licensed under PSF License Version 2 and the Zero-Clause BSD license.

Remark: Some commands listed under GPLv3+ are POSIX and can have different implementations under another license; for example, uutils coreutils is MIT licensed, whilst GNU coreutils is GPLv3+ licensed.

Bash can be found at the following URLs:
`https://www.gnu.org/software/bash/`,
`https://tiswww.case.edu/php/chet/bash/bashtop.html`,
`https://savannah.gnu.org/projects/bash/`.

GNU coreutils can be found at the following URLs:
`https://www.gnu.org/software/coreutils/`,
`https://savannah.gnu.org/projects/coreutils/`.

uutils coreutils can be found at the following URLs:
`https://uutils.github.io/coreutils/`,
`https://github.com/uutils/coreutils`.

diff (GNU diffutils) can be found at the following URLs:
`https://www.gnu.org/software/diffutils/`,
`https://savannah.gnu.org/projects/diffutils/`.

find, xargs (GNU findutils) can be found at the following URLs:
`https://www.gnu.org/software/findutils/`,
`https://savannah.gnu.org/projects/findutils/`.

grep (GNU grep) can be found at the following URLs:
`https://www.gnu.org/software/grep/`,
`https://savannah.gnu.org/projects/grep/`.

sed (GNU sed) can be found at the following URLs:
https://www.gnu.org/software/sed/,
https://savannah.gnu.org/projects/sed/.

ShellCheck can be found at the following URLs:
https://www.shellcheck.net/,
https://github.com/koalaman/shellcheck.

GNU Wget can be found at the following URLs:
https://www.gnu.org/software/wget/,
https://savannah.gnu.org/projects/wget/.

more of util-linux can be found at the following URLs:
https://www.kernel.org/pub/linux/utils/util-linux/,
https://git.kernel.org/pub/scm/utils/util-linux/util-linux.git/,
https://github.com/util-linux/util-linux official?.

pandoc can be found at the following URLs:
https://pandoc.org/,
https://github.com/jgm/pandoc.

pdftex/pdflatex can be found at the following URLs:
https://www.tug.org/applications/pdftex/,
https://tug.org/texlive/svn/.

tree can be found at the following URLs:
https://oldmanprogrammer.net/source.php?dir=projects/tree,
https://gitlab.com/OldManProgrammer/unix-tree,
https://github.com/Old-Man-Programmer/tree.

git can be found at the following URLs:
https://git-scm.com/about,
https://github.com/git/git.

pylint can be found at the following URLs:
https://www.pylint.org/,
https://pylint.readthedocs.io/en/stable/,
https://github.com/pylint-dev/pylint,
https://pypi.org/project/pylint/.

notify-send can be found at the following URLs:
https://gitlab.gnome.org/GNOME/libnotify/,
https://gnome.pages.gitlab.gnome.org/libnotify/.

black can be found at the following URLs:
https://github.com/psf/black,
https://black.readthedocs.io/en/stable/,
https://pypi.org/project/black/.

isort can be found at the following URLs:
https://pycqa.github.io/isort/,
https://github.com/pycqa/isort/,
https://pypi.org/project/isort/.

mypy can be found at the following URLs:
https://mypy-lang.org/,
https://mypy.readthedocs.io/en/stable/,

https://github.com/python/mypy,
https://pypi.org/project/mypy/.

pip can be found at the following URLs:
https://pip.pypa.io/en/stable/,
https://github.com/pypa/pip,
https://pypi.org/project/pip/ (egg and chicken ;)).

ruff can be found at the following URLs:
https://docs.astral.sh/ruff/,
https://github.com/astral-sh/ruff,
https://pypi.org/project/ruff/.

bandit can be found at the following URLs:
https://bandit.readthedocs.io/en/latest/,
https://github.com/PyCQA/bandit,
https://pypi.org/project/bandit/.

phpmd can be found at the following URLs:
https://phpmd.org/,
https://github.com/phpmd/phpmd,
https://packagist.org/packages/phpmd/phpmd.

php can be found at the following URLs:
https://www.php.net/,
https://github.com/php/php-src.

tex can be found at the following URL:
https://ctan.org/tex-archive/systems/knuth/dist/tex.

pcregrep, pcre2grep can be found at the following URLs:
https://www.pcre.org/,
https://pcre2project.github.io/pcre2/,
https://github.com/PCRE2Project/pcre1,
https://github.com/PCRE2Project/pcre2,
https://github.com/PCRE2Project/pcre2/blob/main/src/pcre2grep.c.

python3 can be found at the following URLs:
https://www.python.org/,
https://docs.python.org/3/.

If you see something that I missed regarding my dependencies, please tell me/email me :).

3 Listing of files

3.1 build_and_checks_dependencies/build_and_checks.exec.sh.tpl

```
#!/usr/bin/env bash
# This file is part of DevOrSysAdminScripts library.
#
# DevOrSysAdminScripts is free software:
# you can redistribute it and/or modify it under the terms
# of the GNU General Public License
```

```
# as published by the Free Software Foundation,
# either version 3 of the License,
# or (at your option) any later version.
#
# DevOrSysAdminScripts is distributed in the hope
# that it will be useful,
# but WITHOUT ANY WARRANTY;
# without even the implied warranty of
# MERCHANTABILITY or FITNESS FOR A PARTICULAR PURPOSE.
# See the GNU General Public License for more details.
#
# You should have received a copy of
# the GNU General Public License
# along with DevOrSysAdminScripts.
# If not, see <https://www.gnu.org/licenses/>.
#
# ©Copyright 2023-2026 Laurent Frédéric Bernard François Lyaudet
# The file "build_and_checks.exec.sh.tpl" was renamed from
# "build_and_checks.sh.tpl" to "build_and_checks.exec.sh.tpl".
# The file "build_and_checks.exec.sh" generated from the file
# "build_and_checks.sh.tpl" or "build_and_checks.exec.sh.tpl"
# was renamed from
# "build_and_checks.sh" to "build_and_checks.exec.sh".

build_and_checks(){
  # $1=work_directory
  # Options:
  #   --verbose
  local LFBFL_work_directory="."
  if [[ -n "$1" ]]; then
    LFBFL_work_directory="$1"
  fi
  readonly LFBFL_work_directory

  local LFBFL_verbose=""
  if [[ "$*" == *--verbose* ]]; then
    printf "%s %s\n" "$0" "$*"
    LFBFL_verbose="--verbose"
  fi
  readonly LFBFL_verbose

  source ./wget_sha512.libr.sh

  LFBFL_subdir="build_and_checks_dependencies"
  mkdir -p "${LFBFL_subdir}/licenses_templates"
  LFBFL_variables_directory="${LFBFL_work_directory}/"
```

```
  LFBFL_variables_directory+="build_and_checks_variables"
  mkdir -p "${LFBFL_variables_directory}/temp"
  if ! [[ -f "${LFBFL_variables_directory}/.gitignore" ]]; then
    printf "temp/\n\n"\
      > "${LFBFL_variables_directory}/.gitignore"
  fi

  # LFBFL_dependencies_raw_content_download_URL
  local LFBFL_dependencies_URL
  LFBFL_dependencies_URL="https://raw.githubusercontent.com/LLyaudet/"
  LFBFL_dependencies_URL+="DevOrSysAdminScripts/main/${LFBFL_subdir}"
  readonly LFBFL_dependencies_URL

  declare -r LFBFL_common_file_name="common_build_and_checks.exec.sh"
  # LFBFL_script_download_URL
  declare -r\
    LFBFL_script="${LFBFL_dependencies_URL}/${LFBFL_common_file_name}"
  declare -r\
    LFBFL_file_path="./${LFBFL_subdir}/${LFBFL_common_file_name}"
  @sha512_common_build_and_checks.exec.sh@
  wget_sha512 "${LFBFL_file_path}" "${LFBFL_script}"\
    "${LFBFL_correct_sha512}" "${LFBFL_verbose}"
  chmod +x "./${LFBFL_file_path}"

  "${LFBFL_file_path}" "${LFBFL_work_directory}"\
    "${LFBFL_dependencies_URL}" "${LFBFL_verbose}"
}

if [[ "$*" == *--fixed-point-build* ]]; then
  printf -- "--fixed-point-build\n"
  source "./build_and_checks_dependencies/lines_filters.libr.sh"
  LFBFL_work_directory="."
  if [[ -n "$1" ]]; then
    LFBFL_work_directory="$1"
  fi
  LFBFL_variables_directory="${LFBFL_work_directory}/"
  LFBFL_variables_directory+="build_and_checks_variables"
  LFBFL_data_file_name="${LFBFL_variables_directory}/"
  LFBFL_data_file_name+="repository_data.txt"
  LFBFL_repository_name=""
  grep_variable "${LFBFL_data_file_name}" repository_name\
    --result-variable-prefix="LFBFL_"\
    --replace-line-returns-by=""
  # Touching the 3 following files first let us iterate
  # upgrade_build_and_checks 2 times instead of 3 to reach
  # a fixed point.
```

```
  touch "${LFBFL_variables_directory}/${LFBFL_repository_name}.tex"
  touch "${LFBFL_work_directory}/${LFBFL_repository_name}.pdf"
  touch "${LFBFL_work_directory}/${LFBFL_repository_name}.html"
  # The two iterations of build_and_checks needs to complete
  # during the same minute to have a fixed point.
  build_and_checks "$@"
  # build_and_checks "$@" second iteration below
  # Then calling this script without --fixed-point-build should not
  # yield any new superficial modifications.
fi

build_and_checks "$@"
```

3.2 build_and_checks_dependencies/build_dependencies_notes.exec.php

```
#!/usr/bin/php
<?php
/*
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*/

require_once('string_escaping.libr.php');


function build_dependencies_notes(string $s_working_directory) : void {
```

```
32    $s_subdir1 = $s_working_directory.'/build_and_checks_variables';
33    $s_subdir2 = $s_subdir1.'/temp';
34    $s_data_file_path = $s_subdir1.'/dependencies_data.json';
35    $s_data = file_get_contents($s_data_file_path);
36    $arr_data = json_decode($s_data, true);
37    $s_result_content1 = '';
38    $s_result_content2 = '';
39    $s_result_content3 = '';
40
41    foreach($arr_data as $arr_dependency){
42      $s_URL_word = "URL";
43      if(count($arr_dependency["URLs"]) > 1){
44        $s_URL_word .= "s";
45      }
46
47      $s_result_content3 .= "<p>\n";
48
49      $s_result_content1 .= (
50        $arr_dependency["dependency_name"]
51        ." can be found at the following "
52        .$s_URL_word
53        ":\n"
54      );
55      $s_result_content2 .= (
56        $arr_dependency["dependency_name"]
57        ." can be found at the following "
58        .$s_URL_word
59        ":\\newline\n"
60      );
61      $s_result_content3 .= (
62        $arr_dependency["dependency_name"]
63        ." can be found at the following "
64        .$s_URL_word
65        ":<br>\n"
66      );
67
68      for(
69        $i = 0, $i_max = count($arr_dependency["URLs"]) - 1;
70        $i <= $i_max;
71        ++$i
72      ){
73        $mixed_URL = $arr_dependency["URLs"][$i];
74        $s_comment = '';
75        $s_URL = '';
76        if(is_array($mixed_URL)){
77          $s_URL = $mixed_URL["URL"];
```

```
        $s_comment = ' '.$mixed_URL["comment"];
      }
      else{
        $s_URL = $mixed_URL;
      }

      $s_result_content1 .= $s_URL.$s_comment."\n";

      if(strlen($s_URL) > 60){
        $s_result_content2 .= (
          "{\\small\n"
          .'  \url{'.$s_URL."}\n"
          .'}'
        );
      }
      else{
        $s_result_content2 .= '\url{'.$s_URL.'}';
      }
      if($i === $i_max){
        $s_result_content2 .= $s_comment.".\n";
      }
      else{
        $s_result_content2 .= $s_comment.",\\newline\n";
      }

      $s_result_content3 .= (
        "<a\n"
        .'href="'.string_escaping\HTML\escape_attribute($s_URL)."\"\n"
        .'><span style="font-family:monospace"'."\n"
        .'>'.string_escaping\HTML\escape_text($s_URL)."</span\n"
        .'></a>'
      );
      if($i === $i_max){
        $s_result_content3 .= $s_comment.".\n";
      }
      else{
        $s_result_content3 .= $s_comment.",<br>\n";
      }
    }

    $s_result_content1 .= "\n";
    $s_result_content2 .= "\n";
    $s_result_content3 .= "</p>\n\n";
  }

  // Add a "footer".
```

```
  $s_result_content1 .= (
    "If you see something that I missed regarding my dependencies,\n"
    ."please tell me/email me :).\n"
  );
  $s_result_content2 .= (
    "If you see something that I missed regarding my dependencies,\n"
    ."please tell me/email me :).\n"
  );
  $s_result_content3 .= (
    "<p>\n"
    ."If you see something that I missed regarding my dependencies,\n"
    ."please tell me/email me :).\n"
    ."</p>\n"
  );

  // We overwrite results if needed.
  $s_result_path1 = $s_subdir1.'/dependencies_notes.txt';
  $s_existing_result_content1 = file_get_contents($s_result_path1);
  if($s_result_content1 !== $s_existing_result_content1){
    file_put_contents($s_result_path1, $s_result_content1);
  }
  $s_result_path2 = $s_subdir2.'/dependencies_notes.tex';
  $s_existing_result_content2 = file_get_contents($s_result_path2);
  if($s_result_content2 !== $s_existing_result_content2){
    file_put_contents($s_result_path2, $s_result_content2);
  }
  $s_result_path3 = $s_subdir2.'/dependencies_notes.html';
  $s_existing_result_content3 = file_get_contents($s_result_path3);
  if($s_result_content3 !== $s_existing_result_content3){
    file_put_contents($s_result_path3, $s_result_content3);
  }
}

build_dependencies_notes($argv[1]);
```

3.3 build_and_checks_dependencies/build_md_from_printable_md.exec.sh

```
#!/usr/bin/env bash
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```

```
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# This file was renamed from "build_md_from_printable_md.sh"
# to "build_md_from_printable_md.exec.sh".

LFBFL_subdir="build_and_checks_dependencies"
# shellcheck source=common_options.libr.sh
source "./${LFBFL_subdir}/common_options.libr.sh"
# shellcheck source=overwrite_if_not_equal.libr.sh
source "./${LFBFL_subdir}/overwrite_if_not_equal.libr.sh"

build_md_from_printable_md(){
  # Options:
  #   --verbose
  #   --work-directory=""
  #   --base-name="README"
  declare -i LFBFL_i_verbose=0
  get_verbose_option "$@"
  local LFBFL_work_directory=""
  get_work_directory_option "$@"
  declare -a LFBFL_return_traps_stack
  local LFBFL_previous_return_trap
  init_return_trap
  pushd_to_work_directory --trap-popd
  can_continue_after_enhanced_pushd || return 1

  local LFBFL_base_name="README"
  local LFBFL_arg
  for LFBFL_arg in "$@"; do
    if [[ "${LFBFL_arg}" == --base-name=* ]]; then
      LFBFL_base_name=${LFBFL_arg#--base-name=}
      if [[ LFBFL_i_verbose -eq 1 ]]; then
        printf "Searching md file: %s.\n" "${LFBFL_base_name}"
      fi
```

```
      break
    fi
  done
  readonly LFBFL_base_name

  # Remove line returns here to keep lines short.
  local LFBFL_sed_expression
  LFBFL_sed_expression='s/(\n[ ]*\[[a-zA-Z0-9:-]*\]: [^\n\\]*)\\\n/\1/Mg'
  LFBFL_sed_expression+=';s/(\n[ ]*<http[^\n\\]*)\\\n/\1/Mg'
  LFBFL_sed_expression+=';s/(\n[ ]*- <http[^\n\\]*)\\\n/\1/Mg'
  readonly LFBFL_sed_expression
  # Since Markdown can contain code, it is limited to Markdown URLs.
  # Hence it is way more complicated than 's/\\\n//Mg' expression
  # in other files.
  # Another downside is that more than one pass is needed.

  if [[ -f "${LFBFL_base_name}.md.tpl" ]]; then
    sed -Ez -e "${LFBFL_sed_expression}"\
            -e "${LFBFL_sed_expression}"\
            -e "${LFBFL_sed_expression}"\
            -e "${LFBFL_sed_expression}"\
      "${LFBFL_base_name}.md.tpl"\
      > "${LFBFL_base_name}.md.temp"
    overwrite_if_not_equal "${LFBFL_base_name}.md"\
      "${LFBFL_base_name}.md.temp"
  else
    printf "No file %s.md.tpl\n" "${LFBFL_base_name}"
    # Unfortunately, we cannot return early, since we want to propose
    # HTML file generation with pandoc from .md file,
    # if the file .md.tpl doesn't exist.
  fi

  if [[ ! -f "${LFBFL_base_name}.md" ]]; then
    printf "No file %s.md\n" "${LFBFL_base_name}"
    return 1
  fi

  pandoc --from markdown --to html --standalone\
    --shift-heading-level-by=-1 "${LFBFL_base_name}.md"\
    -o "${LFBFL_base_name}.html.temp"
  # Correcting pandoc HTML "by hand".
  # https://github.com/jgm/pandoc/issues/10957
  # With the most open-minded answer found on Internet ;) XD.
  # This second sed expression is what I need,
  # but it may be unsuitable for you.
  # Other problems:
```

```
  # https://github.com/jgm/pandoc/issues/11484
  local LFBFL_prewrap="s~(code \{(\n|[^}])*)(    \})"
  LFBFL_prewrap+="~\1      white-space: pre-wrap;\n\3~M"
  readonly LFBFL_prewrap
  sed -i -E -e 's~(  <meta .*) />~\1>~g'\
            -e 's~<html .*>~<html lang="en">~'\
            -e '/    code\{white-space: pre-wrap;\}/d'\
    "${LFBFL_base_name}.html.temp"
  sed -i -Ez -e 's~(<img(\n|[^a-z0-9])(\n|[^>])*) />~\1>~Mg'\
             -e "${LFBFL_prewrap}"\
    "${LFBFL_base_name}.html.temp"
  overwrite_if_not_equal "${LFBFL_base_name}.html"\
    "${LFBFL_base_name}.html.temp"

  # java -Xss128M -jar "../jar_venv/vnu.jar" --exit-zero-always\
  #   --verbose ${LFBFL_base_name}.html"
}

build_md_from_printable_md "$@"
```

3.4 build_and_checks_dependencies/check_shell_scripts_beginnings.libr.sh

```
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# This file was renamed from "check_shell_scripts_beginnings.sh"
```

```
# to "check_shell_scripts_beginnings.libr.sh".

LFBFL_subdir="build_and_checks_dependencies"
# shellcheck source=common_options.libr.sh
source "./${LFBFL_subdir}/common_options.libr.sh"
# shellcheck source=lines_filters.libr.sh
source "./${LFBFL_subdir}/lines_filters.libr.sh"

declare -gr LFBFL_SHELL_SCRIPT_BEGINNING="#!/usr/bin/env bash"

check_one_shell_script_beginning(){
  # $1=file_path
  #
  # declare -r LFBFL_file_path=$(basename "$1")
  # if [[ "${LFBFL_file_path}" =~ license_file_header_.*\.sh ]]; then
  #   return 0
  # fi
  # diff <(head -n 1 "$1") <(printf '#!/usr/bin/env bash')
  # ------------------------------------------------------------------
  # is of course slower than
  # ------------------------------------------------------------------
  # diff <(head -n 1 "$1") <(printf '#!/usr/bin/env bash')
  # ------------------------------------------------------------------
  # is of course slower than
  # ------------------------------------------------------------------
  declare -r LFBFL_head=$(head -n 1 "$1")
  [[ "${LFBFL_head}" == "${LFBFL_SHELL_SCRIPT_BEGINNING}" ]]\
    || printf "%s:File has wrong shell script beginning.\n" "$1"
}

check_shell_scripts_beginnings(){
  # Options:
  #   --verbose
  #   --work-directory=""
  declare -i LFBFL_i_verbose=0
  get_verbose_option "$@"
  local LFBFL_work_directory=""
  get_work_directory_option "$@"
  declare -a LFBFL_return_traps_stack
  local LFBFL_previous_return_trap
  init_return_trap
  pushd_to_work_directory --trap-popd
  can_continue_after_enhanced_pushd || return 1

  # shopt -s globstar
  # for LFBFL_file_path in **/*.sh; do
```

```
  local LFBFL_file_path
  local LFBFL_head
  declare -r LFBFL_s_files_paths=$(
    find . -type f -name "*.sh" -printf '%P\n'\
    | relevant_find
  )
  if [[ -z "${LFBFL_s_files_paths}" ]]; then
    [[ LFBFL_i_verbose -eq 1 ]]\
      && printf "check_shell_scripts_beginnings: No file found.\n"
    return
  fi
  declare -a LFBFL_arr_files_paths
  mapfile -t LFBFL_arr_files_paths <<< "${LFBFL_s_files_paths}"
  for LFBFL_file_path in "${LFBFL_arr_files_paths[@]}"; do
    [[ LFBFL_i_verbose -eq 1 ]]\
      && printf "%s:Checking shell script beginning.\n"\
          "${LFBFL_file_path}"
    LFBFL_head=$(head -n 1 "${LFBFL_file_path}")
    [[ "${LFBFL_head}" == "${LFBFL_SHELL_SCRIPT_BEGINNING}" ]]\
      || printf "%s:File has wrong shell script beginning.\n"\
          "${LFBFL_file_path}"
  done
}
```

3.5 build_and_checks_dependencies/check_shell_scripts_indentation.libr.sh

```
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LFBFL_subdir="build_and_checks_dependencies"
# shellcheck source=common_options.libr.sh
source "./${LFBFL_subdir}/common_options.libr.sh"
# shellcheck source=lines_filters.libr.sh
source "./${LFBFL_subdir}/lines_filters.libr.sh"

check_one_shell_script_indentation(){
  # $1=file_path
  declare -i LFBFL_i_file_with_error=0
  local LFBFL_some_line
  local LFBFL_previous_line="workeduntilsomeasshole"
  # created a command named "workeduntilsomeasshole"
  # and called it without neither indentation, nor parameters
  # in a .sh script XD.
  # Manual justification of the lines L-4 and L-3 was a serendipity.
  # I just tried a joke and was lucky ;) :).
  local LFBFL_line_end
  local LFBFL_previous_line_end
  declare -i LFBFL_i_line_length
  declare -i LFBFL_i_line_end_length
  declare -i LFBFL_i_line_start_length
  local LFBFL_spacing
  declare -i LFBFL_i_offset
  local LFBFL_substring1
  local LFBFL_substring2
  declare -r LFBFL_pattern='^ *-e '
  # This grep would have been enough without some sed commands.
  declare -r LFBFL_s_lines=$(
    grep -EHn -B 1 '^(  )* ([^ ]|$)' "$1"
  )
  if [[ -z "${LFBFL_s_lines}" ]]; then
    return
  fi

  declare -a LFBFL_arr_lines
  mapfile -t LFBFL_arr_lines <<< "${LFBFL_s_lines}"
  readonly LFBFL_arr_lines

  for LFBFL_some_line in "${LFBFL_arr_lines[@]}"; do
    if [[ "${LFBFL_some_line}" == "--" ]]; then
      # When switching between code fragments, we reinit.
      LFBFL_previous_line="workeduntilsomeasshole"
```

```
      continue
    fi
    # Note how it's funny that this loop always work because we check
    # shell scripts beginning in another script and thus we always
    # have a line before.
    if [[ "${LFBFL_previous_line}" == "workeduntilsomeasshole" ]];
    then
      LFBFL_previous_line="${LFBFL_some_line}"
      continue
    fi
    # Remove the name of the file and then the line number
    LFBFL_line_end=${LFBFL_some_line#*:}
    LFBFL_line_end=${LFBFL_line_end#*:}
    # But for the previous line '-' or ':' is used instead of ':'
    # and filenames with '-' are much more frequent than with ':'...
    LFBFL_i_line_length=${#LFBFL_some_line}
    LFBFL_i_line_end_length=${#LFBFL_line_end}
    LFBFL_i_line_start_length=$((
      LFBFL_i_line_length - LFBFL_i_line_end_length
    ))
    LFBFL_i_offset=$((LFBFL_i_line_start_length - 2))
    # Checking if increasing number of digits in the line numbers
    # (99 -> 100).
    LFBFL_substring1=${LFBFL_previous_line:${LFBFL_i_offset}:1}
    # Nice puzzle below :) Look for comments above :)
    case ${LFBFL_substring1} in
      -) ((LFBFL_i_offset += 1));;
      :) ((LFBFL_i_offset += 1));;
      [0-9]) ((LFBFL_i_offset += 2));;
      *) printf ":-(\n";;
    esac
    LFBFL_previous_line_end=${LFBFL_previous_line:${LFBFL_i_offset}}
    if [[ "${LFBFL_line_end}" =~ ${LFBFL_pattern} ]]; then
      # Line starts with spacing and -e
      # Keep max spaces
      LFBFL_spacing=${LFBFL_line_end%%-*}
      LFBFL_i_offset=${#LFBFL_spacing}
      LFBFL_substring1=${LFBFL_line_end:${LFBFL_i_offset}:3}
      LFBFL_substring2=${LFBFL_previous_line_end:${LFBFL_i_offset}:3}
      if [[ "${LFBFL_substring1}" != "${LFBFL_substring2}" ]]; then
        printf "%s\n" "${LFBFL_some_line}"
        LFBFL_i_file_with_error=1
      fi
    else
      printf "%s\n" "${LFBFL_some_line}"
      LFBFL_i_file_with_error=1
```

```
    fi
    LFBFL_previous_line="${LFBFL_some_line}"
  done

  if [[ LFBFL_i_file_with_error -eq 1 ]]; then
    printf "%s:File has some lines with odd number of spaces.\n" "$1"
  fi
}

check_shell_scripts_indentation(){
  # Options:
  #   --verbose
  #   --work-directory=""
  declare -i LFBFL_i_verbose=0
  get_verbose_option "$@"
  local LFBFL_work_directory=""
  get_work_directory_option "$@"
  declare -a LFBFL_return_traps_stack
  local LFBFL_previous_return_trap
  init_return_trap
  pushd_to_work_directory --trap-popd
  can_continue_after_enhanced_pushd || return 1

  # shopt -s globstar
  # for LFBFL_file_path in **/*.sh; do
  local LFBFL_file_path
  declare -r LFBFL_s_files_paths=$(
    find . -type f -name "*.sh" -printf '%P\n'\
    | relevant_find
  )
  if [[ -z "${LFBFL_s_files_paths}" ]]; then
    [[ LFBFL_i_verbose -eq 1 ]]\
      && printf "check_shell_scripts_indentation: No file found.\n"
    return
  fi
  declare -a LFBFL_arr_files_paths
  mapfile -t LFBFL_arr_files_paths <<< "${LFBFL_s_files_paths}"
  readonly LFBFL_arr_files_paths
  for LFBFL_file_path in "${LFBFL_arr_files_paths[@]}"; do
    [[ LFBFL_i_verbose -eq 1 ]]\
      && printf "%s:Checking shell script indentation.\n"\
          "${LFBFL_file_path}"
    check_one_shell_script_indentation "${LFBFL_file_path}"
  done
}
```

3.6 build_and_checks_dependencies/check_URLs.libr.sh

```
#!/usr/bin/env bash
# This file is part of DevOrSysAdminScripts library.
#
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#
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# This file was renamed from "check_URLs.sh" to "check_URLs.libr.sh".

LFBFL_subdir="build_and_checks_dependencies"
# shellcheck source=common_options.libr.sh
source "./${LFBFL_subdir}/common_options.libr.sh"
# shellcheck source=get_common_text_glob_patterns.libr.sh
source "./${LFBFL_subdir}/get_common_text_glob_patterns.libr.sh"
# shellcheck source=lines_filters.libr.sh
source "./${LFBFL_subdir}/lines_filters.libr.sh"

check_URLs(){
  # Options:
  #   --verbose
  #   --work-directory=""
  declare -i LFBFL_i_verbose=0
  get_verbose_option "$@"
  local LFBFL_work_directory=""
  get_work_directory_option "$@"
  declare -a LFBFL_return_traps_stack
  local LFBFL_previous_return_trap
  init_return_trap
```

```
  pushd_to_work_directory --trap-popd
  can_continue_after_enhanced_pushd || return 1

  enhanced_set_shell_option pipefail --trap-unset

  get_COMMON_TEXT_FILES_GLOB_PATTERNS

  declare -Ar LFBFL_substitutions=(
    ["http://www.gnu.org/licenses/"]="https://www.gnu.org/licenses/"
  )

  local LFBFL_pattern
  local LFBFL_base_file_name
  local LFBFL_substitution
  local LFBFL_substitution2
  local LFBFL_file_path
  local LFBFL_s_files_paths
  declare -a LFBFL_arr_files_paths
  for LFBFL_pattern in "${COMMON_TEXT_FILES_GLOB_PATTERNS[@]}"; do
    if [[ LFBFL_i_verbose -eq 1 ]]; then
      printf "check_URLs: Iterating on pattern: %s.\n" "${LFBFL_pattern}"
    fi
    find . -type f -name "${LFBFL_pattern}" -printf '%P\n'\
      | xargs grep -H 'http:'\
      | grep -v "| xargs grep -H 'htt"\
      | grep -vP "['\"]http(:[^'\"]*)['\"].*['\"]https\\1['\"]"\
      | grep -v 'http://www.w3.org/1999/xhtml'\
      | grep -v 'http://www.w3.org/TR/xhtml1/DTD/xhtml1-transitional.dtd'
    # Second grep after xargs grep just above will remove false positives
    # from substitutions that fit on one line.

    LFBFL_s_files_paths=$(
      find . -type f -name "${LFBFL_pattern}" -printf '%P\n'\
      | relevant_find\
      | not_main_html_find
    )
    if [[ -z "${LFBFL_s_files_paths}" ]]; then
      [[ LFBFL_i_verbose -eq 1 ]] && printf "check_URLs: No file found.\n"
      continue
    fi
    mapfile -t LFBFL_arr_files_paths <<< "${LFBFL_s_files_paths}"
    for LFBFL_file_path in "${LFBFL_arr_files_paths[@]}"; do
      LFBFL_base_file_name=$(basename "${LFBFL_file_path}")
      [[ "${LFBFL_base_file_name}" != "check_URLs.libr.sh" ]]\
        || continue
      if [[ LFBFL_i_verbose -eq 1 ]]; then
```

```
        printf "check_URLs: Handling the file: %s.\n" "${LFBFL_file_path}"
      fi
      for LFBFL_substitution in "${!LFBFL_substitutions[@]}"; do
        LFBFL_substitution2=${LFBFL_substitutions[${LFBFL_substitution}]}
        if grep -q "${LFBFL_substitution}" "${LFBFL_file_path}"; then
          sed -i "s|${LFBFL_substitution}|${LFBFL_substitution2}|g"\
            "${LFBFL_file_path}"
        fi
      done
    done
  done
}
```

3.7 build_and_checks_dependencies/common_build_and_checks.exec.sh

```
#!/usr/bin/env bash
# This file is part of DevOrSysAdminScripts library.
#
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# The file "common_build_and_checks.exec.sh.tpl" was renamed from
# "common_build_and_checks.sh.tpl" to
# "common_build_and_checks.exec.sh.tpl".
# The file "common_build_and_checks.exec.sh" generated from the file
# "common_build_and_checks.sh.tpl"
# or "common_build_and_checks.exec.sh.tpl"
# was renamed from
# "common_build_and_checks.sh" to "common_build_and_checks.exec.sh".
```

```
32
33 common_build_and_checks(){
34   # $1 LFBFL_work_directory
35   # $2 LFBFL_dependencies_URL
36   # $3 optional --verbose
37   local LFBFL_work_directory="${1:-.}"
38   LFBFL_work_directory=$(realpath "${LFBFL_work_directory}")
39   local LFBFL_work_directory_option="--work-directory="
40   LFBFL_work_directory_option+="${LFBFL_work_directory}"
41   declare -r LFBFL_dependencies_URL="$2"
42   local LFBFL_verbose=""
43   declare -i LFBFL_i_verbose=0
44   if [[ "$*" == *--verbose* ]]; then
45     printf "%s %s\n" "$0" "$*"
46     LFBFL_verbose="--verbose"
47     LFBFL_i_verbose=1
48   fi
49   readonly LFBFL_verbose
50   readonly LFBFL_i_verbose
51   declare -ar LFBFL_some_common_options=(
52     "${LFBFL_verbose}"
53     # "$--work-directory={LFBFL_work_directory}" doesn't work
54     "${LFBFL_work_directory_option}"
55   )
56
57   source ./wget_sha512.libr.sh
58
59   local LFBFL_subdir="build_and_checks_dependencies"
60   declare -r LFBFL_subdir2="${LFBFL_subdir}/licenses_templates"
61   declare -r LFBFL_subdir3="build_and_checks_variables"
62   local LFBFL_file_name
63   local LFBFL_script_download_URL
64   local LFBFL_file_path
65   local LFBFL_correct_sha512
66
67   wrapped_wget_sha512(){
68     wget_sha512 "${LFBFL_file_path}"\
69       "${LFBFL_script_download_URL}"\
70       "${LFBFL_correct_sha512}"\
71       "${LFBFL_verbose}"
72   }
73
74   LFBFL_file_name="build_md_from_printable_md.exec.sh"
75   LFBFL_script_download_URL="${LFBFL_dependencies_URL}/${LFBFL_file_name}"
76   LFBFL_file_path="./${LFBFL_subdir}/${LFBFL_file_name}"
77   LFBFL_correct_sha512='565323e37ba608d97d46b20ec976f087ff52315c3ca24'
```

```
LFBFL_correct_sha512+='5d1d15c587f349f70443a1ac33e6e72008f22cfca5fa'
LFBFL_correct_sha512+='ff2a87d5113c8e9ed2a576b095c35c8995b0447'
wrapped_wget_sha512
chmod +x "./${LFBFL_file_path}"

LFBFL_file_name="build_dependencies_notes.exec.php"
LFBFL_script_download_URL="${LFBFL_dependencies_URL}/${LFBFL_file_name}"
LFBFL_file_path="./${LFBFL_subdir}/${LFBFL_file_name}"
LFBFL_correct_sha512='6b1bdeb49a81366dc3956dc70f381d1b0380e411f79f1'
LFBFL_correct_sha512+='5731da611f16469584469efdf056d3e02fc673864af7'
LFBFL_correct_sha512+='e288b4bb20edba77b8a74afe91209bbb63528a8'
wrapped_wget_sha512
chmod +x "./${LFBFL_file_path}"

LFBFL_file_name="check_shell_scripts_beginnings.libr.sh"
LFBFL_script_download_URL="${LFBFL_dependencies_URL}/${LFBFL_file_name}"
LFBFL_file_path="./${LFBFL_subdir}/${LFBFL_file_name}"
LFBFL_correct_sha512='e5d12684ceb7e143e381743a9eb18e8ecb77bf0a7b8c4'
LFBFL_correct_sha512+='820a511495ef0b6e9fb16db4bec64d7548df2513f6f1'
LFBFL_correct_sha512+='3e13d1494a174a28a23914833e0cceaff6d9e10'
wrapped_wget_sha512

LFBFL_file_name="check_shell_scripts_indentation.libr.sh"
LFBFL_script_download_URL="${LFBFL_dependencies_URL}/${LFBFL_file_name}"
LFBFL_file_path="./${LFBFL_subdir}/${LFBFL_file_name}"
LFBFL_correct_sha512='261b9cd75cde6f837e7b958d565db5ca7318852cb30e7'
LFBFL_correct_sha512+='47ed6a63bbf0bed9673c2cf0f6cae277b371562cf896'
LFBFL_correct_sha512+='baeb52cc1a509c1464e3901206d96b750bfaf94'
wrapped_wget_sha512

LFBFL_file_name="check_URLs.libr.sh"
LFBFL_script_download_URL="${LFBFL_dependencies_URL}/${LFBFL_file_name}"
LFBFL_file_path="./${LFBFL_subdir}/${LFBFL_file_name}"
LFBFL_correct_sha512='2027e3f9cdb847cc71b4b69149486779df666a1b18c30'
LFBFL_correct_sha512+='ab161b7c96ee2c48fed4ef78edb489d566d568fb7b1c'
LFBFL_correct_sha512+='8390a680a5212f5a04c9bd6dd05744374bbc5b3'
wrapped_wget_sha512

LFBFL_file_name="common_options.libr.sh"
LFBFL_script_download_URL="${LFBFL_dependencies_URL}/${LFBFL_file_name}"
LFBFL_file_path="./${LFBFL_subdir}/${LFBFL_file_name}"
LFBFL_correct_sha512='58f125f10f9c10cd38f04cdeec997bb7dc376ba6f0c09'
LFBFL_correct_sha512+='dfc1d2c0927855d709fbf002c027e67370cc0f8864b7'
LFBFL_correct_sha512+='8b23ab7685fc192413cc657df46475e28b989e2'
wrapped_wget_sha512
```

```
LFBFL_file_name="comparisons.libr.sh"
LFBFL_script_download_URL="${LFBFL_dependencies_URL}/${LFBFL_file_name}"
LFBFL_file_path="./${LFBFL_subdir}/${LFBFL_file_name}"
LFBFL_correct_sha512='eb2bfabf11d219a4f7a105c95ea5fc2b1f6ddd668fad7'
LFBFL_correct_sha512+='344963c62d574eb3ecb93b35aad8b5ee4c2558792478'
LFBFL_correct_sha512+='f7ef9da058fba1cfe2195b61c87e1ed8686f200'
wrapped_wget_sha512

LFBFL_file_name="create_PDF.exec.sh"
LFBFL_script_download_URL="${LFBFL_dependencies_URL}/${LFBFL_file_name}"
LFBFL_file_path="./${LFBFL_subdir}/${LFBFL_file_name}"
LFBFL_correct_sha512='e8d57774edc7b98394b039dc7de2867deeece68775f7c'
LFBFL_correct_sha512+='82b5617efc9a3377772a6a1ff78957622ebef77b90d6'
LFBFL_correct_sha512+='7d93f449aebb875fba2ed5c4467b1387a4962ef'
wrapped_wget_sha512
chmod +x "./${LFBFL_file_path}"

LFBFL_file_name="generate_from_template.libr.sh"
LFBFL_script_download_URL="${LFBFL_dependencies_URL}/${LFBFL_file_name}"
LFBFL_file_path="./${LFBFL_subdir}/${LFBFL_file_name}"
LFBFL_correct_sha512='6369d2f6f28986d56398b2a9533fddbd14251fd8c8853'
LFBFL_correct_sha512+='203f3d8561065b2109abc8c328f0045b4dfb47895690'
LFBFL_correct_sha512+='a275d560ac684aed15e41307c1531c2e88a3643'
wrapped_wget_sha512

LFBFL_file_name="get_common_text_glob_patterns.libr.sh"
LFBFL_script_download_URL="${LFBFL_dependencies_URL}/${LFBFL_file_name}"
LFBFL_file_path="./${LFBFL_subdir}/${LFBFL_file_name}"
LFBFL_correct_sha512='fc041abf937df4c7cb1241c73e776e8bd07a5525bc9a9'
LFBFL_correct_sha512+='29c83d597ddd6aa88a6e61a99d22ca88c11561f1cdbc'
LFBFL_correct_sha512+='0fbf5df4794a2384f33e7e7dd5b65a53ab46f4a'
wrapped_wget_sha512

LFBFL_file_name="grammar_and_spelling_check.libr.sh"
LFBFL_script_download_URL="${LFBFL_dependencies_URL}/${LFBFL_file_name}"
LFBFL_file_path="./${LFBFL_subdir}/${LFBFL_file_name}"
LFBFL_correct_sha512='48f0edc5841f59f92b8126240df13e651a887bb399640'
LFBFL_correct_sha512+='589fb4bdc56dfd9006e6908ad952f1ce037e6dd22293'
LFBFL_correct_sha512+='5667301e528a767d0ef00ad43d733990b1ae89f'
wrapped_wget_sha512

# /licenses_templates/ ---------------------------------------------
local LFBFL_dependencies_URL2
LFBFL_dependencies_URL2="${LFBFL_dependencies_URL}/licenses_templates"
readonly LFBFL_dependencies_URL2
LFBFL_file_name="build_licenses_templates.exec.sh"
```

```
170     LFBFL_script_download_URL="${LFBFL_dependencies_URL2}/${LFBFL_file_name}"
171     LFBFL_file_path="./${LFBFL_subdir2}/${LFBFL_file_name}"
172     LFBFL_correct_sha512='b521beeff9a3e45d7e188d0e1b7352689ac91a86b7355'
173     LFBFL_correct_sha512+='a802aabc56d88b57cd357738e8da89e7227aef52acf5'
174     LFBFL_correct_sha512+='50db4b2e67f189e8c64b9d1a2cc64eb1a7b8084'
175     wrapped_wget_sha512
176     chmod +x "./${LFBFL_file_path}"
177
178     LFBFL_file_name="license_file_header_AGPLv3+.tpl"
179     LFBFL_script_download_URL="${LFBFL_dependencies_URL2}/${LFBFL_file_name}"
180     LFBFL_file_path="./${LFBFL_subdir2}/${LFBFL_file_name}"
181     LFBFL_correct_sha512='9f9aa56d9b90d5003a12c5ec3370ac31aab74fc4a2524'
182     LFBFL_correct_sha512+='2c5fbcb0197b0abbb8242f62dafc76eaf60faa5da173'
183     LFBFL_correct_sha512+='05924b31b64b037083532fced9f98445000e2d0'
184     wrapped_wget_sha512
185
186     LFBFL_file_name="license_file_header_GPLv3+.tpl"
187     LFBFL_script_download_URL="${LFBFL_dependencies_URL2}/${LFBFL_file_name}"
188     LFBFL_file_path="./${LFBFL_subdir2}/${LFBFL_file_name}"
189     LFBFL_correct_sha512='2bdef9a729a5a9ca5ed87ef8a518877c5ee6b31b45336'
190     LFBFL_correct_sha512+='08660316a4e595fb31465a4847f4d53a2f7264118bd4'
191     LFBFL_correct_sha512+='eab7c1d1892e00bc1861455ed02898f194db067'
192     wrapped_wget_sha512
193
194     LFBFL_file_name="license_file_header_LGPLv3+.tpl"
195     LFBFL_script_download_URL="${LFBFL_dependencies_URL2}/${LFBFL_file_name}"
196     LFBFL_file_path="./${LFBFL_subdir2}/${LFBFL_file_name}"
197     LFBFL_correct_sha512='265fd0d086b48ab798f35a072335a4976e4eda8c256fe'
198     LFBFL_correct_sha512+='bba6c77c98ca86cdd844c6ac150c3445c80debbb3fc2'
199     LFBFL_correct_sha512+='574b5fceeeaf79ec734a759bdef5fe798067c62'
200     wrapped_wget_sha512
201     # ------------------------------------------------------------------
202
203     LFBFL_file_name="lines_counts.libr.sh"
204     LFBFL_script_download_URL="${LFBFL_dependencies_URL}/${LFBFL_file_name}"
205     LFBFL_file_path="./${LFBFL_subdir}/${LFBFL_file_name}"
206     LFBFL_correct_sha512='ec36e818b0aab4e386e4b01e643b07ad6bb1ef2ec5ccc'
207     LFBFL_correct_sha512+='4cfa697bac206f9a65d3cd710994a18db7397597a403'
208     LFBFL_correct_sha512+='463ec1353bc951ec4d79f54ccee61261dd279d8'
209     wrapped_wget_sha512
210
211     LFBFL_file_name="lines_filters.libr.sh"
212     LFBFL_script_download_URL="${LFBFL_dependencies_URL}/${LFBFL_file_name}"
213     LFBFL_file_path="./${LFBFL_subdir}/${LFBFL_file_name}"
214     LFBFL_correct_sha512='86a32b551709cf0743b69dabd1bbe091bda33a46b0207'
215     LFBFL_correct_sha512+='8aa2aa7c3539d86f8c70dd7fc526c06ed8fcc2511390'
```

```
LFBFL_correct_sha512+='dd9398bf27599405ba5112ffb679591eabb499c'
wrapped_wget_sha512

LFBFL_file_name="lines_maps.libr.sh"
LFBFL_script_download_URL="${LFBFL_dependencies_URL}/${LFBFL_file_name}"
LFBFL_file_path="./${LFBFL_subdir}/${LFBFL_file_name}"
LFBFL_correct_sha512='25c3e5082951260295b5f0e87ba1f2c978efdea1e8926'
LFBFL_correct_sha512+='072ce4d2062dc2a47bb92b9098f081e553eeae5d9154'
LFBFL_correct_sha512+='fdc50c9d88ae5365b776a23eb6e026b884a8374'
wrapped_wget_sha512

LFBFL_file_name="overwrite_if_not_equal.libr.sh"
LFBFL_script_download_URL="${LFBFL_dependencies_URL}/${LFBFL_file_name}"
LFBFL_file_path="./${LFBFL_subdir}/${LFBFL_file_name}"
LFBFL_correct_sha512='2e2e9c804ec8e1879fc96df595f30994a0685934483ce'
LFBFL_correct_sha512+='c9deed4c2fb98f79d4a393e13b6511b3b740ce9f073a'
LFBFL_correct_sha512+='e7ed124f08a60b35c896d60dd710223d2592d16'
wrapped_wget_sha512

LFBFL_file_name="python_black_complement.libr.sh"
LFBFL_script_download_URL="${LFBFL_dependencies_URL}/${LFBFL_file_name}"
LFBFL_file_path="./${LFBFL_subdir}/${LFBFL_file_name}"
LFBFL_correct_sha512='abe0e5701ad6a56937b1a1e3687366e87e8cb2578a179'
LFBFL_correct_sha512+='8e15ce81e0aaafc208c1b325058287c331092765ef73'
LFBFL_correct_sha512+='2c584cadf33e3acbcc195f25b0b632a0ce229d3'
wrapped_wget_sha512

LFBFL_file_name="python_isort_complement.libr.sh"
LFBFL_script_download_URL="${LFBFL_dependencies_URL}/${LFBFL_file_name}"
LFBFL_file_path="./${LFBFL_subdir}/${LFBFL_file_name}"
LFBFL_correct_sha512='70542f00cb59ed52d6576e6c066c5dc701ac90274ab68'
LFBFL_correct_sha512+='475086182b13b8cdf75f6b0263ac813e0010dd6cdb86'
LFBFL_correct_sha512+='9239899c2a6161f912bf1f789ad30965b8c1fe5'
wrapped_wget_sha512

LFBFL_file_name="repository_data.libr.sh"
LFBFL_script_download_URL="${LFBFL_dependencies_URL}/${LFBFL_file_name}"
LFBFL_file_path="./${LFBFL_subdir}/${LFBFL_file_name}"
LFBFL_correct_sha512='e3e8b487c45bf3960c5eb739fd1d29229ca358bfecb23'
LFBFL_correct_sha512+='e585fa7abf65c9cdaa263e799e96a83a22c660323e31'
LFBFL_correct_sha512+='bf2cdbc2f857b43957ef9405d2b8bed95dac7c0'
wrapped_wget_sha512

LFBFL_file_name="shell_checks_complement.libr.sh"
LFBFL_script_download_URL="${LFBFL_dependencies_URL}/${LFBFL_file_name}"
LFBFL_file_path="./${LFBFL_subdir}/${LFBFL_file_name}"
```

```
LFBFL_correct_sha512='e70c9b931d1adb41be165c2d84e17dd9dbb23a3d6d48d'
LFBFL_correct_sha512+='1b609a9187e96a2faa831c78335e440edef26792b307'
LFBFL_correct_sha512+='346fb053c5c0dd1fac493e1db3657950fd6ae08'
wrapped_wget_sha512

LFBFL_file_name="split_score.exec.php"
LFBFL_script_download_URL="${LFBFL_dependencies_URL}/${LFBFL_file_name}"
LFBFL_file_path="./${LFBFL_subdir}/${LFBFL_file_name}"
LFBFL_correct_sha512='2f7ff107a43716109b1acbd7d89870d05577af00a981a'
LFBFL_correct_sha512+='05f162a054c96387fb99544f8225c333b1212f74326c'
LFBFL_correct_sha512+='f9ec6196e796f6257dfbba9347ac8c57b4f09db'
wrapped_wget_sha512

LFBFL_file_name="split_score.libr.php"
LFBFL_script_download_URL="${LFBFL_dependencies_URL}/${LFBFL_file_name}"
LFBFL_file_path="./${LFBFL_subdir}/${LFBFL_file_name}"
LFBFL_correct_sha512='ca60f3621f0808f0ba28bdfb860f6f8d3d872fa26c93b'
LFBFL_correct_sha512+='94a9ce0b1b90f5ecb4fc4c4e1cea30368d99e7e7b3f7'
LFBFL_correct_sha512+='7ec5f80052754e987a51bceffc5bb8ee552c480'
wrapped_wget_sha512

LFBFL_file_name="strings_functions.libr.sh"
LFBFL_script_download_URL="${LFBFL_dependencies_URL}/${LFBFL_file_name}"
LFBFL_file_path="./${LFBFL_subdir}/${LFBFL_file_name}"
LFBFL_correct_sha512='1c5cb59a9ad306ca8cd49b398146dc6621de0e69cb510'
LFBFL_correct_sha512+='b9f41d4412d5b853f26eae7128410419be7dfb0d36cf'
LFBFL_correct_sha512+='1a6272f70f270e551406866b1e1fd209bdcce44'
wrapped_wget_sha512

LFBFL_file_name="too_long_code_lines.libr.sh"
LFBFL_script_download_URL="${LFBFL_dependencies_URL}/${LFBFL_file_name}"
LFBFL_file_path="./${LFBFL_subdir}/${LFBFL_file_name}"
LFBFL_correct_sha512='a235e533d7c34796ee1f70225bab0f1a79d19e8454189'
LFBFL_correct_sha512+='833872276a455629866ca50f3045eeeb8c7213ef3a45'
LFBFL_correct_sha512+='88749523191fd615f27afbcaf3096290d11589a'
wrapped_wget_sha512

LFBFL_file_name="update_or_check_files_names_listing.exec.sh"
LFBFL_script_download_URL="${LFBFL_dependencies_URL}/${LFBFL_file_name}"
LFBFL_file_path="./${LFBFL_subdir}/${LFBFL_file_name}"
LFBFL_correct_sha512='9d9100a8527ea9717aff99f483c81b7921fd079ace4f9'
LFBFL_correct_sha512+='ead5cecd3c89da90f43959297ff78184eadf55bff79e'
LFBFL_correct_sha512+='b716060befe5eff9f00cc3b2f4dc69522e8bf8f'
wrapped_wget_sha512
chmod +x "./${LFBFL_file_path}"
```

```
308     # shellcheck source=check_shell_scripts_beginnings.libr.sh
309     source "./${LFBFL_subdir}/check_shell_scripts_beginnings.libr.sh"
310     # shellcheck source=check_shell_scripts_indentation.libr.sh
311     source "./${LFBFL_subdir}/check_shell_scripts_indentation.libr.sh"
312     # shellcheck source=check_URLs.libr.sh
313     source "./${LFBFL_subdir}/check_URLs.libr.sh"
314     # shellcheck source=common_options.libr.sh
315     source "./${LFBFL_subdir}/common_options.libr.sh"
316     # shellcheck source=comparisons.libr.sh
317     source "./${LFBFL_subdir}/comparisons.libr.sh"
318     # shellcheck source=generate_from_template.libr.sh
319     source "./${LFBFL_subdir}/generate_from_template.libr.sh"
320     # shellcheck source=get_common_text_glob_patterns.libr.sh
321     source "./${LFBFL_subdir}/get_common_text_glob_patterns.libr.sh"
322     # shellcheck source=grammar_and_spelling_check.libr.sh
323     source "./${LFBFL_subdir}/grammar_and_spelling_check.libr.sh"
324     # shellcheck source=lines_counts.libr.sh
325     source "./${LFBFL_subdir}/lines_counts.libr.sh"
326     # shellcheck source=lines_filters.libr.sh
327     source "./${LFBFL_subdir}/lines_filters.libr.sh"
328     # shellcheck source=lines_maps.libr.sh
329     source "./${LFBFL_subdir}/lines_maps.libr.sh"
330     # shellcheck source=overwrite_if_not_equal.libr.sh
331     source "./${LFBFL_subdir}/overwrite_if_not_equal.libr.sh"
332     # shellcheck source=python_black_complement.libr.sh
333     source "./${LFBFL_subdir}/python_black_complement.libr.sh"
334     # shellcheck source=python_isort_complement.libr.sh
335     source "./${LFBFL_subdir}/python_isort_complement.libr.sh"
336     # shellcheck source=shell_checks_complement.libr.sh
337     source "./${LFBFL_subdir}/shell_checks_complement.libr.sh"
338     # shellcheck source=strings_functions.libr.sh
339     source "./${LFBFL_subdir}/strings_functions.libr.sh"
340     # shellcheck source=too_long_code_lines.libr.sh
341     source "./${LFBFL_subdir}/too_long_code_lines.libr.sh"
342
343     enhanced_set_bash_option extglob
344     # shellcheck source=repository_data.libr.sh
345     source "./${LFBFL_subdir}/repository_data.libr.sh"
346     # shellcheck disable=SC2154,SC2309
347     [[ i_enhanced_set_bash_option_extglob_result -eq 0 ]]\
348       && enhanced_unset_bash_option extglob
349
350     declare -a LFBFL_return_traps_stack
351     local LFBFL_previous_return_trap
352     init_return_trap
353
```

```
  enhanced_set_shell_option pipefail --trap-unset

  declare -a LFBFL_some_common_options2=("${LFBFL_some_common_options[@]}")
  keep_options LFBFL_some_common_options2 --verbose

  printf "Building license headers\n"
  "./${LFBFL_subdir2}/build_licenses_templates.exec.sh"\
    "${LFBFL_some_common_options[@]}"

  printf "Building README.md\n"
  "./${LFBFL_subdir}/build_md_from_printable_md.exec.sh"\
    --work-directory="${LFBFL_work_directory}"\
    --base-name=README\
    "${LFBFL_verbose}"

  printf "Building other MarkDown files\n"
  local LFBFL_some_directory
  declare -r LFBFL_readme="${LFBFL_work_directory}/README.md.tpl"
  local LFBFL_s_files_paths
  LFBFL_s_files_paths=$(
    find "${LFBFL_work_directory}" -type f -name "*.md.tpl"\
    | relevant_find
  )
  declare -a LFBFL_arr_files_paths
  if [[ -n "${LFBFL_s_files_paths}" ]]; then
    mapfile -t LFBFL_arr_files_paths <<< "${LFBFL_s_files_paths}"
    for LFBFL_file_path in "${LFBFL_arr_files_paths[@]}"; do
      [[ "${LFBFL_file_path}" == "${LFBFL_readme}" ]] && continue
      printf "Found template %s.\n" "${LFBFL_file_path}"
      LFBFL_some_directory=$(dirname "${LFBFL_file_path}")
      LFBFL_file_name=$(basename "${LFBFL_file_path}")
      LFBFL_file_name=${LFBFL_file_name%.md.tpl}
      "./${LFBFL_subdir}/build_md_from_printable_md.exec.sh"\
        --work-directory="${LFBFL_some_directory}"\
        --base-name="${LFBFL_file_name}"\
        "${LFBFL_verbose}"
    done
  fi

  printf "Building dependencies_notes\n"
  php "./${LFBFL_subdir}/build_dependencies_notes.exec.php"\
    "${LFBFL_work_directory}"

  declare -i LFBFL_i_directory_changed
  pushd_to_work_directory
  LFBFL_i_directory_changed=$?
```

```
  can_continue_after_enhanced_pushd || return 1

  local LFBFL_data_file_name="${LFBFL_subdir3}/repository_data.txt"

  declare -i LFBFL_i_max_line_length
  grep_variable "${LFBFL_data_file_name}" max_line_length\
    --result-variable-prefix=LFBFL_i_

  declare -i LFBFL_i_upgrade_venvs=0
  local LFBFL_upgrade_venvs_ts_file="${LFBFL_subdir3}/upgrade_venvs_ts"
  readonly LFBFL_upgrade_venvs_ts_file
  declare -i LFBFL_i_upgrade_venvs_ts
  declare -i LFBFL_i_current_ts
  local LFBFL_upgrade_venvs_answer

  local LFBFL_upgrade_venvs_time_interval_in_seconds=""
  get_upgrade_venvs_time_interval_in_seconds "${LFBFL_data_file_name}"\
    "${LFBFL_some_common_options2[@]}"

  if [[ -f "${LFBFL_upgrade_venvs_ts_file}" ]]; then
    LFBFL_i_upgrade_venvs_ts=$(
      stat --format %Y "${LFBFL_upgrade_venvs_ts_file}"
    )
    LFBFL_i_current_ts=$(date +%s)
    ((LFBFL_i_current_ts -= LFBFL_i_upgrade_venvs_ts))
    ((LFBFL_i_current_ts -= LFBFL_upgrade_venvs_time_interval_in_seconds))
    if [[ LFBFL_i_current_ts -gt 0 ]]; then
      LFBFL_i_upgrade_venvs=1
    fi
  else
    LFBFL_i_upgrade_venvs=1
  fi
  if [[ LFBFL_i_upgrade_venvs -eq 1 ]]; then
    local LFBFL_upgrade_venvs=""
    grep_variable "${LFBFL_data_file_name}" upgrade_venvs\
      --result-variable-prefix=LFBFL_
    if [[ "${LFBFL_upgrade_venvs}" != "auto" ]]; then
      read -r -n 1 -t 10\
        -p "Upgrade venvs and composer global? [Y/n]"\
        LFBFL_upgrade_venvs_answer
      if [[ "${LFBFL_upgrade_venvs_answer}" == "n" ]]; then
        LFBFL_i_upgrade_venvs=0
      fi
    fi
  fi

```

```
  printf "Running shellcheck\n"
  LFBFL_s_files_paths=$(
    find . -type f -name "*.sh"\
    | relevant_find
  )
  if [[ -n "${LFBFL_s_files_paths}" ]]; then
    mapfile -t LFBFL_arr_files_paths <<< "${LFBFL_s_files_paths}"
    for LFBFL_file_path in "${LFBFL_arr_files_paths[@]}"; do
      shellcheck --rcfile="${LFBFL_subdir3}/shellcheck.ini"\
        "${LFBFL_file_path}"
    done
  fi
  printf "Running shell_checks_complement\n"
  shell_checks_complement "${LFBFL_some_common_options2[@]}"

  printf -- "---Python---\n"

  maybe_deactivate(){
    if command -v deactivate
    then
      deactivate
    fi
  }

  source_without_return_trap(){
    declare -i LFBFL_i_no_return_traps=1
    # shellcheck disable=SC1090,SC1091
    source "$1"
  }

  printf "Running isort\n"
  local LFBFL_isort_venv=""
  grep_variable "${LFBFL_data_file_name}" isort_venv\
    --result-variable-prefix=LFBFL_\
    --replace-line-returns-by=""
  if [[ -n "${LFBFL_isort_venv}" ]]; then
    maybe_deactivate
    source_without_return_trap "${LFBFL_isort_venv}/bin/activate"
  fi
  if [[ LFBFL_i_upgrade_venvs -eq 1 ]]; then
    pip install --upgrade isort
  fi
  isort --profile=black .
  if [[ -n "${LFBFL_isort_venv}" ]]; then
    deactivate
  fi
```

```
  printf "Running python_isort_complement\n"
  python_isort_complement "${LFBFL_some_common_options2[@]}"

  printf "Running black\n"
  # First, we update the configuration file with max_line_length.
  local LFBFL_update_max_length="s/^line-length = [0-9]*$"
  LFBFL_update_max_length+="/line-length = ${LFBFL_i_max_line_length}/"
  sed -E -e "${LFBFL_update_max_length}"\
    "${LFBFL_subdir3}/black.toml"\
    > "${LFBFL_subdir3}/black.toml.temp"
  overwrite_if_not_equal "${LFBFL_subdir3}/black.toml"\
    "${LFBFL_subdir3}/black.toml.temp"

  local LFBFL_black_venv=""
  grep_variable "${LFBFL_data_file_name}" black_venv\
    --result-variable-prefix=LFBFL_\
    --replace-line-returns-by=""
  if [[ -n "${LFBFL_black_venv}" ]]; then
    maybe_deactivate
    source_without_return_trap "${LFBFL_black_venv}/bin/activate"
  fi
  if [[ LFBFL_i_upgrade_venvs -eq 1 ]]; then
    pip install --upgrade black
  fi
  black --config="${LFBFL_subdir3}/black.toml" .
  if [[ -n "${LFBFL_black_venv}" ]]; then
    deactivate
  fi
  printf "Running python_black_complement\n"
  python_black_complement "${LFBFL_some_common_options2[@]}"

  printf "Probing if mypy should be runned\n"
  local LFBFL_mypy_venv=""
  grep_variable "${LFBFL_data_file_name}" mypy_venv\
    --result-variable-prefix=LFBFL_\
    --replace-line-returns-by=""
  if [[ -n "${LFBFL_mypy_venv}" ]]; then
    maybe_deactivate
    source_without_return_trap "${LFBFL_mypy_venv}/bin/activate"
  fi
  if [[ LFBFL_i_upgrade_venvs -eq 1 ]]; then
    pip install --upgrade mypy
  fi
  local LFBFL_directory_path
  declare -i LFBFL_i_no_toml=1
  LFBFL_s_files_paths=$(
```

```
    find . -type f -name "pyproject.toml"\
    | relevant_find
  )
  if [[ -n "${LFBFL_s_files_paths}" ]]; then
    mapfile -t LFBFL_arr_files_paths <<< "${LFBFL_s_files_paths}"
    for LFBFL_file_path in "${LFBFL_arr_files_paths[@]}"; do
      if grep --quiet "Typing :: Typed" "${LFBFL_file_path}"; then
        printf "Running mypy\n"
        LFBFL_directory_path=$(dirname "${LFBFL_file_path}")
        mypy "${LFBFL_directory_path}"
      fi
      LFBFL_i_no_toml=0
    done
  fi
  if [[ LFBFL_i_no_toml -eq 1 ]]; then
    printf "Running mypy\n"
    mypy .
  fi
  if [[ -n "${LFBFL_mypy_venv}" ]]; then
    deactivate
  fi

  printf "Running bandit\n"
  LFBFL_bandit_venv=""
  grep_variable "${LFBFL_data_file_name}" bandit_venv\
    --result-variable-prefix=LFBFL_\
    --replace-line-returns-by=""
  if [[ -n "${LFBFL_bandit_venv}" ]]; then
    maybe_deactivate
    source_without_return_trap "${LFBFL_bandit_venv}/bin/activate"
  fi
  if [[ LFBFL_i_upgrade_venvs -eq 1 ]]; then
    pip install --upgrade bandit
  fi
  bandit --ini="${LFBFL_subdir3}/bandit.ini"\
    --baseline="${LFBFL_subdir3}/bandit_baseline.json"\
    --recursive .
  # Saving new baseline in temp if necessary.
  bandit --ini="${LFBFL_subdir3}/bandit.ini"\
    --format=json\
    --output="${LFBFL_subdir3}/temp/bandit_baseline.json"\
    --recursive .
  if [[ -n "${LFBFL_bandit_venv}" ]]; then
    deactivate
  fi

```

```
  printf "Running pylint\n"
  LFBFL_pylint_venv=""
  grep_variable "${LFBFL_data_file_name}" pylint_venv\
    --result-variable-prefix=LFBFL_\
    --replace-line-returns-by=""
  if [[ -n "${LFBFL_pylint_venv}" ]]; then
    maybe_deactivate
    source_without_return_trap "${LFBFL_pylint_venv}/bin/activate"
  fi
  if [[ LFBFL_i_upgrade_venvs -eq 1 ]]; then
    pip install --upgrade pylint
  fi
  pylint --rcfile="${LFBFL_subdir3}/pylintrc.toml" --recursive=y .
  if [[ -n "${LFBFL_pylint_venv}" ]]; then
    deactivate
  fi

  printf "Running ruff\n"
  LFBFL_ruff_venv=""
  grep_variable "${LFBFL_data_file_name}" ruff_venv\
    --result-variable-prefix=LFBFL_\
    --replace-line-returns-by=""
  if [[ -n "${LFBFL_ruff_venv}" ]]; then
    maybe_deactivate
    source_without_return_trap "${LFBFL_ruff_venv}/bin/activate"
  fi
  if [[ LFBFL_i_upgrade_venvs -eq 1 ]]; then
    pip install --upgrade ruff
  fi
  ruff check --config="${LFBFL_subdir3}/ruff.toml"
  if [[ -n "${LFBFL_ruff_venv}" ]]; then
    deactivate
  fi
  printf -- "---Python end---\n"

  printf -- "---PHP---\n"
  printf "Running PHPMD\n"
  if [[ LFBFL_i_upgrade_venvs -eq 1 ]]; then
    composer global require phpmd/phpmd
  fi

  declare -r LFBFL_phpmd_baseline="${LFBFL_subdir3}/phpmd_baseline.xml"
  local LFBFL_temp_phpmd_baseline="${LFBFL_subdir3}/temp/"
  LFBFL_temp_phpmd_baseline+="phpmd_baseline.xml"
  readonly LFBFL_temp_phpmd_baseline

```

```
630     local LFBFL_phpmd_rulesets="cleancode,codesize,controversial,"
631     LFBFL_phpmd_rulesets+="design,naming,unusedcode"
632     readonly LFBFL_phpmd_rulesets
633
634     phpmd --color --baseline-file="${LFBFL_phpmd_baseline}"\
635       . text "${LFBFL_phpmd_rulesets}"
636     # Saving new baseline in temp if necessary.
637     rm "${LFBFL_temp_phpmd_baseline}"
638     phpmd --color --generate-baseline\
639       --baseline-file="${LFBFL_temp_phpmd_baseline}"\
640       . text "${LFBFL_phpmd_rulesets}"
641     sed -i -e 's/" file="/"\n    file="/g' "${LFBFL_temp_phpmd_baseline}"
642     sed -i -E -e 's~(file=".*)/>~\1\n  />~g' "${LFBFL_temp_phpmd_baseline}"
643     diff "${LFBFL_phpmd_baseline}" "${LFBFL_temp_phpmd_baseline}"
644     printf -- "---PHP end---\n"
645
646     printf -- "---JS---\n"
647     LFBFL_npm_lint_directories=""
648     grep_variable "${LFBFL_data_file_name}" npm_lint_directories\
649       --result-variable-prefix=LFBFL_
650     if [[ -n "${LFBFL_npm_lint_directories}" ]]; then
651       printf "Running ESLint\n"
652       local LFBFL_JS_directory
653       declare -a LFBFL_arr_paths
654       mapfile -t LFBFL_arr_paths <<< "${LFBFL_npm_lint_directories}"
655       for LFBFL_JS_directory in "${LFBFL_arr_paths[@]}"; do
656         (cd "${LFBFL_JS_directory}" && npm run lint)
657       done
658     fi
659     printf -- "---JS end---\n"
660
661     printf "Analyzing too long lines\n"
662     # shellcheck disable=SC2248
663     too_long_code_lines "${LFBFL_some_common_options2[@]}" \
664       --max-line-length=${LFBFL_i_max_line_length}
665
666     printf "Analyzing shell scripts beginnings\n"
667     check_shell_scripts_beginnings "${LFBFL_some_common_options2[@]}" \
668       | relevant_grep
669
670     printf "Analyzing shell scripts indentation\n"
671     check_shell_scripts_indentation "${LFBFL_some_common_options2[@]}" \
672       | relevant_grep
673
674     printf "Analyzing URLs\n"
675     check_URLs "${LFBFL_some_common_options2[@]}" \
```

```
    | relevant_grep

  printf "Analyzing strange characters: hover over in doubt\n"
  # shellcheck disable=SC1111
  LFBFL_usual_characters="\x00-\x7Fàâéèêëîïôûç©‘”└├│«»"
  grep --binary-files=without-match --color=always --exclude-dir .git\
    --line-number --perl-regexp --recursive --\
    "[^${LFBFL_usual_characters}]" .

  printf "Grammar and spelling check\n"
  grammar_and_spelling_check "${LFBFL_data_file_name}"\
    "${LFBFL_some_common_options2[@]}"

  [[ LFBFL_i_directory_changed -eq 0 ]] && popd_from_work_directory

  printf "Checking listed files\n"
  "./${LFBFL_subdir}/update_or_check_files_names_listing.exec.sh"\
    "${LFBFL_some_common_options[@]}"

  printf "Creating the PDF file of the listing of the source code\n"
  "./${LFBFL_subdir}/create_PDF.exec.sh" "${LFBFL_some_common_options[@]}"

  pushd_to_work_directory --trap-popd
  can_continue_after_enhanced_pushd || return 1

  if [[ -f "${LFBFL_subdir3}/post_build.sh" ]]; then
    "./${LFBFL_subdir3}/post_build.sh" "${LFBFL_verbose}"
  fi

  # ------------------------------------------------------------------
  # It is at the end that all the HTML files are generated and that
  # we can check there are no mistakes.
  printf "Running Nu W3C HTML CSS and SVG validator\n"
  local LFBFL_upgrade_vnu_jar_time_interval_in_seconds=""
  grep_variable "${LFBFL_data_file_name}"\
    upgrade_vnu_jar_time_interval_in_seconds\
    --result-variable-prefix=LFBFL_

  if [[ "${LFBFL_upgrade_vnu_jar_time_interval_in_seconds}" == "wat" ]];
  then
    LFBFL_upgrade_vnu_jar_time_interval_in_seconds=${RANDOM}
  fi

  declare -i LFBFL_i_vnu_jar_ts
  local LFBFL_vnu_jar_path=""
  grep_variable "${LFBFL_data_file_name}" vnu_jar_path\
```

```
    --result-variable-prefix=LFBFL_\
    --replace-line-returns-by=""
  if [[ -f "${LFBFL_vnu_jar_path}" ]]; then
    LFBFL_i_vnu_jar_ts=$(stat --format %Y "${LFBFL_vnu_jar_path}")
    LFBFL_i_current_ts=$(date +%s)
    ((LFBFL_i_current_ts -= LFBFL_i_vnu_jar_ts))
    ((
      LFBFL_i_current_ts -= LFBFL_upgrade_vnu_jar_time_interval_in_seconds
    ))
    if [[ LFBFL_i_current_ts -gt 0 ]]; then
      printf $'/!\\ATTENTION: Nu W3C validator is too old./!\\\n'
      printf "Go get the latest version here:\n"
      printf "https://github.com/validator/validator/releases\n"
    fi
    # vnu has false positives for xhtml
    local LFBFL_use_vnu_for_xhtml=""
    grep_variable "${LFBFL_data_file_name}" use_vnu_for_xhtml\
      --result-variable-prefix=LFBFL_
    local LFBFL_find_regexp
    if [[ "${LFBFL_use_vnu_for_xhtml}" == "true" ]]; then
      LFBFL_find_regexp=".*\.\(css\|html\|svg\|xhtml\)"
    else
      LFBFL_find_regexp=".*\.\(css\|html\|svg\)"
    fi
    declare -r LFBFL_files_for_Nu=$(
      find . -iregex "${LFBFL_find_regexp}"\
      | relevant_find
    )
    if [[ -n "${LFBFL_files_for_Nu}" ]]; then
      declare -a LFBFL_files_for_Nu2
      mapfile -t LFBFL_files_for_Nu2 <<< "${LFBFL_files_for_Nu}"
      readonly LFBFL_files_for_Nu2
      java -Xss128M -jar "${LFBFL_vnu_jar_path}" --exit-zero-always\
        --verbose "${LFBFL_files_for_Nu2[@]}"
    fi
  fi
  # ------------------------------------------------------------------

  if [[ LFBFL_i_upgrade_venvs -eq 1 ]]; then
    touch "${LFBFL_upgrade_venvs_ts_file}"
  fi
}

common_build_and_checks "$@"
```

3.8 build_and_checks_dependencies/common_build_and_checks.exec.sh.tpl

```
#!/usr/bin/env bash
# This file is part of DevOrSysAdminScripts library.
#
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# MERCHANTABILITY or FITNESS FOR A PARTICULAR PURPOSE.
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#
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# the GNU General Public License
# along with DevOrSysAdminScripts.
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#
# ©Copyright 2023-2026 Laurent Frédéric Bernard François Lyaudet
# The file "common_build_and_checks.exec.sh.tpl" was renamed from
# "common_build_and_checks.sh.tpl" to
# "common_build_and_checks.exec.sh.tpl".
# The file "common_build_and_checks.exec.sh" generated from the file
# "common_build_and_checks.sh.tpl"
# or "common_build_and_checks.exec.sh.tpl"
# was renamed from
# "common_build_and_checks.sh" to "common_build_and_checks.exec.sh".

common_build_and_checks(){
  # $1 LFBFL_work_directory
  # $2 LFBFL_dependencies_URL
  # $3 optional --verbose
  local LFBFL_work_directory="${1:-.}"
  LFBFL_work_directory=$(realpath "${LFBFL_work_directory}")
  local LFBFL_work_directory_option="--work-directory="
  LFBFL_work_directory_option+="${LFBFL_work_directory}"
  declare -r LFBFL_dependencies_URL="$2"
  local LFBFL_verbose=""
  declare -i LFBFL_i_verbose=0
  if [[ "$*" == *--verbose* ]]; then
```

```
  printf "%s %s\n" "$0" "$*"
  LFBFL_verbose="--verbose"
  LFBFL_i_verbose=1
fi
readonly LFBFL_verbose
readonly LFBFL_i_verbose
declare -ar LFBFL_some_common_options=(
  "${LFBFL_verbose}"
  # "$--work-directory={LFBFL_work_directory}" doesn't work
  "${LFBFL_work_directory_option}"
)

source ./wget_sha512.libr.sh

local LFBFL_subdir="build_and_checks_dependencies"
declare -r LFBFL_subdir2="${LFBFL_subdir}/licenses_templates"
declare -r LFBFL_subdir3="build_and_checks_variables"
local LFBFL_file_name
local LFBFL_script_download_URL
local LFBFL_file_path
local LFBFL_correct_sha512

wrapped_wget_sha512(){
  wget_sha512 "${LFBFL_file_path}"\
    "${LFBFL_script_download_URL}"\
    "${LFBFL_correct_sha512}"\
    "${LFBFL_verbose}"
}

LFBFL_file_name="build_md_from_printable_md.exec.sh"
LFBFL_script_download_URL="${LFBFL_dependencies_URL}/${LFBFL_file_name}"
LFBFL_file_path="./${LFBFL_subdir}/${LFBFL_file_name}"
@sha512_build_md_from_printable_md.exec.sh@
wrapped_wget_sha512
chmod +x "./${LFBFL_file_path}"

LFBFL_file_name="build_dependencies_notes.exec.php"
LFBFL_script_download_URL="${LFBFL_dependencies_URL}/${LFBFL_file_name}"
LFBFL_file_path="./${LFBFL_subdir}/${LFBFL_file_name}"
@sha512_build_dependencies_notes.exec.php@
wrapped_wget_sha512
chmod +x "./${LFBFL_file_path}"

LFBFL_file_name="check_shell_scripts_beginnings.libr.sh"
LFBFL_script_download_URL="${LFBFL_dependencies_URL}/${LFBFL_file_name}"
LFBFL_file_path="./${LFBFL_subdir}/${LFBFL_file_name}"
```

```
@sha512_check_shell_scripts_beginnings.libr.sh@
wrapped_wget_sha512

LFBFL_file_name="check_shell_scripts_indentation.libr.sh"
LFBFL_script_download_URL="${LFBFL_dependencies_URL}/${LFBFL_file_name}"
LFBFL_file_path="./${LFBFL_subdir}/${LFBFL_file_name}"
@sha512_check_shell_scripts_indentation.libr.sh@
wrapped_wget_sha512

LFBFL_file_name="check_URLs.libr.sh"
LFBFL_script_download_URL="${LFBFL_dependencies_URL}/${LFBFL_file_name}"
LFBFL_file_path="./${LFBFL_subdir}/${LFBFL_file_name}"
@sha512_check_URLs.libr.sh@
wrapped_wget_sha512

LFBFL_file_name="common_options.libr.sh"
LFBFL_script_download_URL="${LFBFL_dependencies_URL}/${LFBFL_file_name}"
LFBFL_file_path="./${LFBFL_subdir}/${LFBFL_file_name}"
@sha512_common_options.libr.sh@
wrapped_wget_sha512

LFBFL_file_name="comparisons.libr.sh"
LFBFL_script_download_URL="${LFBFL_dependencies_URL}/${LFBFL_file_name}"
LFBFL_file_path="./${LFBFL_subdir}/${LFBFL_file_name}"
@sha512_comparisons.libr.sh@
wrapped_wget_sha512

LFBFL_file_name="create_PDF.exec.sh"
LFBFL_script_download_URL="${LFBFL_dependencies_URL}/${LFBFL_file_name}"
LFBFL_file_path="./${LFBFL_subdir}/${LFBFL_file_name}"
@sha512_create_PDF.exec.sh@
wrapped_wget_sha512
chmod +x "./${LFBFL_file_path}"

LFBFL_file_name="generate_from_template.libr.sh"
LFBFL_script_download_URL="${LFBFL_dependencies_URL}/${LFBFL_file_name}"
LFBFL_file_path="./${LFBFL_subdir}/${LFBFL_file_name}"
@sha512_generate_from_template.libr.sh@
wrapped_wget_sha512

LFBFL_file_name="get_common_text_glob_patterns.libr.sh"
LFBFL_script_download_URL="${LFBFL_dependencies_URL}/${LFBFL_file_name}"
LFBFL_file_path="./${LFBFL_subdir}/${LFBFL_file_name}"
@sha512_get_common_text_glob_patterns.libr.sh@
wrapped_wget_sha512

```

```
LFBFL_file_name="grammar_and_spelling_check.libr.sh"
LFBFL_script_download_URL="${LFBFL_dependencies_URL}/${LFBFL_file_name}"
LFBFL_file_path="./${LFBFL_subdir}/${LFBFL_file_name}"
@sha512_grammar_and_spelling_check.libr.sh@
wrapped_wget_sha512

# /licenses_templates/ ---------------------------------------------
local LFBFL_dependencies_URL2
LFBFL_dependencies_URL2="${LFBFL_dependencies_URL}/licenses_templates"
readonly LFBFL_dependencies_URL2
LFBFL_file_name="build_licenses_templates.exec.sh"
LFBFL_script_download_URL="${LFBFL_dependencies_URL2}/${LFBFL_file_name}"
LFBFL_file_path="./${LFBFL_subdir2}/${LFBFL_file_name}"
@sha512_build_licenses_templates.exec.sh@
wrapped_wget_sha512
chmod +x "./${LFBFL_file_path}"

LFBFL_file_name="license_file_header_AGPLv3+.tpl"
LFBFL_script_download_URL="${LFBFL_dependencies_URL2}/${LFBFL_file_name}"
LFBFL_file_path="./${LFBFL_subdir2}/${LFBFL_file_name}"
@sha512_license_file_header_AGPLv3+.tpl@
wrapped_wget_sha512

LFBFL_file_name="license_file_header_GPLv3+.tpl"
LFBFL_script_download_URL="${LFBFL_dependencies_URL2}/${LFBFL_file_name}"
LFBFL_file_path="./${LFBFL_subdir2}/${LFBFL_file_name}"
@sha512_license_file_header_GPLv3+.tpl@
wrapped_wget_sha512

LFBFL_file_name="license_file_header_LGPLv3+.tpl"
LFBFL_script_download_URL="${LFBFL_dependencies_URL2}/${LFBFL_file_name}"
LFBFL_file_path="./${LFBFL_subdir2}/${LFBFL_file_name}"
@sha512_license_file_header_LGPLv3+.tpl@
wrapped_wget_sha512
# ------------------------------------------------------------------

LFBFL_file_name="lines_counts.libr.sh"
LFBFL_script_download_URL="${LFBFL_dependencies_URL}/${LFBFL_file_name}"
LFBFL_file_path="./${LFBFL_subdir}/${LFBFL_file_name}"
@sha512_lines_counts.libr.sh@
wrapped_wget_sha512

LFBFL_file_name="lines_filters.libr.sh"
LFBFL_script_download_URL="${LFBFL_dependencies_URL}/${LFBFL_file_name}"
LFBFL_file_path="./${LFBFL_subdir}/${LFBFL_file_name}"
@sha512_lines_filters.libr.sh@
```

```
183     wrapped_wget_sha512
184
185     LFBFL_file_name="lines_maps.libr.sh"
186     LFBFL_script_download_URL="${LFBFL_dependencies_URL}/${LFBFL_file_name}"
187     LFBFL_file_path="./${LFBFL_subdir}/${LFBFL_file_name}"
188     @sha512_lines_maps.libr.sh@
189     wrapped_wget_sha512
190
191     LFBFL_file_name="overwrite_if_not_equal.libr.sh"
192     LFBFL_script_download_URL="${LFBFL_dependencies_URL}/${LFBFL_file_name}"
193     LFBFL_file_path="./${LFBFL_subdir}/${LFBFL_file_name}"
194     @sha512_overwrite_if_not_equal.libr.sh@
195     wrapped_wget_sha512
196
197     LFBFL_file_name="python_black_complement.libr.sh"
198     LFBFL_script_download_URL="${LFBFL_dependencies_URL}/${LFBFL_file_name}"
199     LFBFL_file_path="./${LFBFL_subdir}/${LFBFL_file_name}"
200     @sha512_python_black_complement.libr.sh@
201     wrapped_wget_sha512
202
203     LFBFL_file_name="python_isort_complement.libr.sh"
204     LFBFL_script_download_URL="${LFBFL_dependencies_URL}/${LFBFL_file_name}"
205     LFBFL_file_path="./${LFBFL_subdir}/${LFBFL_file_name}"
206     @sha512_python_isort_complement.libr.sh@
207     wrapped_wget_sha512
208
209     LFBFL_file_name="repository_data.libr.sh"
210     LFBFL_script_download_URL="${LFBFL_dependencies_URL}/${LFBFL_file_name}"
211     LFBFL_file_path="./${LFBFL_subdir}/${LFBFL_file_name}"
212     @sha512_repository_data.libr.sh@
213     wrapped_wget_sha512
214
215     LFBFL_file_name="shell_checks_complement.libr.sh"
216     LFBFL_script_download_URL="${LFBFL_dependencies_URL}/${LFBFL_file_name}"
217     LFBFL_file_path="./${LFBFL_subdir}/${LFBFL_file_name}"
218     @sha512_shell_checks_complement.libr.sh@
219     wrapped_wget_sha512
220
221     LFBFL_file_name="split_score.exec.php"
222     LFBFL_script_download_URL="${LFBFL_dependencies_URL}/${LFBFL_file_name}"
223     LFBFL_file_path="./${LFBFL_subdir}/${LFBFL_file_name}"
224     @sha512_split_score.exec.php@
225     wrapped_wget_sha512
226
227     LFBFL_file_name="split_score.libr.php"
228     LFBFL_script_download_URL="${LFBFL_dependencies_URL}/${LFBFL_file_name}"
```

```
LFBFL_file_path="./${LFBFL_subdir}/${LFBFL_file_name}"
@sha512_split_score.libr.php@
wrapped_wget_sha512

LFBFL_file_name="strings_functions.libr.sh"
LFBFL_script_download_URL="${LFBFL_dependencies_URL}/${LFBFL_file_name}"
LFBFL_file_path="./${LFBFL_subdir}/${LFBFL_file_name}"
@sha512_strings_functions.libr.sh@
wrapped_wget_sha512

LFBFL_file_name="too_long_code_lines.libr.sh"
LFBFL_script_download_URL="${LFBFL_dependencies_URL}/${LFBFL_file_name}"
LFBFL_file_path="./${LFBFL_subdir}/${LFBFL_file_name}"
@sha512_too_long_code_lines.libr.sh@
wrapped_wget_sha512

LFBFL_file_name="update_or_check_files_names_listing.exec.sh"
LFBFL_script_download_URL="${LFBFL_dependencies_URL}/${LFBFL_file_name}"
LFBFL_file_path="./${LFBFL_subdir}/${LFBFL_file_name}"
@sha512_update_or_check_files_names_listing.exec.sh@
wrapped_wget_sha512
chmod +x "./${LFBFL_file_path}"

# shellcheck source=check_shell_scripts_beginnings.libr.sh
source "./${LFBFL_subdir}/check_shell_scripts_beginnings.libr.sh"
# shellcheck source=check_shell_scripts_indentation.libr.sh
source "./${LFBFL_subdir}/check_shell_scripts_indentation.libr.sh"
# shellcheck source=check_URLs.libr.sh
source "./${LFBFL_subdir}/check_URLs.libr.sh"
# shellcheck source=common_options.libr.sh
source "./${LFBFL_subdir}/common_options.libr.sh"
# shellcheck source=comparisons.libr.sh
source "./${LFBFL_subdir}/comparisons.libr.sh"
# shellcheck source=generate_from_template.libr.sh
source "./${LFBFL_subdir}/generate_from_template.libr.sh"
# shellcheck source=get_common_text_glob_patterns.libr.sh
source "./${LFBFL_subdir}/get_common_text_glob_patterns.libr.sh"
# shellcheck source=grammar_and_spelling_check.libr.sh
source "./${LFBFL_subdir}/grammar_and_spelling_check.libr.sh"
# shellcheck source=lines_counts.libr.sh
source "./${LFBFL_subdir}/lines_counts.libr.sh"
# shellcheck source=lines_filters.libr.sh
source "./${LFBFL_subdir}/lines_filters.libr.sh"
# shellcheck source=lines_maps.libr.sh
source "./${LFBFL_subdir}/lines_maps.libr.sh"
# shellcheck source=overwrite_if_not_equal.libr.sh
```

```
source "./${LFBFL_subdir}/overwrite_if_not_equal.libr.sh"
# shellcheck source=python_black_complement.libr.sh
source "./${LFBFL_subdir}/python_black_complement.libr.sh"
# shellcheck source=python_isort_complement.libr.sh
source "./${LFBFL_subdir}/python_isort_complement.libr.sh"
# shellcheck source=shell_checks_complement.libr.sh
source "./${LFBFL_subdir}/shell_checks_complement.libr.sh"
# shellcheck source=strings_functions.libr.sh
source "./${LFBFL_subdir}/strings_functions.libr.sh"
# shellcheck source=too_long_code_lines.libr.sh
source "./${LFBFL_subdir}/too_long_code_lines.libr.sh"

enhanced_set_bash_option extglob
# shellcheck source=repository_data.libr.sh
source "./${LFBFL_subdir}/repository_data.libr.sh"
# shellcheck disable=SC2154,SC2309
[[ i_enhanced_set_bash_option_extglob_result -eq 0 ]]\
  && enhanced_unset_bash_option extglob

declare -a LFBFL_return_traps_stack
local LFBFL_previous_return_trap
init_return_trap

enhanced_set_shell_option pipefail --trap-unset

declare -a LFBFL_some_common_options2=("${LFBFL_some_common_options[@]}")
keep_options LFBFL_some_common_options2 --verbose

printf "Building license headers\n"
"./${LFBFL_subdir2}/build_licenses_templates.exec.sh"\
  "${LFBFL_some_common_options[@]}"

printf "Building README.md\n"
"./${LFBFL_subdir}/build_md_from_printable_md.exec.sh"\
  --work-directory="${LFBFL_work_directory}"\
  --base-name=README\
  "${LFBFL_verbose}"

printf "Building other MarkDown files\n"
local LFBFL_some_directory
declare -r LFBFL_readme="${LFBFL_work_directory}/README.md.tpl"
local LFBFL_s_files_paths
LFBFL_s_files_paths=$(
  find "${LFBFL_work_directory}" -type f -name "*.md.tpl"\
  | relevant_find
)
```

```
  declare -a LFBFL_arr_files_paths
  if [[ -n "${LFBFL_s_files_paths}" ]]; then
    mapfile -t LFBFL_arr_files_paths <<< "${LFBFL_s_files_paths}"
    for LFBFL_file_path in "${LFBFL_arr_files_paths[@]}"; do
      [[ "${LFBFL_file_path}" == "${LFBFL_readme}" ]] && continue
      printf "Found template %s.\n" "${LFBFL_file_path}"
      LFBFL_some_directory=$(dirname "${LFBFL_file_path}")
      LFBFL_file_name=$(basename "${LFBFL_file_path}")
      LFBFL_file_name=${LFBFL_file_name%.md.tpl}
      "./${LFBFL_subdir}/build_md_from_printable_md.exec.sh"\
        --work-directory="${LFBFL_some_directory}"\
        --base-name="${LFBFL_file_name}"\
        "${LFBFL_verbose}"
    done
  fi

  printf "Building dependencies_notes\n"
  php "./${LFBFL_subdir}/build_dependencies_notes.exec.php"\
    "${LFBFL_work_directory}"

  declare -i LFBFL_i_directory_changed
  pushd_to_work_directory
  LFBFL_i_directory_changed=$?
  can_continue_after_enhanced_pushd || return 1

  local LFBFL_data_file_name="${LFBFL_subdir3}/repository_data.txt"

  declare -i LFBFL_i_max_line_length
  grep_variable "${LFBFL_data_file_name}" max_line_length\
    --result-variable-prefix=LFBFL_i_

  declare -i LFBFL_i_upgrade_venvs=0
  local LFBFL_upgrade_venvs_ts_file="${LFBFL_subdir3}/upgrade_venvs_ts"
  readonly LFBFL_upgrade_venvs_ts_file
  declare -i LFBFL_i_upgrade_venvs_ts
  declare -i LFBFL_i_current_ts
  local LFBFL_upgrade_venvs_answer

  local LFBFL_upgrade_venvs_time_interval_in_seconds=""
  get_upgrade_venvs_time_interval_in_seconds "${LFBFL_data_file_name}"\
    "${LFBFL_some_common_options2[@]}"

  if [[ -f "${LFBFL_upgrade_venvs_ts_file}" ]]; then
    LFBFL_i_upgrade_venvs_ts=$(
      stat --format %Y "${LFBFL_upgrade_venvs_ts_file}"
    )
```

```
    LFBFL_i_current_ts=$(date +%s)
    ((LFBFL_i_current_ts -= LFBFL_i_upgrade_venvs_ts))
    ((LFBFL_i_current_ts -= LFBFL_upgrade_venvs_time_interval_in_seconds))
    if [[ LFBFL_i_current_ts -gt 0 ]]; then
      LFBFL_i_upgrade_venvs=1
    fi
  else
    LFBFL_i_upgrade_venvs=1
  fi
  if [[ LFBFL_i_upgrade_venvs -eq 1 ]]; then
    local LFBFL_upgrade_venvs=""
    grep_variable "${LFBFL_data_file_name}" upgrade_venvs\
      --result-variable-prefix=LFBFL_
    if [[ "${LFBFL_upgrade_venvs}" != "auto" ]]; then
      read -r -n 1 -t 10\
        -p "Upgrade venvs and composer global? [Y/n]"\
        LFBFL_upgrade_venvs_answer
      if [[ "${LFBFL_upgrade_venvs_answer}" == "n" ]]; then
        LFBFL_i_upgrade_venvs=0
      fi
    fi
  fi

  printf "Running shellcheck\n"
  LFBFL_s_files_paths=$(
    find . -type f -name "*.sh"\
    | relevant_find
  )
  if [[ -n "${LFBFL_s_files_paths}" ]]; then
    mapfile -t LFBFL_arr_files_paths <<< "${LFBFL_s_files_paths}"
    for LFBFL_file_path in "${LFBFL_arr_files_paths[@]}"; do
      shellcheck --rcfile="${LFBFL_subdir3}/shellcheck.ini"\
        "${LFBFL_file_path}"
    done
  fi
  printf "Running shell_checks_complement\n"
  shell_checks_complement "${LFBFL_some_common_options2[@]}"

  printf -- "---Python---\n"

  maybe_deactivate(){
    if command -v deactivate
    then
      deactivate
    fi
  }
```

```
413
414   source_without_return_trap(){
415     declare -i LFBFL_i_no_return_traps=1
416     # shellcheck disable=SC1090,SC1091
417     source "$1"
418   }
419
420   printf "Running isort\n"
421   local LFBFL_isort_venv=""
422   grep_variable "${LFBFL_data_file_name}" isort_venv\
423     --result-variable-prefix=LFBFL_\
424     --replace-line-returns-by=""
425   if [[ -n "${LFBFL_isort_venv}" ]]; then
426     maybe_deactivate
427     source_without_return_trap "${LFBFL_isort_venv}/bin/activate"
428   fi
429   if [[ LFBFL_i_upgrade_venvs -eq 1 ]]; then
430     pip install --upgrade isort
431   fi
432   isort --profile=black .
433   if [[ -n "${LFBFL_isort_venv}" ]]; then
434     deactivate
435   fi
436   printf "Running python_isort_complement\n"
437   python_isort_complement "${LFBFL_some_common_options2[@]}"
438
439   printf "Running black\n"
440   # First, we update the configuration file with max_line_length.
441   local LFBFL_update_max_length="s/^line-length = [0-9]*$"
442   LFBFL_update_max_length+="/line-length = ${LFBFL_i_max_line_length}/"
443   sed -E -e "${LFBFL_update_max_length}"\
444     "${LFBFL_subdir3}/black.toml"\
445     > "${LFBFL_subdir3}/black.toml.temp"
446   overwrite_if_not_equal "${LFBFL_subdir3}/black.toml"\
447     "${LFBFL_subdir3}/black.toml.temp"
448
449   local LFBFL_black_venv=""
450   grep_variable "${LFBFL_data_file_name}" black_venv\
451     --result-variable-prefix=LFBFL_\
452     --replace-line-returns-by=""
453   if [[ -n "${LFBFL_black_venv}" ]]; then
454     maybe_deactivate
455     source_without_return_trap "${LFBFL_black_venv}/bin/activate"
456   fi
457   if [[ LFBFL_i_upgrade_venvs -eq 1 ]]; then
458     pip install --upgrade black
```

```
  fi
  black --config="${LFBFL_subdir3}/black.toml" .
  if [[ -n "${LFBFL_black_venv}" ]]; then
    deactivate
  fi
  printf "Running python_black_complement\n"
  python_black_complement "${LFBFL_some_common_options2[@]}"

  printf "Probing if mypy should be runned\n"
  local LFBFL_mypy_venv=""
  grep_variable "${LFBFL_data_file_name}" mypy_venv\
    --result-variable-prefix=LFBFL_\
    --replace-line-returns-by=""
  if [[ -n "${LFBFL_mypy_venv}" ]]; then
    maybe_deactivate
    source_without_return_trap "${LFBFL_mypy_venv}/bin/activate"
  fi
  if [[ LFBFL_i_upgrade_venvs -eq 1 ]]; then
    pip install --upgrade mypy
  fi
  local LFBFL_directory_path
  declare -i LFBFL_i_no_toml=1
  LFBFL_s_files_paths=$(
    find . -type f -name "pyproject.toml"\
    | relevant_find
  )
  if [[ -n "${LFBFL_s_files_paths}" ]]; then
    mapfile -t LFBFL_arr_files_paths <<< "${LFBFL_s_files_paths}"
    for LFBFL_file_path in "${LFBFL_arr_files_paths[@]}"; do
      if grep --quiet "Typing :: Typed" "${LFBFL_file_path}"; then
        printf "Running mypy\n"
        LFBFL_directory_path=$(dirname "${LFBFL_file_path}")
        mypy "${LFBFL_directory_path}"
      fi
      LFBFL_i_no_toml=0
    done
  fi
  if [[ LFBFL_i_no_toml -eq 1 ]]; then
    printf "Running mypy\n"
    mypy .
  fi
  if [[ -n "${LFBFL_mypy_venv}" ]]; then
    deactivate
  fi

  printf "Running bandit\n"
```

```
LFBFL_bandit_venv=""
grep_variable "${LFBFL_data_file_name}" bandit_venv\
  --result-variable-prefix=LFBFL_\
  --replace-line-returns-by=""
if [[ -n "${LFBFL_bandit_venv}" ]]; then
  maybe_deactivate
  source_without_return_trap "${LFBFL_bandit_venv}/bin/activate"
fi
if [[ LFBFL_i_upgrade_venvs -eq 1 ]]; then
  pip install --upgrade bandit
fi
bandit --ini="${LFBFL_subdir3}/bandit.ini"\
  --baseline="${LFBFL_subdir3}/bandit_baseline.json"\
  --recursive .
# Saving new baseline in temp if necessary.
bandit --ini="${LFBFL_subdir3}/bandit.ini"\
  --format=json\
  --output="${LFBFL_subdir3}/temp/bandit_baseline.json"\
  --recursive .
if [[ -n "${LFBFL_bandit_venv}" ]]; then
  deactivate
fi

printf "Running pylint\n"
LFBFL_pylint_venv=""
grep_variable "${LFBFL_data_file_name}" pylint_venv\
  --result-variable-prefix=LFBFL_\
  --replace-line-returns-by=""
if [[ -n "${LFBFL_pylint_venv}" ]]; then
  maybe_deactivate
  source_without_return_trap "${LFBFL_pylint_venv}/bin/activate"
fi
if [[ LFBFL_i_upgrade_venvs -eq 1 ]]; then
  pip install --upgrade pylint
fi
pylint --rcfile="${LFBFL_subdir3}/pylintrc.toml" --recursive=y .
if [[ -n "${LFBFL_pylint_venv}" ]]; then
  deactivate
fi

printf "Running ruff\n"
LFBFL_ruff_venv=""
grep_variable "${LFBFL_data_file_name}" ruff_venv\
  --result-variable-prefix=LFBFL_\
  --replace-line-returns-by=""
if [[ -n "${LFBFL_ruff_venv}" ]]; then
```

```
551       maybe_deactivate
552       source_without_return_trap "${LFBFL_ruff_venv}/bin/activate"
553     fi
554     if [[ LFBFL_i_upgrade_venvs -eq 1 ]]; then
555       pip install --upgrade ruff
556     fi
557     ruff check --config="${LFBFL_subdir3}/ruff.toml"
558     if [[ -n "${LFBFL_ruff_venv}" ]]; then
559       deactivate
560     fi
561     printf -- "---Python end---\n"
562
563     printf -- "---PHP---\n"
564     printf "Running PHPMD\n"
565     if [[ LFBFL_i_upgrade_venvs -eq 1 ]]; then
566       composer global require phpmd/phpmd
567     fi
568
569     declare -r LFBFL_phpmd_baseline="${LFBFL_subdir3}/phpmd_baseline.xml"
570     local LFBFL_temp_phpmd_baseline="${LFBFL_subdir3}/temp/"
571     LFBFL_temp_phpmd_baseline+="phpmd_baseline.xml"
572     readonly LFBFL_temp_phpmd_baseline
573
574     local LFBFL_phpmd_rulesets="cleancode,codesize,controversial,"
575     LFBFL_phpmd_rulesets+="design,naming,unusedcode"
576     readonly LFBFL_phpmd_rulesets
577
578     phpmd --color --baseline-file="${LFBFL_phpmd_baseline}"\
579       . text "${LFBFL_phpmd_rulesets}"
580     # Saving new baseline in temp if necessary.
581     rm "${LFBFL_temp_phpmd_baseline}"
582     phpmd --color --generate-baseline\
583       --baseline-file="${LFBFL_temp_phpmd_baseline}"\
584       . text "${LFBFL_phpmd_rulesets}"
585     sed -i -e 's/" file="/"\n    file="/g' "${LFBFL_temp_phpmd_baseline}"
586     sed -i -E -e 's~(file=".*)/>~\1\n  />~g' "${LFBFL_temp_phpmd_baseline}"
587     diff "${LFBFL_phpmd_baseline}" "${LFBFL_temp_phpmd_baseline}"
588     printf -- "---PHP end---\n"
589
590     printf -- "---JS---\n"
591     LFBFL_npm_lint_directories=""
592     grep_variable "${LFBFL_data_file_name}" npm_lint_directories\
593       --result-variable-prefix=LFBFL_
594     if [[ -n "${LFBFL_npm_lint_directories}" ]]; then
595       printf "Running ESLint\n"
596       local LFBFL_JS_directory
```

```
  declare -a LFBFL_arr_paths
  mapfile -t LFBFL_arr_paths <<< "${LFBFL_npm_lint_directories}"
  for LFBFL_JS_directory in "${LFBFL_arr_paths[@]}"; do
    (cd "${LFBFL_JS_directory}" && npm run lint)
  done
fi
printf -- "---JS end---\n"

printf "Analyzing too long lines\n"
# shellcheck disable=SC2248
too_long_code_lines "${LFBFL_some_common_options2[@]}" \
  --max-line-length=${LFBFL_i_max_line_length}

printf "Analyzing shell scripts beginnings\n"
check_shell_scripts_beginnings "${LFBFL_some_common_options2[@]}" \
  | relevant_grep

printf "Analyzing shell scripts indentation\n"
check_shell_scripts_indentation "${LFBFL_some_common_options2[@]}" \
  | relevant_grep

printf "Analyzing URLs\n"
check_URLs "${LFBFL_some_common_options2[@]}" \
  | relevant_grep

printf "Analyzing strange characters: hover over in doubt\n"
# shellcheck disable=SC1111
LFBFL_usual_characters="\x00-\x7Fàâéèêëîïôûç©"”└├│«»"
grep --binary-files=without-match --color=always --exclude-dir .git\
  --line-number --perl-regexp --recursive --\
  "[^${LFBFL_usual_characters}]" .

printf "Grammar and spelling check\n"
grammar_and_spelling_check "${LFBFL_data_file_name}"\
  "${LFBFL_some_common_options2[@]}"

[[ LFBFL_i_directory_changed -eq 0 ]] && popd_from_work_directory

printf "Checking listed files\n"
"./${LFBFL_subdir}/update_or_check_files_names_listing.exec.sh"\
  "${LFBFL_some_common_options[@]}"

printf "Creating the PDF file of the listing of the source code\n"
"./${LFBFL_subdir}/create_PDF.exec.sh" "${LFBFL_some_common_options[@]}"

pushd_to_work_directory --trap-popd
```

```
643     can_continue_after_enhanced_pushd || return 1
644
645     if [[ -f "${LFBFL_subdir3}/post_build.sh" ]]; then
646       "./${LFBFL_subdir3}/post_build.sh" "${LFBFL_verbose}"
647     fi
648
649     # ----------------------------------------------------------------
650     # It is at the end that all the HTML files are generated and that
651     # we can check there are no mistakes.
652     printf "Running Nu W3C HTML CSS and SVG validator\n"
653     local LFBFL_upgrade_vnu_jar_time_interval_in_seconds=""
654     grep_variable "${LFBFL_data_file_name}"\
655       upgrade_vnu_jar_time_interval_in_seconds\
656       --result-variable-prefix=LFBFL_
657
658     if [[ "${LFBFL_upgrade_vnu_jar_time_interval_in_seconds}" == "wat" ]];
659     then
660       LFBFL_upgrade_vnu_jar_time_interval_in_seconds=${RANDOM}
661     fi
662
663     declare -i LFBFL_i_vnu_jar_ts
664     local LFBFL_vnu_jar_path=""
665     grep_variable "${LFBFL_data_file_name}" vnu_jar_path\
666       --result-variable-prefix=LFBFL_\
667       --replace-line-returns-by=""
668     if [[ -f "${LFBFL_vnu_jar_path}" ]]; then
669       LFBFL_i_vnu_jar_ts=$(stat --format %Y "${LFBFL_vnu_jar_path}")
670       LFBFL_i_current_ts=$(date +%s)
671       ((LFBFL_i_current_ts -= LFBFL_i_vnu_jar_ts))
672       ((
673         LFBFL_i_current_ts -= LFBFL_upgrade_vnu_jar_time_interval_in_seconds
674       ))
675       if [[ LFBFL_i_current_ts -gt 0 ]]; then
676         printf $'/!\\ATTENTION: Nu W3C validator is too old./!\\\n'
677         printf "Go get the latest version here:\n"
678         printf "https://github.com/validator/validator/releases\n"
679       fi
680       # vnu has false positives for xhtml
681       local LFBFL_use_vnu_for_xhtml=""
682       grep_variable "${LFBFL_data_file_name}" use_vnu_for_xhtml\
683         --result-variable-prefix=LFBFL_
684       local LFBFL_find_regexp
685       if [[ "${LFBFL_use_vnu_for_xhtml}" == "true" ]]; then
686         LFBFL_find_regexp=".*\.\(css\|html\|svg\|xhtml\)"
687       else
688         LFBFL_find_regexp=".*\.\(css\|html\|svg\)"
```

```
    fi
    declare -r LFBFL_files_for_Nu=$(
      find . -iregex "${LFBFL_find_regexp}"\
      | relevant_find
    )
    if [[ -n "${LFBFL_files_for_Nu}" ]]; then
      declare -a LFBFL_files_for_Nu2
      mapfile -t LFBFL_files_for_Nu2 <<< "${LFBFL_files_for_Nu}"
      readonly LFBFL_files_for_Nu2
      java -Xss128M -jar "${LFBFL_vnu_jar_path}" --exit-zero-always\
        --verbose "${LFBFL_files_for_Nu2[@]}"
    fi
  fi
  # ------------------------------------------------------------------

  if [[ LFBFL_i_upgrade_venvs -eq 1 ]]; then
    touch "${LFBFL_upgrade_venvs_ts_file}"
  fi
}

common_build_and_checks "$@"
```

3.9 build_and_checks_dependencies/common_options.libr.sh

```
#!/usr/bin/env bash
# This file is part of DevOrSysAdminScripts library.
#
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# of the GNU General Public License
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#
```

```
# ©Copyright 2023-2026 Laurent Frédéric Bernard François Lyaudet

get_where_was_i(){
  # Auxiliary function needed below.
  # See usages below if you want to reuse it.
  # $1=offset
  LFBFL_where_was_i="$0"
  if [[ "${BASH_SOURCE[$1]}" != "" && "${BASH_SOURCE[$1]}" != "$0" ]]; then
    LFBFL_where_was_i+=" ${BASH_SOURCE[$1]}"
  fi
  if [[ "${FUNCNAME[$1]}" != "" ]]; then
    LFBFL_where_was_i+=" ${FUNCNAME[$1]}()"
  fi
  readonly LFBFL_where_was_i
  # Comment out line above in case someone would use distinct
  # offsets in the same function.
}

# RETURN traps handling
# Add the following lines at the start of a function.
#   declare -a LFBFL_return_traps_stack
#   local LFBFL_previous_return_trap
#   init_return_trap

init_return_trap(){
  local LFBFL_previous_return_trap2
  LFBFL_previous_return_trap2=$(trap -p RETURN)
  if [[ LFBFL_i_verbose -eq 1 ]]; then
    local LFBFL_where_was_i
    get_where_was_i 2
    printf "%s previous_return_trap is: %s.\n"\
      "${LFBFL_where_was_i}"\
      "${LFBFL_previous_return_trap2}"
  fi
  if [[ -z "${LFBFL_previous_return_trap2}" ]]; then
    LFBFL_previous_return_trap2="trap - RETURN"
  fi
  declare -ir LFBFL_function_depth=$((${#FUNCNAME[@]} - 1))
  LFBFL_previous_return_trap="${LFBFL_function_depth}:"
  LFBFL_previous_return_trap+="${LFBFL_previous_return_trap2}"
  trap 'execute_return_traps' RETURN
}

execute_return_traps(){
  if [[ "${FUNCNAME[1]}" == "init_return_trap" ]]; then
    return
```

```
  fi
  # ------------------------------------------------------------------------
  # See https://lists.gnu.org/archive/html/help-bash/2026-03/msg00052.html
  if [[ "${FUNCNAME[1]}" == "source" ]]; then
    # Not Working
    return
  fi
  if [[ ${LFBFL_i_no_return_traps:-0} -eq 1 ]]; then
    return
  fi
  # ------------------------------------------------------------------------
  local LFBFL_some_return_trap
  local LFBFL_where_was_i
  get_where_was_i 2
  declare -ir LFBFL_function_depth=$((${#FUNCNAME[@]} - 1))
  while [[ -n "${LFBFL_return_traps_stack}" ]]; do
    LFBFL_some_return_trap="${LFBFL_return_traps_stack[-1]}"
    if [[ "${LFBFL_some_return_trap}" != ${LFBFL_function_depth}:* ]]; then
      # We only treat the top of the stack that should correspond to the
      # current function.
      break
    fi
    LFBFL_some_return_trap=${LFBFL_some_return_trap#*:}
    if [[ LFBFL_i_verbose -eq 1 ]]; then
      printf "%s executing some return trap: %s.\n"\
        "${LFBFL_where_was_i}"\
        "${LFBFL_some_return_trap}"
    fi
    eval "${LFBFL_some_return_trap}"
    unset 'LFBFL_return_traps_stack[-1]'
  done
  if [[ "${LFBFL_previous_return_trap}" != ${LFBFL_function_depth}:* ]];
  then
    # We only set the previous return trap if such a trap was replaced
    # by an init_return_trap.
    return
  fi
  LFBFL_previous_return_trap=${LFBFL_previous_return_trap#*:}
  if [[ LFBFL_i_verbose -eq 1 ]]; then
    printf "%s setting back to previous_return_trap: %s.\n"\
      "${LFBFL_where_was_i}"\
      "${LFBFL_previous_return_trap}"
  fi
  eval "${LFBFL_previous_return_trap}"
}

```

```

is_top_dirstack_directory(){
  # This function should not be needed since when testing "${DIRSTACK[0]}"
  # is already a realpath. But I keep it, in case it is needed.
  # https://lists.gnu.org/archive/html/help-bash/2026-03/msg00005.html
  # $1=to_directory needs to be "realpathed" already
  if [[ -z "$1" ]]; then
    return 0
  fi
  local LFBFL_real_path
  LFBFL_real_path=$(realpath "${DIRSTACK[0]}")
  readonly LFBFL_real_path
  [[ "${LFBFL_real_path}" == "$1" ]]
}

enhanced_pushd(){
  # Auxiliary function needed below.
  # See usages below if you want to reuse it.
  # $1=to_directory must be sanitized with realpath before call
  # $2=offset_for_where_was_i
  # $3=error_message_intermediate_complement
  # returns an error code whenever no pushd happened
  [[ LFBFL_i_verbose -eq 1 ]]\
    && printf "enhanced_pushd requested: %s\n" "$1"
  declare -gi i_enhanced_pushd_result=0
  if [[ -z "$1" ]]; then
    i_enhanced_pushd_result=110
    # shellcheck disable=SC2248
    return ${i_enhanced_pushd_result}
  fi
  [[ LFBFL_i_verbose -eq 1 ]] && printf "enhanced_pushd validated1\n"
  is_top_dirstack_directory "$1" && {
    i_enhanced_pushd_result=111
    # shellcheck disable=SC2248
    return ${i_enhanced_pushd_result}
  }
  [[ LFBFL_i_verbose -eq 1 ]] && printf "enhanced_pushd validated2\n"

  pushd "$1" || {
    i_enhanced_pushd_result=$?
    local LFBFL_where_was_i
    get_where_was_i "$2"
    printf "%s %s%s no such directory.\n" "${LFBFL_where_was_i}" "$3" "$1"
    # shellcheck disable=SC2248
    return ${i_enhanced_pushd_result}
  }
```

```
  [[ LFBFL_i_verbose -eq 1 ]] && printf "enhanced_pushd executed\n"
  # shellcheck disable=SC2248
  return ${i_enhanced_pushd_result}
}

enhanced_popd(){
  # Auxiliary function needed below.
  # See usages below if you want to reuse it.
  # $1=to_directory
  # $2=offset_for_where_was_i
  [[ LFBFL_i_verbose -eq 1 ]]\
    && printf "enhanced_popd requested: %s\n" "$1"
  if [[ -z "$1" ]]; then
    return 0
  fi

  declare -i LFBFL_i_popd_result
  popd || {
    LFBFL_i_popd_result=$?
    readonly LFBFL_i_popd_result
    local LFBFL_where_was_i
    get_where_was_i "$2"
    printf "%s popd failed.\n" "${LFBFL_where_was_i}"
    # shellcheck disable=SC2248
    return ${LFBFL_i_popd_result}
  }
  [[ LFBFL_i_verbose -eq 1 ]] && printf "enhanced_popd executed\n"
  return 0
}

can_continue_after_enhanced_pushd(){
  [[ i_enhanced_pushd_result -eq 0
  || i_enhanced_pushd_result -eq 110
  || i_enhanced_pushd_result -eq 111
  ]]
}

# Main usage for the next functions relative to verbose mode:
# Add the two following lines at the start of a function.
#   declare -i LFBFL_i_verbose=0
#   get_verbose_option "$@"
# /!\ You really need all of the two lines. /!\
# Otherwise, the call to get_verbose_option may impact a function above;
# in case you didn't copy all of "$@" in nested function calls...
# /!\ The same warning applies for other functions below. /!\
# /!\ I will not repeat it below. /!\
```

```
# Use get_verbose_options_array, if needed.

get_verbose_options_array(){
  # Use this constant with code like:
  # "${LFBFL_VERBOSE_OPTIONS[LFBFL_i_verbose]}"
  # It could be extended if there is more than one level of verboseness.
  if [[ -n "${LFBFL_VERBOSE_OPTIONS}" ]]; then
    return
  fi

  declare -agr LFBFL_VERBOSE_OPTIONS=(
    ""
    "--verbose"
  )
}

get_verbose_option(){
  # This command is to be called in another one with same arguments.
  # Options:
  #   --verbose
  # This test is faster than:
  #   for LFBFL_arg in "$@"; do
  #     if [[ "${LFBFL_arg}" == "--verbose" ]]; then
  # But it is not bug free.
  if [[ "$*" == *--verbose* ]]; then
    local LFBFL_where_was_i
    get_where_was_i 2
    printf "%s %s\n" "${LFBFL_where_was_i}" "$*"
    LFBFL_i_verbose=1
  fi
  # shellcheck disable=SC2034
  readonly LFBFL_i_verbose
}


get_some_option(){
  # $1=variable_name
  # $2=some_option_long_name
  # $3=option_default_value
  # $4=read_only 0 or 1 to make the variable readonly
  # This command is to be called in another one with same arguments on top
  # of the three previous.
  local LFBFL_arg
  for LFBFL_arg in "$@"; do
    if [[ "${LFBFL_arg}" == "$2"=* ]]; then
      printf -v "$1" "%s" "${LFBFL_arg#"$2"=}"
```

```
      printf -v "$1" "%s" "${!1:-$3}"
      break
    fi
  done
  if [[ $4 -eq 1 ]]; then
    readonly "$1"
  fi
}

keep_options(){
  # $1=variable_name of array of options
  # Other arguments: the options to keep.
  declare -a LFBFL_array_of_filtered_options
  declare -n LFBFL_array_of_options=$1
  local LFBFL_some_argument
  local LFBFL_some_argument2
  for LFBFL_some_argument in "${LFBFL_array_of_options[@]}"; do
    for LFBFL_some_argument2 in "$@"; do
      if [[
        "${LFBFL_some_argument}" == "${LFBFL_some_argument2}"
        || "${LFBFL_some_argument}" == "${LFBFL_some_argument2}"=*
      ]]; then
        LFBFL_array_of_filtered_options+=("${LFBFL_some_argument}")
      fi
    done
  done
  LFBFL_array_of_options=("${LFBFL_array_of_filtered_options[@]}")
}


# Main usage for the next functions relative to work_directory:
# Add the five following lines at the start of a function.
#   local LFBFL_work_directory=""
#   get_work_directory_option "$@"
#   pushd_to_work_directory\
#     && trap 'popd_from_work_directory' RETURN
#   can_continue_after_enhanced_pushd || return 1
# Or:
#   local LFBFL_work_directory=""
#   get_work_directory_option "$@"
#   pushd_to_work_directory --trap-popd
#   can_continue_after_enhanced_pushd || return 1

get_work_directory_option(){
  # This command is to be called in another one with same arguments.
  # Options:
```

```
  #   --work-directory=""
  get_some_option LFBFL_work_directory --work-directory "." 0 "$@"
  # LFBFL_work_directory=${LFBFL_work_directory/#~/${HOME}} line below
  if [[ -n "${LFBFL_work_directory}" ]]; then
    LFBFL_work_directory=$(realpath "${LFBFL_work_directory}")
  fi
  readonly LFBFL_work_directory
}

pushd_to_work_directory(){
  # Options:
  #   --trap-popd
  enhanced_pushd "${LFBFL_work_directory}" 3 " --work-directory="
  if [[ i_enhanced_pushd_result -eq 0 ]]; then
    local LFBFL_arg
    for LFBFL_arg in "$@"; do
      if [[ "${LFBFL_arg}" == --trap-popd ]]; then
        declare -ir LFBFL_function_depth=$((${#FUNCNAME[@]} - 1))
        local LFBFL_some_return_trap
        LFBFL_some_return_trap="${LFBFL_function_depth}:"
        LFBFL_some_return_trap+="popd_from_work_directory 4"
        readonly LFBFL_some_return_trap
        LFBFL_return_traps_stack+=("${LFBFL_some_return_trap}")
        break
      fi
    done
  fi
  # shellcheck disable=SC2248
  return ${i_enhanced_pushd_result}
}

popd_from_work_directory(){
  # $1=offset for where_was_i
  enhanced_popd "${LFBFL_work_directory}" "${1:-3}"
}

work_directory_is_top_dirstack_directory(){
  is_top_dirstack_directory "${LFBFL_work_directory}"
}


get_offset_option(){
  # This command is to be called in another one with same arguments on top
  # of the default value.
  # $1=option_default_value
  # Options:
```

```
  #   --offset=2
  get_some_option LFBFL_i_offset --offset "$1" 1 "$@"
}


# Functions relative to shell and bash options
# (See SHELLOPTS and BASHOPTS, set and shopt):
# - shell options usage, example with pipefail:
#     enhanced_set_shell_option pipefail\
#       && trap 'enhanced_unset_shell_option pipefail' RETURN
#   or maybe:
#     enhanced_unset_shell_option pipefail\
#       && trap 'enhanced_set_shell_option pipefail' RETURN
#   if you need to unset it in some function.
#   or better if you use my RETURN traps handler:
#     enhanced_set_shell_option pipefail --trap-unset
#     enhanced_unset_shell_option pipefail --trap-set
# - bash options usage, example with globstar:
#     enhanced_set_bash_option globstar\
#       && trap 'enhanced_unset_bash_option globstar' RETURN
#   or maybe:
#     enhanced_unset_bash_option globstar\
#       && trap 'enhanced_set_bash_option globstar' RETURN
#   or better if you use my RETURN traps handler:
#     enhanced_set_bash_option globstar --trap-unset
#     enhanced_unset_bash_option globstar --trap-set

enhanced_set_shell_option(){
  # $1=optname some shell option for set
  # Options:
  #   --offset=2
  #   --trap-unset
  declare -r LFBFL_result_name="i_enhanced_set_shell_option_$1_result"
  declare -gi "${LFBFL_result_name}"=0
  if [[ -o "$1" ]]; then
    printf -v "${LFBFL_result_name}" "%d" "1"
    # shellcheck disable=SC2086
    return ${!LFBFL_result_name}
  fi
  set -o "$1"
  declare -i LFBFL_i_offset=2
  get_offset_option 2 "$@"
  if [[ LFBFL_i_verbose -eq 1 ]]; then
    local LFBFL_where_was_i
    # shellcheck disable=SC2248
    get_where_was_i ${LFBFL_i_offset}
```

```
    printf "%s %s shell option activated.\n" "${LFBFL_where_was_i}" "$1"
  fi
  local LFBFL_arg
  for LFBFL_arg in "$@"; do
    if [[ "${LFBFL_arg}" == --trap-unset ]]; then
      declare -ir LFBFL_function_depth=$((${#FUNCNAME[@]} - 1))
      local LFBFL_some_return_trap
      LFBFL_some_return_trap="${LFBFL_function_depth}:"
      LFBFL_some_return_trap+="enhanced_unset_shell_option '$1' --offset=3"
      readonly LFBFL_some_return_trap
      LFBFL_return_traps_stack+=("${LFBFL_some_return_trap}")
      break
    fi
  done
  # shellcheck disable=SC2086
  return ${!LFBFL_result_name}
}

enhanced_unset_shell_option(){
  # $1=optname some shell option for set
  # Options:
  #   --offset=2
  #   --trap-set
  declare -r LFBFL_result_name="i_enhanced_unset_shell_option_$1_result"
  declare -gi "${LFBFL_result_name}"=0
  if [[ ! -o "$1" ]]; then
    printf -v "${LFBFL_result_name}" "%d" "1"
    # shellcheck disable=SC2086
    return ${!LFBFL_result_name}
  fi
  set +o "$1"
  declare -i LFBFL_i_offset=2
  get_offset_option 2 "$@"
  if [[ LFBFL_i_verbose -eq 1 ]]; then
    local LFBFL_where_was_i
    # shellcheck disable=SC2248
    get_where_was_i ${LFBFL_i_offset}
    printf "%s %s shell option unactivated.\n" "${LFBFL_where_was_i}" "$1"
  fi
  local LFBFL_arg
  for LFBFL_arg in "$@"; do
    if [[ "${LFBFL_arg}" == --trap-set ]]; then
      declare -ir LFBFL_function_depth=$((${#FUNCNAME[@]} - 1))
      local LFBFL_some_return_trap
      LFBFL_some_return_trap="${LFBFL_function_depth}:"
      LFBFL_some_return_trap+="enhanced_set_shell_option '$1' --offset=3"
```

```
      readonly LFBFL_some_return_trap
      LFBFL_return_traps_stack+=("${LFBFL_some_return_trap}")
      break
    fi
  done
  # shellcheck disable=SC2086
  return ${!LFBFL_result_name}
}

enhanced_set_bash_option(){
  # $1=optname some bash option for shopt
  # Options:
  #   --offset=2
  #   --trap-unset
  declare -r LFBFL_result_name="i_enhanced_set_bash_option_$1_result"
  declare -gi "${LFBFL_result_name}"=0
  # declare -r LFBFL_regexp="^(.*:)?$1(:.*)?$" funny but brainfucked
  # if [[ "${BASHOPTS}" =~ ${LFBFL_regexp} ]]; then
  if shopt -q "$1"; then
    printf -v "${LFBFL_result_name}" "%d" "1"
    # shellcheck disable=SC2086
    return ${!LFBFL_result_name}
  fi
  shopt -s "$1"
  declare -i LFBFL_i_offset=2
  get_offset_option 2 "$@"
  if [[ LFBFL_i_verbose -eq 1 ]]; then
    local LFBFL_where_was_i
    # shellcheck disable=SC2248
    get_where_was_i ${LFBFL_i_offset}
    printf "%s %s bash option activated.\n" "${LFBFL_where_was_i}" "$1"
  fi
  local LFBFL_arg
  for LFBFL_arg in "$@"; do
    if [[ "${LFBFL_arg}" == --trap-unset ]]; then
      declare -ir LFBFL_function_depth=$((${#FUNCNAME[@]} - 1))
      local LFBFL_some_return_trap
      LFBFL_some_return_trap="${LFBFL_function_depth}:"
      LFBFL_some_return_trap+="enhanced_unset_bash_option '$1' --offset=3"
      readonly LFBFL_some_return_trap
      LFBFL_return_traps_stack+=("${LFBFL_some_return_trap}")
      break
    fi
  done
  # shellcheck disable=SC2086
  return ${!LFBFL_result_name}
```

```
}

enhanced_unset_bash_option(){
  # $1=optname some bash option for shopt
  # Options:
  #   --offset=2
  #   --trap-set
  declare -r LFBFL_result_name="i_enhanced_unset_bash_option_$1_result"
  declare -gi "${LFBFL_result_name}"=0
  if ! shopt -q "$1"; then
    printf -v "${LFBFL_result_name}" "%d" "1"
    # shellcheck disable=SC2086
    return ${!LFBFL_result_name}
  fi
  shopt -u "$1"
  declare -i LFBFL_i_offset=2
  get_offset_option 2 "$@"
  if [[ LFBFL_i_verbose -eq 1 ]]; then
    local LFBFL_where_was_i
    # shellcheck disable=SC2248
    get_where_was_i ${LFBFL_i_offset}
    printf "%s %s bash option unactivated.\n" "${LFBFL_where_was_i}" "$1"
  fi
  local LFBFL_arg
  for LFBFL_arg in "$@"; do
    if [[ "${LFBFL_arg}" == --trap-set ]]; then
      declare -ir LFBFL_function_depth=$((${#FUNCNAME[@]} - 1))
      local LFBFL_some_return_trap
      LFBFL_some_return_trap="${LFBFL_function_depth}:"
      LFBFL_some_return_trap+="enhanced_set_bash_option '$1' --offset=3"
      readonly LFBFL_some_return_trap
      LFBFL_return_traps_stack+=("${LFBFL_some_return_trap}")
      break
    fi
  done
  # shellcheck disable=SC2086
  return ${!LFBFL_result_name}
}
```

3.10 build_and_checks_dependencies/comparisons.libr.sh

```
#!/usr/bin/env bash
# This file is part of DevOrSysAdminScripts library.
#
# DevOrSysAdminScripts is free software:
# you can redistribute it and/or modify it under the terms
```

```
# of the GNU General Public License
# as published by the Free Software Foundation,
# either version 3 of the License,
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#
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# The file "comparisons.sh" was renamed to "comparisons.libr.sh".

# shellcheck source=common_options.libr.sh
source "./build_and_checks_dependencies/common_options.libr.sh"

equal(){
  # Returns 1 if all the arguments strings are equal.
  # Renvoie 1 dans $? si toutes les chaînes en arguments sont égales.
  # J'ai rerepensé au 3-way-merge
  # et à mon article sur le principe de première différence
  # appliqué à la largeur modulaire, de clique ou de rang avec
  # l'idée d'un principe de première différence ternaire.
  declare -i LFBFL_i_first=1
  local LFBFL_current
  local LFBFL_arg
  for LFBFL_arg in "$@"; do
    if [[ LFBFL_i_first -eq 1 ]]; then
      LFBFL_current="${LFBFL_arg}"
      LFBFL_i_first=0
      continue
    fi
    if [[ "${LFBFL_current}" != "${LFBFL_arg}" ]]; then
      return 0
    fi
  done
  return 1
}

```

```
some_distinct(){
  ! equal "$@"
}

all_distinct(){
  # Returns 1 if all the arguments strings are distinct.
  declare -ir LFBFL_ij_max=$#
  declare -i LFBFL_i
  declare -i LFBFL_j
  for ((LFBFL_i=1; LFBFL_i<LFBFL_ij_max; ++LFBFL_i)); do
    for ((LFBFL_j = LFBFL_i + 1; LFBFL_j<=LFBFL_ij_max; ++LFBFL_j)); do
      [[ "${!LFBFL_i}" == "${!LFBFL_j}" ]] && return 0
    done
  done
  return 1
}

all_distinct2(){
  # Returns 1 if all the arguments strings are distinct.
  local LFBFL_element
  local LFBFL_previous_element
  if [[ $# -le 1 ]]; then
    return 1
  fi
  declare -r LFBFL_s_sorted_arguments=$(
    printf "%s\n" "$@" \
    | sort
  )
  declare -a LFBFL_arr_sorted_arguments
  mapfile -t LFBFL_arr_sorted_arguments <<< "${LFBFL_s_sorted_arguments}"
  declare -i LFBFL_i
  declare -ir LFBFL_i_max=${#LFBFL_arr_sorted_arguments[@]}
  LFBFL_previous_element=${LFBFL_arr_sorted_arguments[0]}
  for ((LFBFL_i=1; LFBFL_i<LFBFL_i_max; ++LFBFL_i)); do
    LFBFL_element="${LFBFL_arr_sorted_arguments[${LFBFL_i}]}"
    if [[ "${LFBFL_previous_element}" == "${LFBFL_element}" ]]; then
      return 0
    fi
    LFBFL_previous_element="${LFBFL_element}"
  done
  return 1
}

all_distinct3(){
  # Returns 1 if all the arguments strings are distinct.
  # all_distinct3 is slower than old all_distinct2 since a call to sort is
```

```
  # needed:
  # the manual of uniq says it doesn't detect non-adjacent repeated lines.
  # A few weeks ago, when testing without sort, I had no bug...
  # I may have tested all_distinct3 a a b and not all_distinct3 a b a
  # There is a sort also in all_distinct2, but uniq could have an option to
  # sort before "applying -d", or sort could have an option for printing
  # duplicates only. It would complement -u option of sort.
  # Both solutions would avoid one pipe and one more shell command call.
  # Note that old all_distinct2 was faster with pipes and read -r commands,
  # but read -r has small drawbacks. Using mapfile is a cleaner way to do
  # it, since it handles arguments with spaces from scratch.
  # Arguments with line-returns are still not handled.
  # Use C, PHP or Python for example for that.
  # Somehow Bash was made to push people to stop being lazy and do actual C
  # code, even if it needs compiling something, adding a makefile, etc. XD
  # Note that C code would be faster with all_distinct2 blueprint,
  # that's the reason I kept all 3 versions, see the (asymptotic)
  # complexity and simple algorithms ideas on this toy problem.
  if [[ $# -le 1 ]]; then
    return 1
  fi
  declare -r LFBFL_s_duplicated_arguments=$(
    printf "%s\n" "$@" \
    | sort\
    | uniq -d
  )
  [[ -n "${LFBFL_s_duplicated_arguments}" ]]
}

some_equal(){
  ! all_distinct "$@"
}

some_equal2(){
  ! all_distinct2 "$@"
}

some_equal3(){
  ! all_distinct3 "$@"
}

max(){
  # Returns the maximum string among the arguments.
  # Renvoie la chaîne la plus grande parmi celles proposées.
  # $1=sort_command with the flags you want to be applied.
  # $ max sort a b c
```

```
  # c
  if [[ $# -eq 0 ]]; then
    return 1
  fi
  enhanced_set_shell_option pipefail\
    && trap 'enhanced_unset_shell_option pipefail' RETURN
  printf "%s\n" "${@:2}" \
    | eval "$1 -r"\
    | head -1
}

min(){
  # Returns the minimum string among the arguments.
  # Renvoie la chaîne la plus petite parmi celles proposées.
  # $1=sort_command with the flags you want to be applied.
  # $ min sort a b c
  # a
  if [[ $# -eq 0 ]]; then
    return 1
  fi
  enhanced_set_shell_option pipefail\
    && trap 'enhanced_unset_shell_option pipefail' RETURN
  printf "%s\n" "${@:2}" \
    | eval "$1"\
    | head -1
}

is_substring(){
  # $1=string
  # $2=substring
  declare -r LFBFL_var_1=$(
    printf "%s" "$1"\
    | sed -e 's/\\/\\&/g' -e "s/'/\\\&/g"
  )
  declare -r LFBFL_var_2=$(
    printf "%s" "$2"\
    | sed -e 's/\\/\\&/g' -e "s/'/\\\&/g"
  )
  local LFBFL_var_3="\$a = '${LFBFL_var_1}';"
  LFBFL_var_3+=" \$b = preg_quote('${LFBFL_var_2}');"
  LFBFL_var_3+=" \$c = '/'.addcslashes(\$b, '/').'/';"
  LFBFL_var_3+=" die(1 - preg_match(\$c, \$a));"
  readonly LFBFL_var_3
  php -r "${LFBFL_var_3}"
  return $?
}
```

```

is_subfile(){
  # $1=file_path
  # $2=sub_file_path
  local LFBFL_var_1="\$a = file_get_contents('$1');"
  LFBFL_var_1+=" \$b = preg_quote(file_get_contents('$2'));"
  LFBFL_var_1+=" \$c = '/'.addcslashes(\$b, '/').'/';"
  LFBFL_var_1+=" die(1 - preg_match(\$c, \$a));"
  readonly LFBFL_var_1
  php -r "${LFBFL_var_1}"
  return $?
}

alias min_from_int_sort="min 'sort --numeric-sort'"
alias max_from_int_sort="max 'sort --numeric-sort'"
```

3.11 build_and_checks_dependencies/create_PDF.exec.sh

```
#!/usr/bin/env bash
# This file is part of DevOrSysAdminScripts library.
#
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# of the GNU General Public License
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#
# ©Copyright 2023-2026 Laurent Frédéric Bernard François Lyaudet
# This file was renamed from "create_PDF.sh" to "create_PDF.exec.sh".

LFBFL_subdir="build_and_checks_dependencies"
# shellcheck source=common_options.libr.sh
source "./${LFBFL_subdir}/common_options.libr.sh"
```

```
29 # shellcheck source=generate_from_template.libr.sh
30 source "./${LFBFL_subdir}/generate_from_template.libr.sh"
31 # shellcheck source=lines_counts.libr.sh
32 source "./${LFBFL_subdir}/lines_counts.libr.sh"
33 # shellcheck source=lines_filters.libr.sh
34 source "./${LFBFL_subdir}/lines_filters.libr.sh"
35 # shellcheck source=lines_maps.libr.sh
36 source "./${LFBFL_subdir}/lines_maps.libr.sh"
37 # shellcheck source=overwrite_if_not_equal.libr.sh
38 source "./${LFBFL_subdir}/overwrite_if_not_equal.libr.sh"
39 # shellcheck source=strings_functions.libr.sh
40 source "./${LFBFL_subdir}/strings_functions.libr.sh"
41
42 create_PDF(){
43   # Options:
44   #   --verbose
45   #   --work-directory=""
46   declare -i LFBFL_i_verbose=0
47   get_verbose_option "$@"
48   local LFBFL_work_directory=""
49   get_work_directory_option "$@"
50   declare -a LFBFL_return_traps_stack
51   local LFBFL_previous_return_trap
52   init_return_trap
53   pushd_to_work_directory --trap-popd
54   can_continue_after_enhanced_pushd || return 1
55
56   enhanced_set_shell_option pipefail --trap-unset
57
58   declare -a LFBFL_common_options=("$@")
59   keep_options LFBFL_common_options --verbose
60
61   local LFBFL_subdir2="build_and_checks_variables"
62   local LFBFL_data_file_name="${LFBFL_subdir2}/repository_data.txt"
63   local LFBFL_repository_name=""
64   grep_variable "${LFBFL_data_file_name}" repository_name\
65     --result-variable-prefix="LFBFL_"\
66     --replace-line-returns-by=""
67
68   local LFBFL_repository_web_url=""
69   grep_variable "${LFBFL_data_file_name}" repository_web_url\
70     --result-variable-prefix="LFBFL_"\
71     --replace-line-returns-by=""
72
73   local LFBFL_repository_git_url=""
74   grep_variable "${LFBFL_data_file_name}" repository_git_url\
```

```
75      --result-variable-prefix="LFBFL_"\
76      --replace-line-returns-by=""
77
78    local LFBFL_abstract=""
79    grep_variable "${LFBFL_data_file_name}" abstract\
80      --result-variable-prefix="LFBFL_"
81
82    local LFBFL_acknowledgments=""
83    grep_variable "${LFBFL_data_file_name}" acknowledgments\
84      --result-variable-prefix="LFBFL_"
85
86    local LFBFL_author_email=""
87    grep_variable "${LFBFL_data_file_name}" author_email\
88      --result-variable-prefix="LFBFL_"\
89      --replace-line-returns-by=""
90
91    local LFBFL_author_full_name=""
92    grep_variable "${LFBFL_data_file_name}" author_full_name\
93      --result-variable-prefix="LFBFL_"\
94      --replace-line-returns-by=" "
95
96    local LFBFL_author_website=""
97    grep_variable "${LFBFL_data_file_name}" author_website\
98      --result-variable-prefix="LFBFL_"\
99      --replace-line-returns-by=""
100
101   declare -i LFBFL_i_max_line_length
102   grep_variable "${LFBFL_data_file_name}" max_line_length\
103     --result-variable-prefix="LFBFL_i_"
104
105   declare -r LFBFL_current_date=$(date -I"date")
106
107   declare -r LFBFL_current_git_SHA1=$(git rev-parse HEAD)
108
109   declare -ir LFBFL_i_number_of_commits=$(
110     git shortlog\
111     | space_starting_lines\
112     | wc -l
113   )
114
115   local LFBFL_redirect="/dev/stdout"
116   if [[ LFBFL_i_verbose -eq 0 ]]; then
117     LFBFL_redirect="/dev/null"
118   fi
119   {
120     code_lines_count_all "${LFBFL_common_options[@]}"
```

```
    code_lines_count_not_empty "${LFBFL_common_options[@]}"
    code_lines_count_empty "${LFBFL_common_options[@]}"
  } &> "${LFBFL_redirect}"

  local LFBFL_number_of_lines="${i_code_lines_count_all_result}"
  LFBFL_number_of_lines+=" total lines,"
  LFBFL_number_of_lines+=" ${i_code_lines_count_not_empty_result}"
  LFBFL_number_of_lines+=" not empty lines,"
  LFBFL_number_of_lines+=" ${i_code_lines_count_empty_result}"
  LFBFL_number_of_lines+=" empty lines."
  readonly LFBFL_number_of_lines

  declare -r LFBFL_temp_path="./${LFBFL_subdir2}/temp"

  # see https://gitlab.com/OldManProgrammer/unix-tree/-/issues
  # ?show=eyJpaWQiOiI0MyIsImZ1bGxfcGF0aCI6Ik9sZE1hblByb2dyYW1t
  # ZXIvdW5peC10cmVlIiwiaWQiOjE4NTE0NTgzOH0%3D
  tree -a --gitignore\
    -I "node_modules/"\
    -I "__pycache__/"\
    -I ".mypy_cache/"\
    -I ".ruff_cache/"\
    -I ".git/"\
    | replace_non_ascii_spaces\
    > "${LFBFL_temp_path}/current_tree_light.txt"

  tree -a -DFh --gitignore\
    -I "node_modules/"\
    -I "__pycache__/"\
    -I ".mypy_cache/"\
    -I ".ruff_cache/"\
    -I ".git/"\
    | replace_non_ascii_spaces\
    > "${LFBFL_temp_path}/current_tree.txt.temp"

  local LFBFL_s_files_paths
  # Remove line returns that are here to keep lines short.
  LFBFL_s_files_paths=$(
    sed -Ez 's/(.)\\\n/\1/Mg' "./${LFBFL_subdir2}/files_names_listing.txt"\
    | grep -v '^//'
  )
  readonly LFBFL_s_files_paths
  declare -a LFBFL_arr_files_paths
  if [[ -n "${LFBFL_s_files_paths}" ]]; then
    mapfile -t LFBFL_arr_files_paths <<< "${LFBFL_s_files_paths}"
  fi
```

```
167     readonly LFBFL_arr_files_paths
168
169     local LFBFL_license=""
170     grep_variable "${LFBFL_data_file_name}" license\
171       --result-variable-prefix="LFBFL_"
172
173     local LFBFL_temp_files_listing="${LFBFL_temp_path}/"
174     LFBFL_temp_files_listing+="files_listing.tex.tpl.temp"
175     readonly LFBFL_temp_files_listing
176     cp "${LFBFL_temp_path}/license_file_header_${LFBFL_license}.tex"\
177       "${LFBFL_temp_files_listing}"
178     local LFBFL_temp_files_listing2="${LFBFL_temp_path}/"
179     LFBFL_temp_files_listing2+="files_listing.html.tpl.temp"
180     readonly LFBFL_temp_files_listing2
181     : > "${LFBFL_temp_files_listing2}"
182     # HTML <li> elements, hence "lis".
183     local LFBFL_temp_files_lis
184     LFBFL_temp_files_lis="${LFBFL_temp_path}/files_lis.html.tpl"
185     readonly LFBFL_temp_files_lis
186     : > "${LFBFL_temp_files_lis}"
187     # shellcheck disable=SC2248
188     get_split_score_simple 1 ${LFBFL_i_max_line_length} /
189     declare -r LFBFL_score_command="${get_split_score_result}"
190     declare -i LFBFL_i_score_command_properties
191     LFBFL_i_score_command_properties=${i_get_split_score_result2}
192     readonly LFBFL_i_score_command_properties
193     # shellcheck disable=SC2248
194     get_split_score_simple 1 ${LFBFL_i_max_line_length} ':'
195     declare -r LFBFL_score_command2="${get_split_score_result}"
196     declare -i LFBFL_i_score_command_properties2
197     LFBFL_i_score_command_properties2=${i_get_split_score_result2}
198     readonly LFBFL_i_score_command_properties2
199     declare -r LFBFL_suffix='%'
200     declare -i LFBFL_i=0
201     local LFBFL_cleaned_path1
202     local LFBFL_cleaned_path2
203     local LFBFL_new_lines
204     local LFBFL_new_lines2
205     local LFBFL_new_lines3
206     local LFBFL_file_path
207     local LFBFL_s_format
208     for LFBFL_file_path in "${LFBFL_arr_files_paths[@]}"; do
209       printf "Listing file for tex/HTML : %s\n" "${LFBFL_file_path}"
210       # printf "Listing file for tex : %s\n" "${LFBFL_file_path}"
211       # LFBFL_base_file_name=$(basename "${LFBFL_file_path}")
212       LFBFL_cleaned_path1="${LFBFL_file_path//_/\\_}"
```

```
    LFBFL_cleaned_path2=$(
      printf "%s" "${LFBFL_file_path}"\
      | sed -e 's/\//:/g' -e 's/\.//g'
    )
    LFBFL_new_lines="  ${LFBFL_cleaned_path1}"
    if [[ ${#LFBFL_new_lines} -gt LFBFL_i_max_line_length ]]; then
      # shellcheck disable=SC2248
      repeated_split_last_line "${LFBFL_new_lines}"\
        ""\
        ${LFBFL_i_max_line_length}\
        "${LFBFL_suffix}"\
        "${LFBFL_score_command}"\
        ${LFBFL_i_score_command_properties}\
        "\\"
      LFBFL_new_lines=${repeated_split_last_line_result}
    fi
    LFBFL_new_lines2="  ${LFBFL_cleaned_path2}"
    if [[ ${#LFBFL_new_lines2} -gt LFBFL_i_max_line_length ]]; then
      # shellcheck disable=SC2248
      repeated_split_last_line "${LFBFL_new_lines2}"\
        ""\
        ${LFBFL_i_max_line_length}\
        "${LFBFL_suffix}"\
        "${LFBFL_score_command2}"\
        ${LFBFL_i_score_command_properties2}
      LFBFL_new_lines2=${repeated_split_last_line_result}
    fi
    LFBFL_new_lines3="${LFBFL_file_path}"
    if [[ ${#LFBFL_new_lines3} -gt LFBFL_i_max_line_length ]]; then
      # shellcheck disable=SC2248
      repeated_split_last_line "${LFBFL_new_lines3}"\
        ""\
        ${LFBFL_i_max_line_length}\
        "${LFBFL_suffix}"\
        "${LFBFL_score_command}"\
        ${LFBFL_i_score_command_properties}
      LFBFL_new_lines3=${repeated_split_last_line_result}
    fi
    LFBFL_s_format="\\subsection{\n%s\n}\n\\label{\n%s\n}\n\n"
    LFBFL_s_format+="\\VerbatimInput[numbers=left,xleftmargin=-5mm]"
    LFBFL_s_format+="{\n%s\n}\n\n\n"
    # shellcheck disable=SC2059
    printf "${LFBFL_s_format}"\
      "${LFBFL_new_lines}"\
      "${LFBFL_new_lines2}"\
      "${LFBFL_new_lines3}"\
```

```
      >> "${LFBFL_temp_files_listing}"

    # printf "Listing file for HTML : %s\n" "${LFBFL_file_path}"
    ((++LFBFL_i))
    LFBFL_new_lines="${LFBFL_file_path}"
    if [[ ${#LFBFL_file_path} -gt LFBFL_i_max_line_length ]]; then
      # shellcheck disable=SC2248
      repeated_split_last_line "${LFBFL_new_lines}"\
        "-->"\
        ${LFBFL_i_max_line_length}\
        "<!--"\
        "${LFBFL_score_command}"\
        ${LFBFL_i_score_command_properties}
      LFBFL_new_lines=${repeated_split_last_line_result}
    fi
    {
      LFBFL_s_format="<h3 id=\"subsection3.%s\">3.%s\n%s\n</h3>\n"
      LFBFL_s_format+="<pre class=\"numbered_lines\">\n"
      # shellcheck disable=SC2059
      printf "${LFBFL_s_format}"\
        "${LFBFL_i}"\
        "${LFBFL_i}"\
        "${LFBFL_new_lines}"
      sed -e 's/&/\&amp;/g' -e 's/</\&lt;/g'\
        < "${LFBFL_file_path}"
      printf "</pre>\n\n\n"
    } >> "${LFBFL_temp_files_listing2}"
    LFBFL_s_format="      <li><a href=\"#subsection3.%s\">\n%s\n"
    LFBFL_s_format+="      </a></li>\n"
    # shellcheck disable=SC2059
    printf "${LFBFL_s_format}" "${LFBFL_i}" "${LFBFL_new_lines}"\
      >> "${LFBFL_temp_files_lis}"
  done
  overwrite_if_not_equal "./${LFBFL_subdir2}/files_listing.tex.tpl"\
    "${LFBFL_temp_files_listing}"
  sed -i -e "s|\\n</pre>|</pre>|g" "${LFBFL_temp_files_listing2}"
  overwrite_if_not_equal\
    "${LFBFL_temp_path}/files_listing.html.tpl"\
    "${LFBFL_temp_files_listing2}"

  # We verify if some lines are beyond max_line_length characters
  # in current_tree_light.txt and current_tree.txt.
  declare -ar LFBFL_trees=(
    "current_tree_light.txt"
    "current_tree.txt.temp"
  )
```

```
  local LFBFL_tree_path
  local LFBFL_line
  local LFBFL_prefix
  declare -i LFBFL_i_prefix_last_position
  local LFBFL_char
  local LFBFL_line_start
  declare -ir LFBFL_i_overlength=$((LFBFL_i_max_line_length+1))
  local LFBFL_file_name
  local LFBFL_s_lines
  declare -a LFBFL_arr_lines
  declare -i LFBFL_i_keep_length
  for LFBFL_tree in "${LFBFL_trees[@]}"; do
    LFBFL_tree_path="${LFBFL_temp_path}/${LFBFL_tree}"
    # shellcheck disable=SC2248
    LFBFL_s_lines=$(
      grep '.\{'${LFBFL_i_overlength}'\}' "${LFBFL_tree_path}"
    )
    if [[ -z "${LFBFL_s_lines}" ]]; then
      continue
    fi
    mapfile -t LFBFL_arr_lines <<< "${LFBFL_s_lines}"
    for LFBFL_line in "${LFBFL_arr_lines[@]}"; do
      # printf "LFBFL_line: %s\n" "${LFBFL_line}"
      LFBFL_prefix=$(
        printf "%s" "${LFBFL_line}"\
        | sed -E -e 's/(.*)——[^—]+$/\1/g'
      )
      # printf "LFBFL_prefix: %s\n" "${LFBFL_prefix}"
      LFBFL_i_prefix_last_position=$((${#LFBFL_prefix}-1))
      LFBFL_char="${LFBFL_prefix:${LFBFL_i_prefix_last_position}:1}"
      # printf "LFBFL_char: %s\n" "${LFBFL_char}"
      LFBFL_prefix=$(
        printf "%s" "${LFBFL_prefix}"\
        | sed -E -e 's/[^ ]+$//g'
      )
      if [[ "${LFBFL_char}" == "└" ]]; then
        LFBFL_prefix+="  "
      else
        LFBFL_prefix+="│ "
      fi
      # printf "LFBFL_prefix: %s\n" "${LFBFL_prefix}"
      LFBFL_file_name=$(
        printf "%s" "${LFBFL_line}"\
        | sed -E 's|.* ([^ ]+)$|\1|g'
      )
      # printf "LFBFL_file_name: %s\n" "${LFBFL_file_name}"
```

```
      LFBFL_i_keep_length=$((${#LFBFL_line} - ${#LFBFL_file_name}))
      LFBFL_line_start=${LFBFL_line:0:${LFBFL_i_keep_length}}
      LFBFL_line_start=$(
        printf "%s" "${LFBFL_line_start}"\
        | sed -e 's/ *$//g'
      )
      # printf "LFBFL_line_start: %s\n" "${LFBFL_line_start}"
      LFBFL_new_lines="${LFBFL_prefix}${LFBFL_file_name}"
      if [[ ${#LFBFL_new_lines} -gt LFBFL_i_max_line_length ]]; then
        # shellcheck disable=SC2248
        repeated_split_last_line "${LFBFL_new_lines}"\
          "${LFBFL_prefix}"\
          ${LFBFL_i_max_line_length}\
          ""\
          ""\
          ""
        LFBFL_new_lines=${repeated_split_last_line_result}
      fi
      printf "%s\n%s\n" "${LFBFL_line_start}" "${LFBFL_new_lines}"\
        > "${LFBFL_tree_path}.some_line.temp"
      insert_file_at_token "${LFBFL_tree_path}"\
        "${LFBFL_line}"\
        "${LFBFL_tree_path}.some_line.temp"\
        ""\
        --quiet
    done
    rm -f "${LFBFL_tree_path}.some_line.temp"
  done

  overwrite_if_not_equal "${LFBFL_temp_path}/current_tree.txt"\
    "${LFBFL_temp_path}/current_tree.txt.temp" 1 1

  declare -r LFBFL_tex_path_start=\
"${LFBFL_temp_path}/${LFBFL_repository_name}.tex"
  declare -r LFBFL_html_path_start=\
"${LFBFL_temp_path}/${LFBFL_repository_name}.html"

  if [[ -f "./${LFBFL_subdir2}/${LFBFL_repository_name}.tex.tpl" ]];
  then
    # If there is a template, we init the process from it.
    cp "./${LFBFL_subdir2}/${LFBFL_repository_name}.tex.tpl"\
      "${LFBFL_tex_path_start}.1"
  elif [[ -f "./${LFBFL_subdir2}/${LFBFL_repository_name}.tex" ]];
  then
    # Otherwise if there is a tex file, we init the process from it.
    cp "./${LFBFL_subdir2}/${LFBFL_repository_name}.tex"\
```

```
      "${LFBFL_tex_path_start}.1"
  else
    printf "Neither .tex.tpl, nor .tex in ./%s/\n" "${LFBFL_subdir2}"
  fi

  # Same logic with repository HTML file.
  if [[ -f "./${LFBFL_subdir2}/${LFBFL_repository_name}.html.tpl" ]];
  then
    cp "./${LFBFL_subdir2}/${LFBFL_repository_name}.html.tpl"\
      "${LFBFL_html_path_start}.1"
  elif [[ -f "./${LFBFL_repository_name}.html" ]]; then
    cp "./${LFBFL_repository_name}.html" "${LFBFL_html_path_start}.1"
  else
    printf "Neither .html.tpl in ./%s/, nor .html in ./\n"\
      "${LFBFL_subdir2}"
  fi

  # shellcheck disable=SC2001
  printf "%s\n" "${LFBFL_abstract}"\
    | sed -e 's/\\n/\n/g'\
    > "${LFBFL_temp_path}/abstract_temp"

  # shellcheck disable=SC2001
  printf "%s\n" "${LFBFL_acknowledgments}"\
    | sed -e 's/\\n/\n/g'\
    > "${LFBFL_temp_path}/acknowledgments_temp"

  # But the filling still occurs, in case the dev want to refill
  # part of the tex/html file that he modified with @token@
  # using some computed results.
  # Tex filling:
  if [[ -f "${LFBFL_tex_path_start}.1" ]]; then
    sed -i -e "s|@repository_name@|${LFBFL_repository_name}|g"\
           -e "s|@repository_web_url@|${LFBFL_repository_web_url}|g"\
           -e "s|@repository_git_url@|${LFBFL_repository_git_url}|g"\
           -e "s|@author_email@|${LFBFL_author_email}|g"\
           -e "s|@author_full_name@|${LFBFL_author_full_name}|g"\
           -e "s|@author_website@|${LFBFL_author_website}|g"\
           -e "s|@current_date@|${LFBFL_current_date}|g"\
           -e "s|@current_git_SHA1@|${LFBFL_current_git_SHA1}|g"\
           -e "s|@number_of_commits@|${LFBFL_i_number_of_commits}|g"\
           -e "s|@number_of_lines@|${LFBFL_number_of_lines}|g"\
      "${LFBFL_tex_path_start}.1"

    insert_file_at_token "${LFBFL_tex_path_start}.1" @abstract@\
      "${LFBFL_temp_path}/abstract_temp"\
```

```
      "${LFBFL_tex_path_start}.2"

    insert_file_at_token "${LFBFL_tex_path_start}.2"\
      @acknowledgments@\
      "${LFBFL_temp_path}/acknowledgments_temp"\
      "${LFBFL_tex_path_start}.3"

    insert_file_at_token "${LFBFL_tex_path_start}.3"\
      @current_tree_light@\
      "${LFBFL_temp_path}/current_tree_light.txt"\
      "${LFBFL_tex_path_start}.4"

    insert_file_at_token "${LFBFL_tex_path_start}.4"\
      @current_tree@\
      "${LFBFL_temp_path}/current_tree.txt"\
      "${LFBFL_tex_path_start}.5"

    insert_file_at_token "${LFBFL_tex_path_start}.5"\
      @dependencies_notes@\
      "./${LFBFL_temp_path}/dependencies_notes.tex"\
      "${LFBFL_tex_path_start}.6"

    insert_file_at_token "${LFBFL_tex_path_start}.6"\
      @files_listing_VerbatimInput@\
      "./${LFBFL_subdir2}/files_listing.tex.tpl"\
      "${LFBFL_tex_path_start}.7"

    overwrite_if_not_equal\
      "./${LFBFL_subdir2}/${LFBFL_repository_name}.tex"\
      "${LFBFL_tex_path_start}.7" 1
  fi

  # HTML filling:
  declare -i LFBFL_i_HTML_updated=0
  if [[ -f "${LFBFL_html_path_start}.1" ]]; then
    sed -i -e "s|@repository_name@|${LFBFL_repository_name}|g"\
           -e "s|@repository_web_url@|${LFBFL_repository_web_url}|g"\
           -e "s|@repository_git_url@|${LFBFL_repository_git_url}|g"\
           -e "s|@author_email@|${LFBFL_author_email}|g"\
           -e "s|@author_full_name@|${LFBFL_author_full_name}|g"\
           -e "s|@author_website@|${LFBFL_author_website}|g"\
           -e "s|@current_date@|${LFBFL_current_date}|g"\
           -e "s|@current_git_SHA1@|${LFBFL_current_git_SHA1}|g"\
           -e "s|@number_of_commits@|${LFBFL_i_number_of_commits}|g"\
           -e "s|@number_of_lines@|${LFBFL_number_of_lines}|g"\
      "${LFBFL_html_path_start}.1"
```

```
    insert_file_at_token "${LFBFL_html_path_start}.1" @abstract@\
      "${LFBFL_temp_path}/abstract_temp"\
      "${LFBFL_html_path_start}.2"

    insert_file_at_token "${LFBFL_html_path_start}.2"\
      @acknowledgments@\
      "${LFBFL_temp_path}/acknowledgments_temp"\
      "${LFBFL_html_path_start}.3"

    insert_file_at_token "${LFBFL_html_path_start}.3"\
      @files_lis@ "${LFBFL_temp_files_lis}"\
      "${LFBFL_html_path_start}.4"

    insert_file_at_token "${LFBFL_html_path_start}.4"\
      @current_tree_light@\
      "${LFBFL_temp_path}/current_tree_light.txt"\
      "${LFBFL_html_path_start}.5"

    insert_file_at_token "${LFBFL_html_path_start}.5"\
      @current_tree@\
      "${LFBFL_temp_path}/current_tree.txt"\
      "${LFBFL_html_path_start}.6"

    insert_file_at_token "${LFBFL_html_path_start}.6"\
      @dependencies_notes@\
      "./${LFBFL_temp_path}/dependencies_notes.html"\
      "${LFBFL_html_path_start}.7"

    insert_file_at_token "${LFBFL_html_path_start}.7"\
      @files_listing_HTMLPreInput@\
      "${LFBFL_temp_path}/files_listing.html.tpl"\
      "${LFBFL_html_path_start}.8"

    overwrite_if_not_equal "./${LFBFL_repository_name}.html"\
      "${LFBFL_html_path_start}.8" 1
    LFBFL_i_HTML_updated=$?
  fi

  # The HTML contains everything including listings with source code.
  # If it doesn't get updated,
  # then it is useless to recompute the PDF.
  if [[ LFBFL_i_HTML_updated -eq 0 ]]; then
    return
  fi

```

```
  local LFBFL_latex_command=""
  grep_variable "${LFBFL_data_file_name}" latex_command\
    --result-variable-prefix="LFBFL_"\
    --replace-line-returns-by=""

  case ${LFBFL_latex_command} in
    pdflatex) : ;;
    lualatex) : ;;
    *) printf "Unsupported latex command" ; return 1 ;;
  esac

  local LFBFL_command
  LFBFL_command="${LFBFL_latex_command}"
  LFBFL_command+=" './${LFBFL_subdir2}/${LFBFL_repository_name}.tex'"
  if [[ LFBFL_i_verbose -eq 0 ]]; then
    LFBFL_command+=" > /dev/null"
  fi
  readonly LFBFL_command

  for ((i=0; i<3; i++)); do
    eval "${LFBFL_command}"
  done

  LFBFL_files_to_temp=(\
    "${LFBFL_repository_name}.aux"\
    "${LFBFL_repository_name}.log"\
    "${LFBFL_repository_name}.out"\
    "${LFBFL_repository_name}.toc"
  )
  for LFBFL_file_name in "${LFBFL_files_to_temp[@]}"; do
    mv "${LFBFL_file_name}" "${LFBFL_temp_path}/"
  done
}

create_PDF "$@"
```

3.12 build_and_checks_dependencies/generate_from_template.libr.sh

```
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```

```
# or (at your option) any later version.
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# This file was renamed from "generate_from_template.sh" to
# "generate_from_template.libr.sh".

LFBFL_subdir="build_and_checks_dependencies"
# shellcheck source=lines_filters.libr.sh
source "./${LFBFL_subdir}/lines_filters.libr.sh"
# shellcheck source=overwrite_if_not_equal.libr.sh
source "./${LFBFL_subdir}/overwrite_if_not_equal.libr.sh"

generate_from_template_with_block_comments(){
  # $1=base_file_path
  # $2=target_file_path
  # $3=enter_block_comment
  # $4=exit_block_comment
  local LFBFL_temp
  LFBFL_temp="$2.generate_from_template_with_block_comments.temp"
  readonly LFBFL_temp
  {
    printf "%s\n" "$3"
    cat "$1"
    printf "%s\n" "$4"
  } > "${LFBFL_temp}"
  overwrite_if_not_equal "$2" "${LFBFL_temp}"
}

generate_from_template_with_line_comments(){
  # $1=base_file_path
  # $2=target_file_path
  # $3=line_comment_prefix
  # $4=optional_post_processing
  # $5=file_prefix
```

```
55    local LFBFL_temp
56    LFBFL_temp="$2.generate_from_template_with_line_comments.temp"
57    readonly LFBFL_temp
58    if [[ -n "$5" ]]; then
59      printf "%s\n" "$5"\
60        > "${LFBFL_temp}"
61    fi
62    sed -e "s/^/$3/g" "$1"\
63      >> "${LFBFL_temp}"
64    if [[ -n "$4" ]]; then
65      eval "$4"
66    fi
67    overwrite_if_not_equal "$2" "${LFBFL_temp}"
68  }
69
70  split_file_in_two(){
71    # $1=file_path
72    # $2=token assume that token is the only thing on his line,
73    #    except for a possible indentation.
74    # $3=path_of_part1_of_file
75    # $4=path_of_part2_of_file
76    # We cannot add --line-regexp with --fixed-strings because of the
77    # possible indentation in file.
78    declare -ir LFBFL_i_line_number=$(
79      grep_fixed_string_with_anchor "$1" "$2"\
80        --enforce-line-ends-with-fixed-string --fixed-strings --line-number\
81      | head --lines=1\
82      | cut -f 1 -d ':'
83    )
84    declare -ir LFBFL_i_line_count=$(ll_wc -l -n "$1")
85    declare -ir LFBFL_i_lines_after=$((
86      LFBFL_i_line_count - LFBFL_i_line_number
87    ))
88    head --lines="$((LFBFL_i_line_number - 1))" "$1"\
89      > "$3"
90    tail --lines="${LFBFL_i_lines_after}" "$1"\
91      > "$4"
92  }
93
94  insert_file_at_token(){
95    # $1=file_path
96    # $2=token assume that token is the only thing on his line,
97    #    except for a possible indentation.
98    # $3=path_of_file_to_insert
99    # $4=path_of_result_file
100   # Options:
```

```
  #   $5=--quiet to suppress diff output in overwrite_if_not_equal
  declare -r LFBFL_start_file_name="$1.insert_file_at_token1.temp"
  declare -r LFBFL_end_file_name="$1.insert_file_at_token2.temp"
  local LFBFL_result_file_name
  if [[ -n "$4" ]]; then
    LFBFL_result_file_name="$4"
  else
    LFBFL_result_file_name="$1.insert_file_at_token3.temp"
  fi
  readonly LFBFL_result_file_name
  split_file_in_two "$1" "$2" "${LFBFL_start_file_name}"\
    "${LFBFL_end_file_name}"
  cat "${LFBFL_start_file_name}" "$3" "${LFBFL_end_file_name}"\
    > "${LFBFL_result_file_name}"
  rm "${LFBFL_start_file_name}" "${LFBFL_end_file_name}"
  if [[ -z "$4" ]]; then
    overwrite_if_not_equal "$1" "${LFBFL_result_file_name}" "" "" "$5"
  fi
}
```

3.13 build_and_checks_dependencies/get_common_text_glob_patterns.libr.sh

```
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# This file was renamed from "get_common_text_glob_patterns.sh"
```

```
# to "get_common_text_glob_patterns.libr.sh".

get_COMMON_TEXT_PATHS_GLOB_PATTERNS(){
  if [[ -n "${COMMON_TEXT_PATHS_GLOB_PATTERNS}" ]]; then
    return
  fi

  declare -agr COMMON_TEXT_PATHS_GLOB_PATTERNS=(
    '**/*.c'
    '**/*.css'
    '**/*.h'
    '**/*.htm'
    '**/*.html'
    '**/*.ini'
    '**/*.js'
    '**/*.json'
    '**/*.md'
    '**/*.php'
    '**/*.py'
    '**/*.sh'
    '**/*.sql'
    '**/*.tex'
    '**/*.toml'
    '**/*.tpl'
    '**/*.ts'
    '**/*.txt'
    '**/*.xml'
    '**/*.yml'
    '**/COPYING'
    '**/COPYING.LESSER'
    '**/pre-commit'
    '**/post-commit'
    '**/.gitignore'
    '**/py.typed'
  )
}


get_COMMON_TEXT_FILES_GLOB_PATTERNS(){
  if [[ -n "${COMMON_TEXT_FILES_GLOB_PATTERNS}" ]]; then
    return
  fi

  declare -agr COMMON_TEXT_FILES_GLOB_PATTERNS=(
    '*.c'
    '*.css'
```

```
    '*.h'
    '*.htm'
    '*.html'
    '*.ini'
    '*.js'
    '*.json'
    '*.md'
    '*.php'
    '*.py'
    '*.sh'
    '*.sql'
    '*.tex'
    '*.toml'
    '*.tpl'
    '*.ts'
    '*.txt'
    '*.xml'
    '*.yml'
    'COPYING'
    'COPYING.LESSER'
    'pre-commit'
    'post-commit'
    '.gitignore'
    'py.typed'
  )
}
```

3.14 build_and_checks_dependencies/grammar_and_spelling_check.libr.sh

```
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#
```

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# ©Copyright 2023-2026 Laurent Frédéric Bernard François Lyaudet
# 2026-03-22T15:30+01:00: This file was renamed from
# "grammar_and_spell_check.libr.sh"
# to
# "grammar_and_spelling_check.libr.sh".
# This file was renamed from "grammar_and_spell_check.sh"
# to "grammar_and_spell_check.libr.sh".

LFBFL_subdir="build_and_checks_dependencies"
# shellcheck source=common_options.libr.sh
source "./${LFBFL_subdir}/common_options.libr.sh"
# shellcheck source=get_common_text_glob_patterns.libr.sh
source "./${LFBFL_subdir}/get_common_text_glob_patterns.libr.sh"
# shellcheck source=lines_filters.libr.sh
source "./${LFBFL_subdir}/lines_filters.libr.sh"

grammar_and_spelling_check(){
  # $1=configuration_path build_and_checks_variables/repository_data.txt
  # Options:
  #   --verbose
  #   --work-directory=""
  declare -i LFBFL_i_verbose=0
  get_verbose_option "$@"
  local LFBFL_work_directory=""
  get_work_directory_option "$@"
  declare -a LFBFL_return_traps_stack
  local LFBFL_previous_return_trap
  init_return_trap
  pushd_to_work_directory --trap-popd
  can_continue_after_enhanced_pushd || return 1

  enhanced_set_shell_option pipefail --trap-unset

  get_COMMON_TEXT_FILES_GLOB_PATTERNS
  local LFBFL_grammar_and_spelling_checker_command=""
  grep_variable "$1" grammar_and_spelling_checker_command\
    --result-variable-prefix=LFBFL_\
    --replace-line-returns-by=""
  declare -r LFBFL_command="${LFBFL_grammar_and_spelling_checker_command}"
  if [[ -z "${LFBFL_command}" ]]; then
    if [[ LFBFL_i_verbose -eq 1 ]]; then
```

```
      printf "grammar_and_spelling_check: No command in configuration.\n"
    fi
    return
  fi
  if [[ LFBFL_i_verbose -eq 1 ]]; then
    printf "grammar_and_spelling_check: Command found: %s.\n"\
      "${LFBFL_command}"
  fi
  local LFBFL_file_path
  local LFBFL_s_files_paths
  declare -a LFBFL_arr_files_paths
  local LFBFL_pattern
  local LFBFL_eval_string
  for LFBFL_pattern in "${COMMON_TEXT_FILES_GLOB_PATTERNS[@]}"; do
    if [[ LFBFL_i_verbose -eq 1 ]]; then
      printf "grammar_and_spelling_check: Iterating on pattern: %s.\n"\
        "${LFBFL_pattern}"
    fi
    LFBFL_s_files_paths=$(
      find . -type f -name "${LFBFL_pattern}" -printf '%P\n'\
      | relevant_find
    )
    if [[ -z "${LFBFL_s_files_paths}" ]]; then
      [[ LFBFL_i_verbose -eq 1 ]]\
        && printf "grammar_and_spelling_check: No file found.\n"
      continue
    fi
    mapfile -t LFBFL_arr_files_paths <<< "${LFBFL_s_files_paths}"
    for LFBFL_file_path in "${LFBFL_arr_files_paths[@]}"; do
      LFBFL_eval_string="${LFBFL_command} ${LFBFL_file_path}"
      eval "${LFBFL_eval_string}"
    done
  done
}
```

3.15 build_and_checks_dependencies/licenses_templates/build_licenses_templates.exec.sh

```
#!/usr/bin/env bash
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```

```
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# This file was renamed from "build_licenses_templates.sh"
# to "build_licenses_templates.exec.sh".

LFBFL_subdir="build_and_checks_dependencies"
# shellcheck source=check_shell_scripts_beginnings.libr.sh
source "./${LFBFL_subdir}/check_shell_scripts_beginnings.libr.sh"
# shellcheck source=common_options.libr.sh
source "./${LFBFL_subdir}/common_options.libr.sh"
# shellcheck source=comparisons.libr.sh
source "./${LFBFL_subdir}/comparisons.libr.sh"
# shellcheck source=generate_from_template.libr.sh
source "./${LFBFL_subdir}/generate_from_template.libr.sh"
# shellcheck source=lines_filters.libr.sh
source "./${LFBFL_subdir}/lines_filters.libr.sh"
# shellcheck source=overwrite_if_not_equal.libr.sh
source "./${LFBFL_subdir}/overwrite_if_not_equal.libr.sh"

build_licenses_templates(){
  # Options:
  #   --verbose
  #   --work-directory=""
  declare -i LFBFL_i_verbose=0
  get_verbose_option "$@"
  local LFBFL_work_directory=""
  get_work_directory_option "$@"
  declare -a LFBFL_return_traps_stack
  local LFBFL_previous_return_trap
  init_return_trap

  enhanced_set_shell_option pipefail --trap-unset

  local LFBFL_s_format
```

```
56
57    # Source
58    local LFBFL_license_source_subdir
59    LFBFL_license_source_subdir="./${LFBFL_subdir}/licenses_templates/"
60    readonly LFBFL_license_source_subdir
61    local LFBFL_source_license_prefix="${LFBFL_license_source_subdir}"
62    LFBFL_source_license_prefix+="license_file_header_"
63    readonly LFBFL_source_license_prefix
64    # Target
65    local LFBFL_license_target_subdir
66    LFBFL_license_target_subdir="${LFBFL_work_directory}/"
67    LFBFL_license_target_subdir="build_and_checks_variables/temp/"
68    readonly LFBFL_license_target_subdir
69    local LFBFL_target_license_prefix="${LFBFL_license_target_subdir}"
70    LFBFL_target_license_prefix+="license_file_header_"
71    readonly LFBFL_target_license_prefix
72
73    declare -ar LFBFL_licenses=(
74      "AGPLv3+"
75      "GPLv3+"
76      "LGPLv3+"
77    )
78
79    # ------------------------------------------------------------------
80    # Various language definitions for text file formats
81    declare -ar LFBFL_block_comment_languages=(
82      "c"
83      "py"
84    )
85    declare -ar LFBFL_block_comment_enters=(
86      '/*'
87      '"""'
88    )
89    declare -ar LFBFL_block_comment_exits=(
90      '*/'
91      '"""'
92    )
93    equal "${#LFBFL_block_comment_languages[@]}"\
94          "${#LFBFL_block_comment_enters[@]}"\
95          "${#LFBFL_block_comment_exits[@]}"
96    declare -i LFBFL_i_equal_result=$?
97    if [[ LFBFL_i_equal_result -eq 0 ]]; then
98      printf $'/!\\Problem with the definitions of languages arrays 1/!\\\n'
99      return 1
100   fi
101
```

```
102   declare -ar LFBFL_line_comment_languages=(
103     "sh"
104     "sql"
105     "tex"
106   )
107   declare -ar LFBFL_line_comment_prefixes=(
108     '# '
109     '-- '
110     '% '
111   )
112   equal "${#LFBFL_line_comment_languages[@]}"\
113         "${#LFBFL_line_comment_prefixes[@]}"
114   LFBFL_i_equal_result=$?
115   if [[ LFBFL_i_equal_result -eq 0 ]]; then
116     printf $'/!\\Problem with the definitions of languages arrays 2/!\\\n'
117     return 1
118   fi
119   # ------------------------------------------------------------------
120
121   # ------------------------------------------------------------------
122   # We generate the license header templates adapted to various text
123   # files formats.
124   # Kamoulox do endfor, Jacques Beauregard XD
125   # (C'est toujours mieux que l'inverse :) !)
126   # for do endfor -> for do done (Fort Doux Donne).
127   local LFBFL_some_license
128   local LFBFL_license_prefix
129   local LFBFL_license_prefix2
130   local LFBFL_extension
131   local LFBFL_license_file_name
132   local LFBFL_enter_string
133   local LFBFL_exit_string
134   local LFBFL_temp2
135   local LFBFL_prefix_string
136   local LFBFL_intermediate_file_name
137   local LFBFL_file_prefix
138   declare -i LFBFL_i_generate_from_template_result
139   declare -i LFBFL_i
140   for LFBFL_some_license in "${LFBFL_licenses[@]}"; do
141     LFBFL_license_prefix="${LFBFL_source_license_prefix}"
142     LFBFL_license_prefix+="${LFBFL_some_license}"
143     LFBFL_license_prefix2="${LFBFL_target_license_prefix}"
144     LFBFL_license_prefix2+="${LFBFL_some_license}"
145     for ((
146       LFBFL_i = 0;
147       LFBFL_i < ${#LFBFL_block_comment_languages[@]};
```

```
      ++LFBFL_i
    )); do
      LFBFL_extension=${LFBFL_block_comment_languages[LFBFL_i]}
      LFBFL_license_file_name="${LFBFL_license_prefix2}"
      LFBFL_license_file_name+=".${LFBFL_extension}"
      LFBFL_enter_string=${LFBFL_block_comment_enters[LFBFL_i]}
      LFBFL_exit_string=${LFBFL_block_comment_exits[LFBFL_i]}
      generate_from_template_with_block_comments\
        "${LFBFL_license_prefix}.tpl"\
        "${LFBFL_license_file_name}.tpl"\
        "${LFBFL_enter_string}" "${LFBFL_exit_string}"
      LFBFL_i_generate_from_template_result=$?
      if [[ LFBFL_i_verbose -eq 1 ]]; then
        if [[ LFBFL_i_generate_from_template_result -eq 1 ]]; then
          printf "License template %s.tpl updated.\n"\
            "${LFBFL_license_file_name}"
        elif [[ LFBFL_i_generate_from_template_result -eq 2 ]]; then
          printf "License template %s.tpl created.\n"\
            "${LFBFL_license_file_name}"
        fi
      fi
    done
    LFBFL_temp2=".generate_from_template_with_line_comments.temp"
    for ((
      LFBFL_i = 0;
      LFBFL_i < ${#LFBFL_line_comment_languages[@]};
      ++LFBFL_i
    )); do
      LFBFL_extension=${LFBFL_line_comment_languages[LFBFL_i]}
      LFBFL_license_file_name="${LFBFL_license_prefix2}"
      LFBFL_license_file_name+=".${LFBFL_extension}"
      LFBFL_prefix_string=${LFBFL_line_comment_prefixes[LFBFL_i]}
      LFBFL_intermediate_file_name="${LFBFL_license_file_name}.tpl"
      LFBFL_intermediate_file_name+="${LFBFL_temp2}"
      LFBFL_file_prefix=""
      if [[ "${LFBFL_extension}" == "sh" ]]; then
        LFBFL_file_prefix="${LFBFL_SHELL_SCRIPT_BEGINNING}"
      fi
      # The sed command is here to remove trailing spaces of previously
      # empty lines after comment prefix is added to the line.
      generate_from_template_with_line_comments\
        "${LFBFL_license_prefix}.tpl"\
        "${LFBFL_license_file_name}.tpl"\
        "${LFBFL_prefix_string}"\
        "sed -Ei -e 's/\s*$//g' '${LFBFL_intermediate_file_name}'"\
        "${LFBFL_file_prefix}"
```

```
      LFBFL_i_generate_from_template_result=$?
      if [[ LFBFL_i_verbose -eq 1 ]]; then
        if [[ LFBFL_i_generate_from_template_result -eq 1 ]]; then
          printf "License template %s.tpl updated.\n"\
            "${LFBFL_license_file_name}"
        elif [[ LFBFL_i_generate_from_template_result -eq 2 ]]; then
          printf "License template %s.tpl created.\n"\
            "${LFBFL_license_file_name}"
        fi
      fi
    done
  done
  # ------------------------------------------------------------------

  # Now that base license templates generated languages license
  # templates in working directory, we can go there.
  pushd_to_work_directory --trap-popd
  can_continue_after_enhanced_pushd || return 1

  local LFBFL_data_file_name="build_and_checks_variables/"
  LFBFL_data_file_name+="repository_data.txt"
  readonly LFBFL_data_file_name
  local LFBFL_repository_name=""
  grep_variable "${LFBFL_data_file_name}" repository_name\
    --result-variable-prefix="LFBFL_"\
    --replace-line-returns-by=""
  local LFBFL_license=""
  grep_variable "${LFBFL_data_file_name}" license\
    --result-variable-prefix="LFBFL_"
  local LFBFL_license2=""
  grep_variable "${LFBFL_data_file_name}" license2\
    --result-variable-prefix="LFBFL_"
  local LFBFL_author_full_name=""
  grep_variable "${LFBFL_data_file_name}" author_full_name\
    --result-variable-prefix="LFBFL_"\
    --replace-line-returns-by=" "

  LFBFL_license_prefix="build_and_checks_variables/temp/"
  LFBFL_license_prefix+="license_file_header_"
  readonly LFBFL_license_prefix
  LFBFL_license_prefix2="${LFBFL_license_prefix}${LFBFL_license}"
  readonly LFBFL_license_prefix2
  declare -r\
    LFBFL_license_prefix3="${LFBFL_license_prefix}${LFBFL_license2}"

  # First year according to current state of git repository.
```

```
240   declare -ir LFBFL_i_first_year=$(
241     git log\
242     | grep 'Date:'\
243     | tail -1\
244     | cut -f 8 -d ' '
245   )
246   # Last year according to current state of git repository.
247   declare -ir LFBFL_i_last_year=$(
248     git log\
249     | grep 'Date:'\
250     | head -1\
251     | cut -f 8 -d ' '
252   )
253   local LFBFL_copyright_string
254   LFBFL_copyright_string="${LFBFL_i_first_year}-${LFBFL_i_last_year}"
255   LFBFL_copyright_string+=" ${LFBFL_author_full_name}"
256   readonly LFBFL_copyright_string
257
258   prepare_filled_license_file_block(){
259     # $1=license_file_name
260     sed -e "s/@repository_name@/${LFBFL_repository_name}/g"\
261         -e "s/@copyright_string@/${LFBFL_copyright_string}/g"\
262         "$1.tpl"\
263       > "$1.temp"
264     overwrite_if_not_equal "$1" "$1.temp"
265     declare -ir LFBFL_i_result=$?
266     if [[ LFBFL_i_verbose -eq 1 ]]; then
267       if [[ LFBFL_i_result -eq 1 ]]; then
268         printf "License template %s updated.\n" "$1"
269       elif [[ LFBFL_i_result -eq 2 ]]; then
270         printf "License template %s created.\n" "$1"
271       fi
272     fi
273     head --lines=-1 "$1"\
274       | tail --lines=+2\
275       > "$1.temp"
276     # Strange world: tail --lines=-1 was working
277     # to exclude first line.
278     # I tested it months ago, and saw with my eyes the start and end
279     # block comments removed with:
280     # head --lines=-1 "$1" | tail --lines=-1.
281     # Now, man page says to use --lines=+2 instead.
282     # Welcome in the world where AI change your environment and your
283     # documentation on the fly...
284   }
285
```

```
286   declare -Ar LFBFL_all_block_comment_languages=(
287     ["c"]="c"
288     ["php"]="c"
289     ["py"]="py"
290     ["js"]="c"
291   )
292
293   local LFBFL_key
294   local LFBFL_dest
295   local LFBFL_license_file_name2
296   declare -i LFBFL_i_not_subfile
297   declare -i LFBFL_i_not_subfile2
298   local LFBFL_file_path
299   local LFBFL_s_files_paths
300   declare -a LFBFL_arr_files_paths
301   for LFBFL_key in "${!LFBFL_all_block_comment_languages[@]}"; do
302     LFBFL_dest=${LFBFL_all_block_comment_languages[${LFBFL_key}]}
303     LFBFL_license_file_name="${LFBFL_license_prefix2}.${LFBFL_dest}"
304     prepare_filled_license_file_block "${LFBFL_license_file_name}"
305     if [[ -n "${LFBFL_license2}" ]]; then
306       LFBFL_license_file_name2="${LFBFL_license_prefix3}"
307       LFBFL_license_file_name2+=".${LFBFL_dest}"
308       prepare_filled_license_file_block "${LFBFL_license_file_name2}"
309     fi
310
311     LFBFL_s_files_paths=$(
312       find . -type f -iregex ".*\.\(${LFBFL_key}\|${LFBFL_key}\.tpl\)"\
313         -printf '%P\n'\
314       | relevant_find
315     )
316     if [[ -z "${LFBFL_s_files_paths}" ]]; then
317       if [[ LFBFL_i_verbose -eq 1 ]]; then
318         LFBFL_s_format="build_licenses_templates: "
319         LFBFL_s_format+="No file found with extension .%s or .%s.tpl.\n"
320         # shellcheck disable=SC2059
321         printf "${LFBFL_s_format}" "${LFBFL_key}" "${LFBFL_key}"
322       fi
323       continue
324     fi
325     mapfile -t LFBFL_arr_files_paths <<< "${LFBFL_s_files_paths}"
326     for LFBFL_file_path in "${LFBFL_arr_files_paths[@]}"; do
327       is_subfile "${LFBFL_file_path}"\
328         "${LFBFL_license_file_name}.temp"
329       LFBFL_i_not_subfile=$?
330       LFBFL_i_not_subfile2=1
331       if [[ -n "${LFBFL_license2}" ]]; then
```

```
332         is_subfile "${LFBFL_file_path}"\
333           "${LFBFL_license_file_name2}.temp"
334         LFBFL_i_not_subfile2=$?
335       fi
336       if [[ LFBFL_i_not_subfile -ge 1 && LFBFL_i_not_subfile2 -ge 1 ]];
337       then
338         printf "File %s has no/wrong license header?\n"\
339           "${LFBFL_file_path}"
340       fi
341     done
342     rm "${LFBFL_license_file_name}.temp"
343     if [[ -n "${LFBFL_license2}" ]]; then
344       rm "${LFBFL_license_file_name2}.temp"
345     fi
346   done
347
348   prepare_filled_license_file_line(){
349     # $1=LFBFL_license_file_name
350     sed -e "s/@repository_name@/${LFBFL_repository_name}/g"\
351         -e "s/@copyright_string@/${LFBFL_copyright_string}/g"\
352         "$1.tpl"\
353       > "$1.temp"
354     overwrite_if_not_equal "$1" "$1.temp"
355     declare -ir LFBFL_i_result=$?
356     if [[ LFBFL_i_verbose -eq 1 ]]; then
357       if [[ LFBFL_i_result -eq 1 ]]; then
358         printf "License template %s updated.\n" "$1"
359       elif [[ LFBFL_i_result -eq 2 ]]; then
360         printf "License template %s created.\n" "$1"
361       fi
362     fi
363   }
364
365   declare -Ar LFBFL_all_line_comment_languages=(
366     ["sh"]="sh"
367     ["sql"]="sql"
368     ["tex"]="tex"
369   )
370
371   for LFBFL_key in "${!LFBFL_all_line_comment_languages[@]}"; do
372     LFBFL_dest=${LFBFL_all_line_comment_languages[${LFBFL_key}]}
373     LFBFL_license_file_name="${LFBFL_license_prefix2}.${LFBFL_dest}"
374     prepare_filled_license_file_line "${LFBFL_license_file_name}"
375     if [[ -n "${LFBFL_license2}" ]]; then
376       LFBFL_license_file_name2="${LFBFL_license_prefix3}"
377       LFBFL_license_file_name2+=".${LFBFL_dest}"
```

```
      prepare_filled_license_file_line "${LFBFL_license_file_name2}"
    fi

    LFBFL_s_files_paths=$(
      find . -type f -iregex ".*\.\(${LFBFL_key}\|${LFBFL_key}\.tpl\)"\
        -printf '%P\n'\
      | relevant_find
    )
    if [[ -z "${LFBFL_s_files_paths}" ]]; then
      if [[ LFBFL_i_verbose -eq 1 ]]; then
        LFBFL_s_format="build_licenses_templates: "
        LFBFL_s_format+="No file found with extension .%s or .%s.tpl.\n"
        # shellcheck disable=SC2059
        printf "${LFBFL_s_format}" "${LFBFL_key}" "${LFBFL_key}"
      fi
      continue
    fi
    mapfile -t LFBFL_arr_files_paths <<< "${LFBFL_s_files_paths}"
    for LFBFL_file_path in "${LFBFL_arr_files_paths[@]}"; do
      is_subfile "${LFBFL_file_path}" "${LFBFL_license_file_name}"
      LFBFL_i_not_subfile=$?
      LFBFL_i_not_subfile2=1
      if [[ -n "${LFBFL_license2}" ]]; then
        is_subfile "${LFBFL_file_path}" "${LFBFL_license_file_name2}"
        LFBFL_i_not_subfile2=$?
      fi
      if [[ LFBFL_i_not_subfile -ge 1 && LFBFL_i_not_subfile2 -ge 1 ]];
      then
        printf "File %s has no/wrong license header?\n"\
          "${LFBFL_file_path}"
      fi
    done
  done
}

build_licenses_templates "$@"
```

3.16 build_and_checks_dependencies/licenses_templates/license_file_header_AGPLv3+.tpl

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```

3.17 build_and_checks_dependencies/licenses_templates/license_file_header_GPLv3+.tpl

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```

3.18 build_and_checks_dependencies/licenses_templates/license_file_header_LGPLv3+.tpl

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```

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```

3.19 build_and_checks_dependencies/lines_counts.libr.sh

```
#!/usr/bin/env bash
# This file is part of DevOrSysAdminScripts library.
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# This file was renamed from "lines_counts.sh" to
```

```
25 # "lines_counts.libr.sh".
26
27 LFBFL_subdir="build_and_checks_dependencies"
28 # shellcheck source=common_options.libr.sh
29 source "./${LFBFL_subdir}/common_options.libr.sh"
30 # shellcheck source=get_common_text_glob_patterns.libr.sh
31 source "./${LFBFL_subdir}/get_common_text_glob_patterns.libr.sh"
32 # shellcheck source=lines_filters.libr.sh
33 source "./${LFBFL_subdir}/lines_filters.libr.sh"
34
35 all_code_lines(){
36   # Options:
37   #   --verbose
38   #   --work-directory=""
39   declare -i LFBFL_i_verbose=0
40   get_verbose_option "$@"
41   local LFBFL_work_directory=""
42   get_work_directory_option "$@"
43   declare -a LFBFL_return_traps_stack
44   local LFBFL_previous_return_trap
45   init_return_trap
46   pushd_to_work_directory --trap-popd
47   can_continue_after_enhanced_pushd || return 1
48
49   enhanced_set_shell_option pipefail --trap-unset
50
51   get_COMMON_TEXT_FILES_GLOB_PATTERNS
52   local LFBFL_pattern
53   for LFBFL_pattern in "${COMMON_TEXT_FILES_GLOB_PATTERNS[@]}"; do
54     if [[ LFBFL_i_verbose -eq 1 ]]; then
55       printf "all_code_lines: Iterating on pattern: %s.\n"\
56         "${LFBFL_pattern}"
57     fi
58     find . -type f -name "${LFBFL_pattern}" -printf '%P\n'\
59       | xargs grep --invert-match --perl-regexp --with-filename 'a(?!a)a'
60   done
61 }
62
63 all_self_code_lines(){
64   # Options:
65   #   --verbose
66   #   --work-directory=""
67   declare -i LFBFL_i_verbose=0
68   get_verbose_option "$@"
69   local LFBFL_work_directory=""
70   get_work_directory_option "$@"
```

```
  declare -a LFBFL_return_traps_stack
  local LFBFL_previous_return_trap
  init_return_trap
  pushd_to_work_directory --trap-popd
  can_continue_after_enhanced_pushd || return 1

  enhanced_set_shell_option pipefail --trap-unset

  declare -a LFBFL_common_options=("$@")
  keep_options LFBFL_common_options --verbose

  # below and another lines, see SC2145 false positive
  all_code_lines "${LFBFL_common_options[@]}" \
    | relevant_grep\
    | not_license_grep\
    | not_main_tex_grep\
    | not_main_html_grep
}

# shellcheck disable=SC2120
all_self_empty_code_lines(){
  # Options:
  #   --verbose
  #   --work-directory=""
  declare -i LFBFL_i_verbose=0
  get_verbose_option "$@"
  local LFBFL_work_directory=""
  get_work_directory_option "$@"
  declare -a LFBFL_return_traps_stack
  local LFBFL_previous_return_trap
  init_return_trap
  pushd_to_work_directory --trap-popd
  can_continue_after_enhanced_pushd || return 1

  enhanced_set_shell_option pipefail --trap-unset

  declare -a LFBFL_common_options=("$@")
  keep_options LFBFL_common_options --verbose

  all_self_code_lines "${LFBFL_common_options[@]}" \
    | empty_lines_after_file_name
}

# shellcheck disable=SC2120
all_self_not_empty_code_lines(){
  # Options:
```

```
  #   --verbose
  #   --work-directory=""
  declare -i LFBFL_i_verbose=0
  get_verbose_option "$@"
  local LFBFL_work_directory=""
  get_work_directory_option "$@"
  declare -a LFBFL_return_traps_stack
  local LFBFL_previous_return_trap
  init_return_trap
  pushd_to_work_directory --trap-popd
  can_continue_after_enhanced_pushd || return 1

  enhanced_set_shell_option pipefail --trap-unset

  declare -a LFBFL_common_options=("$@")
  keep_options LFBFL_common_options --verbose

  all_self_code_lines "${LFBFL_common_options[@]}" \
    | not_empty_lines_after_file_name
}

code_lines_count_all(){
  # Options:
  #   --verbose
  #   --work-directory=""
  declare -i LFBFL_i_verbose=0
  get_verbose_option "$@"
  local LFBFL_work_directory=""
  get_work_directory_option "$@"
  declare -a LFBFL_return_traps_stack
  local LFBFL_previous_return_trap
  init_return_trap
  pushd_to_work_directory --trap-popd
  can_continue_after_enhanced_pushd || return 1

  enhanced_set_shell_option pipefail --trap-unset

  declare -gi i_code_lines_count_all_result
  i_code_lines_count_all_result=$(
    all_self_code_lines\
    | wc --lines
  )
  # The printf is for command line use.
  # A script should use the result value instead.
  # shellcheck disable=SC2248
  printf "%s\n" ${i_code_lines_count_all_result}
```

```
}

code_lines_count_empty(){
  # Options:
  #   --verbose
  #   --work-directory=""
  declare -i LFBFL_i_verbose=0
  get_verbose_option "$@"
  local LFBFL_work_directory=""
  get_work_directory_option "$@"
  declare -a LFBFL_return_traps_stack
  local LFBFL_previous_return_trap
  init_return_trap
  pushd_to_work_directory --trap-popd
  can_continue_after_enhanced_pushd || return 1

  enhanced_set_shell_option pipefail --trap-unset

  declare -gi i_code_lines_count_empty_result
  # shellcheck disable=SC2119
  i_code_lines_count_empty_result=$(
    all_self_empty_code_lines\
    | wc --lines
  )
  # The printf is for command line use.
  # A script should use the result value instead.
  # shellcheck disable=SC2248
  printf "%s\n" ${i_code_lines_count_empty_result}
}

code_lines_count_not_empty(){
  # Options:
  #   --verbose
  #   --work-directory=""
  declare -i LFBFL_i_verbose=0
  get_verbose_option "$@"
  local LFBFL_work_directory=""
  get_work_directory_option "$@"
  declare -a LFBFL_return_traps_stack
  local LFBFL_previous_return_trap
  init_return_trap
  pushd_to_work_directory --trap-popd
  can_continue_after_enhanced_pushd || return 1

  enhanced_set_shell_option pipefail --trap-unset

```

```
  declare -gi i_code_lines_count_not_empty_result
  # shellcheck disable=SC2119
  i_code_lines_count_not_empty_result=$(
    all_self_not_empty_code_lines\
    | wc --lines
  )
  # The printf is for command line use.
  # A script should use the result value instead.
  # shellcheck disable=SC2248
  printf "%s\n" ${i_code_lines_count_not_empty_result}
}
```

3.20 build_and_checks_dependencies/lines_filters.libr.sh

```
#!/usr/bin/env bash
# This file is part of DevOrSysAdminScripts library.
#
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# of the GNU General Public License
# as published by the Free Software Foundation,
# either version 3 of the License,
# or (at your option) any later version.
#
# DevOrSysAdminScripts is distributed in the hope
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# but WITHOUT ANY WARRANTY;
# without even the implied warranty of
# MERCHANTABILITY or FITNESS FOR A PARTICULAR PURPOSE.
# See the GNU General Public License for more details.
#
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# the GNU General Public License
# along with DevOrSysAdminScripts.
# If not, see <https://www.gnu.org/licenses/>.
#
# ©Copyright 2023-2026 Laurent Frédéric Bernard François Lyaudet
# This file was renamed from "lines_filters.sh" to
# "lines_filters.libr.sh".

# The goal of this file is to have some common line filters
# for bash processing and pipes.
# As is the case for other .sh files in this repository,
# it can be copied somewhere in your home directory
# and sourced in your .bashrc.

```

```
# shellcheck source=common_options.libr.sh
source "./build_and_checks_dependencies/common_options.libr.sh"

ll_wc(){
  # current options to wc:
  # -c --bytes
  # -m --chars
  # -l --lines
  # --files0-from # Incompatible with cat,
  #               # another thing intentional by ?
  #               # I don't see the meaning of "files0".
  # -L --max-line-length
  # -w --words
  # --help
  # --version
  # New option by Laurent Lyaudet (and probably other before):
  # -n --no-filenames display only number(s)
  # I think I did that the first time many years ago (Teliae ?).
  # Since then, they added it as ignored... (-n)
  enhanced_set_shell_option pipefail\
    && trap 'enhanced_unset_shell_option pipefail' RETURN
  declare -a ll_wc_var_args=()
  declare -i ll_wc_var_i_number_only=0
  declare -i ll_wc_var_i=0
  for ll_wc_var_arg in "$@"; do
    # printf "%s\n" "${ll_wc_var_arg}"
    if [[ "${ll_wc_var_arg}" == "-n" ]]; then
      ll_wc_var_i_number_only=1
    elif [[ "${ll_wc_var_arg}" == "--no-filenames" ]]; then
      ll_wc_var_i_number_only=1
    elif [[ ll_wc_var_i_number_only -eq 0 ]]; then
      ll_wc_var_args[ll_wc_var_i]="${ll_wc_var_arg}"
      ll_wc_var_i=$((ll_wc_var_i + 1))
    elif [[ "${ll_wc_var_arg}" =~ -.* ]]; then
      ll_wc_var_args[ll_wc_var_i]="${ll_wc_var_arg}"
      ll_wc_var_i=$((ll_wc_var_i + 1))
    fi
  done
  # typeset -p ll_wc_var_args
  # printf "%s\n" "${ll_wc_var_i_number_only}"
  if [[ ll_wc_var_i_number_only -gt 0 ]]; then
    # Too simple code, only the base use case is handled now.
    # Convenient but incomplete.
    # With more time,
    # the simpler would be to submit a patch to GNU coreutils,
    # unless blockade from assholes.
```

```
    # Otherwise, it may be more simply handled in PHP or Python
    # (argparse, etc.).
    # You'll need to start by improving cat first,
    # with --files0-from handling.
    # wc does not display file_name when stream...
    # hence cat of last arg "${!#}"
    cat "${!#}"\
      | wc "${ll_wc_var_args[@]}"
  else
    # printf "normal\n"
    wc "${ll_wc_var_args[@]}"
  fi
}

# I submitted a patch for wc :).
# printf "https://domain/list-path/2024-05/msg00013.html" | \
#   sed --expression='s|domain|lists.gnu.org|' \
#       --expression='s|list-path|archive/html/bug-coreutils|'
# I found another way to have a long URL fit on 70 characters.
# If it is merged, the following would be better:
# If --no-filenames option available,
# we redefine ll_wc command as an alias.
wc_help_text=$(wc --help)
if [[ "${wc_help_text}" == *" --no-filenames "* ]]; then
  alias ll_wc='wc'
fi

# Trace avant le (2e ?) renommage
# de wget_sha512.sh à wget_sha512.libr.sh.
# $ ll_wc wget_sha512.sh
#  41  189 1224 wget_sha512.sh
# $ ll_wc -n wget_sha512.sh
#     41     189    1224
# $ ll_wc -l wget_sha512.sh
# 41 wget_sha512.sh
# $ ll_wc -l -n wget_sha512.sh
# 41

in_place_grep(){
  # Same arguments and options as grep.
  declare -r LFBFL_temp="${!#}.in_place_grep.temp"
  grep "$@" \
    > "${LFBFL_temp}"
  # declare -ir LFBFL_i_lines_before=$(ll_wc -l -n "${!#}")
  # declare -ir LFBFL_i_lines_after=$(ll_wc -l -n "${LFBFL_temp}")
  # printf "%s\n" "${LFBFL_i_lines_before}"
```

```
  # printf "%s\n" "${LFBFL_i_lines_after}"
  # if [[ LFBFL_i_lines_before -eq LFBFL_i_lines_after ]]; then
  # Almost as fast (on large files) and correct in all cases :).
  # Thanks Paul Eggert for the suggestion :).
  # https://debbugs.gnu.org/cgi/bugreport.cgi?bug=80547
  if cmp --silent "${LFBFL_temp}" "${!#}"; then
    rm "${LFBFL_temp}"
    return
  fi
  mv "${LFBFL_temp}" "${!#}"
}

grep_fixed_string_with_anchor(){
  # $1=file_path
  # $2=some_fixed_string
  # Options:
  #   --enforce-line-starts-with-fixed-string
  #   --enforce-line-ends-with-fixed-string
  #   Apart from $1 and $2 and the options above,
  #   all other arguments are considered grep_options like --quiet
  #   for example.
  #   But note that --fixed-strings will be added below, you don't need to
  #   supply it.
  declare -a LFBFL_grep_options=()
  declare -a LFBFL_grep_options2=()
  declare -i LFBFL_i=0
  declare -i LFBFL_i_enforce_line_starts=0
  declare -i LFBFL_i_enforce_line_ends=0
  declare -i LFBFL_i_quiet=0
  local LFBFL_arg
  for LFBFL_arg in "$@"; do
    ((++LFBFL_i))
    if [[ LFBFL_i -lt 3 ]]; then
      continue
    fi
    if [[ "${LFBFL_arg}" == "--enforce-line-starts-with-fixed-string" ]];
    then
      LFBFL_i_enforce_line_starts=1
      continue
    fi
    if [[ "${LFBFL_arg}" == "--enforce-line-ends-with-fixed-string" ]];
    then
      LFBFL_i_enforce_line_ends=1
      continue
    fi
    if [[ "${LFBFL_arg}" == "-q" || "${LFBFL_arg}" == "--quiet" ]];
```

```
171       then
172         LFBFL_i_quiet=1
173         LFBFL_grep_options+=("${LFBFL_arg}")
174         continue
175       fi
176       LFBFL_grep_options2+=("${LFBFL_arg}")
177     done
178
179     if [[
180       LFBFL_i_enforce_line_starts -eq 0
181       && LFBFL_i_enforce_line_ends -eq 0
182     ]]; then
183       grep "${LFBFL_grep_options[@]}" --fixed-strings -- "$2" "$1"
184       return
185     fi
186     declare -r LFBFL_s_lines=$(
187       grep "${LFBFL_grep_options2[@]}" --color=never --fixed-strings --\
188         "$2" "$1"
189     )
190     declare -i LFBFL_i_result=1
191     if [[ -z "${LFBFL_s_lines}" ]]; then
192       # shellcheck disable=SC2248
193       return ${LFBFL_i_result}
194     fi
195     declare -a LFBFL_arr_lines
196     mapfile -t LFBFL_arr_lines <<< "${LFBFL_s_lines}"
197     readonly LFBFL_arr_lines
198
199     declare -i LFBFL_i_fixed_string_length=${#2}
200     local LFBFL_some_line
201     local LFBFL_line_part
202     for LFBFL_some_line in "${LFBFL_arr_lines[@]}"; do
203       if [[ "${LFBFL_some_line}" == "$2" ]]; then
204         LFBFL_i_result=0
205         if [[ LFBFL_i_quiet -eq 1 ]]; then
206           break
207         else
208           printf "%s\n" "${LFBFL_some_line}"
209         fi
210         continue
211       fi
212       if [[
213         LFBFL_i_enforce_line_starts -eq 1
214         && LFBFL_i_enforce_line_ends -eq 1
215       ]]; then
216         continue
```

```
    fi
    if [[ LFBFL_i_enforce_line_starts -eq 1 ]]; then
      LFBFL_line_part=${LFBFL_some_line:0:${LFBFL_i_fixed_string_length}}
    fi
    if [[ LFBFL_i_enforce_line_ends -eq 1 ]]; then
      LFBFL_line_part=${LFBFL_some_line: -${LFBFL_i_fixed_string_length}}
    fi
    if [[ "${LFBFL_line_part}" == "$2" ]]; then
      LFBFL_i_result=0
      if [[ LFBFL_i_quiet -eq 1 ]]; then
        break
      else
        printf "%s\n" "${LFBFL_some_line}"
      fi
    fi
  done
  LFBFL_i_result=$((LFBFL_i_result | $?))
  # shellcheck disable=SC2248
  return ${LFBFL_i_result}
}

grep_variable(){
  # $1=file_path
  # $2=variable_name
  # Options :
  #   --replace-line-returns-by=""
  #   --result-variable-prefix="LFBFL_" for example
  declare -r LFBFL_regexp="(?<=^$2=).*$"
  local LFBFL_arg
  # printf "%s\n" "${LFBFL_regexp}"
  local LFBFL_variable_value
  if [[ "$*" == *--replace-line-returns-by=* ]]; then
    local LFBFL_s_replace=""
    for LFBFL_arg in "$@"; do
      if [[ "${LFBFL_arg}" == --replace-line-returns-by=* ]]; then
        LFBFL_s_replace=${LFBFL_arg#--replace-line-returns-by=}
        break
      fi
    done
    # sed -Ez -e...
    LFBFL_variable_value=$(
      grep --only-matching --perl-regexp "${LFBFL_regexp}" "$1"\
      | sed --null-data --regexp-extended\
        --expression='s/\n/'"${LFBFL_s_replace}"'/Mg'
    )
    # We remove the trailing replace string,
```

```
    # since the intent is only to replace the intermediate \n.
    if [[ -n "${LFBFL_s_replace}" ]]; then
    LFBFL_variable_value=${LFBFL_variable_value%"${LFBFL_s_replace}"}
    fi
  else
    LFBFL_variable_value=$(
      grep --only-matching --perl-regexp "${LFBFL_regexp}" "$1"
    )
  fi
  local LFBFL_prefix=""
  for LFBFL_arg in "$@"; do
    if [[ "${LFBFL_arg}" == --result-variable-prefix=* ]]; then
      LFBFL_prefix=${LFBFL_arg#--result-variable-prefix=}
      break
    fi
  done
  declare -r LFBFL_variable_name="${LFBFL_prefix}$2"
  # printf "%s : %s\n " "${LFBFL_variable_name}" "${LFBFL_variable_value}"
  # The declare line may be dropped by a local or declare
  # in outer scope.
  # declare -g "${LFBFL_variable_name}"
  # So the assignment must be done on another line.
  printf -v "${LFBFL_variable_name}" "%s" "${LFBFL_variable_value}"
  # printf "%s\n" "${!LFBFL_variable_name}"
}

empty_lines(){
  grep '^$'
}

not_empty_lines(){
  grep --invert-match '^$'
}

space_starting_lines(){
  grep '^ '
}

not_space_starting_lines(){
  grep --invert-match '^ '
}

# Space starting or empty lines = spampty
spampty_lines(){
  grep --extended-regexp '^( |$)'
}
```

```
309
310 not_spampty_lines(){
311   grep '^[^ ]'
312   # grep --extended-regexp --invert-match '^( |$)'
313 }
314
315 all_space_lines(){
316   grep '^[ ]\+$'
317 }
318
319 not_all_space_lines(){
320   grep --invert-match '^[ ]\+$'
321 }
322
323 # All space or empty lines = aspampty
324 aspampty_lines(){
325   grep '^[ ]*$'
326 }
327
328 not_aspampty_lines(){
329   grep '[^ ]'
330   # grep --invert-match '^[ ]*$'
331 }
332
333 empty_lines_after_file_name(){
334   grep '^[^:]\+:$'
335 }
336
337 not_empty_lines_after_file_name(){
338   grep --invert-match '^[^:]\+:$'
339 }
340
341 space_starting_lines_after_file_name(){
342   grep '^[^:]\+: '
343 }
344
345 not_space_starting_lines_after_file_name(){
346   grep --invert-match '^[^:]\+: '
347 }
348
349 spampty_lines_after_file_name(){
350   grep --extended-regexp '^[^:]+:( |$)'
351 }
352
353 not_spampty_lines_after_file_name(){
354   # The two variants results differ on lines without filename before.
```

```
  # Since the intent of the function is only on lines "blabla.sh:whatever",
  # use the fastest one. I didn't do extensive testing and just assumed
  # that the "positive" one is faster as usual when querying database.
  # But there is no index here ;) XD.
  grep '^[^:]\+:[^ ]'
  # grep --extended-regexp --invert-match '^[^:]+:( |$)'
}

all_space_lines_after_file_name(){
  grep '^[^:]\+:[ ]\+$'
}

not_all_space_lines_after_file_name(){
  grep --invert-match '^[^:]\+:[ ]\+$'
}

aspampty_lines_after_file_name(){
  grep '^[^:]\+:[ ]*$'
}

not_aspampty_lines_after_file_name(){
  grep --invert-match '^[^:]\+:[ ]*$'
}

# If you need to test the previous 20 functions,
# look at file spampty_data.txt in miscellaneous.
# And give a try at something like:
# cat -E <(cat miscellaneous/spampty_data.txt | empty_lines)
# -E = --show-ends :). Nice command :).

not_JS_dependencies_find(){
  grep --extended-regexp --invert-match\
    --regexp="(^|/)node_modules/"\
    --regexp="(^|/)package-lock\.json$"
}

not_JS_dependencies_grep(){
  grep --extended-regexp --invert-match\
    --regexp="^([^:]+/)?node_modules/"\
    --regexp="^([^:]+/)?package-lock\.json:"
}

not_dependencies_find(){
  not_JS_dependencies_find
}
```

```
not_dependencies_grep(){
  not_JS_dependencies_grep
}

not_python_cache_find(){
  grep --extended-regexp --invert-match\
    --regexp="(^|/)__pycache__/"\
    --regexp="(^|/)\.mypy_cache/"\
    --regexp="(^|/)\.ruff_cache/"
}

not_python_cache_grep(){
  grep --extended-regexp --invert-match\
    --regexp="^([^:]+/)?__pycache__/"\
    --regexp="^([^:]+/)?\.mypy_cache/"\
    --regexp="^([^:]+/)?\.ruff_cache/"
}

not_cache_find(){
  not_python_cache_find
}

not_cache_grep(){
  not_python_cache_grep
}

not_git_find(){
  grep --extended-regexp --invert-match "(^|/)\.git/"
}

not_git_grep(){
  grep --extended-regexp --invert-match "^([^:]+/)?\.git/"
}

not_archive_find(){
  grep --extended-regexp --invert-match "(\.gz|\.rar|\.tar|\.tgz|\.whl)$"
}

not_archive_grep(){
  grep --extended-regexp --invert-match\
    --regexp="^[^:]*(\.gz|\.rar|\.tar|\.tgz|\.whl):"
}

not_license_find(){
  grep --extended-regexp --invert-match\
    --regexp="(^|/)COPYING$"\
```

```
    --regexp="(^|/)COPYING\.LESSER$"
}

not_license_grep(){
  grep --extended-regexp --invert-match\
    --regexp="^([^:]+/)?COPYING:"\
    --regexp="^([^:]+/)?COPYING\.LESSER:"
}

not_main_tex_find(){
  local LFBFL_repository_name=""
  grep_variable build_and_checks_variables/repository_data.txt\
    repository_name\
    --result-variable-prefix="LFBFL_"\
    --replace-line-returns-by=""
  grep --extended-regexp --invert-match\
    --regexp="(^|/)${LFBFL_repository_name}\.tex$"
}

not_main_tex_grep(){
  local LFBFL_repository_name=""
  grep_variable build_and_checks_variables/repository_data.txt\
    repository_name\
    --result-variable-prefix="LFBFL_"\
    --replace-line-returns-by=""
  grep --extended-regexp --invert-match\
    --regexp="^([^:]+/)?${LFBFL_repository_name}\.tex:"
}

not_main_html_find(){
  local LFBFL_repository_name=""
  grep_variable build_and_checks_variables/repository_data.txt\
    repository_name\
    --result-variable-prefix="LFBFL_"\
    --replace-line-returns-by=""
  grep --extended-regexp --invert-match\
    --regexp="(^|/)${LFBFL_repository_name}\.html$"
}

not_main_html_grep(){
  local LFBFL_repository_name=""
  grep_variable build_and_checks_variables/repository_data.txt\
    repository_name\
    --result-variable-prefix="LFBFL_"\
    --replace-line-returns-by=""
  grep --extended-regexp --invert-match\
```

```
    --regexp="^([^:]+/)?${LFBFL_repository_name}\.html:"
}

not_temp_file_find(){
  grep --extended-regexp --invert-match\
    --regexp="(^|/)build_and_checks_variables/temp/"
}

not_temp_file_grep(){
  grep --extended-regexp --invert-match\
    --regexp="^([^:]+/)?build_and_checks_variables/temp/[^:]*:"
}

relevant_find(){
  enhanced_set_shell_option pipefail\
    && trap 'enhanced_unset_shell_option pipefail' RETURN
  not_dependencies_find\
    | not_cache_find\
    | not_git_find\
    | not_archive_find\
    | not_temp_file_find
}

relevant_grep(){
  enhanced_set_shell_option pipefail\
    && trap 'enhanced_unset_shell_option pipefail' RETURN
  not_dependencies_grep\
    | not_cache_grep\
    | not_git_grep\
    | not_archive_grep\
    | not_temp_file_grep
}
```

3.21 build_and_checks_dependencies/lines_maps.libr.sh

```
#!/usr/bin/env bash
# This file is part of DevOrSysAdminScripts library.
#
# DevOrSysAdminScripts is free software:
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# of the GNU General Public License
# as published by the Free Software Foundation,
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# or (at your option) any later version.
#
# DevOrSysAdminScripts is distributed in the hope
```

```
# that it will be useful,
# but WITHOUT ANY WARRANTY;
# without even the implied warranty of
# MERCHANTABILITY or FITNESS FOR A PARTICULAR PURPOSE.
# See the GNU General Public License for more details.
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# the GNU General Public License
# along with DevOrSysAdminScripts.
# If not, see <https://www.gnu.org/licenses/>.
#
# ©Copyright 2023-2026 Laurent Frédéric Bernard François Lyaudet
# This file was renamed from "lines_maps.sh" to "lines_maps.libr.sh".

# https://en.wikibooks.org/wiki/Unicode/Character_reference/2000-2FFF
# Some invisible unicode characters that are a problem:
# 2000-200F 2028-202F 205F 2060-206F
# [
#   '2000', '2001', '2002', '2003', '2004', '2005', '2006', '2007',
#   '2008', '2009', '200A', '200B', '200C', '200D', '200E', '200F',
#   '2028', '2029', '202A', '202B', '202C', '202D', '202E', '202F',
#   '205F',
#   '2060', '2061', '2062', '2063', '2064', '2065', '2066', '2067',
#   '2068', '2069', '206A', '206B', '206C', '206D', '206E', '206F',
# ]
replace_non_ascii_spaces(){
  sed -e 's/\xc2\xa0/ /g'\
      -e 's/\xe2\x80\x80/ /g'\
      -e 's/\xe2\x80\x81/ /g'\
      -e 's/\xe2\x80\x82/ /g'\
      -e 's/\xe2\x80\x83/ /g'\
      -e 's/\xe2\x80\x84/ /g'\
      -e 's/\xe2\x80\x85/ /g'\
      -e 's/\xe2\x80\x86/ /g'\
      -e 's/\xe2\x80\x87/ /g'\
      -e 's/\xe2\x80\x88/ /g'\
      -e 's/\xe2\x80\x89/ /g'\
      -e 's/\xe2\x80\x8a/ /g'\
      -e 's/\xe2\x80\x8b/ /g'\
      -e 's/\xe2\x80\x8c/ /g'\
      -e 's/\xe2\x80\x8d/ /g'\
      -e 's/\xe2\x80\x8e/ /g'\
      -e 's/\xe2\x80\x8f/ /g'\
      -e 's/\xe2\x80\xa8/ /g'\
      -e 's/\xe2\x80\xa9/ /g'\
      -e 's/\xe2\x80\xaa/ /g'\
```

```
        -e 's/\xe2\x80\xab/ /g'\
        -e 's/\xe2\x80\xac/ /g'\
        -e 's/\xe2\x80\xad/ /g'\
        -e 's/\xe2\x80\xae/ /g'\
        -e 's/\xe2\x80\xaf/ /g'\
        -e 's/\xe2\x81\x9f/ /g'\
        -e 's/\xe2\x81\xa0/ /g'\
        -e 's/\xe2\x81\xa1/ /g'\
        -e 's/\xe2\x81\xa2/ /g'\
        -e 's/\xe2\x81\xa3/ /g'\
        -e 's/\xe2\x81\xa4/ /g'\
        -e 's/\xe2\x81\xa5/ /g'\
        -e 's/\xe2\x81\xa6/ /g'\
        -e 's/\xe2\x81\xa7/ /g'\
        -e 's/\xe2\x81\xa8/ /g'\
        -e 's/\xe2\x81\xa9/ /g'\
        -e 's/\xe2\x81\xaa/ /g'\
        -e 's/\xe2\x81\xab/ /g'\
        -e 's/\xe2\x81\xac/ /g'\
        -e 's/\xe2\x81\xad/ /g'\
        -e 's/\xe2\x81\xae/ /g'\
        -e 's/\xe2\x81\xaf/ /g'
}
```

3.22 build_and_checks_dependencies/overwrite_if_not_equal.libr.sh

```
#!/usr/bin/env bash
# This file is part of DevOrSysAdminScripts library.
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# This file was renamed from "overwrite_if_not_equal.sh"
# to "overwrite_if_not_equal.libr.sh".

overwrite_if_not_equal(){
  # $1=target_file_path
  # $2=temp_file_path
  # $3=keep_temp
  # $4=tree_mode : when the two files are the output of tree command,
  # we ignore if all the differences are only on directories timestamps.
  # $5=--quiet : we even don't want the message of diff saying that
  #              files are distinct
  # Return 0 target was the same
  #        1 target was different
  #        2 target did not already exist
  if [[ ! -f "$1" ]]; then
    if [[ -n "$3" ]]; then
      cp -p "$2" "$1"
    else
      mv "$2" "$1"
    fi
    return 2
  fi
  declare -i LFBFL_i_is_equal
  if [[ -n "$4" ]]; then
    # Not setting pipefail since the result would still be incorrect.
    # shellcheck disable=SC2312
    diff "$1" "$2"\
      | grep --extended-regexp "^(>|<)"\
      | grep --invert-match "/$"
    LFBFL_i_is_equal=${PIPESTATUS[2]}
    if [[ LFBFL_i_is_equal -eq 1 ]]; then
      grep --only-matching "[^ ]*/$" "$1"\
        > "overwrite_if_not_equal.file1.temp"
      grep --only-matching "[^ ]*/$" "$2"\
        > "overwrite_if_not_equal.file2.temp"
      if [[ -n "$5" ]]; then
        diff "overwrite_if_not_equal.file1.temp"\
          "overwrite_if_not_equal.file2.temp"\
          > /dev/null
      else
        diff "overwrite_if_not_equal.file1.temp"\
          "overwrite_if_not_equal.file2.temp"
      fi
```

```
      LFBFL_i_is_equal=1-$?
      rm "overwrite_if_not_equal.file1.temp"\
        "overwrite_if_not_equal.file2.temp"
    fi
  else
    if [[ -n "$5" ]]; then
      diff --brief "$1" "$2" > /dev/null
    else
      diff --brief "$1" "$2"
    fi
    LFBFL_i_is_equal=1-$?
  fi
  if [[ LFBFL_i_is_equal -eq 1 ]]; then
    if [[ -z "$3" ]]; then
      rm "$2"
    fi
    return 0
  fi
  if [[ -n "$3" ]]; then
    cp -p "$2" "$1"
  else
    mv "$2" "$1"
  fi
  return 1
}
```

3.23 build_and_checks_dependencies/python_black_complement.libr.sh

```
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# This file was renamed from "python_black_complement.sh"
# to "python_black_complement.libr.sh".

# shellcheck source=common_options.libr.sh
source "./build_and_checks_dependencies/common_options.libr.sh"

# It was rejected to enhance black with this currently.
# https://github.com/psf/black/issues/2370
# And moreover, at some point, I saw the reverse rule
# in black or pylint to add an empty line...
# This regexp may give false positives,
# but that's not the end of the world.
check_no_empty_line_after_python_function_docstring(){
  # Options:
  #   --verbose
  #   --work-directory=""
  declare -i LFBFL_i_verbose=0
  get_verbose_option "$@"
  local LFBFL_work_directory=""
  get_work_directory_option "$@"
  declare -a LFBFL_return_traps_stack
  local LFBFL_previous_return_trap
  init_return_trap
  pushd_to_work_directory --trap-popd
  can_continue_after_enhanced_pushd || return 1

  enhanced_set_bash_option globstar --trap-unset

  [[ LFBFL_i_verbose -eq 1 ]]\
    && printf "Checking empty lines after Python function docstrings\n"
  pcre2grep -M --\
    $'def ([^"]|"(?!""))*"""([^"]|"(?!""))*"""\n\n(?!\s*def)'\
    **/*.py
}

python_black_complement(){
  # Options:
  #   --verbose
  #   --work-directory=""
  declare -i LFBFL_i_verbose=0
  get_verbose_option "$@"
```

```
  local LFBFL_work_directory=""
  get_work_directory_option "$@"
  declare -a LFBFL_return_traps_stack
  local LFBFL_previous_return_trap
  init_return_trap
  pushd_to_work_directory --trap-popd
  can_continue_after_enhanced_pushd || return 1

  declare -a LFBFL_common_options=("$@")
  keep_options LFBFL_common_options --verbose

  check_no_empty_line_after_python_function_docstring\
    "${LFBFL_common_options[@]}"
}
```

3.24 build_and_checks_dependencies/python_isort_complement.libr.sh

```
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# This file was renamed from "python_isort_complement.sh"
# to "python_isort_complement.libr.sh".

LFBFL_subdir="build_and_checks_dependencies"
# shellcheck source=common_options.libr.sh
source "./${LFBFL_subdir}/common_options.libr.sh"
```

```
30 # shellcheck source=lines_filters.libr.sh
31 source "./${LFBFL_subdir}/lines_filters.libr.sh"
32 # shellcheck source=overwrite_if_not_equal.libr.sh
33 source "./${LFBFL_subdir}/overwrite_if_not_equal.libr.sh"
34
35 check_collections_abc_place(){
36   # Options:
37   #   --verbose
38   #   --work-directory=""
39   declare -i LFBFL_i_verbose=0
40   get_verbose_option "$@"
41   local LFBFL_work_directory=""
42   get_work_directory_option "$@"
43   declare -a LFBFL_return_traps_stack
44   local LFBFL_previous_return_trap
45   init_return_trap
46   pushd_to_work_directory --trap-popd
47   can_continue_after_enhanced_pushd || return 1
48
49   [[ LFBFL_i_verbose -eq 1 ]]\
50     && printf "Checking import of _collections_abc is at the right place\n"
51
52   enhanced_set_shell_option pipefail --trap-unset
53
54   declare -r LFBFL_temp=".check_collections_abc_place.temp"
55   local LFBFL_file_path
56   declare -r LFBFL_s_files_paths=$(
57     find . -type f -name "*.py" -printf '%P\n'\
58     | relevant_find
59   )
60   if [[ -z "${LFBFL_s_files_paths}" ]]; then
61     [[ LFBFL_i_verbose -eq 1 ]]\
62       && printf "check_collections_abc_place: No file found.\n"
63     return
64   fi
65   declare -a LFBFL_arr_files_paths
66   mapfile -t LFBFL_arr_files_paths <<< "${LFBFL_s_files_paths}"
67   readonly LFBFL_arr_files_paths
68   for LFBFL_file_path in "${LFBFL_arr_files_paths[@]}"; do
69     sed -Ez 's/\n(\nfrom _collections_abc[^\n]*)/\1\n/Mg'\
70       "${LFBFL_file_path}"\
71       > "${LFBFL_file_path}${LFBFL_temp}"
72     overwrite_if_not_equal "${LFBFL_file_path}"\
73       "${LFBFL_file_path}${LFBFL_temp}"
74   done
75 }
```

```

python_isort_complement(){
  # Options:
  #   --verbose
  #   --work-directory=""
  declare -i LFBFL_i_verbose=0
  get_verbose_option "$@"
  local LFBFL_work_directory=""
  get_work_directory_option "$@"
  declare -a LFBFL_return_traps_stack
  local LFBFL_previous_return_trap
  init_return_trap
  pushd_to_work_directory --trap-popd
  can_continue_after_enhanced_pushd || return 1

  declare -a LFBFL_common_options=("$@")
  keep_options LFBFL_common_options --verbose

  check_collections_abc_place "${LFBFL_common_options[@]}"
}
```

3.25 build_and_checks_dependencies/repository_data.libr.sh

```
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```

```

LFBFL_subdir="build_and_checks_dependencies"
# shellcheck source=common_options.libr.sh
source "./${LFBFL_subdir}/common_options.libr.sh"
# shellcheck source=lines_filters.libr.sh
source "./${LFBFL_subdir}/lines_filters.libr.sh"

# /!\ You need to activate extglob before parsing this file. /!\
# And I'm kind enough to activate it inside functions that need it below.

get_upgrade_venvs_time_interval_in_seconds(){
  # $1=data_file_path
  # Options:
  #   --verbose
  declare -i LFBFL_i_verbose=0
  get_verbose_option "$@"
  declare -a LFBFL_return_traps_stack
  local LFBFL_previous_return_trap
  init_return_trap

  enhanced_set_bash_option extglob --trap-unset

  grep_variable "$1"\
    upgrade_venvs_time_interval_in_seconds\
    --result-variable-prefix="LFBFL_"

  [[ LFBFL_i_verbose -eq 1 ]]\
    && printf "You say \"%s\".\n"\
        "${LFBFL_upgrade_venvs_time_interval_in_seconds}"

  case ${LFBFL_upgrade_venvs_time_interval_in_seconds} in
    +([0-9])) : ;;
    wat) LFBFL_upgrade_venvs_time_interval_in_seconds=${RANDOM};;
    watyouwant\?) LFBFL_upgrade_venvs_time_interval_in_seconds=${SRANDOM};;
    *) printf "No wat for you?\n";; #TempsDeCerveauDisponible XD SC2249 ;)
  esac

  [[ LFBFL_i_verbose -eq 1 ]]\
    && printf "I say \"%s\".\n"\
        "${LFBFL_upgrade_venvs_time_interval_in_seconds}"
}
```

3.26 build_and_checks_dependencies/shell_checks_complement.libr.sh

```
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```

```
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# This file was renamed from "python_black_complement.sh"
# to "python_black_complement.libr.sh".

# shellcheck source=common_options.libr.sh
source "./build_and_checks_dependencies/common_options.libr.sh"

check_no_size_of_array_first_element(){
  # Options:
  #   --verbose
  #   --work-directory=""
  declare -i LFBFL_i_verbose=0
  get_verbose_option "$@"
  local LFBFL_work_directory=""
  get_work_directory_option "$@"
  declare -a LFBFL_return_traps_stack
  local LFBFL_previous_return_trap
  init_return_trap
  pushd_to_work_directory --trap-popd
  can_continue_after_enhanced_pushd || return 1

  enhanced_set_bash_option globstar --trap-unset

  local LFBFL_s="Checking that shell scripts do not"
  LFBFL_s+=" contain wrong array size expansion."
  [[ LFBFL_i_verbose -eq 1 ]]\
```

```
    && printf "%s\n" "${LFBFL_s}"

  pcre2grep -M -- $'\$\{#.*(\b|_)arr_(?!.*\[@\])' **/*.sh
}

check_no_misplaced_then(){
  # Options:
  #   --verbose
  #   --work-directory=""
  #   --max-line-length
  declare -i LFBFL_i_verbose=0
  get_verbose_option "$@"
  local LFBFL_work_directory=""
  get_work_directory_option "$@"
  declare -a LFBFL_return_traps_stack
  local LFBFL_previous_return_trap
  init_return_trap
  pushd_to_work_directory --trap-popd
  can_continue_after_enhanced_pushd || return 1

  local LFBFL_arg
  declare -i LFBFL_i_max_line_length=70
  for LFBFL_arg in "$@"; do
    if [[ "${LFBFL_arg}" == --max-line-length=* ]]; then
      LFBFL_arg=${LFBFL_arg#--max-line-length=}
      LFBFL_i_max_line_length=$((LFBFL_arg))
      break
    fi
  done
  declare -i LFBFL_i_max_previous_line_length
  LFBFL_i_max_previous_line_length=$((LFBFL_i_max_line_length - 5))

  enhanced_set_bash_option globstar --trap-unset

  [[ LFBFL_i_verbose -eq 1 ]]\
    && printf\
        "Checking that shell scripts do not contain misplaced 'then'.\n"

  local LFBFL_regexp
  LFBFL_regexp="^[^\n]{1,${LFBFL_i_max_previous_line_length}}"
  LFBFL_regexp+="(?<=\]\];)"
  LFBFL_regexp+="(?<!^\s{0,${LFBFL_i_max_previous_line_length}}\]\];)"
  LFBFL_regexp+="\n\s*then"

  pcre2grep -M -- "${LFBFL_regexp}" **/*.sh
}
```

```

shell_checks_complement(){
  # Options:
  #   --verbose
  #   --work-directory=""
  declare -i LFBFL_i_verbose=0
  get_verbose_option "$@"
  local LFBFL_work_directory=""
  get_work_directory_option "$@"
  declare -a LFBFL_return_traps_stack
  local LFBFL_previous_return_trap
  init_return_trap
  pushd_to_work_directory --trap-popd
  can_continue_after_enhanced_pushd || return 1

  declare -a LFBFL_common_options=("$@")
  keep_options LFBFL_common_options --verbose

  check_no_size_of_array_first_element "${LFBFL_common_options[@]}"
  check_no_misplaced_then "${LFBFL_common_options[@]}"
}
```

3.27 build_and_checks_dependencies/split_score.exec.php

```
#!/usr/bin/php
<?php
/*
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2026-01-04T13:38+01:00: This file was renamed from
"split_line_at_most.exec.php"
to
"split_score.exec.php".
This file was renamed from
"call_split_score_after_before_or_before_after.exec.php" to
"split_line_at_most.exec.php".
This file was renamed from
"call_split_score_after_before_or_before_after.php" to
"call_split_score_after_before_or_before_after.exec.php".
*/

require_once('./build_and_checks_dependencies/split_score.libr.php');

/*
php -r\
 $'require_once(\'./build_and_checks_dependencies/'\
$'split_score.libr.php\');'\
$' echo(generate_split_score'\
$'(true, 70, [\'/\'])(\'/\', 60, true));'
*/

// The first cast "(int) generate_split_score" is to avoid false
// positive errors on possible XSS if output is unescaped.
echo(
  (int) generate_split_score(
    (bool) $argv[1], (int) $argv[2], explode(',', $argv[3]),
  )($argv[4], (int) $argv[5], (bool) $argv[6])
);
```

3.28 build_and_checks_dependencies/split_score.libr.php

```
<?php
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2026-01-04T13:38+01:00: This file was renamed from
"split_line_at_most.libr.php"
to
"split_score.libr.php".
This file was renamed from "split_line_at_most.php" to
"split_line_at_most.libr.php".
*/

function generate_split_score(
  bool $b_after_before,
  int $i_max_length,
  array $a_delimiter_strings_domain,
) : Closure {
  return function(
    string $s_delimiter_string,
    int $i_cut_position,
    bool $b_is_cut_after,
  ) use ($b_after_before, $i_max_length, $a_delimiter_strings_domain)
  : int {
    if(
      !in_array(
        $s_delimiter_string, $a_delimiter_strings_domain, true
      )
    ){
      // I do not handle regexps for the moment.
      // I'm too brainfucked. I just want to finish this feature fast
      // for HTML and Tex/PDF listings generation with HTML and Tex on
      // at most 70 characters.
      return 0;
    }
    if($b_is_cut_after === $b_after_before){
      // When $b_after_before,
      // always way better to cut after '/' than before it.
```

```
      // When !$b_after_before (aka before_after),
      // always way better to cut before '/' than after it.
      return 1 + $i_max_length + $i_cut_position;
    }
    return 1 + $i_cut_position;
  };
}
```

3.29 build_and_checks_dependencies/strings_functions.libr.sh

```
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# I think I created a file strings_functions.sh or renamed to
# strings_functions.sh at some point, before committing.
# Hence, someone must have changed or reversed the name of this file
# to string_functions.sh at some point.
# This file was renamed from "string_functions.sh" to
# "strings_functions.libr.sh".

LFBFL_subdir="build_and_checks_dependencies"
# shellcheck source=get_common_text_glob_patterns.libr.sh
source "./${LFBFL_subdir}/get_common_text_glob_patterns.libr.sh"
# shellcheck source=comparisons.libr.sh
source "./${LFBFL_subdir}/comparisons.libr.sh"
# shellcheck source=too_long_code_lines.libr.sh
```

```
source "./${LFBFL_subdir}/too_long_code_lines.libr.sh"

split_line_at(){
  # $1=line
  # $2=position : number of characters in split_line_at_result_beginning
  declare -g split_line_at_result_beginning="${1:0:$2}"
  declare -g split_line_at_result_end="${1:$2}"
  # printf "%s\n" "$split_line_at_result_beginning"
  # printf "%s\n" "$split_line_at_result_end"
}

#-----------------------------------------------------------------------
#-----------------------------------------------------------------------
# Properties for the score for a possible split/cut in the string between
# two characters.
#-----------------------------------------------------------------------
# Some preliminary remarks:
# a) If the character before the split position is a forbidden previous
#   character, the score is -1.
#   (Usually, either we have no or one forbidden previous character that is
#   a backslash.)
# b) If the previous character is not forbidden,
#   and either the character before the position is among the delimiters,
#   or the character after the position is among the delimiters,
#   then the score is computed according to the delimiter found,
#   the position, and the fact that the position is after or before the
#   delimiter.
# c) If two consecutive delimiters are found in the string,
#   the maximum score between the one after the first delimiter
#   and before the second delimiter is taken for the score between the
#   two delimiters.
# d) Otherwise, by default,
#   the score for cutting/splitting at some position is 0.
# e) A substring is given by the goal length given for the resulting
#   strings/lines after split.
# f) When some scores are equal,
#   we split at the rightmost position with maximum score in the substring.
# g) Remark f) implies that below we do not need to consider strict
#    inequalities for the score.
# h) The score function SF(position) which will dictate our choice for
#   split/cut position is in reality
#   SF1a(
#     position, previous_character, next_character, delimiters, forbiddens
#   )
#   but at the cost of expliciting its internal and adding verbosity,
#   we can see that the score really depends on the following arguments
```

```
#   SF1b(
#     position, matched_delimiter_before, matched_delimiter_after,
#     previous_character_is_forbidden
#   )
#   If it is needed,
#   then matched_delimiter_before or matched_delimiter_after is NULL.
#   Somehow previous_character can be seen as position.previous_character,
#   and next_character can be seen as position.next_character.
#   We can see that the previous or next character without
#   knowing that it matches a delimiter is irrelevant.
#   The fact that there is 1 or more delimiters may be interesting.
#   The fact that there is 0 or more forbidden previous characters
#   may be interesting, but we haven't integrated it in the split_score
#   functions.
#   SF1a when the previous character is not forbidden
#   needs a score function SF2a with the following parameters
#   SF2a(position, current_character, delimiter, is_after):
#   SF1a(
#     position, previous_character, next_character, delimiters, forbiddens
#   ) = max(
#     For all delimiter in delimiters,
#     SF2a(position, previous_character, delimiter, TRUE),
#     SF2a(position, next_character, delimiter, FALSE),
#   )
#   =  max(
#     For all delimiter in delimiters,
#     For all couple C in {
#       (position.previous_character, TRUE),
#       (position.next_character, FALSE)
#     }
#     SF2a(position, C.1, delimiter, C.2),
#   )
#   Similarly,
#   SF1b(
#     position, matched_delimiter_before, matched_delimiter_after,
#     previous_character_is_forbidden
#   ) = max(
#     For all delimiter in delimiters,
#     SF2b(position, matched_delimiter_before, TRUE),
#     SF2b(position, matched_delimiter_after, FALSE),
#   )
#   where the score function SF2b has the following parameters
#   SF2b(cut_position, matched_delimiter, is_after).
#   We have also:
#   SF1b(
#     position, matched_delimiter_before, matched_delimiter_after,
```

```
#     previous_character_is_forbidden
#   ) = max(
#     For all delimiter in delimiters,
#     SF2c(position-1, matched_delimiter_before, TRUE),
#     SF2c(position, matched_delimiter_after, FALSE),
#   )
#   where the score function SF2c has the following parameters
#   SF2c(character_position, matched_delimiter, is_after).
#   current_character is only used to check if it is equal to the
#   delimiter, otherwise the score is zero;
#   but when expressing properties it makes sense to compare after
#   and before the same character position.
#   position, delimiter, and is_after may all three impact the score
#   once the current character matches the delimiter.
#   Position is an integer from 1 to goal length.
#   Cutting at position 0 would be stupid,
#   and moreover there would be no previous character.
#   Since we ask for a split, the current length is above goal length,
#   and there is a next character at the last considered position.
#   So, in search of properties that may have an impact on the algorithm,
#   we discard the characters to only keep matched delimiters.
#   Let us use p1 and p2 for position1 and position2,
#   mdb1 and mdb2 for matched_delimiter_before1, etc.,
#   mda1 and mda2 for matched_delimiter_after1, etc.,
#   pcif for previous_character_is_forbidden, etc.,
#   md1 and md2 for matched_delimiter1, etc.,
#   ia1 and ia2 for is_after1, etc.
#
#   What we consider as properties are of 3 forms.
#   The first form is like:
#     For all p1 in [1, goal_length], for all p2 in [1, goal_length],
#     for all mdb1 in delimiters union {NULL},
#     for all mdb2 in delimiters union {NULL},
#     for all mda1 in delimiters union {NULL},
#     for all mda2 in delimiters union {NULL},
#     RelationP(p1, p2) AND RelationM(mdb1, mdb2, mda1, mda2)
#     implies RelationS(
#       SF1b(p1, mdb1, mda1, false),
#       SF1b(p2, mdb2, mda2, false),
#     )
#   There are many choices for RelationP, RelationM and RelationS.
#   We can apply the without loss of generality argument only once when
#   breaking symmetry between equivalent properties.
#   Breaking symmetry on another relation that RelationS first
#   is a challenge that we do not tackle.
#   Indeed RelationS is the only relation where all variables appear.
```

```
#   And breaking symmetry on RelationP would not allow to break symmetry
#   on RelationM freely since some variables are correlated by the
#   suffixes 1 or 2.
#   But some choices for some relation may allow to break symmetry on
#   another relation.
#   Breaking symmetry on the result, there are only 2 choices for
#   RelationS:
#   RS= : SF1b(p1, mdb1, mda1, false) = SF1b(p2, mdb2, mda2, false),
#   RS<= : SF1b(p1, mdb1, mda1, false) <= SF1b(p2, mdb2, mda2, false).
#   But then there are 5+1 choices for RelationP:
#   RP1 : p1 = p2,
#   RP2 : p1 < p2,
#   RP3 : p1 <= p2,
#   RP4 : p1 > p2,
#   RP5 : p1 >= p2,
#   plus the choice that p1 and p2 are not necessarily related
#   (empty RP or RPe).
#   We can add RP6 : p1 = p2 + 1 and RP7 : p2 = p1 + 1, since it implies
#   some equalities between matched delimiters.
#   Note that  p1 = p2 + i or p2 = p1 + i with i >= 2 brings no additional
#   logical relations compared to p1 < p2 or p2 < p1;
#   it only yields subslices of other properties and open a slightly silly
#   door on infinity.
#   If we have RS=, by symmetry we can consider only RP1, RP2, RP3 and RPe,
#   and maybe RP6.
#   Note that RSe or empty RS would have no interest at all.
#   There are many choices for RelationM:
#   mdb1 is NULL AND mdb2 is NULL AND mda1 is NULL AND mda2 is NULL is not
#   interesting at all since both scores are 0 or -1.
#   mdb1 is NULL AND mdb2 is NULL AND mda1 is NULL AND mda2 is not NULL
#   implies that second score is always larger, that's not an additional
#   property, that is a theorem.
#   Hence, at least one of mdb1 or mda1, resp. mdb2 or mda2,
#   must (have the possibility to) be non-NULL.
#   If we start listing the remaining relations, we have:
#   1) mdb1 is NULL AND mdb2 is NULL AND mda1 is not NULL
#     AND mda2 is not NULL
#   RMA1A2
#   2) mdb1 is NULL AND mdb2 is NULL AND mda1 is not NULL
#     AND mda2 is not NULL
#     AND mda1 = mda2
#   RMA1A2=
#   3) mdb1 is NULL AND mdb2 is NULL AND mda1 is not NULL
#     AND mda2 is not NULL
#     AND mda1 != mda2 -> This case has no meaning:
#     If we only know they are different, we cannot say anything on
```

```
#     properties related to delimiters that are different unless the score
#     doesn't depend on the delimiter, and that case is already handled by
#     1). The same remark applies to some other choices.
#     The only thing that could be said would be if we had an order on
#     delimiters scores that are uniform for the position, or the score
#     before some delimiter would always be =, >=, <=, etc. the score
#     before or after another one. So it would only make sense with RPe,
#     and mda1 <= mda2 for example, but that would trivially imply RS<=.
#     It would end up with max(mda1, mdb1) <= max(mda2, mdb2) where max
#     ignores NULL value. You do not deduce or specify anything new here.
#   4) mdb1 is not NULL AND mdb2 is not NULL AND mda1 is NULL
#     AND mda2 is NULL
#   RMB1B2
#   and so on. So if we use the short notation RM followed by A1, resp. A2,
#   resp. B1, resp. B2, when mda1, resp. mda2, resp. mdb1, resp. mdb2,
#   is not null, we have the following possibilities:
#   RMA1A2, RMB1B2, RMA1B2, RMA2B1, all 4 with variant with = like RMA1A2=,
#   RMA1A2B1, RMA1A2B2, RMA1B1B2, RMA2B1B2, all 4 (RMXXYYZZ) with variants
#   with suffixes _1 (XX = YY), _2 (XX = ZZ), _3 (YY = ZZ),
#   and _4 (XX = YY = ZZ)
#   for equalities on the 1st, 2nd or 3rd couple or between the three
#   variables by transitivity,
#   RMALL where none is null, with suffixes A1A2 (A1 = A2), etc.
#   RMALL_A1A2, RMALL_A1B1, RMALL_A1B2, RMALL_A2B1, RMALL_A2B2, RMALL_B1B2,
#   for one edge;
#   RMALL_A1A2B1, RMALL_A1A2B2, RMALL_A1B1B2, RMALL_A2B1B2, one triangle;
#   RMALL_A1A2_B1B2, RMALL_A1B1_A2B2, RMALL_A1B2_A2B1, 2 disjoint edges;
#   RMALL_ALL square with diagonals.
#   So we see that we listed at least 39 possibilities only for RM.
#   We did not list yet the possibilities where the variables not present
#   in some equality can be NULL or not NULL, etc.
#   We see 9 of them : RMe (40), and the following 8:
#   RMA1A2+ = RMA1A2= OR RMA1A2B1_1 OR RMA1A2B2_1 OR RMALL_A1A2
#   RMB1B2+ = RMB1B2= OR RMA1B1B2_3 OR RMA2B1B2_3 OR RMALL_B1B2
#   RMA1B2+ = RMA1B2= OR RMA1A2B2_2 OR RMA1B1B2_2 OR RMALL_A1B2
#   RMA2B1+ = RMA2B1= OR RMA1A2B1_3 OR RMA2B1B2_1 OR RMALL_A2B1
#   RMA1A2B1+ = RMA1A2B1_4 OR RMALL_A1A2B1
#   RMA1A2B2+ = RMA1A2B2_4 OR RMALL_A1A2B2
#   RMA1B1B2+ = RMA1B1B2_4 OR RMALL_A1B1B2
#   RMA2B1B2+ = RMA2B1B2_4 OR RMALL_A2B1B2
#   We don't do complete analysis at the moment.
#   If we wanted to filter out some possibilities,
#   we would keep only the RM possibilities, among the 38,
#   that are symmetric between 1 and 2:
#   RMA1A2, RMB1B2, RMA1A2=, RMB1B2=,
#   RMALL, RMALL_A1A2, RMALL_A2B2, RMALL_A1A2_B1B2, RMALL_A1B1_A2B2,
```

```
#   RMALL_ALL. Still 10 possibilities.
#   We have 1 (RS=) times 5 times 10 + 1 (RS<=) times 8 times 10 = 130
#   combinations (life is too short ;) :) );
#   if we keep 48 RM possibilities, there is 624 combinations,
#   but we did not analyze totally which RM relations are useless.
#
#   The second form is like:
#     For all p1 in [1, goal_length], for all p2 in [1, goal_length],
#     for all md1 in delimiters union {NULL},
#     for all md2 in delimiters union {NULL},
#     for all ia1 in {TRUE,FALSE}, for all ia2 in {TRUE,FALSE},
#     RelationP(p1, p2) AND RelationMM(md1, md2) AND RelationIA(ia1, ia2)
#     implies RelationS(
#       SF2b(p1, md1, ia1)
#       SF2b(p2, md2, ia2)
#     )
#   The third form is like:
#     For all p1 in [1, goal_length], for all p2 in [1, goal_length],
#     for all md1 in delimiters union {NULL},
#     for all md2 in delimiters union {NULL},
#     for all ia1 in {TRUE,FALSE}, for all ia2 in {TRUE,FALSE},
#     RelationPP(p1, p2) AND RelationMM(md1, md2) AND RelationIA(ia1, ia2)
#     implies RelationS(
#       SF2c(p1, md1, ia1)
#       SF2c(p2, md2, ia2)
#     )
#   This time the combinatorics is much more pleasant :).
#   As above, we have RS= and RS<=, RPe, RP1 to RP7;
#   For RelationMM, assume that the possible delimiters correspond to
#   integer numbers 1 to i; hence md1, resp. md2, can be seen as an integer
#   in [0,i], either 0 for NULL or the delimiter index.
#   Clearly, if md1 = k in [1,i] and md2 = l in [1,i], then the only
#   information we can use is k = l or k != l.
#   If we allow that md1 != 0 and md2 = 0,
#   then both RS= and RS<= are false.
#   Thus RMMe is not possible.
#   If we allow that md1 = 0 and md2 != 0, then RS<= is always true,
#   whilst RS= is false.
#   Thus properties with RS<= can trivially be extended with
#   RMM-"allow-md1-0", not useful to explicit this for the moment.
#   If both md1 = 0 and md2 = 0, then both scores are 0 and we also learn
#   nothing more.
#   Thus we can focus on md1 and md2 are not NULL,
#   and are equal RMM= or not RMM!=. It seems that a property with RMM!=
#   could be extended to a property RMM=or!=/RMMNotNULL.
#   and finally RIA0 to RIA8 where TRUE = 1, FALSE = 0 and the digit is
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#   3 * ia1 + ia2, and both variables can be 2 if they can be either TRUE
#   or FALSE. Wrong ! Not pleasant ! XD
#   So for RIA, we keep either: empty RIAe (RIA8),
#   ia1 = ia2 RIA= (RIA0 or RIA4),
#   ia1 = ia2 = TRUE RIA=T (RIA4), ia1 = ia2 = FALSE RIA=F (RIA0),
#   ia1 != ia2 RIA!= (RIA1 or RIA3), ia1 != ia2 = FALSE RIA!=T1 (RIA3),
#   ia1 != ia2 = TRUE RIA!=T2 (RIA1),
#   ia1 = TRUE (RIA5), ia1=FALSE (RIA2),
#   ia2 = TRUE (RIA7), ia2=FALSE (RIA6),
#   even less pleasant XD : 11 possibilities.
#   In total, we have now 1 (RS=) times 5 times 2 times 11
#   + 1 (RS<=) times 8 times 2 times 11 = 286
#   combinations (life is even shorter ;) :) ).
#   We only listed below properties of the third form.
# i) By the previous remark, there is a good chance that we missed some
#   possible property.
#   We hope we did found the properties that had a simple meaning,
#   in particular related to the algorithm where either you
#   loop on the string forward and keep last position with maximum score,
#   or you loop backward and abort as soon as possible as soon you know
#   that you found some position at the end that must have maximum score.
# j) These properties are "boolean" flags (integers that are powers of 2).
#   We first put the exact integers,
#   then wondered about switching everything to exponentiation 2**i,
#   then we took the solution that minimizes the character length between
#   [1-9][0-9]* and 2**i, then reverted to initial before committing.
#   Most efficient interpreted code is what we did first,
#   if there is no opcode cache for constants.
#   Most safe code is when the language hides the details and handles
#   families of flags,
#   but beware if some flag goes into the database.
#   Most readable code is a mix: 1 is easier to read than 2**0 ;) :P,
#   or $((2<<0))...
#   What I would love to see in many languages is a flag type, as a subtype
#   of integers, flag32, flag64 for C for example.
#   I think there is no such thing because sooner or later you play with
#   pseudo-flags that are combination of flags, hence integers.
#   But just for the declaration to let the type checker check that the
#   integer is a power of 2: flag32 MY_FLAG = 4;
#   Or better flag32 MY_FLAG = 2; as an equivalent of int32 MY_FLAG = 1<<2;
#   Another fact is that practice makes perfect, so working with the powers
#   of 2 written out in base 10 gets you accustomed to them.
#   Safe code should be favored since (pre-)compilation optimizes it,
#   but for this project, I prefer to keep the integers in decimal and
#   check that it doubles when reading my listing, because I'm familiar
#   with that and the powers of 2 doesn't go beyond what I can check fast
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#   enough in my brain.
# k) Some of the properties below are union of other properties,
#   and are here to simplify not only code but understanding.
#   split_score_properties_logical_closure() function below checks that
#   all flags that are consequences of already activated flags are set.
# l) We use SSP_ prefix for SPLIT_SCORE_PROPERTY_.
# Some of these remarks are repeated below in the corresponding functions,
# and some remarks are added just after the constants.
#-----------------------------------------------------------------------
# Any score after a character that matches a delimiter
# is larger than or equal to
# any score before a character that matches a delimiter.
# RPPe RMMNotNull RIA!=T2 RS<=
declare -gir SSP_ALWAYS_LARGER_AFTER=1

# Any score before a character that matches a delimiter
# is larger than or equal to
# any score after a character that matches a delimiter.
# RPPe RMMNotNull RIA!=T1 RS<=
declare -gir SSP_ALWAYS_LARGER_BEFORE=2

# The score doesn't depend on the delimiter that is found.
# RPP1 RMMNotNull RIA= RS=
declare -gir SSP_DELIMITER_UNIFORM=4

# Whenever a character that matches a delimiter is found,
# the score after is equal to the score before.
# Or equivalently:
# For any delimiter, and for any character position,
# if the character at this position matches this delimiter,
# the score after the character
# is equal to
# the score before the character.
# RPP1 RMM= RIA!= RS=
declare -gir SSP_EQUAL_AFTER_OR_BEFORE=8
# declare -gir SSP_SAME_DELIMITER_AND_POSITION_EQUAL_AFTER_OR_BEFORE=

# Whenever a character that matches a delimiter is found,
# the score after is larger than or equal to the score before.
# Or equivalently:
# For any delimiter, and for any character position,
# if the character at this position matches this delimiter,
# the score after the character
# is larger than or equal to
# the score before the character.
# RPP1 RMM= RIA!=T2 RS<=
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declare -gir SSP_LARGER_AFTER=16
# declare -gir SSP_SAME_DELIMITER_AND_POSITION_LARGER_AFTER=

# Whenever a character that matches a delimiter is found,
# the score before is larger than or equal to the score after.
# Or equivalently:
# For any delimiter, and for any character position,
# if the character at this position matches this delimiter,
# the score before the character
# is larger than or equal to
# the score after the character.
# RPP1 RMM= RIA!=T1 RS<=
declare -gir SSP_LARGER_BEFORE=32
# declare -gir SSP_SAME_DELIMITER_AND_POSITION_LARGER_BEFORE=

# The score after, resp. before, a character that matches a delimiter
# doesn't decrease when the position in the string increase.
# (RPP2 or RPP3) RMM= RIA= RS<=
declare -gir SSP_POSITION_NOT_DECREASING=64

# The score after a character that matches a delimiter doesn't decrease
# when the position in the string increases.
# (RPP2 or RPP3) RMM= RIA=T RS<=
declare -gir SSP_POSITION_NOT_DECREASING_AFTER=128

# The score before a character that matches a delimiter doesn't decrease
# when the position in the string increases.
# (RPP2 or RPP3) RMM= RIA=F RS<=
declare -gir SSP_POSITION_NOT_DECREASING_BEFORE=256

# The score after, resp. before, a character that matches a delimiter
# doesn't depend on the position in the string.
# (RPP2 or RPP3) RMM= RIA= RS=
declare -gir SSP_POSITION_UNIFORM=512

# The score after a character that matches a delimiter doesn't depend
# on the position in the string.
# (RPP2 or RPP3) RMM= RIA=T RS=
declare -gir SSP_POSITION_UNIFORM_AFTER=1024

# The score before a character that matches a delimiter doesn't depend
# on the position in the string.
# (RPP2 or RPP3) RMM= RIA=F RS=
declare -gir SSP_POSITION_UNIFORM_BEFORE=2048

# For any delimiter,
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# any score after a character that matches this delimiter
# is equal to
# any score before a character that matches this delimiter.
# RPPe RMM= RIA!= RS=
# <=> RPPe RMM= RIAe RS=
declare -gir SSP_SAME_DELIMITER_EQUAL_AFTER_OR_BEFORE=4096

# For any delimiter,
# any score after a character that matches this delimiter
# is larger than or equal to
# any score before a character that matches this delimiter.
# RPPe RMM= RIA!=T2 RS<=
declare -gir SSP_SAME_DELIMITER_LARGER_AFTER=8192

# For any delimiter,
# any score before a character that matches this delimiter
# is larger than or equal to
# any score after a character that matches this delimiter.
# RPPe RMM= RIA!=T1 RS<=
declare -gir SSP_SAME_DELIMITER_LARGER_BEFORE=16384

# For any character position,
# assuming that a character at this position matches some delimiter,
# the score after a character at this position that matches a delimiter
# is equal to
# the score before a character at this position that matches a delimiter.
# RPP1 RMMNotNull RIA!= RS=
declare -gir SSP_SAME_POSITION_EQUAL_AFTER_OR_BEFORE=32768

# For any character position,
# assuming that a character at this position matches some delimiter,
# the score after a character at this position that matches a delimiter
# is larger than or equal to
# the score before a character at this position that matches a delimiter.
# RPP1 RMMNotNull RIA!=T2 RS<=
declare -gir SSP_SAME_POSITION_LARGER_AFTER=65536

# For any character position,
# assuming that a character at this position matches some delimiter,
# the score before a character at this position that matches a delimiter
# is larger than or equal to
# the score after a character at this position that matches a delimiter.
# RPP1 RMMNotNull RIA!=T1 RS<=
declare -gir SSP_SAME_POSITION_LARGER_BEFORE=131072

# For all possible split/cut positions, the score is equal as long as
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# at least one of the previous and next characters is a delimiter.
# Or equivalently:
# Any score after a character that matches a delimiter
# is equal to
# any score before a character that matches a delimiter.
# is_after doesn't change anything.
# So we cut on the last position adjacent to a delimiter without
# forbidden previous character (edge case, on last position).
# RPPe RMMNotNull RIAe RS=
declare -gir SSP_UNIFORM=262144
# declare -gir SSP_ALWAYS_EQUAL_AFTER_OR_BEFORE=1

#-----------------------------------------------------------------------
# m) Note that a property telling that the score is always the same
#   even if the current character isn't among delimiters is not needed,
#   since, in that case, a null split score command will split at the goal
#   length.
#-----------------------------------------------------------------------
#-----------------------------------------------------------------------

split_score_properties_logical_closure(){
  # $1=split_score_properties
  # Since this code is not performance critical,
  # we followed lexicographic order on constants,
  # instead of an order to minimize the operations.
  # It is also safer to avoid forgetting some relation.
  declare -i LFBFL_i_result=$(($1))
  declare -i LFBFL_i_result_old=0
  declare -i LFBFL_i_all_flags=$((
    SSP_ALWAYS_LARGER_AFTER
    | SSP_ALWAYS_LARGER_BEFORE
    | SSP_DELIMITER_UNIFORM
    | SSP_EQUAL_AFTER_OR_BEFORE
    | SSP_LARGER_AFTER
    | SSP_LARGER_BEFORE
    | SSP_POSITION_NOT_DECREASING
    | SSP_POSITION_NOT_DECREASING_AFTER
    | SSP_POSITION_NOT_DECREASING_BEFORE
    | SSP_POSITION_UNIFORM
    | SSP_POSITION_UNIFORM_AFTER
    | SSP_POSITION_UNIFORM_BEFORE
    | SSP_SAME_DELIMITER_EQUAL_AFTER_OR_BEFORE
    | SSP_SAME_DELIMITER_LARGER_AFTER
    | SSP_SAME_DELIMITER_LARGER_BEFORE
    | SSP_SAME_POSITION_EQUAL_AFTER_OR_BEFORE
    | SSP_SAME_POSITION_LARGER_AFTER
```

```
      | SSP_SAME_POSITION_LARGER_BEFORE
      | SSP_UNIFORM
    ))
    until [[ LFBFL_i_result_old -eq LFBFL_i_result ]]; do
      LFBFL_i_result_old=$((LFBFL_i_result))
      if ((LFBFL_i_result & SSP_ALWAYS_LARGER_AFTER)); then
        LFBFL_i_result=$((
          LFBFL_i_result
          | SSP_LARGER_AFTER
          | SSP_SAME_DELIMITER_LARGER_AFTER
          | SSP_SAME_POSITION_LARGER_AFTER
        ))
      fi
      if ((LFBFL_i_result & SSP_ALWAYS_LARGER_BEFORE)); then
        LFBFL_i_result=$((
          LFBFL_i_result
          | SSP_LARGER_BEFORE
          | SSP_SAME_DELIMITER_LARGER_BEFORE
          | SSP_SAME_POSITION_LARGER_BEFORE
        ))
      fi
      if ((LFBFL_i_result & SSP_POSITION_NOT_DECREASING)); then
        LFBFL_i_result=$((
          LFBFL_i_result
          | SSP_POSITION_NOT_DECREASING_AFTER
          | SSP_POSITION_NOT_DECREASING_BEFORE
        ))
      fi
      if ((LFBFL_i_result & SSP_POSITION_UNIFORM)); then
        LFBFL_i_result=$((
          LFBFL_i_result
          | SSP_POSITION_UNIFORM_AFTER
          | SSP_POSITION_UNIFORM_BEFORE
          | SSP_POSITION_NOT_DECREASING
          | SSP_POSITION_NOT_DECREASING_AFTER
          | SSP_POSITION_NOT_DECREASING_BEFORE
        ))
      fi
      if ((LFBFL_i_result & SSP_POSITION_UNIFORM_AFTER)); then
        LFBFL_i_result=$((
          LFBFL_i_result
          | SSP_POSITION_NOT_DECREASING_AFTER
        ))
      fi
      if ((LFBFL_i_result & SSP_POSITION_UNIFORM_BEFORE)); then
        LFBFL_i_result=$((
```

```
589         LFBFL_i_result
590         | SSP_POSITION_NOT_DECREASING_BEFORE
591       ))
592     fi
593     if ((LFBFL_i_result & SSP_SAME_DELIMITER_EQUAL_AFTER_OR_BEFORE)); then
594       LFBFL_i_result=$((
595         LFBFL_i_result
596         | SSP_EQUAL_AFTER_OR_BEFORE
597         | SSP_LARGER_AFTER
598         | SSP_LARGER_BEFORE
599         | SSP_SAME_DELIMITER_LARGER_AFTER
600         | SSP_SAME_DELIMITER_LARGER_BEFORE
601       ))
602     fi
603     if ((LFBFL_i_result & SSP_SAME_DELIMITER_LARGER_AFTER)); then
604       LFBFL_i_result=$((
605         LFBFL_i_result
606         | SSP_LARGER_AFTER
607       ))
608     fi
609     if ((LFBFL_i_result & SSP_SAME_DELIMITER_LARGER_BEFORE)); then
610       LFBFL_i_result=$((
611         LFBFL_i_result
612         | SSP_LARGER_BEFORE
613       ))
614     fi
615     if ((LFBFL_i_result & SSP_SAME_POSITION_EQUAL_AFTER_OR_BEFORE)); then
616       LFBFL_i_result=$((
617         LFBFL_i_result
618         | SSP_DELIMITER_UNIFORM
619         | SSP_EQUAL_AFTER_OR_BEFORE
620         | SSP_LARGER_AFTER
621         | SSP_LARGER_BEFORE
622         | SSP_SAME_POSITION_LARGER_AFTER
623         | SSP_SAME_POSITION_LARGER_BEFORE
624       ))
625     fi
626     if ((LFBFL_i_result & SSP_SAME_POSITION_LARGER_AFTER)); then
627       LFBFL_i_result=$((
628         LFBFL_i_result
629         | SSP_LARGER_AFTER
630       ))
631     fi
632     if ((LFBFL_i_result & SSP_SAME_POSITION_LARGER_BEFORE)); then
633       LFBFL_i_result=$((
634         LFBFL_i_result
```

```
635         | SSP_LARGER_BEFORE
636       ))
637     fi
638     if ((LFBFL_i_result & SSP_UNIFORM)); then
639       LFBFL_i_result=$((LFBFL_i_all_flags))
640       break
641     fi
642     if ((LFBFL_i_result & SSP_ALWAYS_LARGER_AFTER))\
643     && ((
644       LFBFL_i_result & (
645         SSP_ALWAYS_LARGER_BEFORE
646         | SSP_EQUAL_AFTER_OR_BEFORE
647         | SSP_LARGER_BEFORE
648         | SSP_SAME_DELIMITER_EQUAL_AFTER_OR_BEFORE
649         | SSP_SAME_DELIMITER_LARGER_BEFORE
650         | SSP_SAME_POSITION_EQUAL_AFTER_OR_BEFORE
651         | SSP_SAME_POSITION_LARGER_BEFORE
652       )
653     )); then
654       LFBFL_i_result=$((LFBFL_i_all_flags))
655       break
656     fi
657     if ((LFBFL_i_result & SSP_ALWAYS_LARGER_BEFORE))\
658     && ((
659       LFBFL_i_result & (
660         SSP_EQUAL_AFTER_OR_BEFORE
661         | SSP_LARGER_AFTER
662         | SSP_SAME_DELIMITER_EQUAL_AFTER_OR_BEFORE
663         | SSP_SAME_DELIMITER_LARGER_AFTER
664         | SSP_SAME_POSITION_EQUAL_AFTER_OR_BEFORE
665         | SSP_SAME_POSITION_LARGER_AFTER
666       )
667     )); then
668       LFBFL_i_result=$((LFBFL_i_all_flags))
669       break
670     fi
671     if ((LFBFL_i_result & SSP_DELIMITER_UNIFORM))\
672     && ((LFBFL_i_result & SSP_EQUAL_AFTER_OR_BEFORE)); then
673       LFBFL_i_result=$((
674         LFBFL_i_result
675         | SSP_SAME_POSITION_EQUAL_AFTER_OR_BEFORE
676       ))
677     fi
678     if ((LFBFL_i_result & SSP_DELIMITER_UNIFORM))\
679     && ((LFBFL_i_result & SSP_LARGER_AFTER)); then
680       LFBFL_i_result=$((
```

```
681         LFBFL_i_result
682         | SSP_SAME_POSITION_LARGER_AFTER
683       ))
684     fi
685     if ((LFBFL_i_result & SSP_DELIMITER_UNIFORM))\
686     && ((LFBFL_i_result & SSP_LARGER_BEFORE)); then
687       LFBFL_i_result=$((
688         LFBFL_i_result
689         | SSP_SAME_POSITION_LARGER_BEFORE
690       ))
691     fi
692     if ((LFBFL_i_result & SSP_DELIMITER_UNIFORM))\
693     && ((LFBFL_i_result & SSP_SAME_DELIMITER_EQUAL_AFTER_OR_BEFORE)); then
694       LFBFL_i_result=$((LFBFL_i_all_flags))
695       break
696     fi
697     if ((LFBFL_i_result & SSP_DELIMITER_UNIFORM))\
698     && ((LFBFL_i_result & SSP_SAME_DELIMITER_LARGER_AFTER)); then
699       LFBFL_i_result=$((
700         LFBFL_i_result
701         | SSP_ALWAYS_LARGER_AFTER
702       ))
703     fi
704     if ((LFBFL_i_result & SSP_DELIMITER_UNIFORM))\
705     && ((LFBFL_i_result & SSP_SAME_DELIMITER_LARGER_BEFORE)); then
706       LFBFL_i_result=$((
707         LFBFL_i_result
708         | SSP_ALWAYS_LARGER_BEFORE
709       ))
710     fi
711     if ((LFBFL_i_result & SSP_LARGER_AFTER))\
712     && ((LFBFL_i_result & SSP_LARGER_BEFORE)); then
713       LFBFL_i_result=$((
714         LFBFL_i_result
715         | SSP_EQUAL_AFTER_OR_BEFORE
716       ))
717     fi
718     if ((LFBFL_i_result & SSP_LARGER_AFTER))\
719     && ((
720       LFBFL_i_result & (
721         SSP_POSITION_UNIFORM
722         | SSP_POSITION_UNIFORM_AFTER
723       )
724     )); then
725       LFBFL_i_result=$((
726         LFBFL_i_result
```

```
727         | SSP_SAME_DELIMITER_LARGER_AFTER
728       ))
729     fi
730     if ((LFBFL_i_result & SSP_LARGER_BEFORE))\
731     && ((
732       LFBFL_i_result & (
733         SSP_POSITION_UNIFORM
734         | SSP_POSITION_UNIFORM_BEFORE
735       )
736     )); then
737       LFBFL_i_result=$((
738         LFBFL_i_result
739         | SSP_SAME_DELIMITER_LARGER_BEFORE
740       ))
741     fi
742     if ((LFBFL_i_result & SSP_POSITION_NOT_DECREASING_AFTER))\
743     && ((LFBFL_i_result & SSP_POSITION_NOT_DECREASING_BEFORE)); then
744       LFBFL_i_result=$((
745         LFBFL_i_result
746         | SSP_POSITION_NOT_DECREASING
747       ))
748     fi
749     if ((LFBFL_i_result & SSP_POSITION_UNIFORM))\
750     && ((LFBFL_i_result & SSP_EQUAL_AFTER_OR_BEFORE)); then
751       LFBFL_i_result=$((
752         LFBFL_i_result
753         | SSP_SAME_DELIMITER_EQUAL_AFTER_OR_BEFORE
754       ))
755     fi
756     if ((LFBFL_i_result & SSP_POSITION_UNIFORM))\
757     && ((LFBFL_i_result & SSP_SAME_POSITION_EQUAL_AFTER_OR_BEFORE)); then
758       LFBFL_i_result=$((LFBFL_i_all_flags))
759       break
760     fi
761     if ((LFBFL_i_result & SSP_POSITION_UNIFORM))\
762     && ((LFBFL_i_result & SSP_SAME_POSITION_LARGER_AFTER)); then
763       LFBFL_i_result=$((
764         LFBFL_i_result
765         | SSP_ALWAYS_LARGER_AFTER
766       ))
767     fi
768     if ((LFBFL_i_result & SSP_POSITION_UNIFORM))\
769     && ((LFBFL_i_result & SSP_SAME_POSITION_LARGER_BEFORE)); then
770       LFBFL_i_result=$((
771         LFBFL_i_result
772         | SSP_ALWAYS_LARGER_BEFORE
```

```
773         ))
774       fi
775       if ((LFBFL_i_result & SSP_POSITION_UNIFORM_AFTER))\
776       && ((LFBFL_i_result & SSP_POSITION_UNIFORM_BEFORE)); then
777         LFBFL_i_result=$((
778           LFBFL_i_result
779           | SSP_POSITION_UNIFORM
780         ))
781       fi
782       if ((LFBFL_i_result & SSP_POSITION_UNIFORM_AFTER))\
783       && ((LFBFL_i_result & SSP_SAME_POSITION_LARGER_AFTER)); then
784         LFBFL_i_result=$((
785           LFBFL_i_result
786           | SSP_ALWAYS_LARGER_AFTER
787         ))
788       fi
789       if ((LFBFL_i_result & SSP_POSITION_UNIFORM_BEFORE))\
790       && ((LFBFL_i_result & SSP_SAME_POSITION_LARGER_BEFORE)); then
791         LFBFL_i_result=$((
792           LFBFL_i_result
793           | SSP_ALWAYS_LARGER_BEFORE
794         ))
795       fi
796       if ((LFBFL_i_result & SSP_SAME_DELIMITER_EQUAL_AFTER_OR_BEFORE))\
797       && ((
798         LFBFL_i_result & (
799           SSP_SAME_POSITION_EQUAL_AFTER_OR_BEFORE
800           | SSP_SAME_POSITION_LARGER_AFTER
801           | SSP_SAME_POSITION_LARGER_BEFORE
802         )
803       )); then
804         LFBFL_i_result=$((LFBFL_i_all_flags))
805         break
806       fi
807       if ((LFBFL_i_result & SSP_SAME_DELIMITER_LARGER_AFTER))\
808       && ((
809         LFBFL_i_result & (
810           SSP_SAME_POSITION_EQUAL_AFTER_OR_BEFORE
811           | SSP_SAME_POSITION_LARGER_BEFORE
812         )
813       )); then
814         LFBFL_i_result=$((LFBFL_i_all_flags))
815         break
816       fi
817       if ((LFBFL_i_result & SSP_SAME_DELIMITER_LARGER_AFTER))\
818       && ((LFBFL_i_result & SSP_SAME_POSITION_LARGER_AFTER)); then
```

```
      LFBFL_i_result=$((
        LFBFL_i_result
        | SSP_ALWAYS_LARGER_AFTER
      ))
    fi
    if ((LFBFL_i_result & SSP_SAME_DELIMITER_LARGER_BEFORE))\
    && ((
      LFBFL_i_result & (
        SSP_SAME_POSITION_EQUAL_AFTER_OR_BEFORE
        |  SSP_SAME_POSITION_LARGER_AFTER
      )
    )); then
      LFBFL_i_result=$((LFBFL_i_all_flags))
      break
    fi
    if ((LFBFL_i_result & SSP_SAME_DELIMITER_LARGER_BEFORE))\
    && ((LFBFL_i_result & SSP_SAME_POSITION_LARGER_BEFORE)); then
      LFBFL_i_result=$((
        LFBFL_i_result
        | SSP_ALWAYS_LARGER_BEFORE
      ))
    fi
    if ((LFBFL_i_result & SSP_SAME_POSITION_LARGER_AFTER))\
    && ((LFBFL_i_result & SSP_SAME_POSITION_LARGER_BEFORE)); then
      LFBFL_i_result=$((
        LFBFL_i_result
        | SSP_SAME_POSITION_EQUAL_AFTER_OR_BEFORE
      ))
    fi
  done
  declare -gi i_split_score_properties_logical_closure_result=$((
    LFBFL_i_result
  ))
}

split_score_simple(){
  # $1=after_before
  # $2=delimiters_strings_domain concatenated characters/delimiters
  # $3=delimiter_string a single character
  # $4=is_cut_after
  declare -gi i_split_score_result
  if [[ "$2" != *$3* ]]; then
    i_split_score_result=0
    return
  fi
  if [[ "$4" == "$1" ]]; then
```

```
865     i_split_score_result=2
866     return
867   fi
868   i_split_score_result=1
869 }
870
871 get_split_score_simple(){
872   # $1=after_before
873   # $2=max_length
874   # $3=delimiters_strings_domain
875   declare -g get_split_score_result="split_score_simple $1 '$3'"
876   declare -gi i_get_split_score_result2
877   # i_get_split_score_result2=$((11-$1*4))
878   # if [[ $1 -eq 1 ]]; then
879   #   i_get_split_score_result2=7
880   # else
881   #   i_get_split_score_result2=11
882   # fi
883   i_get_split_score_result2=$((
884     SSP_DELIMITER_UNIFORM
885     + SSP_POSITION_UNIFORM
886     + SSP_LARGER_BEFORE
887     + (
888       $1 * (
889         SSP_LARGER_AFTER
890         - SSP_LARGER_BEFORE
891       )
892     )
893   ))
894   # shellcheck disable=SC2248
895   split_score_properties_logical_closure ${i_get_split_score_result2}
896   i_get_split_score_result2=$((
897     i_split_score_properties_logical_closure_result
898   ))
899 }
900
901 get_split_score_exec(){
902   declare -g SPLIT_SCORE_EXEC="./build_and_checks_dependencies"
903   SPLIT_SCORE_EXEC+="/split_score.exec.php"
904 }
905
906 split_score(){
907   # $1=after_before
908   # $2=max_length
909   # $3=delimiters_strings_domain concatenated characters/delimiters
910   # $4=delimiter_string a single character
```

```
  # $5=cut_position
  # $6=is_cut_after
  get_split_score_exec
  declare -r LFBFL_command="${SPLIT_SCORE_EXEC} $1 $2 '$3' '$4' $5 $6"
  declare -gi i_split_score_result
  i_split_score_result=$(eval "${LFBFL_command}")
}

# This function is currently not called,
# hence we need this shellcheck disable.
# shellcheck disable=SC2034
get_split_score(){
  # $1=after_before
  # $1=max_length
  # $2=delimiters_strings_domain
  declare -g get_split_score_result="split_score $1 $2 '$3'"
  declare -gi i_get_split_score_result2
  # i_get_split_score_result2=$((9-$1*4))
  i_get_split_score_result2=$((
    SSP_DELIMITER_UNIFORM
    + SSP_LARGER_BEFORE
    + (
      $1 * (
        SSP_LARGER_AFTER
        - SSP_LARGER_BEFORE
      )
    )
  ))
  # shellcheck disable=SC2248
  split_score_properties_logical_closure ${i_get_split_score_result2}
  i_get_split_score_result2=$((
    i_split_score_properties_logical_closure_result
  ))
}

split_line_at_most(){
  # $1=line
  # $2=max_length of beginning result string
  # $3=split_score_command
  #   negative score means do not consider as a possible split
  # $4=split_score_command_properties :
  #   These properties are flags.
  #   If someday some property is not a flag,
  #   then create split_score_command_properties2.
  #   Old values to get a simple idea without reading everything at the
  #   beginning of file:
```

```
#   - null or 0 no property;
#   - 1 split score doesn't depend on which delimiter matches;
#   - 2 split score doesn't depend on the current position;
#   - 4 any split score after is always larger
#     than any split score before;
#   - 8 any split score before is always larger
#     than any split score after;
# $5=forbidden_previous_character
# $6=split_score_command_delimiter_strings_domain
#   optimisation if small domain
#   - this argument can be "null", then we will use $3 only instead,
#     in that case, only mono-character strings are considered,
#   - this argument can be an array of characters,
#     optimize accordingly,
#   - this argument can be an array of strings,
#     optimize accordingly,
#   - this argument can be an array of regexps,
#     optimize accordingly.
#     Not sure an array of regexps makes more sense than a single
#     one. Maybe they could be both cases with corresponding
#     optimizations. Same for a single character or substring.
#
# Note that we're always looking for the last position to split
# among the positions with maximum score;
# if some position gets a score from 2 delimiters,
# its score is always the maximum of the two corresponding scores.
declare -g split_line_at_most_result_start
declare -g split_line_at_most_result_end
declare -A LFBFL_positions
# printf "%s %s %s %s\n" "$1" "$2" "$3" "$4"
# For my use case in Bash scripts, I will need only an array of
# characters. See get_split_score().
if [[ -n "$6" ]]; then
  printf $'split_line_at_most() $6 NOT IMPLEMENTED YET\n'
fi
if [[ $2 -ge ${#1} ]]; then
  split_line_at_most_result_start=$1
  split_line_at_most_result_end=""
  return
fi
# We know that ${#1} > $2, hence ${#1} - 1 >= $2
# Edge cases:
# There is always at least one character that is overlength,
# but we need to look only at the score before that character that is
# overlength, the score after that character is invalid,
# whatever ${#1} - 1 = $2, or ${#1} - 1 > $2.
```

```
1003    # That first character that is overlength has index $2.
1004    # The cut position after that character has index $2 + 1.
1005    # Looping from character position going from 0 to $2, we need to drop
1006    # score before character 0 and drop score after character $2.
1007    # We do this with forbidden score -1.
1008    declare -ir LFBFL_i_max=$(($2))
1009    declare -i LFBFL_i
1010    declare -i LFBFL_j
1011    local LFBFL_current_char
1012
1013    # At the beginning, each position has score 0.
1014    LFBFL_positions["0"]="-1"
1015    for ((LFBFL_i = 1; LFBFL_i <= LFBFL_i_max; ++LFBFL_i)) do
1016      LFBFL_positions["${LFBFL_i}"]="0"
1017    done
1018    LFBFL_positions["${LFBFL_i}"]="-1"
1019
1020    # If we have a forbidden previous character,
1021    # we mark the forbidden positions with -1.
1022    if [[ -n "$5" ]]; then
1023      for ((LFBFL_i = 0; LFBFL_i < LFBFL_i_max;)) do
1024        LFBFL_current_char="${1:${LFBFL_i}:1}"
1025        ((++LFBFL_i))
1026        if [[ "${LFBFL_current_char}" == "$5" ]]; then
1027          LFBFL_positions["${LFBFL_i}"]="-1"
1028        fi
1029      done
1030    fi
1031
1032    declare -r LFBFL_sort_command="sort --numeric-sort"
1033
1034    if (($4 & SSP_DELIMITER_UNIFORM))\
1035    && (($4 & SSP_LARGER_AFTER))\
1036    && (($4 & SSP_POSITION_NOT_DECREASING_AFTER)); then
1037      for ((LFBFL_i = LFBFL_i_max; LFBFL_i >= 0; --LFBFL_i)) do
1038        LFBFL_current_char="${1:${LFBFL_i}:1}"
1039        eval "$3 '${LFBFL_current_char}' 1"
1040        # needs j=i+1 printf "%s|%s\n" "${i_split_score_result}" "${LFBFL_j}"
1041        if [[ i_split_score_result -ge 1 ]]; then
1042          LFBFL_j=$((LFBFL_i + 1))
1043          # printf "%s|%s\n" "${i_split_score_result}" "${LFBFL_j}"
1044          if [[ ${LFBFL_positions["${LFBFL_j}"]} != "-1" ]]; then
1045            LFBFL_positions["${LFBFL_j}"]=$(
1046              max "${LFBFL_sort_command}"\
1047                "${LFBFL_positions["${LFBFL_j}"]}"\
1048                "${i_split_score_result}"
```

```
1049          )
1050          break
1051        fi
1052      fi
1053      eval "$3 '${LFBFL_current_char}' 0"
1054      # printf "%s|%s\n" "${i_split_score_result}" "${LFBFL_i}"
1055      if [[ i_split_score_result -ge 1 ]]; then
1056        # printf "%s|%s\n" "${i_split_score_result}" "${LFBFL_i}"
1057        if [[ ${LFBFL_positions["${LFBFL_i}"]} != "-1" ]]; then
1058          LFBFL_positions["${LFBFL_i}"]=$(
1059            max "${LFBFL_sort_command}"\
1060              "${LFBFL_positions["${LFBFL_i}"]}"\
1061              "${i_split_score_result}"
1062          )
1063        fi
1064      fi
1065    done
1066  elif (($4 & SSP_DELIMITER_UNIFORM))\
1067  && (($4 & SSP_LARGER_BEFORE))\
1068  && (($4 & SSP_POSITION_NOT_DECREASING_BEFORE)); then
1069    for ((LFBFL_i = LFBFL_i_max; LFBFL_i >= 0; --LFBFL_i)) do
1070      LFBFL_current_char="${1:${LFBFL_i}:1}"
1071      # We still need to break in second if, after computing score after,
1072      # because SSP_LARGER_BEFORE implies large inequality instead of
1073      # strict inequality.
1074      # And since the algorithm is to take the largest position with
1075      # maximum score, there is a possibility that we'll break on a found
1076      # score before but for splitting on a score after.
1077      eval "$3 '${LFBFL_current_char}' 1"
1078      # needs j=i+1 printf "%s|%s\n" "${i_split_score_result}" "${LFBFL_j}"
1079      if [[ i_split_score_result -ge 1 ]]; then
1080        LFBFL_j=$((LFBFL_i + 1))
1081        # printf "%s|%s\n" "${i_split_score_result}" "${LFBFL_j}"
1082        if [[ ${LFBFL_positions["${LFBFL_j}"]} != "-1" ]]; then
1083          LFBFL_positions["${LFBFL_j}"]=$(
1084            max "${LFBFL_sort_command}"\
1085              "${LFBFL_positions["${LFBFL_j}"]}"\
1086              "${i_split_score_result}"
1087          )
1088        fi
1089      fi
1090      eval "$3 '${LFBFL_current_char}' 0"
1091      # printf "%s|%s\n" "${i_split_score_result}" "${LFBFL_i}"
1092      if [[ i_split_score_result -ge 1 ]]; then
1093        # printf "%s|%s\n" "${i_split_score_result}" "${LFBFL_i}"
1094        if [[ ${LFBFL_positions["${LFBFL_i}"]} != "-1" ]]; then
```

```
1095          LFBFL_positions["${LFBFL_i}"]=$(
1096            max "${LFBFL_sort_command}"\
1097              "${LFBFL_positions["${LFBFL_i}"]}"\
1098              "${i_split_score_result}"
1099          )
1100          break
1101        fi
1102      fi
1103    done
1104  else
1105    for ((LFBFL_i = 0; LFBFL_i <= LFBFL_i_max;)) do
1106      LFBFL_current_char="${1:${LFBFL_i}:1}"
1107      eval "$3 '${LFBFL_current_char}' ${LFBFL_i} 0"
1108      # printf "%s|%s\n" "${i_split_score_result}" "${LFBFL_i}"
1109      if [[ i_split_score_result -ge 1 ]]; then
1110        # printf "%s|%s\n" "${i_split_score_result}" "${LFBFL_i}"
1111        if [[ ${LFBFL_positions["${LFBFL_i}"]} != "-1" ]]; then
1112          LFBFL_positions["${LFBFL_i}"]=$(
1113            max "${LFBFL_sort_command}"\
1114              "${LFBFL_positions["${LFBFL_i}"]}"\
1115              "${i_split_score_result}"
1116          )
1117        fi
1118      fi
1119      ((++LFBFL_i))
1120      eval "$3 '${LFBFL_current_char}' ${LFBFL_i} 1"
1121      # printf "%s|%s\n" "${i_split_score_result}" "${LFBFL_i}"
1122      if [[ i_split_score_result -ge 1 ]]; then
1123        # printf "%s|%s\n" "${i_split_score_result}" "${LFBFL_i}"
1124        if [[ ${LFBFL_positions["${LFBFL_i}"]} != "-1" ]]; then
1125          LFBFL_positions["${LFBFL_i}"]=$(
1126            max "${LFBFL_sort_command}"\
1127              "${LFBFL_positions["${LFBFL_i}"]}"\
1128              "${i_split_score_result}"
1129          )
1130        fi
1131      fi
1132    done
1133  fi
1134  declare -ir LFBFL_i_max_score=$(
1135    max 'sort --numeric-sort' "${LFBFL_positions[@]}"
1136  )
1137  local LFBFL_value
1138  declare -i LFBFL_i_best_position
1139  for ((LFBFL_i = LFBFL_i_max; LFBFL_i >= 0; --LFBFL_i)) do
1140    if [[ LFBFL_i -eq 0 ]]; then
```

```
      printf $'/!\\split_line_at_most bug: no split position found./!\\\n'
    fi
    local LFBFL_value=${LFBFL_positions[${LFBFL_i}]}
    # printf "%s %s\n" "${LFBFL_i} ${LFBFL_value}"
    if [[ "${LFBFL_value}" == "${LFBFL_i_max_score}" ]]; then
      LFBFL_i_best_position=$((LFBFL_i))
      break
    fi
  done
  # shellcheck disable=SC2248
  split_line_at "$1" ${LFBFL_i_best_position}
  split_line_at_most_result_start="${split_line_at_result_beginning}"
  split_line_at_most_result_end="${split_line_at_result_end}"
  # printf "%s\n" "$split_line_at_most_result_start"
  # printf "%s\n" "$split_line_at_most_result_end"
}

split_last_line(){
  # $1=new_lines
  # $2=prefix
  # $3=max_length
  # $4=suffix
  # $5=split_score_command
  # $6=split_score_command_properties
  # $7=forbidden_previous_character ('\' usually)
  # Options:
  #   --verbose
  # In general, a prefix/suffix can be a required line prefix/suffix
  # for all final lines like a comment prefix,
  # or it can be a continuation line prefix/suffix applied only
  # when splitting a line in 2 lines,
  # or it can be a prefix/suffix for the string made of all the lines.
  # Clearly, there is no gain to couple handling a prefix/suffix for
  # the string made of all the lines in this function.
  # Since we add a prefix/suffix when splitting,
  # it is easy to see that we do not need to distinguish between
  # required line prefix/suffix and continuation line prefix/suffix.
  # The $2 and $4 arguments are given as concatenation of the
  # required and continuation line prefixes/suffixes as needed by the
  # user.
  # Thus lines arguments should be given already prefixed
  # and suffixed.
  # This is logic, since it would be cumbersome to handle prefixing
  # or not already splitted lines in this function
  # (see repeated_split_last_line).
  # The given max_length is the goal.
```

```
1187    # This function adapts the effective max position for the split by
1188    # taking into account the length of the suffix.
1189    declare -i LFBFL_i_verbose=0
1190    get_verbose_option "$@"
1191    declare -a LFBFL_return_traps_stack
1192    local LFBFL_previous_return_trap
1193    init_return_trap
1194
1195    enhanced_set_shell_option pipefail --trap-unset
1196
1197    declare -g split_last_line_result="$1"
1198    declare -ir LFBFL_i_overlength=$(($3 + 1))
1199    declare -ir LFBFL_i_length2=$(($3 - ${#4}))
1200    # shellcheck disable=SC2248
1201    get_overlength_regexp ${LFBFL_i_overlength}
1202
1203    # Testing if some line is too long.
1204    printf "%s" "$1"\
1205      | grep -q "${get_overlength_regexp_result}";
1206    if [[ ${PIPESTATUS[1]} -eq 1 ]]; then
1207      return
1208    fi
1209
1210    declare -r LFBFL_start=$(
1211      printf "%s" "$1"\
1212      | head --lines=-1
1213    )
1214    # printf "start: %s\n" "${LFBFL_start}"
1215    declare -r LFBFL_last_line=$(
1216      printf "%s" "$1"\
1217      | tail --lines=1
1218    )
1219    # printf "last_line: %s\n" "${LFBFL_last_line}"
1220    split_last_line_result=""
1221    if [[ -n "${LFBFL_start}" ]]; then
1222      split_last_line_result="${LFBFL_start}"
1223      # Last line-return of LFBFL_start seems dropped by $() and not ="${}".
1224      # printf "%s %s\n" "${#split_last_line_result}"
1225      #   "${split_last_line_result}"
1226      split_last_line_result+=$'\n'
1227      # something="${split_last_line_result}"
1228      # printf "%s %s\n" "${#something}" "${something}"
1229    fi
1230    if [[ -n "$5" ]]; then
1231      # shellcheck disable=SC2248
1232      split_line_at_most "${LFBFL_last_line}" ${LFBFL_i_length2}\
```

```
      "$5" "$6" "$7"
    split_last_line_result+="${split_line_at_most_result_start}"
    split_last_line_result+="$4"
    split_last_line_result+=$'\n'
    split_last_line_result+="$2${split_line_at_most_result_end}"
  else
    split_last_line_result+="${LFBFL_last_line:0:${LFBFL_i_length2}}"
    split_last_line_result+="$4"
    split_last_line_result+=$'\n'
    split_last_line_result+="$2${LFBFL_last_line:${LFBFL_i_length2}}"
  fi
  # printf "result: %s\n" "${split_last_line_result}"
}

repeated_split_last_line(){
  # See split_last_line for the first arguments.
  # $8=final_suffix to handle when the end of the original string
  #    is the suffix.
  declare -g repeated_split_last_line_result="$1"
  local LFBFL_result="$1x"
  until
    [[ "${LFBFL_result}" == "${repeated_split_last_line_result}" ]];
  do
    split_last_line "${repeated_split_last_line_result}" "$2" "$3"\
      "$4" "$5" "$6" "$7"
    LFBFL_result="${repeated_split_last_line_result}"
    repeated_split_last_line_result="${split_last_line_result}"
    # printf "%s\n" "${repeated_split_last_line_result}"
  done
  if [[ -n "$4" && -n "$8" ]]; then
    declare -ir LFBFL_i_offset=$((-${#4}))
    local LFBFL_lines_end
    LFBFL_lines_end="${repeated_split_last_line_result: ${LFBFL_i_offset}}"
    if [[ "${LFBFL_lines_end}" == "$4" ]]; then
      repeated_split_last_line_result+="$8"
    fi
  fi
}
```

3.30 build_and_checks_dependencies/too_long_code_lines.libr.sh

```
#!/usr/bin/env bash
# This file is part of DevOrSysAdminScripts library.
#
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```

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#
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# This file was renamed from "too_long_code_lines.sh"
# to "too_long_code_lines.libr.sh".

LFBFL_subdir="build_and_checks_dependencies"
# shellcheck source=common_options.libr.sh
source "./${LFBFL_subdir}/common_options.libr.sh"
# shellcheck source=get_common_text_glob_patterns.libr.sh
source "./${LFBFL_subdir}/get_common_text_glob_patterns.libr.sh"
# shellcheck source=lines_filters.libr.sh
source "./${LFBFL_subdir}/lines_filters.libr.sh"

get_overlength_regexp(){
  declare -g get_overlength_regexp_result=".\{$1\}"
  # declare -g get_overlength_regexp_result="^.\{$1\}"
  # declare -g get_overlength_regexp_result=".\{$1\}\$"
  # First one seems to be faster by a few percents when testing like this:
  # time for ((i = 0; i < 1000; ++i)); do grep '.{70}' COPYING; done
  # time for ((i = 0; i < 1000; ++i)); do grep '^.{70}' COPYING; done
  # time for ((i = 0; i < 1000; ++i)); do grep '.{70}$' COPYING; done
}

too_long_code_lines(){
  # Options:
  #   --verbose
  #   --work-directory=""
  #   --max-line-length
  declare -i LFBFL_i_verbose=0
  get_verbose_option "$@"
```

```
52    local LFBFL_work_directory=""
53    get_work_directory_option "$@"
54    declare -a LFBFL_return_traps_stack
55    local LFBFL_previous_return_trap
56    init_return_trap
57    pushd_to_work_directory --trap-popd
58    can_continue_after_enhanced_pushd || return 1
59
60    local LFBFL_arg
61    declare -i LFBFL_i_max_line_length=70
62    for LFBFL_arg in "$@"; do
63      if [[ "${LFBFL_arg}" == --max-line-length=* ]]; then
64        LFBFL_arg=${LFBFL_arg#--max-line-length=}
65        LFBFL_i_max_line_length=$((LFBFL_arg))
66        break
67      fi
68    done
69
70    declare -ir LFBFL_i_overlength=$((LFBFL_i_max_line_length+1))
71    # shellcheck disable=SC2248
72    get_overlength_regexp ${LFBFL_i_overlength}
73
74    enhanced_set_shell_option pipefail --trap-unset
75
76    get_COMMON_TEXT_FILES_GLOB_PATTERNS
77    local LFBFL_pattern
78    local LFBFL_long_line
79    local LFBFL_file_path
80    local LFBFL_line
81    local LFBFL_extension
82    local LFBFL_base_name
83    local LFBFL_s_long_lines
84    declare -a LFBFL_arr_long_lines
85    for LFBFL_pattern in "${COMMON_TEXT_FILES_GLOB_PATTERNS[@]}"; do
86      if [[ LFBFL_i_verbose -eq 1 ]]; then
87        printf "too_long_code_lines: Iterating on pattern: %s.\n"\
88          "${LFBFL_pattern}"
89      fi
90      # shellcheck disable=SC2248
91      LFBFL_s_long_lines=$(
92        find . -type f -name "${LFBFL_pattern}" -printf '%P\n'\
93        | relevant_find\
94        | not_license_find\
95        | xargs grep --line-number --with-filename\
96          "${get_overlength_regexp_result}"
97      )
```

```
    if [[ -z "${LFBFL_s_long_lines}" ]]; then
      [[ LFBFL_i_verbose -eq 1 ]]\
        && printf "too_long_code_lines: No file found with long lines.\n"
      continue
    fi
    mapfile -t LFBFL_arr_long_lines <<< "${LFBFL_s_long_lines}"
    for LFBFL_long_line in "${LFBFL_arr_long_lines[@]}"; do
      LFBFL_file_path=${LFBFL_long_line%%:*} # Drop long ':*' suffix
      # Remove file_path and line_number
      LFBFL_line=${LFBFL_long_line#*:} # Drop short '*:' prefix
      LFBFL_line=${LFBFL_line#*:} # Drop short '*:' prefix
      # LFBFL_line="${LFBFL_line/%\\/}" # Drop/Replace '\' suffix by ''
      LFBFL_extension=${LFBFL_file_path##*.} # Drop long '*.' prefix
      LFBFL_base_name=${LFBFL_file_path%.*} # Drop short '.*' suffix
      if [[ "${LFBFL_extension}" == "html" ]]; then
        if [[ -f "${LFBFL_base_name}.md" ]]; then
          continue
        fi
        if grep --quiet --fixed-strings -- "${LFBFL_line}"\
          "./build_and_checks_variables/temp/files_listing.html.tpl"
        then
          continue
        fi
        printf "%s\n" "${LFBFL_long_line}"
      elif [[ "${LFBFL_extension}" == "md" ]]; then
        if ! [[ -f "${LFBFL_base_name}.md.tpl" ]]; then
          printf "%s\n" "${LFBFL_long_line}"
        fi
      else
        printf "%s\n" "${LFBFL_long_line}"
      fi
    done
  done
}
```

3.31 build_and_checks_dependencies/update_common_build_and_checks.exec.sh

```
#!/usr/bin/env bash
# This file is part of DevOrSysAdminScripts library.
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```

```
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#
# ©Copyright 2023-2026 Laurent Frédéric Bernard François Lyaudet
# This file was renamed from "update_common_build_and_checks.sh"
# to "update_common_build_and_checks.exec.sh".

source "overwrite_if_not_equal.libr.sh"

update_common_build_and_checks(){
  declare -i LFBFL_i_verbose=0
  if [[ "$*" == *--verbose* ]]; then
    printf "%s %s\n" "$0" "$*"
    LFBFL_i_verbose=1
  fi
  readonly LFBFL_i_verbose

  if [[ ! -o pipefail ]]; then
    [[ LFBFL_i_verbose -eq 1 ]] && printf "pipefail option activated\n"
    set -o pipefail
    trap 'set +o pipefail' RETURN
  fi

  declare -r LFBFL_common_file_name="common_build_and_checks.exec.sh"
  cp "./${LFBFL_common_file_name}.tpl"\
    "./${LFBFL_common_file_name}.temp"

  declare -ar LFBFL_files_names=(
    "build_md_from_printable_md.exec.sh"
    "build_dependencies_notes.exec.php"
    "check_shell_scripts_beginnings.libr.sh"
    "check_shell_scripts_indentation.libr.sh"
    "check_URLs.libr.sh"
    "common_options.libr.sh"
    "comparisons.libr.sh"
    "create_PDF.exec.sh"
```

```
    "generate_from_template.libr.sh"
    "get_common_text_glob_patterns.libr.sh"
    "grammar_and_spelling_check.libr.sh"
    "licenses_templates/build_licenses_templates.exec.sh"
    "licenses_templates/license_file_header_AGPLv3+.tpl"
    "licenses_templates/license_file_header_GPLv3+.tpl"
    "licenses_templates/license_file_header_LGPLv3+.tpl"
    "lines_counts.libr.sh"
    "lines_filters.libr.sh"
    "lines_maps.libr.sh"
    "update_or_check_files_names_listing.exec.sh"
    "overwrite_if_not_equal.libr.sh"
    "python_black_complement.libr.sh"
    "python_isort_complement.libr.sh"
    "repository_data.libr.sh"
    "shell_checks_complement.libr.sh"
    "split_score.exec.php"
    "split_score.libr.php"
    "strings_functions.libr.sh"
    "too_long_code_lines.libr.sh"
  )

  local LFBFL_file_name
  local LFBFL_file_sha512
  local LFBFL_file_sha512_all
  local LFBFL_base_file_name
  for LFBFL_file_name in "${LFBFL_files_names[@]}"; do
    LFBFL_file_sha512=$(
      sha512sum "./${LFBFL_file_name}"\
      | cut -f1 -d' '
    )
    LFBFL_file_sha512_all="LFBFL_correct_sha512="
    LFBFL_file_sha512_all+="'${LFBFL_file_sha512:0:45}'\n"
    LFBFL_file_sha512_all+="  LFBFL_correct_sha512\+="
    LFBFL_file_sha512_all+="'${LFBFL_file_sha512:45:44}'\n"
    LFBFL_file_sha512_all+="  LFBFL_correct_sha512\+="
    LFBFL_file_sha512_all+="'${LFBFL_file_sha512:89}'"
    LFBFL_base_file_name=$(basename "${LFBFL_file_name}")
    sed -i\
      "s|@sha512_${LFBFL_base_file_name}@|${LFBFL_file_sha512_all}|g"\
      "./${LFBFL_common_file_name}.temp"
  done

  overwrite_if_not_equal "./${LFBFL_common_file_name}"\
    "./${LFBFL_common_file_name}.temp"

```

```
  declare -r LFBFL_main_file_name="build_and_checks.exec.sh"
  cp "./${LFBFL_main_file_name}.tpl" "../${LFBFL_main_file_name}.temp"

  LFBFL_file_sha512=$(
    sha512sum "./${LFBFL_common_file_name}"\
    | cut -f1 -d' '
  )
  LFBFL_file_sha512_all="local LFBFL_correct_sha512\n"
  LFBFL_file_sha512_all+="  LFBFL_correct_sha512="
  LFBFL_file_sha512_all+="'${LFBFL_file_sha512:0:45}'\n"
  LFBFL_file_sha512_all+="  LFBFL_correct_sha512\+="
  LFBFL_file_sha512_all+="'${LFBFL_file_sha512:45:44}'\n"
  LFBFL_file_sha512_all+="  LFBFL_correct_sha512\+="
  LFBFL_file_sha512_all+="'${LFBFL_file_sha512:89}'"
  sed -i\
    "s|@sha512_${LFBFL_common_file_name}@|${LFBFL_file_sha512_all}|g"\
    "../${LFBFL_main_file_name}.temp"

  overwrite_if_not_equal "../${LFBFL_main_file_name}"\
    "../${LFBFL_main_file_name}.temp"
}

update_common_build_and_checks "$@"
```

3.32 build_and_checks_dependencies/update_or_check_files_names_listing.exec.sh

```
#!/usr/bin/env bash
# This file is part of DevOrSysAdminScripts library.
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# 2025-12-30T21:33+01:00: This file was moved from
# "./build_and_checks_dependencies/listings"
# to
# "./build_and_checks_dependencies".
# This file was renamed from "update_or_check_files_names_listing.sh"
# to "update_or_check_files_names_listing.exec.sh".
# This file was renamed from "update_or_check_file_names_listing.sh"
# to "update_or_check_files_names_listing.sh"?

LFBFL_subdir="build_and_checks_dependencies"
# shellcheck source=common_options.libr.sh
source "./${LFBFL_subdir}/common_options.libr.sh"
# shellcheck source=lines_filters.libr.sh
source "./${LFBFL_subdir}/lines_filters.libr.sh"
# shellcheck source=strings_functions.libr.sh
source "./${LFBFL_subdir}/strings_functions.libr.sh"

update_or_check_files_names_listing(){
  # Options:
  #   --verbose
  #   --work-directory=""
  declare -i LFBFL_i_verbose=0
  get_verbose_option "$@"
  local LFBFL_work_directory=""
  get_work_directory_option "$@"
  declare -a LFBFL_return_traps_stack
  local LFBFL_previous_return_trap
  init_return_trap
  pushd_to_work_directory --trap-popd
  can_continue_after_enhanced_pushd || return 1

  declare -i LFBFL_i_write=0
  if [[ "$*" == *--write* ]]; then
    LFBFL_i_write=1
  fi
  readonly LFBFL_i_write

  declare -i LFBFL_i_append=0
  if [[ "$*" == *--append* ]]; then
    LFBFL_i_append=1
  fi
  readonly LFBFL_i_append
```

```
67
68    enhanced_set_shell_option pipefail --trap-unset
69
70    LFBFL_subdir2="build_and_checks_variables"
71    local LFBFL_listing="./${LFBFL_subdir2}/files_names_listing.txt"
72    readonly LFBFL_listing
73    local LFBFL_data_file_name="./${LFBFL_subdir2}/repository_data.txt"
74    readonly LFBFL_data_file_name
75
76    declare -i LFBFL_i_max_line_length
77    grep_variable "${LFBFL_data_file_name}" max_line_length\
78      --result-variable-prefix="LFBFL_i_"
79
80    if [[ LFBFL_i_write -eq 1 ]]; then
81      : > "${LFBFL_listing}"
82    fi
83    # Remove line returns that are here to keep lines short.
84    sed -Ez 's/(.)\\\n/\1/Mg' "${LFBFL_listing}"\
85      > "${LFBFL_listing}.temp"
86    # shopt -s dotglob was needed at some point but I don't see why now.
87    # shellcheck disable=SC2248
88    get_split_score_simple 1 ${LFBFL_i_max_line_length} /
89    declare -r LFBFL_split_score_command="${get_split_score_result}"
90    declare -i LFBFL_i_split_score_command_properties
91    LFBFL_i_split_score_command_properties=${i_get_split_score_result2}
92    readonly LFBFL_i_split_score_command_properties
93    local LFBFL_suffix=\\ # instead of '\' to avoid shellcheck SC1003
94    readonly LFBFL_suffix
95    # There is always a $'\n' final suffix from >>.
96    # But when origin string ends with the suffix we must add something.
97    declare -r LFBFL_final_suffix=$'\n\\'
98
99    local LFBFL_s_files_paths
100   LFBFL_s_files_paths=$(
101     find . -type f -printf '%P\n'\
102     | relevant_find\
103     | sort
104   )
105   # printf "%s\n" "${LFBFL_s_files_paths}"
106   declare -a LFBFL_arr_files_paths
107   if [[ -n "${LFBFL_s_files_paths}" ]]; then
108     if [[ LFBFL_i_verbose -eq 1 ]]; then
109       printf\
110         "update_or_check_files_names_listing: Some files to process.\n"
111     fi
112     mapfile -t LFBFL_arr_files_paths <<< "${LFBFL_s_files_paths}"
```

```
113     fi
114     # printf "%s\n" "${LFBFL_arr_files_paths[@]}"
115
116     local LFBFL_file_path
117     local LFBFL_base_file_name
118     for LFBFL_file_path in "${LFBFL_arr_files_paths[@]}"; do
119       # if [[ LFBFL_i_verbose -eq 1 ]]; then
120       #   printf\
121       #     "update_or_check_files_names_listing: Maybe ignored file: %s\n"\
122       #     "${LFBFL_file_path}"
123       # fi
124       git check-ignore -q "${LFBFL_file_path}" && continue
125       LFBFL_base_file_name=$(basename "${LFBFL_file_path}")
126       [[ "${LFBFL_base_file_name}" != files_names_listing.txt* ]]\
127         || continue
128       if [[ "${LFBFL_base_file_name}" == *.md ]]; then
129         if [[ -f "${LFBFL_file_path}.tpl" ]]; then
130           continue
131         fi
132       fi
133       if [[ LFBFL_i_verbose -eq 1 ]]; then
134         printf "update_or_check_files_names_listing: Non-ignored file: %s\n"\
135           "${LFBFL_file_path}"
136       fi
137       # shellcheck disable=SC2248
138       repeated_split_last_line "${LFBFL_file_path}"\
139         ""\
140         ${LFBFL_i_max_line_length}\
141         "${LFBFL_suffix}"\
142         "${LFBFL_split_score_command}"\
143         ${LFBFL_i_split_score_command_properties}\
144         ""\
145         "${LFBFL_final_suffix}"
146       if [[ LFBFL_i_write -eq 1 ]]; then
147         # shellcheck disable=SC2001
148         printf "%s\n" "${repeated_split_last_line_result}"\
149           >> "${LFBFL_listing}"
150         continue
151       fi
152       if grep_fixed_string_with_anchor "${LFBFL_listing}.temp"\
153           "${LFBFL_file_path}" --quiet --enforce-line-ends-with-fixed-string;
154       then
155         :
156       else
157         printf "The file %s is not listed in %s.\n"\
158           "${LFBFL_file_path}"\
```

```
        "${LFBFL_listing}"
      if [[ LFBFL_i_append -eq 1 ]]; then
        # shellcheck disable=SC2001
        printf "%s\n" "${repeated_split_last_line_result}"\
          >> "${LFBFL_listing}"
      fi
    fi
  done

  LFBFL_s_files_paths=$(cat "${LFBFL_listing}.temp")
  if [[ -n "${LFBFL_s_files_paths}" ]]; then
    mapfile -t LFBFL_arr_files_paths <<< "${LFBFL_s_files_paths}"
  fi

  for LFBFL_file_path in "${LFBFL_arr_files_paths[@]}"; do
    [[ "${LFBFL_file_path}" != '//'* ]] || continue
    if [[ ! -f "${LFBFL_file_path}" ]]; then
      printf "The non-file %s is listed in %s.\n"\
        "${LFBFL_file_path}"\
        "${LFBFL_listing}"
    fi
  done

  rm "${LFBFL_listing}.temp"

  if [[ LFBFL_i_verbose -eq 1 ]]; then
    cat "${LFBFL_listing}"
  fi
}

update_or_check_files_names_listing "$@"
```

3.33 build_and_checks_variables/.gitignore

```
temp/
upgrade_venvs_ts

```

3.34 build_and_checks_variables/bandit.ini

```
# FILE: .bandit
[bandit]
```

3.35 build_and_checks_variables/bandit_baseline.json

```
{"results":[]}
```

3.36 build_and_checks_variables/black.toml

```
[tool.black]
line-length = 75
```

3.37 build_and_checks_variables/dependencies_data.json

```
[
  { "dependency_name": "Bash", "URLs": [
      "https://www.gnu.org/software/bash/",
      "https://tiswww.case.edu/php/chet/bash/bashtop.html",
      "https://savannah.gnu.org/projects/bash/"
  ] },
  { "dependency_name": "GNU coreutils", "URLs": [
      "https://www.gnu.org/software/coreutils/",
      "https://savannah.gnu.org/projects/coreutils/"
  ] },
  { "dependency_name": "uutils coreutils", "URLs": [
      "https://uutils.github.io/coreutils/",
      "https://github.com/uutils/coreutils"
  ] },
  { "dependency_name": "diff (GNU diffutils)", "URLs": [
      "https://www.gnu.org/software/diffutils/",
      "https://savannah.gnu.org/projects/diffutils/"
  ] },
  { "dependency_name": "find, xargs (GNU findutils)", "URLs": [
      "https://www.gnu.org/software/findutils/",
      "https://savannah.gnu.org/projects/findutils/"
  ] },
  { "dependency_name": "grep (GNU grep)", "URLs": [
      "https://www.gnu.org/software/grep/",
      "https://savannah.gnu.org/projects/grep/"
  ] },
  { "dependency_name": "sed (GNU sed)", "URLs": [
      "https://www.gnu.org/software/sed/",
      "https://savannah.gnu.org/projects/sed/"
  ] },
  { "dependency_name": "ShellCheck", "URLs": [
      "https://www.shellcheck.net/",
      "https://github.com/koalaman/shellcheck"
  ] },
  { "dependency_name": "GNU Wget", "URLs": [
      "https://www.gnu.org/software/wget/",
      "https://savannah.gnu.org/projects/wget/"
  ] },
  { "dependency_name": "more of util-linux", "URLs": [
```

```
40          "https://www.kernel.org/pub/linux/utils/util-linux/",
41          "https://git.kernel.org/pub/scm/utils/util-linux/util-linux.git/",
42          {
43            "URL": "https://github.com/util-linux/util-linux",
44            "comment": "official?"
45          }
46      ] },
47      { "dependency_name": "pandoc", "URLs": [
48          "https://pandoc.org/",
49          "https://github.com/jgm/pandoc"
50      ] },
51      { "dependency_name": "pdftex/pdflatex", "URLs": [
52          "https://www.tug.org/applications/pdftex/",
53          "https://tug.org/texlive/svn/"
54      ] },
55      { "dependency_name": "tree", "URLs": [
56          "https://oldmanprogrammer.net/source.php?dir=projects/tree",
57          "https://gitlab.com/OldManProgrammer/unix-tree",
58          "https://github.com/Old-Man-Programmer/tree"
59      ] },
60      { "dependency_name": "git", "URLs": [
61          "https://git-scm.com/about",
62          "https://github.com/git/git"
63      ] },
64      { "dependency_name": "pylint", "URLs": [
65          "https://www.pylint.org/",
66          "https://pylint.readthedocs.io/en/stable/",
67          "https://github.com/pylint-dev/pylint",
68          "https://pypi.org/project/pylint/"
69      ] },
70      { "dependency_name": "notify-send", "URLs": [
71          "https://gitlab.gnome.org/GNOME/libnotify/",
72          "https://gnome.pages.gitlab.gnome.org/libnotify/"
73      ] },
74      { "dependency_name": "black", "URLs": [
75          "https://github.com/psf/black",
76          "https://black.readthedocs.io/en/stable/",
77          "https://pypi.org/project/black/"
78      ] },
79      { "dependency_name": "isort", "URLs": [
80          "https://pycqa.github.io/isort/",
81          "https://github.com/pycqa/isort/",
82          "https://pypi.org/project/isort/"
83      ] },
84      { "dependency_name": "mypy", "URLs": [
85          "https://mypy-lang.org/",
```

```
86          "https://mypy.readthedocs.io/en/stable/",
87          "https://github.com/python/mypy",
88          "https://pypi.org/project/mypy/"
89      ] },
90      { "dependency_name": "pip", "URLs": [
91          "https://pip.pypa.io/en/stable/",
92          "https://github.com/pypa/pip",
93          {
94            "URL": "https://pypi.org/project/pip/",
95            "comment": "(egg and chicken ;) )"
96          }
97      ] },
98      { "dependency_name": "ruff", "URLs": [
99          "https://docs.astral.sh/ruff/",
100         "https://github.com/astral-sh/ruff",
101         "https://pypi.org/project/ruff/"
102     ] },
103     { "dependency_name": "bandit", "URLs": [
104         "https://bandit.readthedocs.io/en/latest/",
105         "https://github.com/PyCQA/bandit",
106         "https://pypi.org/project/bandit/"
107     ] },
108     { "dependency_name": "phpmd", "URLs": [
109         "https://phpmd.org/",
110         "https://github.com/phpmd/phpmd",
111         "https://packagist.org/packages/phpmd/phpmd"
112     ] },
113     { "dependency_name": "php", "URLs": [
114         "https://www.php.net/",
115         "https://github.com/php/php-src"
116     ] },
117     { "dependency_name": "tex", "URLs": [
118         "https://ctan.org/tex-archive/systems/knuth/dist/tex"
119     ] },
120     { "dependency_name": "pcregrep, pcre2grep", "URLs": [
121         "https://www.pcre.org/",
122         "https://pcre2project.github.io/pcre2/",
123         "https://github.com/PCRE2Project/pcre1",
124         "https://github.com/PCRE2Project/pcre2",
125         "https://github.com/PCRE2Project/pcre2/blob/main/src/pcre2grep.c"
126     ] },
127     { "dependency_name": "python3", "URLs": [
128         "https://www.python.org/",
129         "https://docs.python.org/3/"
130     ] }
131 ]
```

3.38 build_and_checks_variables/dependencies_notes.txt

```
Bash can be found at the following URLs:
https://www.gnu.org/software/bash/
https://tiswww.case.edu/php/chet/bash/bashtop.html
https://savannah.gnu.org/projects/bash/

GNU coreutils can be found at the following URLs:
https://www.gnu.org/software/coreutils/
https://savannah.gnu.org/projects/coreutils/

uutils coreutils can be found at the following URLs:
https://uutils.github.io/coreutils/
https://github.com/uutils/coreutils

diff (GNU diffutils) can be found at the following URLs:
https://www.gnu.org/software/diffutils/
https://savannah.gnu.org/projects/diffutils/

find, xargs (GNU findutils) can be found at the following URLs:
https://www.gnu.org/software/findutils/
https://savannah.gnu.org/projects/findutils/

grep (GNU grep) can be found at the following URLs:
https://www.gnu.org/software/grep/
https://savannah.gnu.org/projects/grep/

sed (GNU sed) can be found at the following URLs:
https://www.gnu.org/software/sed/
https://savannah.gnu.org/projects/sed/

ShellCheck can be found at the following URLs:
https://www.shellcheck.net/
https://github.com/koalaman/shellcheck

GNU Wget can be found at the following URLs:
https://www.gnu.org/software/wget/
https://savannah.gnu.org/projects/wget/

more of util-linux can be found at the following URLs:
https://www.kernel.org/pub/linux/utils/util-linux/
https://git.kernel.org/pub/scm/utils/util-linux/util-linux.git/
https://github.com/util-linux/util-linux official?

pandoc can be found at the following URLs:
https://pandoc.org/
```

```
https://github.com/jgm/pandoc

pdftex/pdflatex can be found at the following URLs:
https://www.tug.org/applications/pdftex/
https://tug.org/texlive/svn/

tree can be found at the following URLs:
https://oldmanprogrammer.net/source.php?dir=projects/tree
https://gitlab.com/OldManProgrammer/unix-tree
https://github.com/Old-Man-Programmer/tree

git can be found at the following URLs:
https://git-scm.com/about
https://github.com/git/git

pylint can be found at the following URLs:
https://www.pylint.org/
https://pylint.readthedocs.io/en/stable/
https://github.com/pylint-dev/pylint
https://pypi.org/project/pylint/

notify-send can be found at the following URLs:
https://gitlab.gnome.org/GNOME/libnotify/
https://gnome.pages.gitlab.gnome.org/libnotify/

black can be found at the following URLs:
https://github.com/psf/black
https://black.readthedocs.io/en/stable/
https://pypi.org/project/black/

isort can be found at the following URLs:
https://pycqa.github.io/isort/
https://github.com/pycqa/isort/
https://pypi.org/project/isort/

mypy can be found at the following URLs:
https://mypy-lang.org/
https://mypy.readthedocs.io/en/stable/
https://github.com/python/mypy
https://pypi.org/project/mypy/

pip can be found at the following URLs:
https://pip.pypa.io/en/stable/
https://github.com/pypa/pip
https://pypi.org/project/pip/ (egg and chicken ;) )

```

```
ruff can be found at the following URLs:
https://docs.astral.sh/ruff/
https://github.com/astral-sh/ruff
https://pypi.org/project/ruff/

bandit can be found at the following URLs:
https://bandit.readthedocs.io/en/latest/
https://github.com/PyCQA/bandit
https://pypi.org/project/bandit/

phpmd can be found at the following URLs:
https://phpmd.org/
https://github.com/phpmd/phpmd
https://packagist.org/packages/phpmd/phpmd

php can be found at the following URLs:
https://www.php.net/
https://github.com/php/php-src

tex can be found at the following URL:
https://ctan.org/tex-archive/systems/knuth/dist/tex

pcregrep, pcre2grep can be found at the following URLs:
https://www.pcre.org/
https://pcre2project.github.io/pcre2/
https://github.com/PCRE2Project/pcre1
https://github.com/PCRE2Project/pcre2
https://github.com/PCRE2Project/pcre2/blob/main/src/pcre2grep.c

python3 can be found at the following URLs:
https://www.python.org/
https://docs.python.org/3/

If you see something that I missed regarding my dependencies,
please tell me/email me :).
```

3.39 build_and_checks_variables/DevOrSysAdminScripts.html.tpl

```
<!DOCTYPE html>
<html lang="en">
<!--
This file is part of DevOrSysAdminScripts library.

DevOrSysAdminScripts is free software:
you can redistribute it and/or modify it under the terms
of the GNU Lesser General Public License
```

```
as published by the Free Software Foundation,
either version 3 of the License,
or (at your option) any later version.

DevOrSysAdminScripts is distributed in the hope
that it will be useful,
but WITHOUT ANY WARRANTY;
without even the implied warranty of
MERCHANTABILITY or FITNESS FOR A PARTICULAR PURPOSE.
See the GNU Lesser General Public License for more details.

You should have received a copy of
the GNU Lesser General Public License
along with DevOrSysAdminScripts.
If not, see <https://www.gnu.org/licenses/>.

©Copyright 2023-2026 Laurent Frédéric Bernard François Lyaudet
-->
<head>
<meta http-equiv="Content-Type" content="text/html; charset=UTF-8">
<meta name="generator" content="DevOrSysAdminScripts">
<style>
.title{
  margin: 2ex auto;
  text-align: center;
}
.titlemain{
  margin: 1ex 2ex 2ex 1ex;
}
.titlerest{
  font-size: 1.17em;
  margin: 0ex 2ex;
}
table{
  border-collapse: collapse;
}
td{
  padding: 0;
}
pre{
  text-align: left;
  margin-left: 0ex;
  margin-right: auto;
}
pre.numbered_lines:before{
  background: snow;
```

```
  border-right: 1px black solid;
  color: #010101;
  content: attr(line-numbers);
  float: left;
  padding: 0 1ch;
  text-align: right;
}
blockquote{
  margin-left: 4ex;
  margin-right: 4ex;
  text-align: left;
}
ol.level1, ol.level2{
  list-style-type: none;
  /*
  list-style-type: decimal;
  Previous rule is shorter and more efficient if you need only level1.
  But you don't have an accessible counter if you use decimal.
  */
}
ol.level1{
  counter-reset: level1;
}
ol.level2{
  counter-reset: level2;
}
ol.level1 li:before{
  content: counter(level1) ". ";
  counter-increment: level1;
}
ol.level2 li:before{
  content: counter(level1) "." counter(level2) " ";
  counter-increment: level2;
}
</style>
<title>@repository_name@</title>
</head>
<body >
<table class="title">
  <tr>
    <td style="padding:1ex">
      <h1 class="titlemain">@repository_name@</h1>
      <h2 class="titlerest">
        @author_full_name@
        <br>
        <a href="@author_website@">
```

```
          <span style="font-family:monospace">
          @author_website@
          </span>
        </a>
        <br>
        <a href="mailto:@author_email@">
          <span style="font-family:monospace">
          @author_email@
          </span>
        </a>
      </h2>
    </td>
  </tr>
</table>

<blockquote class="abstract">
  <span style="font-weight:bold">Abstract:</span>
  @abstract@
</blockquote>
<p>Current version: @current_date@</p>
<p>Current number of commits: @number_of_commits@</p>
<p>Current git SHA1: @current_git_SHA1@</p>
<p>
Code lines: @number_of_lines@
</p>


<h2>Contents</h2>

<ol class="level1">
  <li><a href="#section1">Files Tree</a></li>
  <li><a href="#section2">Notes on licenses and dependencies</a></li>
  <li><a href="#section3">Listing of files</a><br>
    <ol class="level2">
      @files_lis@
    </ol>
  </li>
</ol>


<h2 id="section1">1 Files tree</h2>
<pre class="verbatim">
@current_tree_light@
</pre>
<pre class="verbatim">
@current_tree@
```

```
</pre>


<h2 id="section2">2 Notes on licenses and dependencies</h2>

<p>
This repository is dual licensed.
All the Bash scripts source code is covered by GPLv3+.
All the PHP scripts source code is covered by LGPLv3+.
All the Python scripts source code is covered by LGPLv3+.
</p>

<p>
The text of the licenses is available at:
<a href="https://www.gnu.org/licenses/"><span
  style="font-family:monospace"
>https://www.gnu.org/licenses/</span></a>.
The git repository of this source code is also available at:
<a href="@repository_web_url@"><span
 style="font-family:monospace"
>@repository_web_url@</span></a>.
Clone it with this URL:
<span style="font-family:monospace"
>@repository_git_url@</span>.
</p>

<div>
<p>
Here is a listing of the dependencies of the code in this repository:
</p>
<ul>
<li> The following dependencies are GPLv3+ licensed:<br>
basename, bash (and its builtins), cat, chmod, cp, cut, date, diff,
dirname, env, find, grep, head, mkdir, mv, realpath, rm, sed,
sha512sum, shellcheck, sort, tail, touch, wc, wget, xargs;
<li> The following dependencies are GPLv2+ licensed:<br>
more, pandoc, pdftex, pdflatex, tree;
<li> The following dependencies are GPLv2 licensed:<br>
git, pylint;
<li> The following dependencies are LGPLv2.1+ licensed:<br>
notify-send;
<li> The following dependencies are MIT licensed:<br>
black, isort, mypy, pip, ruff;
<li> The following dependencies are Apache-2.0 licensed:<br>
bandit;
<li> The following dependencies are BSD-3-Clause licensed:<br>
```

```
phpmd;
<li> php is licensed under PHP License v3.01;
<li> tex is licensed under Knuth license;
<li> pcregrep, pcre2grep has a custom license from Cambridge
University, but PCRE2 has BSD-3-Clause WITH PCRE2-exception.
<li> python3 is licensed under PSF License Version 2 and the Zero-Clause
BSD license.
</ul>
</div>

<p>
Remark: Some commands listed under GPLv3+ are POSIX
and can have different implementations under another license;
for example, uutils coreutils is MIT licensed,
whilst GNU coreutils is GPLv3+ licensed.
</p>

@dependencies_notes@


<h2 id="section3">3 Listing of files</h2>

@files_listing_HTMLPreInput@

<p>
@acknowledgments@
</p>

<script>
"use strict";
const pres = Array.from(
  document.getElementsByClassName('numbered_lines')
);
pres.forEach(
  (pre) => {
    const nb = Array.from(pre.innerText.matchAll("\n")).length + 1;
    let numbers = "";
    for(let i = 1; i <= nb; ++i){
      numbers += i.toString();
      if(i < nb){ numbers += "\n "; }
    }
    pre.setAttribute('line-numbers', numbers);
  }
);
</script>
</html>
```

3.40 build_and_checks_variables/DevOrSysAdminScripts.tex.tpl

```
% This file is part of DevOrSysAdminScripts library.
%
% DevOrSysAdminScripts is free software:
% you can redistribute it and/or modify it under the terms
% of the GNU Lesser General Public License
% as published by the Free Software Foundation,
% either version 3 of the License,
% or (at your option) any later version.
%
% DevOrSysAdminScripts is distributed in the hope
% that it will be useful,
% but WITHOUT ANY WARRANTY;
% without even the implied warranty of
% MERCHANTABILITY or FITNESS FOR A PARTICULAR PURPOSE.
% See the GNU Lesser General Public License for more details.
%
% You should have received a copy of
% the GNU Lesser General Public License
% along with DevOrSysAdminScripts.
% If not, see <https://www.gnu.org/licenses/>.
%
% ©Copyright 2023-2026 Laurent Frédéric Bernard François Lyaudet

% 2025-12-29T17:08+01:00: This file was moved from
% "./latex"
% to
% "./build_and_checks_variables".

\documentclass{article}

\usepackage[utf8]{inputenc}
\usepackage[T1]{fontenc}
\usepackage{subfigure}
\usepackage{amsmath}
\usepackage{amssymb}
\usepackage{amsthm}
\usepackage[pdftex]{hyperref}
\usepackage{tikz}
\usepackage{caption}
\usepackage[round]{natbib}
\usepackage{fancyhdr}
\usepackage{amsfonts}
\usepackage{times}
\usepackage{ifpdf}
```

```
\usepackage{latexsym}
\usepackage{graphicx}
\usepackage{enumerate}
\usepackage{pmboxdraw}
\usepackage{fancyvrb}
\usepackage[french,english]{babel}

% *** les environnements ***
%\theoremstyle{break}
\newtheorem{definition}{Definition}[section]
\newtheorem{proposition}[definition]{Proposition}
\newtheorem{theorem}[definition]{Theorem}
\newtheorem{lemma}[definition]{Lemma}
\newtheorem{corollary}[definition]{Corollary}
\newtheorem{remark}[definition]{Remark}
\newtheorem{openproblem}[definition]{Open problem}

\begin{document}

\author{
  @author_full_name@\\
  \url{@author_website@}\\
  \href{mailto:@author_email@}{@author_email@}
}
\title{@repository_name@}

\maketitle
\begin{abstract}
@abstract@
\end{abstract}

Current version: @current_date@

Current number of commits: @number_of_commits@

Current git SHA1: @current_git_SHA1@

Code lines: @number_of_lines@

\tableofcontents
\newpage

\section{Files tree}
\label{section:tree}

\begin{verbatim}
```

```
@current_tree_light@
\end{verbatim}

\begin{verbatim}
@current_tree@
\end{verbatim}

\section{Notes on licenses and dependencies}
\label{section:notes_on}

This repository is dual licensed.
All the Bash scripts source code is covered by GPLv3+.
All the PHP scripts source code is covered by LGPLv3+.
All the Python scripts source code is covered by LGPLv3+.

The text of the licenses is available at:
\url{https://www.gnu.org/licenses/}.
The git repository of this source code is also available at:
\url{@repository_web_url@}.
Clone it with this URL:
\url{@repository_git_url@}.

Here is a listing of the dependencies of the code in this repository:
\begin{itemize}
\item The following dependencies are GPLv3+ licensed:\newline
basename, bash (and its builtins), cat, chmod, cp, cut, date, diff,
dirname, env, find, grep, head, mkdir, mv, realpath, rm, sed,
sha512sum, shellcheck, sort, tail, touch, wc, wget, xargs;
\item The following dependencies are GPLv2+ licensed:\newline
more, pandoc, pdftex, pdflatex, tree;
\item The following dependencies are GPLv2 licensed:\newline
git, pylint;
\item The following dependencies are LGPLv2.1+ licensed:\newline
notify-send;
\item The following dependencies are MIT licensed:\newline
black, isort, mypy, pip, ruff;
\item The following dependencies are Apache-2.0 licensed:\newline
bandit;
\item The following dependencies are BSD-3-Clause licensed:\newline
phpmd;
\item php is licensed under PHP License v3.01;
\item tex is licensed under Knuth license;
\item pcregrep, pcre2grep has a custom license from Cambridge
University, but PCRE2 has BSD-3-Clause WITH PCRE2-exception,
\item python3 is licensed under PSF License Version 2 and the Zero-Clause
BSD license.
```

```
\end{itemize}

Remark: Some commands listed under GPLv3+ are POSIX
and can have different implementations under another license;
for example, uutils coreutils is MIT licensed,
whilst GNU coreutils is GPLv3+ licensed.

@dependencies_notes@

\section{Listing of files}
\label{section:listing}

@files_listing_VerbatimInput@

@acknowledgments@

\end{document}
```

3.41 build_and_checks_variables/files_listing.tex.tpl

```
% This file is part of DevOrSysAdminScripts library.
%
% DevOrSysAdminScripts is free software:
% you can redistribute it and/or modify it under the terms
% of the GNU General Public License
% as published by the Free Software Foundation,
% either version 3 of the License,
% or (at your option) any later version.
%
% DevOrSysAdminScripts is distributed in the hope
% that it will be useful,
% but WITHOUT ANY WARRANTY;
% without even the implied warranty of
% MERCHANTABILITY or FITNESS FOR A PARTICULAR PURPOSE.
% See the GNU General Public License for more details.
%
% You should have received a copy of
% the GNU General Public License
% along with DevOrSysAdminScripts.
% If not, see <https://www.gnu.org/licenses/>.
%
% ©Copyright 2023-2026 Laurent Frédéric Bernard François Lyaudet
\subsection{
  build\_and\_checks\_dependencies/build\_and\_checks.exec.sh.tpl
}
\label{
```

```
  build_and_checks_dependencies:build_and_checksexecshtpl
}

\VerbatimInput[numbers=left,xleftmargin=-5mm]{
build_and_checks_dependencies/build_and_checks.exec.sh.tpl
}


\subsection{
  build\_and\_checks\_dependencies/build\_dependencies\_notes.exec.php
}
\label{
  build_and_checks_dependencies:build_dependencies_notesexecphp
}

\VerbatimInput[numbers=left,xleftmargin=-5mm]{
build_and_checks_dependencies/build_dependencies_notes.exec.php
}


\subsection{
  build\_and\_checks\_dependencies/build\_md\_from\_printable\_md.exec.sh
}
\label{
  build_and_checks_dependencies:build_md_from_printable_mdexecsh
}

\VerbatimInput[numbers=left,xleftmargin=-5mm]{
build_and_checks_dependencies/build_md_from_printable_md.exec.sh
}


\subsection{
  build\_and\_checks\_dependencies/%
check\_shell\_scripts\_beginnings.libr.sh
}
\label{
  build_and_checks_dependencies:check_shell_scripts_beginningslibrsh
}

\VerbatimInput[numbers=left,xleftmargin=-5mm]{
build_and_checks_dependencies/check_shell_scripts_beginnings.libr.sh
}


\subsection{
```

```
  build\_and\_checks\_dependencies/%
check\_shell\_scripts\_indentation.libr.sh
}
\label{
  build_and_checks_dependencies:check_shell_scripts_indentationlibrsh
}

\VerbatimInput[numbers=left,xleftmargin=-5mm]{
build_and_checks_dependencies/check_shell_scripts_indentation.libr.sh
}


\subsection{
  build\_and\_checks\_dependencies/check\_URLs.libr.sh
}
\label{
  build_and_checks_dependencies:check_URLslibrsh
}

\VerbatimInput[numbers=left,xleftmargin=-5mm]{
build_and_checks_dependencies/check_URLs.libr.sh
}


\subsection{
  build\_and\_checks\_dependencies/common\_build\_and\_checks.exec.sh
}
\label{
  build_and_checks_dependencies:common_build_and_checksexecsh
}

\VerbatimInput[numbers=left,xleftmargin=-5mm]{
build_and_checks_dependencies/common_build_and_checks.exec.sh
}


\subsection{
  build\_and\_checks\_dependencies/common\_build\_and\_checks.exec.sh.tpl
}
\label{
  build_and_checks_dependencies:common_build_and_checksexecshtpl
}

\VerbatimInput[numbers=left,xleftmargin=-5mm]{
build_and_checks_dependencies/common_build_and_checks.exec.sh.tpl
}
```

```latex


\subsection{
  build\_and\_checks\_dependencies/common\_options.libr.sh
}
\label{
  build_and_checks_dependencies:common_optionslibrsh
}

\VerbatimInput[numbers=left,xleftmargin=-5mm]{
build_and_checks_dependencies/common_options.libr.sh
}


\subsection{
  build\_and\_checks\_dependencies/comparisons.libr.sh
}
\label{
  build_and_checks_dependencies:comparisonslibrsh
}

\VerbatimInput[numbers=left,xleftmargin=-5mm]{
build_and_checks_dependencies/comparisons.libr.sh
}


\subsection{
  build\_and\_checks\_dependencies/create\_PDF.exec.sh
}
\label{
  build_and_checks_dependencies:create_PDFexecsh
}

\VerbatimInput[numbers=left,xleftmargin=-5mm]{
build_and_checks_dependencies/create_PDF.exec.sh
}


\subsection{
  build\_and\_checks\_dependencies/generate\_from\_template.libr.sh
}
\label{
  build_and_checks_dependencies:generate_from_templatelibrsh
}

\VerbatimInput[numbers=left,xleftmargin=-5mm]{
```

```
build_and_checks_dependencies/generate_from_template.libr.sh
}


\subsection{
  build\_and\_checks\_dependencies/%
get\_common\_text\_glob\_patterns.libr.sh
}
\label{
  build_and_checks_dependencies:get_common_text_glob_patternslibrsh
}

\VerbatimInput[numbers=left,xleftmargin=-5mm]{
build_and_checks_dependencies/get_common_text_glob_patterns.libr.sh
}


\subsection{
  build\_and\_checks\_dependencies/grammar\_and\_spelling\_check.libr.sh
}
\label{
  build_and_checks_dependencies:grammar_and_spelling_checklibrsh
}

\VerbatimInput[numbers=left,xleftmargin=-5mm]{
build_and_checks_dependencies/grammar_and_spelling_check.libr.sh
}


\subsection{
  build\_and\_checks\_dependencies/licenses\_templates/%
build\_licenses\_templates.exec.sh
}
\label{
  build_and_checks_dependencies:licenses_templates:%
build_licenses_templatesexecsh
}

\VerbatimInput[numbers=left,xleftmargin=-5mm]{
build_and_checks_dependencies/licenses_templates/%
build_licenses_templates.exec.sh
}


\subsection{
  build\_and\_checks\_dependencies/licenses\_templates/%
```

```
license\_file\_header\_AGPLv3+.tpl
}
\label{
  build_and_checks_dependencies:licenses_templates:%
license_file_header_AGPLv3+tpl
}

\VerbatimInput[numbers=left,xleftmargin=-5mm]{
build_and_checks_dependencies/licenses_templates/%
license_file_header_AGPLv3+.tpl
}


\subsection{
  build\_and\_checks\_dependencies/licenses\_templates/%
license\_file\_header\_GPLv3+.tpl
}
\label{
  build_and_checks_dependencies:licenses_templates:%
license_file_header_GPLv3+tpl
}

\VerbatimInput[numbers=left,xleftmargin=-5mm]{
build_and_checks_dependencies/licenses_templates/%
license_file_header_GPLv3+.tpl
}


\subsection{
  build\_and\_checks\_dependencies/licenses\_templates/%
license\_file\_header\_LGPLv3+.tpl
}
\label{
  build_and_checks_dependencies:licenses_templates:%
license_file_header_LGPLv3+tpl
}

\VerbatimInput[numbers=left,xleftmargin=-5mm]{
build_and_checks_dependencies/licenses_templates/%
license_file_header_LGPLv3+.tpl
}


\subsection{
  build\_and\_checks\_dependencies/lines\_counts.libr.sh
}
```

```
\label{
  build_and_checks_dependencies:lines_countslibrsh
}

\VerbatimInput[numbers=left,xleftmargin=-5mm]{
build_and_checks_dependencies/lines_counts.libr.sh
}


\subsection{
  build\_and\_checks\_dependencies/lines\_filters.libr.sh
}
\label{
  build_and_checks_dependencies:lines_filterslibrsh
}

\VerbatimInput[numbers=left,xleftmargin=-5mm]{
build_and_checks_dependencies/lines_filters.libr.sh
}


\subsection{
  build\_and\_checks\_dependencies/lines\_maps.libr.sh
}
\label{
  build_and_checks_dependencies:lines_mapslibrsh
}

\VerbatimInput[numbers=left,xleftmargin=-5mm]{
build_and_checks_dependencies/lines_maps.libr.sh
}


\subsection{
  build\_and\_checks\_dependencies/overwrite\_if\_not\_equal.libr.sh
}
\label{
  build_and_checks_dependencies:overwrite_if_not_equallibrsh
}

\VerbatimInput[numbers=left,xleftmargin=-5mm]{
build_and_checks_dependencies/overwrite_if_not_equal.libr.sh
}


\subsection{
```

```
  build\_and\_checks\_dependencies/python\_black\_complement.libr.sh
}
\label{
  build_and_checks_dependencies:python_black_complementlibrsh
}

\VerbatimInput[numbers=left,xleftmargin=-5mm]{
build_and_checks_dependencies/python_black_complement.libr.sh
}


\subsection{
  build\_and\_checks\_dependencies/python\_isort\_complement.libr.sh
}
\label{
  build_and_checks_dependencies:python_isort_complementlibrsh
}

\VerbatimInput[numbers=left,xleftmargin=-5mm]{
build_and_checks_dependencies/python_isort_complement.libr.sh
}


\subsection{
  build\_and\_checks\_dependencies/repository\_data.libr.sh
}
\label{
  build_and_checks_dependencies:repository_datalibrsh
}

\VerbatimInput[numbers=left,xleftmargin=-5mm]{
build_and_checks_dependencies/repository_data.libr.sh
}


\subsection{
  build\_and\_checks\_dependencies/shell\_checks\_complement.libr.sh
}
\label{
  build_and_checks_dependencies:shell_checks_complementlibrsh
}

\VerbatimInput[numbers=left,xleftmargin=-5mm]{
build_and_checks_dependencies/shell_checks_complement.libr.sh
}

```

```
\subsection{
  build\_and\_checks\_dependencies/split\_score.exec.php
}
\label{
  build_and_checks_dependencies:split_scoreexecphp
}

\VerbatimInput[numbers=left,xleftmargin=-5mm]{
build_and_checks_dependencies/split_score.exec.php
}


\subsection{
  build\_and\_checks\_dependencies/split\_score.libr.php
}
\label{
  build_and_checks_dependencies:split_scorelibrphp
}

\VerbatimInput[numbers=left,xleftmargin=-5mm]{
build_and_checks_dependencies/split_score.libr.php
}


\subsection{
  build\_and\_checks\_dependencies/strings\_functions.libr.sh
}
\label{
  build_and_checks_dependencies:strings_functionslibrsh
}

\VerbatimInput[numbers=left,xleftmargin=-5mm]{
build_and_checks_dependencies/strings_functions.libr.sh
}


\subsection{
  build\_and\_checks\_dependencies/too\_long\_code\_lines.libr.sh
}
\label{
  build_and_checks_dependencies:too_long_code_lineslibrsh
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\VerbatimInput[numbers=left,xleftmargin=-5mm]{
build_and_checks_dependencies/too_long_code_lines.libr.sh
```

```
}


\subsection{
  build\_and\_checks\_dependencies/%
update\_common\_build\_and\_checks.exec.sh
}
\label{
  build_and_checks_dependencies:update_common_build_and_checksexecsh
}

\VerbatimInput[numbers=left,xleftmargin=-5mm]{
build_and_checks_dependencies/update_common_build_and_checks.exec.sh
}


\subsection{
  build\_and\_checks\_dependencies/%
update\_or\_check\_files\_names\_listing.exec.sh
}
\label{
  build_and_checks_dependencies:update_or_check_files_names_listingexecsh
}

\VerbatimInput[numbers=left,xleftmargin=-5mm]{
build_and_checks_dependencies/update_or_check_files_names_listing.exec.sh
}


\subsection{
  build\_and\_checks\_variables/.gitignore
}
\label{
  build_and_checks_variables:gitignore
}

\VerbatimInput[numbers=left,xleftmargin=-5mm]{
build_and_checks_variables/.gitignore
}


\subsection{
  build\_and\_checks\_variables/bandit.ini
}
\label{
  build_and_checks_variables:banditini
```

```
}

\VerbatimInput[numbers=left,xleftmargin=-5mm]{
build_and_checks_variables/bandit.ini
}


\subsection{
  build\_and\_checks\_variables/bandit\_baseline.json
}
\label{
  build_and_checks_variables:bandit_baselinejson
}

\VerbatimInput[numbers=left,xleftmargin=-5mm]{
build_and_checks_variables/bandit_baseline.json
}


\subsection{
  build\_and\_checks\_variables/black.toml
}
\label{
  build_and_checks_variables:blacktoml
}

\VerbatimInput[numbers=left,xleftmargin=-5mm]{
build_and_checks_variables/black.toml
}


\subsection{
  build\_and\_checks\_variables/dependencies\_data.json
}
\label{
  build_and_checks_variables:dependencies_datajson
}

\VerbatimInput[numbers=left,xleftmargin=-5mm]{
build_and_checks_variables/dependencies_data.json
}


\subsection{
  build\_and\_checks\_variables/dependencies\_notes.txt
}
```

```
\label{
  build_and_checks_variables:dependencies_notestxt
}

\VerbatimInput[numbers=left,xleftmargin=-5mm]{
build_and_checks_variables/dependencies_notes.txt
}


\subsection{
  build\_and\_checks\_variables/DevOrSysAdminScripts.html.tpl
}
\label{
  build_and_checks_variables:DevOrSysAdminScriptshtmltpl
}

\VerbatimInput[numbers=left,xleftmargin=-5mm]{
build_and_checks_variables/DevOrSysAdminScripts.html.tpl
}


\subsection{
  build\_and\_checks\_variables/DevOrSysAdminScripts.tex.tpl
}
\label{
  build_and_checks_variables:DevOrSysAdminScriptstextpl
}

\VerbatimInput[numbers=left,xleftmargin=-5mm]{
build_and_checks_variables/DevOrSysAdminScripts.tex.tpl
}


\subsection{
  build\_and\_checks\_variables/files\_listing.tex.tpl
}
\label{
  build_and_checks_variables:files_listingtextpl
}

\VerbatimInput[numbers=left,xleftmargin=-5mm]{
build_and_checks_variables/files_listing.tex.tpl
}


\subsection{
```

```
  build\_and\_checks\_variables/files\_names\_listing.txt
}
\label{
  build_and_checks_variables:files_names_listingtxt
}

\VerbatimInput[numbers=left,xleftmargin=-5mm]{
build_and_checks_variables/files_names_listing.txt
}


\subsection{
  build\_and\_checks\_variables/licenses\_notes.txt
}
\label{
  build_and_checks_variables:licenses_notestxt
}

\VerbatimInput[numbers=left,xleftmargin=-5mm]{
build_and_checks_variables/licenses_notes.txt
}


\subsection{
  build\_and\_checks\_variables/phpmd\_baseline.xml
}
\label{
  build_and_checks_variables:phpmd_baselinexml
}

\VerbatimInput[numbers=left,xleftmargin=-5mm]{
build_and_checks_variables/phpmd_baseline.xml
}


\subsection{
  build\_and\_checks\_variables/pylintrc.toml
}
\label{
  build_and_checks_variables:pylintrctoml
}

\VerbatimInput[numbers=left,xleftmargin=-5mm]{
build_and_checks_variables/pylintrc.toml
}

```

```latex

\subsection{
  build\_and\_checks\_variables/repository\_data.txt
}
\label{
  build_and_checks_variables:repository_datatxt
}

\VerbatimInput[numbers=left,xleftmargin=-5mm]{
build_and_checks_variables/repository_data.txt
}


\subsection{
  build\_and\_checks\_variables/repository\_data.txt.tpl
}
\label{
  build_and_checks_variables:repository_datatxttpl
}

\VerbatimInput[numbers=left,xleftmargin=-5mm]{
build_and_checks_variables/repository_data.txt.tpl
}


\subsection{
  build\_and\_checks\_variables/ruff.toml
}
\label{
  build_and_checks_variables:rufftoml
}

\VerbatimInput[numbers=left,xleftmargin=-5mm]{
build_and_checks_variables/ruff.toml
}


\subsection{
  build\_and\_checks\_variables/shellcheck.ini
}
\label{
  build_and_checks_variables:shellcheckini
}

\VerbatimInput[numbers=left,xleftmargin=-5mm]{
build_and_checks_variables/shellcheck.ini
```

```
}


\subsection{
  build\_and\_checks\_variables/shellcheck\_notes.txt
}
\label{
  build_and_checks_variables:shellcheck_notestxt
}

\VerbatimInput[numbers=left,xleftmargin=-5mm]{
build_and_checks_variables/shellcheck_notes.txt
}


\subsection{
  build\_and\_checks.exec.sh
}
\label{
  build_and_checksexecsh
}

\VerbatimInput[numbers=left,xleftmargin=-5mm]{
build_and_checks.exec.sh
}


\subsection{
  COPYING
}
\label{
  COPYING
}

\VerbatimInput[numbers=left,xleftmargin=-5mm]{
COPYING
}


\subsection{
  COPYING.LESSER
}
\label{
  COPYINGLESSER
}

```

```
\VerbatimInput[numbers=left,xleftmargin=-5mm]{
COPYING.LESSER
}


\subsection{
  git\_helpers/git\_checks.libr.sh
}
\label{
  git_helpers:git_checkslibrsh
}

\VerbatimInput[numbers=left,xleftmargin=-5mm]{
git_helpers/git_checks.libr.sh
}


\subsection{
  git\_helpers/post-commit
}
\label{
  git_helpers:post-commit
}

\VerbatimInput[numbers=left,xleftmargin=-5mm]{
git_helpers/post-commit
}


\subsection{
  git\_helpers/post\_commit\_renaming\_comment.libr.sh
}
\label{
  git_helpers:post_commit_renaming_commentlibrsh
}

\VerbatimInput[numbers=left,xleftmargin=-5mm]{
git_helpers/post_commit_renaming_comment.libr.sh
}


\subsection{
  git\_helpers/pre-commit
}
\label{
  git_helpers:pre-commit
```

```
}

\VerbatimInput[numbers=left,xleftmargin=-5mm]{
git_helpers/pre-commit
}


\subsection{
  IMAP\_transfert.py
}
\label{
  IMAP_transfertpy
}

\VerbatimInput[numbers=left,xleftmargin=-5mm]{
IMAP_transfert.py
}


\subsection{
  LiensUtiles.txt
}
\label{
  LiensUtilestxt
}

\VerbatimInput[numbers=left,xleftmargin=-5mm]{
LiensUtiles.txt
}


\subsection{
  miscellaneous/funny.sh
}
\label{
  miscellaneous:funnysh
}

\VerbatimInput[numbers=left,xleftmargin=-5mm]{
miscellaneous/funny.sh
}


\subsection{
  miscellaneous/spampty\_data.txt
}
```

```
\label{
  miscellaneous:spampty_datatxt
}

\VerbatimInput[numbers=left,xleftmargin=-5mm]{
miscellaneous/spampty_data.txt
}


\subsection{
  miscellaneous/string\_escaping.html
}
\label{
  miscellaneous:string_escapinghtml
}

\VerbatimInput[numbers=left,xleftmargin=-5mm]{
miscellaneous/string_escaping.html
}


\subsection{
  miscellaneous/string\_escaping.xhtml
}
\label{
  miscellaneous:string_escapingxhtml
}

\VerbatimInput[numbers=left,xleftmargin=-5mm]{
miscellaneous/string_escaping.xhtml
}


\subsection{
  README.md.tpl
}
\label{
  READMEmdtpl
}

\VerbatimInput[numbers=left,xleftmargin=-5mm]{
README.md.tpl
}


\subsection{
```

```
  string\_escaping.libr.php
}
\label{
  string_escapinglibrphp
}

\VerbatimInput[numbers=left,xleftmargin=-5mm]{
string_escaping.libr.php
}


\subsection{
  unicode.libr.php
}
\label{
  unicodelibrphp
}

\VerbatimInput[numbers=left,xleftmargin=-5mm]{
unicode.libr.php
}


\subsection{
  upgrade\_build\_and\_checks.exec.sh
}
\label{
  upgrade_build_and_checksexecsh
}

\VerbatimInput[numbers=left,xleftmargin=-5mm]{
upgrade_build_and_checks.exec.sh
}


\subsection{
  wget\_sha512.libr.sh
}
\label{
  wget_sha512librsh
}

\VerbatimInput[numbers=left,xleftmargin=-5mm]{
wget_sha512.libr.sh
}

```

855

3.42 build_and_checks_variables/files_names_listing.txt

```
// This file is part of DevOrSysAdminScripts library.
//
// DevOrSysAdminScripts is free software:
// you can redistribute it and/or modify it under the terms
// of the GNU Lesser General Public License
// as published by the Free Software Foundation,
// either version 3 of the License,
// or (at your option) any later version.
//
// DevOrSysAdminScripts is distributed in the hope
// that it will be useful,
// but WITHOUT ANY WARRANTY;
// without even the implied warranty of
// MERCHANTABILITY or FITNESS FOR A PARTICULAR PURPOSE.
// See the GNU Lesser General Public License for more details.
//
// You should have received a copy of
// the GNU Lesser General Public License
// along with DevOrSysAdminScripts.
// If not, see <https://www.gnu.org/licenses/>.
//
// ©Copyright 2023-2026 Laurent Frédéric Bernard François Lyaudet
// 2024-10-31T21:37+01:00: This file was moved from
// "./build_and_checks_dependencies/listings"
// to
// "./build_and_checks_variables".
// Blasé20240517Résultat.txt
// Blasé20240517.sh
// Blasé20240518_2.txt
// Blasé20240518_3.txt
build_and_checks_dependencies/build_and_checks.exec.sh.tpl
build_and_checks_dependencies/build_dependencies_notes.exec.php
build_and_checks_dependencies/build_md_from_printable_md.exec.sh
build_and_checks_dependencies/check_shell_scripts_beginnings.libr.sh
build_and_checks_dependencies/check_shell_scripts_indentation.libr.sh
build_and_checks_dependencies/check_URLs.libr.sh
build_and_checks_dependencies/common_build_and_checks.exec.sh
build_and_checks_dependencies/common_build_and_checks.exec.sh.tpl
build_and_checks_dependencies/common_options.libr.sh
build_and_checks_dependencies/comparisons.libr.sh
build_and_checks_dependencies/create_PDF.exec.sh
build_and_checks_dependencies/generate_from_template.libr.sh
```

```
build_and_checks_dependencies/get_common_text_glob_patterns.libr.sh
build_and_checks_dependencies/grammar_and_spelling_check.libr.sh
build_and_checks_dependencies/licenses_templates/\
build_licenses_templates.exec.sh
build_and_checks_dependencies/licenses_templates/\
license_file_header_AGPLv3+.tpl
build_and_checks_dependencies/licenses_templates/\
license_file_header_GPLv3+.tpl
build_and_checks_dependencies/licenses_templates/\
license_file_header_LGPLv3+.tpl
build_and_checks_dependencies/lines_counts.libr.sh
build_and_checks_dependencies/lines_filters.libr.sh
build_and_checks_dependencies/lines_maps.libr.sh
// build_and_checks_dependencies/listings/files_listing.tex.tpl
build_and_checks_dependencies/overwrite_if_not_equal.libr.sh
build_and_checks_dependencies/python_black_complement.libr.sh
build_and_checks_dependencies/python_isort_complement.libr.sh
build_and_checks_dependencies/repository_data.libr.sh
build_and_checks_dependencies/shell_checks_complement.libr.sh
build_and_checks_dependencies/split_score.exec.php
build_and_checks_dependencies/split_score.libr.php
build_and_checks_dependencies/strings_functions.libr.sh
build_and_checks_dependencies/too_long_code_lines.libr.sh
build_and_checks_dependencies/update_common_build_and_checks.exec.sh
build_and_checks_dependencies/\
update_or_check_files_names_listing.exec.sh
build_and_checks_variables/.gitignore
build_and_checks_variables/bandit.ini
build_and_checks_variables/bandit_baseline.json
build_and_checks_variables/black.toml
build_and_checks_variables/dependencies_data.json
build_and_checks_variables/dependencies_notes.txt
build_and_checks_variables/DevOrSysAdminScripts.html.tpl
// build_and_checks_variables/DevOrSysAdminScripts.tex
build_and_checks_variables/DevOrSysAdminScripts.tex.tpl
build_and_checks_variables/files_listing.tex.tpl
build_and_checks_variables/files_names_listing.txt
build_and_checks_variables/licenses_notes.txt
build_and_checks_variables/phpmd_baseline.xml
build_and_checks_variables/pylintrc.toml
build_and_checks_variables/repository_data.txt
build_and_checks_variables/repository_data.txt.tpl
build_and_checks_variables/ruff.toml
build_and_checks_variables/shellcheck.ini
build_and_checks_variables/shellcheck_notes.txt
// build_and_checks_variables/upgrade_venvs_ts
```

```
build_and_checks.exec.sh
COPYING
COPYING.LESSER
// DevOrSysAdminScripts.html
// DevOrSysAdminScripts.pdf
// EncoreBlasé20240517-20240518.txt
git_helpers/git_checks.libr.sh
git_helpers/post-commit
git_helpers/post_commit_renaming_comment.libr.sh
git_helpers/pre-commit
IMAP_transfert.py
LiensUtiles.txt
// Uncomment some files in miscellaneous to do some testing.
// miscellaneous/\
1234567890123456789012345678901234567890123456789012345678901234567890123456789012345
// miscellaneous/\
1234567890123456789012345678901234567890123456789012345678901234567890123456789012345\
56
// miscellaneous/\
1234567890123456789012345678901234567890123456789012345678901234567890123456789012345\
56\
\
// miscellaneous/\
1234567890123456789012345678901234567890123456789012345678901234567890123456789012345\
5\
\
// miscellaneous/\
1234567890123456789012345678901234567890123456789012345678901234567890123456789012345\
\
// miscellaneous/\
1234567890123456789012345678901234567890123456789012345678901234567890123456789012345\
\
// miscellaneous/\
a_long_long_long_long_long_long_long_long_long_filename_so_that_you_can_se\
e_that_it_works.txt
miscellaneous/funny.sh
miscellaneous/spampty_data.txt
miscellaneous/string_escaping.html
miscellaneous/string_escaping.xhtml
// miscellaneous/\
yet_another_long_long_long_long_long_long_long_long_long_filename_so_that_\
you_can_see_that_it_works.txt
// README.html
// README.md
README.md.tpl
string_escaping.libr.php
```

```
unicode.libr.php
upgrade_build_and_checks.exec.sh
wget_sha512.libr.sh
```

3.43 build_and_checks_variables/licenses_notes.txt

```
This repository is dual licensed.
All the Bash scripts source code is covered by GPLv3+.
All the PHP scripts source code is covered by LGPLv3+.
All the Python scripts source code is covered by LGPLv3+.

Here is a listing of the dependencies of the code in this repository:
- The following dependencies are GPLv3+ licensed:
basename, bash (and its builtins), cat, chmod, cp, cut, date, diff,
dirname, env, find, grep, head, mkdir, mv, realpath, rm, sed,
sha512sum, shellcheck, sort, tail, touch, wc, wget, xargs;
- The following dependencies are GPLv2+ licensed:
more, pandoc, pdftex, pdflatex, tree;
- The following dependencies are GPLv2 licensed:
git, pylint;
- The following dependencies are LGPLv2.1+ licensed:
notify-send;
- The following dependencies are MIT licensed:
black, isort, mypy, pip, ruff;
- The following dependencies are Apache-2.0 licensed:
bandit;
- The following dependencies are BSD-3-Clause licensed:
phpmd;
- php is licensed under PHP License v3.01;
- tex is licensed under Knuth license;
- pcregrep, pcre2grep has a custom license from Cambridge University,
but PCRE2 has BSD-3-Clause WITH PCRE2-exception,
- python3 is licensed under PSF License Version 2 and the Zero-Clause BSD
license.

Remark: Some commands listed under GPLv3+ are POSIX
and can have different implementations under another license;
for example, uutils coreutils is MIT licensed,
whilst GNU coreutils is GPLv3+ licensed.
```

3.44 build_and_checks_variables/phpmd_baseline.xml

```
<?xml version="1.0"?>
<phpmd-baseline>
  <violation rule="PHPMD\Rule\CyclomaticComplexity"
    file="build_and_checks_dependencies/build_dependencies_notes.exec.php"
```

```
5     />
6     <violation rule="PHPMD\Rule\Design\NpathComplexity"
7       file="build_and_checks_dependencies/build_dependencies_notes.exec.php"
8     />
9     <violation rule="PHPMD\Rule\Design\LongMethod"
10      file="build_and_checks_dependencies/build_dependencies_notes.exec.php"
11    />
12    <violation rule="PHPMD\Rule\Controversial\CamelCaseParameterName"
13      file="build_and_checks_dependencies/build_dependencies_notes.exec.php"
14    />
15    <violation rule="PHPMD\Rule\Controversial\CamelCaseVariableName"
16      file="build_and_checks_dependencies/build_dependencies_notes.exec.php"
17    />
18    <violation rule="PHPMD\Rule\CleanCode\ElseExpression"
19      file="build_and_checks_dependencies/build_dependencies_notes.exec.php"
20    />
21    <violation rule="PHPMD\Rule\Naming\LongVariable"
22      file="build_and_checks_dependencies/build_dependencies_notes.exec.php"
23    />
24    <violation rule="PHPMD\Rule\Controversial\CamelCaseParameterName"
25      file="build_and_checks_dependencies/split_score.libr.php"
26    />
27    <violation rule="PHPMD\Rule\Naming\LongVariable"
28      file="build_and_checks_dependencies/split_score.libr.php"
29    />
30    <violation rule="PHPMD\Rule\Controversial\CamelCaseVariableName"
31      file="build_and_checks_dependencies/split_score.libr.php"
32    />
33    <violation rule="PHPMD\Rule\Controversial\CamelCaseParameterName"
34      file="string_escaping.libr.php"
35    />
36    <violation rule="PHPMD\Rule\Controversial\CamelCaseVariableName"
37      file="string_escaping.libr.php"
38    />
39    <violation rule="PHPMD\Rule\Design\LongMethod"
40      file="unicode.libr.php"
41    />
42    <violation rule="PHPMD\Rule\Controversial\CamelCaseParameterName"
43      file="unicode.libr.php"
44    />
45    <violation rule="PHPMD\Rule\Naming\LongVariable"
46      file="unicode.libr.php"
47    />
48    <violation rule="PHPMD\Rule\Controversial\CamelCaseVariableName"
49      file="unicode.libr.php"
50    />
```

```
  <violation rule="PHPMD\Rule\CleanCode\MissingImport"
    file="unicode.libr.php"
  />
  <violation rule="PHPMD\Rule\CyclomaticComplexity"
    file="unicode.libr.php"
  />
  <violation rule="PHPMD\Rule\Design\NpathComplexity"
    file="unicode.libr.php"
  />
</phpmd-baseline>
```

3.45 build_and_checks_variables/pylintrc.toml

```
[tool.pylint.main]
exit-zero = true
ignore = ["node_modules"]
jobs = 0
[tool.pylint.basic]
# We use uppercase acronyms like HTML, IMAP, etc.
module-rgx = "^[_a-zA-Z0-9+]+$"
function-rgx = "^[_a-zA-Z0-9+]+$"
[tool.pylint."messages control"]
# Code like R0912 is better to order disables by category.
disable = [
  "C1804",  # use-implicit-booleaness-not-comparison-to-string
  "I0011",  # locally-disabled
  "I0020",  # suppressed-message
  "R0912",  # too-many-branches
  "R0913",  # too-many-arguments
  "R0914",  # too-many-locals
  "R0915",  # too-many-statements
  "R0917",  # too-many-positional-arguments
]
enable = "all"
[tool.pylint.reports]
output-format = "colorized"
```

3.46 build_and_checks_variables/repository_data.txt

```
Look at grep_variable to understand the structure of this file.
Basically, it has almost no structure apart from some lines
starting with a "variable_name=".
Anything else is comments.

repository_name=DevOrSysAdminScripts
repository_web_url=https://github.com/LLyaudet/DevOrSysAdminScripts/
```

```
repository_git_url=git@github.com:LLyaudet/DevOrSysAdminScripts.git

abstract=Some “small” useful scripts
abstract=for enhancing development or system-administrators tasks

license=GPLv3+
license2=LGPLv3+

author_email=Laurent.Lyaudet@gmail.com
author_full_name=Laurent Frédéric Bernard François Lyaudet
author_last_name=Lyaudet
author_name=Laurent Lyaudet
author_website=https://lyaudet.eu/laurent/

acknowledgments=Merci Dieu !
acknowledgments=Merci Père ! Merci Seigneur ! Merci Saint Esprit !

Great day! Now I allow myself lines of 75 characters instead of 70 :).
I checked how many characters fit in the PDF file generated by tex with
default paper size,
when using verbatim environment with line numbers for source code listing.
With a screen capture in GIMP,
I mesured 197px between two-digits line numbers and the left border of
the page.
(It is less than 197px for three-digits or more line numbers.)
The same distance from the right border yields 75 characters allowed.
Since when printing the HTML file, the number of allowed characters in
<pre> tags is way larger, 75 is the right limit with the tools I use.

max_line_length=75

latex_command=pdflatex

grammar_and_spelling_checker_command=


Why one venv for each Python dependency ?
Because no language of my knowledge did it right to handle all
possible dependency conflicts.
So in case of such a conflict, one specialized venv can get you out
of any problem.
The best solution would have been to allow install of different
versions of the same package, and have a mechanism:
- first intent, use the latest available version of a package among
those that satisfy the declared dependency constraints of the package
that has this dependency;
```

```
- second intent, check a dependency overload configuration file
with tree structure: if calling python package is A version 1.1.1
with dependency B itself with dependency C, use B 2.2.2 and C 3.3.3;
you could have along that another rule saying that when called
directly B 2.2.2 should use C 4.4.4;
and even more complicated:
if calling python package is A version 1.1.1 with dependency B and D
both themselves with dependency C, use B 2.2.2, D 0.1.0,
and C 3.3.3 for B, and C 4.4.4 for D;
once you've seen that it can never be more complicated than that once
the same version number guarantees that the code is the same,
you create a config language "à la CSS" for cascading dependencies ;)
XD fit into JSON the dumbest possible way and you stop bothering
about dependency conflicts for the rest of history of informatics.
{
  "A":{
    "1.1.1": {
      "B": {
        "2.2.2": {
          "C": {
            // Dumb, you did not found it by yourself? Failed exam.
            "3.3.3" : {}
          }
        }
      },
      "D": {
        "0.1.0": {
          "C": {
            // Dumb, you did not found it by yourself? Failed exam.
            "4.4.4" : {}
          }
        }
      }
    }
  },
  "B": {
    "2.2.2": {
      "C": {
        // Dumb, you did not found it by yourself? Failed exam.
        "4.4.4" : {}
      }
    }
  }
}
Wonderful someone rediscovered nested structures XD
Seriously so many engineers or computer scientists are dumb to cry.
```

```

bandit_venv=../venv_bandit
black_venv=../venv_black
isort_venv=../venv_isort
mypy_venv=../venv_mypy
pylint_venv=../venv_pylint
ruff_venv=../venv_ruff


upgrade venvs can be auto or ask, any other value is interpreted
as ask.

upgrade_venvs=ask


time interval can be 0 to always upgrade or ask to upgrade,
or a number at most bash max int
https://superuser.com/questions/1030122/
what-is-the-maximum-value-of-a-numeric-bash-shell-variable
it should be enough to simulate never "upgrade or ask for upgrade"
unless you claim to live many billions years ;)
If it is wat, this is a random int XD (see RANDOM).
Note that the probability that it upgrades your dependencies is not
1/2, it increases over time, and is 1 after slightly
less than one day.
If it is watyouwant?, see SRANDOM XD.

upgrade_venvs_time_interval_in_seconds=wat


For the Nu HTML, CSS, and SVG checker.

vnu_jar_path=../jar_venv/vnu.jar

Not fully implemented yet, it will only warn that vnu jar was not
updated since some time.
"wat" works, but "watyouwant?" joke was not repeated ;).

upgrade_vnu_jar_time_interval_in_seconds=604800

use_vnu_for_xhtml=false
```

3.47 build_and_checks_variables/repository_data.txt.tpl

```
Look at grep_variable to understand the structure of this file.
Basically, it has almost no structure apart from some lines
```

```
starting with a "variable_name=".
Anything else is comments.
The goal of this template is to list all possible variables.

repository_name=
repository_web_url=
repository_git_url=

abstract=
abstract=

license=
license2=

author_email=
author_full_name=
author_last_name=
author_name=
author_website=

acknowledgments=
acknowledgments=

max_line_length=

latex_command=

grammar_and_spelling_checker_command=

bandit_venv=
black_venv=
isort_venv=
mypy_venv=
pylint_venv=
ruff_venv=

upgrade_venvs=

upgrade_venvs_time_interval_in_seconds=

vnu_jar_path=

upgrade_vnu_jar_time_interval_in_seconds=

use_vnu_for_xhtml=

```

```
npm_lint_directories=
npm_lint_directories=
```

3.48 build_and_checks_variables/ruff.toml

```
[lint]
select = [
  "ALL",
  "D211",  # blank-line-before-class no...before-class-docstring!
]
# ignore = ["D203", "D212", "D404"]
ignore = [
  "C901",  # complex-structure AKA give-me-toy-code
  "D203",  # (want-)?incorrect-blank-line-before-class
  "D205",  # missing-blank-line-after-summary So so cute|true ;) XD.
  "D212",  # multi-line-summary-first-line
  "D404",  # docstring-starts-with-this Shame! XD

  # raw-string-in-exception may be useful for a real project
  # with "dictionnaries" of errors made of constants.
  "EM101",

  # commented-out-code Of course git log is always faster...
  # Note that if you're tired or someone else in your team is,
  # it may catch some commit that lacked some review,
  # or automated tests: "Last manual test succeeded, ah but when I
  # debugged at some point I commented a part that was needed in
  # another case."...
  "ERA001",

  "N802",  # invalid-function-name IMAP, HTML, etc.
  "PLR0912",  # too-many-branches AKA ...
  "PLR0913",  # too-many-arguments AKA ...
  "PLR0915",  # too-many-statements AKA ...
  # Not enough reading when young makes brain slow to read and
  # understand lengthy functions.
  # If the project is finished, you may play with AKA rules.
]
[lint.per-file-ignores]
"build_and_checks_dependencies/licenses_templates/*.py" = [
  "INP001"  # implicit-namespace-package
]
"IMAP_transfert.py" = [
  "T201",  # print Use log instead in full applications.
  "TRY003",  # raise-vanilla-args
]
```

3.49 build_and_checks_variables/shellcheck.ini

```
enable=all
check-sourced=true
external-sources=true
source-path="."
source-path="SCRIPTDIR"
source-path="./build_and_checks_dependencies"
source-path="./git_helpers"

```

3.50 build_and_checks_variables/shellcheck_notes.txt

```
A listing of the current suppressed warnings/errors from shellcheck,
and also some of the warnings that were suppressed in some previous
state of the git repository (in that case, we use X suffix, see SC2162X):
- SC1090 : Can't follow non-constant source.
           Use a directive to specify location
           # shellcheck source=bla/bli.sh
           Only remaining cases are in common_build_and_checks
           when sourcing some Python venv
           source venv/bin/active
- SC1091 : Not following (file not found most of the time).
           Only remaining cases are in common_build_and_checks
           when sourcing some Python venv
           source venv/bin/active
- SC1111 : Use of slanted double quotes (valid inside single quotes)
           False positive, but may catch some problem.
- SC2001 : Replace some sed with bash builtin ${a//b/c}
           False positive when c contains \n.
- SC2034 : Unused variable
           False positive for globals that are declared in functions
           that are not currently called.
           Somehow it can be used to detect dead code.
- SC2059 : Don't use variables in the printf format string.
           Use printf '..%s..' "$foo".
           False positive when using a variable to build the format string.
- SC2086 : Double quote to prevent globbing and word splitting.
           False positive when returning integer variable
           via an indirection like return ${!foo}.
           See SC2248 below and the corresponding issue on GitHub.
- SC2119 : Use foo "$@" if function's $1 should mean script's $1.
           False positive when passing arguments would yield
           a wrong result.
- SC2120 : foo references arguments, but none are ever passed.
           False positives when some arguments are meant for debug in CLI.
```

```
- SC2125 : Brace expansions and globs are literal.
           False positive when assigning a variable declared with -i.
           https://github.com/koalaman/shellcheck/issues/914
- SC2145 : Argument mixes string and array.
           False positive when using "$@" line return and pipes.
           https://github.com/koalaman/shellcheck/issues/3192
           Example:
             # shellcheck disable=SC2145
             all_self_code_lines "$@"\
               | empty_lines_after_file_name
- SC2154 : var is referenced but not assigned.
           Very useful but false positive when I play with
           variable variable names.
- SC2162X: read without -r will mangle backslashes.
           False positive when read is used without variable:
           https://github.com/koalaman/shellcheck/issues/623
- SC2248 : Prefer double quoting
           False positive on returning integer values.
           https://github.com/koalaman/shellcheck/issues/3405
- SC2309 : -eq treats this as a variable.
             Use = to compare as string (or expand explicitly with $var).
           Very useful but false positive when I play with
           variable variable names.
- SC2312 : Some command is not invoked separately and thus its return
           value is masked. Frequent with pipes.
- SCNNNNX: Use printf instead of echo.
           https://github.com/koalaman/shellcheck/issues/3419

The command to check if there is anything else:
grep -r "shellcheck disable" . | grep -v -e '.git' -e "=SC1090$"\
  -e "=SC1091$" -e "=SC1090,SC1091$" -e "=SC1111$" -e "=SC2001$"\
  -e "=SC2034$" -e "=SC2059$" -e "=SC2086$" -e "=SC2119$" -e "=SC2120$"\
  -e "=SC2125$" -e "=SC2145$" -e "=SC2154$" -e "=SC2248$" -e "=SC2309$"\
  -e "=SC2312$" -e "=SC2154,SC2309$"
```

3.51 build_and_checks.exec.sh

```
#!/usr/bin/env bash
# This file is part of DevOrSysAdminScripts library.
#
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# of the GNU General Public License
# as published by the Free Software Foundation,
# either version 3 of the License,
# or (at your option) any later version.
```

```
#
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#
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# The file "build_and_checks.exec.sh.tpl" was renamed from
# "build_and_checks.sh.tpl" to "build_and_checks.exec.sh.tpl".
# The file "build_and_checks.exec.sh" generated from the file
# "build_and_checks.sh.tpl" or "build_and_checks.exec.sh.tpl"
# was renamed from
# "build_and_checks.sh" to "build_and_checks.exec.sh".

build_and_checks(){
  # $1=work_directory
  # Options:
  #   --verbose
  local LFBFL_work_directory="."
  if [[ -n "$1" ]]; then
    LFBFL_work_directory="$1"
  fi
  readonly LFBFL_work_directory

  local LFBFL_verbose=""
  if [[ "$*" == *--verbose* ]]; then
    printf "%s %s\n" "$0" "$*"
    LFBFL_verbose="--verbose"
  fi
  readonly LFBFL_verbose

  source ./wget_sha512.libr.sh

  LFBFL_subdir="build_and_checks_dependencies"
  mkdir -p "${LFBFL_subdir}/licenses_templates"
  LFBFL_variables_directory="${LFBFL_work_directory}/"
  LFBFL_variables_directory+="build_and_checks_variables"
  mkdir -p "${LFBFL_variables_directory}/temp"
  if ! [[ -f "${LFBFL_variables_directory}/.gitignore" ]]; then
```

```
    printf "temp/\n\n"\
      > "${LFBFL_variables_directory}/.gitignore"
  fi

  # LFBFL_dependencies_raw_content_download_URL
  local LFBFL_dependencies_URL
  LFBFL_dependencies_URL="https://raw.githubusercontent.com/LLyaudet/"
  LFBFL_dependencies_URL+="DevOrSysAdminScripts/main/${LFBFL_subdir}"
  readonly LFBFL_dependencies_URL

  declare -r LFBFL_common_file_name="common_build_and_checks.exec.sh"
  # LFBFL_script_download_URL
  declare -r\
    LFBFL_script="${LFBFL_dependencies_URL}/${LFBFL_common_file_name}"
  declare -r\
    LFBFL_file_path="./${LFBFL_subdir}/${LFBFL_common_file_name}"
  local LFBFL_correct_sha512
  LFBFL_correct_sha512='72bd314bc8f71fd79a201681a1aa7c46bf691452f5019'
  LFBFL_correct_sha512+='d5671d70f0cc29c28402fe84976785a2363b2eb9277a'
  LFBFL_correct_sha512+='783414b079a503a71304ed6ca0f0b0fbd6f52a5'
  wget_sha512 "${LFBFL_file_path}" "${LFBFL_script}"\
    "${LFBFL_correct_sha512}" "${LFBFL_verbose}"
  chmod +x "./${LFBFL_file_path}"

  "${LFBFL_file_path}" "${LFBFL_work_directory}"\
    "${LFBFL_dependencies_URL}" "${LFBFL_verbose}"
}

if [[ "$*" == *--fixed-point-build* ]]; then
  printf -- "--fixed-point-build\n"
  source "./build_and_checks_dependencies/lines_filters.libr.sh"
  LFBFL_work_directory="."
  if [[ -n "$1" ]]; then
    LFBFL_work_directory="$1"
  fi
  LFBFL_variables_directory="${LFBFL_work_directory}/"
  LFBFL_variables_directory+="build_and_checks_variables"
  LFBFL_data_file_name="${LFBFL_variables_directory}/"
  LFBFL_data_file_name+="repository_data.txt"
  LFBFL_repository_name=""
  grep_variable "${LFBFL_data_file_name}" repository_name\
    --result-variable-prefix="LFBFL_"\
    --replace-line-returns-by=""
  # Touching the 3 following files first let us iterate
  # upgrade_build_and_checks 2 times instead of 3 to reach
  # a fixed point.
```

```
  touch "${LFBFL_variables_directory}/${LFBFL_repository_name}.tex"
  touch "${LFBFL_work_directory}/${LFBFL_repository_name}.pdf"
  touch "${LFBFL_work_directory}/${LFBFL_repository_name}.html"
  # The two iterations of build_and_checks needs to complete
  # during the same minute to have a fixed point.
  build_and_checks "$@"
  # build_and_checks "$@" second iteration below
  # Then calling this script without --fixed-point-build should not
  # yield any new superficial modifications.
fi

build_and_checks "$@"
```

3.52 COPYING

```
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                       Version 3, 29 June 2007

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```

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3.54 git_helpers/git_checks.libr.sh

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# "git_checks.libr.sh"
# to
# "git_checks.libr.sh".
# 2024-05-31T01:57+02:00: This file was moved from
# "."
# to
# "./git_helpers".
# 2024-05-31T01:49+02:00: This file was renamed from
# "git_checks.libr.sh"
# to
# "git_checks.libr.sh".
# 2024-05-31T01:49+02:00: This file was moved from
# "."
# to
# "./git_helpers".
# This file was renamed from "git_checks.sh" to "git_checks.libr.sh".

check_files(){
  # Checks used on Django projects.
  # Use `git commit --no-verify` after check.
  declare -r LFBFL_send_summary_1="ATTENTION : models modifiés"
  declare -r LFBFL_send_body_1="Vérifiez si besoin de migrations"
  local LFBFL_send_body_2="Pensez aussi aux contraintes d'unicités"
  LFBFL_send_body_2+=" dès la création du modèle"
  readonly LFBFL_send_body_2
  declare -r LFBFL_send_summary_2="ATTENTION : serializers modifiés"
  declare -r LFBFL_send_body_3=\
"Les nouveaux preloadings sont interdits -> #Prefetch()"
  declare -i LFBFL_i_error=0
  git diff --cached --name-only\
    > /tmp/DOSAS_django_git_check1.temp
  if grep models /tmp/DOSAS_django_git_check1.temp;
  then
    notify-send "${LFBFL_send_summary_1}" "${LFBFL_send_body_1}"
    notify-send "${LFBFL_send_summary_1}" "${LFBFL_send_body_2}"
    LFBFL_i_error=1
```

```
  fi
  if grep serializer /tmp/DOSAS_django_git_check1.temp;
  then
    notify-send "${LFBFL_send_summary_2}" "${LFBFL_send_body_3}"
    LFBFL_i_error=1
  fi
  rm /tmp/DOSAS_django_git_check1.temp
  # shellcheck disable=SC2248
  return ${LFBFL_i_error}
}


check_no_abusive_trailing_comma(){
  # Useful for Python code, although it regularly yields
  # false positives in Python code
  # (but it is easy to distinguish by hand).
  # Use `git commit --no-verify` after check.
  declare -r LFBFL_send_summary_1="ATTENTION"
  local LFBFL_send_body_1="Il semblerait que vous affectiez un tuple"
  LFBFL_send_body_1+=" au lieu d'une autre valeur dans une variable."
  readonly LFBFL_send_body_1
  declare -i LFBFL_i_error=0
  git diff --cached -r\
    > /tmp/DOSAS_django_git_check2.temp
  if grep ' = .*,\s*$' /tmp/DOSAS_django_git_check2.temp;
  then
    notify-send "${LFBFL_send_summary_1}" "${LFBFL_send_body_1}"
    LFBFL_i_error=1
  fi
  rm /tmp/DOSAS_django_git_check2.temp
  # shellcheck disable=SC2248
  return ${LFBFL_i_error}
}


check_black_code_formatting(){
  # attention ça ne marche que sur les fichiers "stagés" avec git add
  declare -r LFBFL_s_files_paths=$(
    git diff --cached --name-only\
    | grep '\.py'
  )
  printf "%s\n" "${LFBFL_s_files_paths}"
  declare -a LFBFL_arr_files_paths
  mapfile -t LFBFL_arr_files_paths <<< "${LFBFL_s_files_paths}"
  readonly LFBFL_arr_files_paths
  declare -i LFBFL_i_error=0
```

```
  # Not very useful since LFBFL_file should be local to the loop by
  # default.
  local LFBFL_file_path
  for LFBFL_file_path in "${LFBFL_arr_files_paths[@]}"; do
    printf "Black will check formatting on file %s\n" "${LFBFL_file_path}"
    if ! black --check --diff "${LFBFL_file_path}";
    then
      LFBFL_i_error=1
    fi
  done
  # shellcheck disable=SC2248
  return ${LFBFL_i_error}
}
```

3.55 git_helpers/post-commit

```
#!/usr/bin/env bash
# This file is part of DevOrSysAdminScripts library.
#
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# of the GNU General Public License
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#
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#
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# the GNU General Public License
# along with DevOrSysAdminScripts.
# If not, see <https://www.gnu.org/licenses/>.
#
# ©Copyright 2023-2024 Laurent Frédéric Bernard François Lyaudet
# 2024-05-31T01:49+02:00: This file was renamed from
# "post-commit"
# to
# "post-commit".
# 2024-05-31T01:49+02:00: This file was moved from
# "."
# to
```

```
# "./git_helpers".

LFBFL_some_directory="../DevOrSysAdminScripts"
declare -i LFBFL_i_pushd_result3
pushd "${LFBFL_some_directory}" || {
  LFBFL_i_pushd_result3=$?
  printf "post-commit no such directory\n"
  # shellcheck disable=SC2248
  exit ${LFBFL_i_pushd_result3}
}
# shellcheck source=post_commit_renaming_comment.libr.sh
source "./git_helpers/post_commit_renaming_comment.libr.sh"
# shellcheck source=lines_filters.libr.sh
source "./build_and_checks_dependencies/lines_filters.libr.sh"
declare -i LFBFL_i_popd_result3
popd || {
  LFBFL_i_popd_result3=$?
  printf "post-commit no popd\n"
  # shellcheck disable=SC2248
  exit ${LFBFL_i_popd_result3}
}

LFBFL_data_file_name="build_and_checks_variables/repository_data.txt"
LFBFL_max_comment_line_length=""
if [[ -f "${LFBFL_data_file_name}" ]]; then
  LFBFL_max_line_length=""
  grep_variable "${LFBFL_data_file_name}" "max_line_length"\
    --result-variable-prefix="LFBFL_"
  LFBFL_max_comment_line_length="--max-comment-line-length="
  LFBFL_max_comment_line_length+="${LFBFL_max_line_length}"
fi

commit_a_file_renamed_comment --verbose --log-directory-change\
  "${LFBFL_max_comment_line_length}"
exit 0
```

3.56 git_helpers/post_commit_renaming_comment.libr.sh

```
#!/usr/bin/env bash
# This file is part of DevOrSysAdminScripts library.
#
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# of the GNU General Public License
# as published by the Free Software Foundation,
# either version 3 of the License,
```

```
# or (at your option) any later version.
#
# DevOrSysAdminScripts is distributed in the hope
# that it will be useful,
# but WITHOUT ANY WARRANTY;
# without even the implied warranty of
# MERCHANTABILITY or FITNESS FOR A PARTICULAR PURPOSE.
# See the GNU General Public License for more details.
#
# You should have received a copy of
# the GNU General Public License
# along with DevOrSysAdminScripts.
# If not, see <https://www.gnu.org/licenses/>.
#
# ©Copyright 2023-2026 Laurent Frédéric Bernard François Lyaudet
# 2024-05-31T01:49+02:00: This file was renamed from
# "post_commit_renaming_comment.libr.sh"
# to
# "post_commit_renaming_comment.libr.sh".
# 2024-05-31T01:49+02:00: This file was moved from
# "."
# to
# "./git_helpers".
# This file was renamed from "post_commit_renaming_comment"
# to "post_commit_renaming_comment.libr.sh".
# (Before first commit it seems. But I had the old name in listing.)

LFBFL_this_file_path=$(realpath "${BASH_SOURCE[0]}")
LFBFL_this_file_directory=$(dirname "${LFBFL_this_file_path}")
LFBFL_some_directory="${LFBFL_this_file_directory}/../"
declare -i LFBFL_i_pushd_result2
pushd "${LFBFL_some_directory}" || {
  LFBFL_i_pushd_result2=$?
  printf "post_commit_renaming_comment.libr.sh no such directory\n"
  # shellcheck disable=SC2248
  exit ${LFBFL_i_pushd_result2}
}
# shellcheck source=common_options.libr.sh
source "build_and_checks_dependencies/common_options.libr.sh"
# shellcheck source=strings_functions.libr.sh
source "build_and_checks_dependencies/strings_functions.libr.sh"
declare -i LFBFL_i_popd_result2
popd || {
  LFBFL_i_popd_result2=$?
  printf "post_commit_renaming_comment.libr.sh no popd\n"
  # shellcheck disable=SC2248
```

```
  exit ${LFBFL_i_popd_result2}
}

# Never forget that the worse security is the illusion of security
# (Kevin Mitnick). So, do not expect to have an accurate trackability
# of code history just because you use this post-commit hook or
# something else more evolved. This is useful but not a guarantee.
# For this reason, this function will only handle when all files
# impacted in the preceding commit are only renamed.
# It assumes that `diff` is the current default difference utility
# parametered with `git`. This point could be enhanced.
commit_a_file_renamed_comment(){
  # Options:
  #   --verbose
  #   --log-directory-change
  #   --max-comment-line-length
  declare -i LFBFL_i_verbose=0
  get_verbose_option "$@"
  local LFBFL_arg
  local LFBFL_s_format

  declare -i LFBFL_i_log_directory_change=0
  if [[ "$*" == *--log-directory-change* ]]; then
    LFBFL_i_log_directory_change=1
  fi
  readonly LFBFL_i_log_directory_change

  declare -i LFBFL_i_max_comment_line_length=70
  for LFBFL_arg in "$@"; do
    if [[ "${LFBFL_arg}" == --max-comment-line-length=* ]]; then
      LFBFL_arg=${LFBFL_arg#--max-comment-line-length=}
      LFBFL_i_max_comment_line_length=$((LFBFL_arg))
      break
    fi
  done
  readonly LFBFL_i_max_comment_line_length

  if [[ LFBFL_i_verbose -eq 1 ]]; then
    LFBFL_s_format="LFBFL_i_log_directory_change: %s\n"
    LFBFL_s_format+="LFBFL_i_max_comment_line_length: %s\n"
    # shellcheck disable=SC2059,SC2248
    printf "${LFBFL_s_format}"\
      "${LFBFL_i_log_directory_change}"\
      ${LFBFL_i_max_comment_line_length}
  fi
  #-----------------------------------------------------------------------
```

```
101
102     enhanced_set_shell_option pipefail\
103       && trap 'enhanced_unset_shell_option pipefail' RETURN
104
105     declare -r LFBFL_renaming_lines=$(
106       git log -p -1\
107       | grep '^diff --git' -A 1
108     )
109     if [[ LFBFL_i_verbose -eq 1 ]]; then
110       printf "LFBFL_renaming_lines: %s\n" "${LFBFL_renaming_lines}"
111     fi
112
113     declare -a LFBFL_renaming_lines_array
114     mapfile -t LFBFL_renaming_lines_array <<< "${LFBFL_renaming_lines}"
115     readonly LFBFL_renaming_lines_array
116
117     declare -r LFBFL_timestamp=$(date --iso-8601="minute")
118     if [[ LFBFL_i_verbose -eq 1 ]]; then
119       printf "LFBFL_timestamp: %s\n" "${LFBFL_timestamp}"
120     fi
121     local LFBFL_split_score_command1
122     declare -i LFBFL_i_split_score_command_properties1
123     # shellcheck disable=SC2248
124     get_split_score_simple 1 ${LFBFL_i_max_comment_line_length} _
125     LFBFL_split_score_command1="${get_split_score_result}"
126     LFBFL_i_split_score_command_properties1=${i_get_split_score_result2}
127     readonly LFBFL_split_score_command1
128     readonly LFBFL_i_split_score_command_properties1
129     local LFBFL_split_score_command2
130     declare -i LFBFL_i_split_score_command_properties2
131     # shellcheck disable=SC2248
132     get_split_score_simple 1 ${LFBFL_i_max_comment_line_length} /
133     LFBFL_split_score_command2="${get_split_score_result}"
134     LFBFL_i_split_score_command_properties2=${i_get_split_score_result2}
135     readonly LFBFL_split_score_command2
136     readonly LFBFL_i_split_score_command_properties2
137     declare -i LFBFL_i_renaming_happened=0
138     declare -i LFBFL_i
139     declare -i LFBFL_j
140     # declare -r LFBFL_special_file_name="files_names_listing.txt"
141     declare -r LFBFL_sfn="files_names_listing.txt"
142     local LFBFL_diff_line
143     local LFBFL_similarity_line
144     local LFBFL_old_file_path
145     local LFBFL_old_file_name
146     local LFBFL_old_file_directory
```

```
  local LFBFL_new_file_path
  local LFBFL_new_file_name
  local LFBFL_new_file_directory
  local LFBFL_useful_file_name
  local LFBFL_extension
  local LFBFL_comment_prefix
  local LFBFL_comment_prefix2
  local LFBFL_old_file_name2
  local LFBFL_old_file_directory2
  local LFBFL_new_file_name2
  local LFBFL_new_file_directory2
  local LFBFL_new_comment
  declare -i LFBFL_i_copyright_line_number
  declare -r LFBFL_send_summary_2="ATTENTION: No line with copyright"
  declare -i LFBFL_i_line_count
  declare -i LFBFL_i_lines_after
  local LFBFL_temp_file_path
  for ((LFBFL_i=0; LFBFL_i<${#LFBFL_renaming_lines_array[@]};)); do
    LFBFL_diff_line="${LFBFL_renaming_lines_array[${LFBFL_i}]}"
    LFBFL_i=$((LFBFL_i + 1))
    LFBFL_similarity_line="${LFBFL_renaming_lines_array[${LFBFL_i}]}"
    LFBFL_i=$((LFBFL_i + 1))
    # We need to increment 3 times since grep adds "--" lines between.
    LFBFL_i=$((LFBFL_i + 1))
    if [[ LFBFL_i_verbose -eq 1 ]]; then
      printf "LFBFL_diff_line: %s\nLFBFL_similarity_line: %s\n"\
        "${LFBFL_diff_line}"\
        "${LFBFL_similarity_line}"
    fi
    if [[ "${LFBFL_similarity_line:17:4}" != "100%" ]]; then
      continue
    fi

    # Extract old file name from diff line. -------------------------
    LFBFL_old_file_path=$(
      printf "%s" "${LFBFL_diff_line}"\
      | cut -d ' ' -f 3
    )
    # It starts by "a/".
    LFBFL_old_file_path=".${LFBFL_old_file_path:1}"
    LFBFL_old_file_name=$(basename "${LFBFL_old_file_path}")
    LFBFL_old_file_directory=$(dirname "${LFBFL_old_file_path}")
    # Extract new file name from diff line. -------------------------
    LFBFL_new_file_path=$(
      printf "%s" "${LFBFL_diff_line}"\
      | cut -d ' ' -f 4
```

```
    )
    # It starts by "b/".
    LFBFL_new_file_path=".${LFBFL_new_file_path:1}"
    LFBFL_new_file_name=$(basename "${LFBFL_new_file_path}")
    LFBFL_new_file_directory=$(dirname "${LFBFL_new_file_path}")
    if [[ LFBFL_i_verbose -eq 1 ]]; then
      LFBFL_s_format="LFBFL_old_file_path: %s\n"
      LFBFL_s_format+="LFBFL_old_file_name: %s\n"
      LFBFL_s_format+="LFBFL_old_file_directory: %s\n"
      LFBFL_s_format+="LFBFL_new_file_path: %s\n"
      LFBFL_s_format+="LFBFL_new_file_name: %s\n"
      LFBFL_s_format+="LFBFL_new_file_directory: %s\n"
      # shellcheck disable=SC2059
      printf "${LFBFL_s_format}"\
        "${LFBFL_old_file_path}"\
        "${LFBFL_old_file_name}"\
        "${LFBFL_old_file_directory}"\
        "${LFBFL_new_file_path}"\
        "${LFBFL_new_file_name}"\
        "${LFBFL_new_file_directory}"
    fi
    # Extract "LFBFL_extension" with smart guessing for .tpl ---------
    # and set prefix for line with ©Copyright. -----------------------
    LFBFL_useful_file_name="${LFBFL_new_file_name}"
    LFBFL_extension="${LFBFL_useful_file_name##*.}"
    if [[ LFBFL_i_verbose -eq 1 ]]; then
      printf "LFBFL_useful_file_name: %s\nLFBFL_extension: %s\n"\
        "${LFBFL_useful_file_name}"\
        "${LFBFL_extension}"
    fi
    for ((LFBFL_j=0; LFBFL_j<3; ++LFBFL_j)); do
      if [[ "${LFBFL_extension}" != "tpl" ]]; then
        break
      fi
      LFBFL_useful_file_name="${LFBFL_useful_file_name:0:-4}"
      LFBFL_extension="${LFBFL_useful_file_name##*.}"
      if [[ LFBFL_i_verbose -eq 1 ]]; then
        printf "LFBFL_useful_file_name: %s\nLFBFL_extension: %s\n"\
          "${LFBFL_useful_file_name}"\
          "${LFBFL_extension}"
      fi
    done
    LFBFL_comment_prefix=""
    if [[ "${LFBFL_extension}" == "sh" ]]; then
      LFBFL_comment_prefix="# "
    fi
```

```
    if [[ "${LFBFL_extension}" == "sql" ]]; then
      LFBFL_comment_prefix="-- "
    fi
    if [[ "${LFBFL_extension}" == "tex" ]]; then
      LFBFL_comment_prefix="% "
    fi
    if [[ "${LFBFL_useful_file_name}" == "${LFBFL_sfn}" ]]; then
      LFBFL_comment_prefix="// "
    fi
    if [[ LFBFL_i_verbose -eq 1 ]]; then
      printf "LFBFL_comment_prefix: %s\n" "${LFBFL_comment_prefix}"
    fi
    #-----------------------------------------------------------------
    # Split if necessary old_file_name, new_file_name, etc. to have
    # comment lines below max_comment_line_length.
    LFBFL_comment_prefix2="${LFBFL_comment_prefix}"'"'
    # -----
    LFBFL_old_file_name2="${LFBFL_comment_prefix2}"
    LFBFL_old_file_name2+="${LFBFL_old_file_name}"
    # shellcheck disable=SC2248
    repeated_split_last_line "${LFBFL_old_file_name2}"\
      "${LFBFL_comment_prefix2}"\
      ${LFBFL_i_max_comment_line_length}\
      '"'\
      "${LFBFL_split_score_command1}"\
      ${LFBFL_i_split_score_command_properties1}
    LFBFL_old_file_name2="${repeated_split_last_line_result}"
    # -----
    LFBFL_old_file_directory2="${LFBFL_comment_prefix2}"
    LFBFL_old_file_directory2+="${LFBFL_old_file_directory}"
    # shellcheck disable=SC2248
    repeated_split_last_line "${LFBFL_old_file_directory2}"\
      "${LFBFL_comment_prefix2}"\
      ${LFBFL_i_max_comment_line_length}\
      '"'\
      "${LFBFL_split_score_command2}"\
      ${LFBFL_i_split_score_command_properties2}
    LFBFL_old_file_directory2="${repeated_split_last_line_result}"
    # -----
    LFBFL_new_file_name2="${LFBFL_comment_prefix2}"
    LFBFL_new_file_name2+="${LFBFL_new_file_name}"
    # shellcheck disable=SC2248
    repeated_split_last_line "${LFBFL_new_file_name2}"\
      "${LFBFL_comment_prefix2}"\
      ${LFBFL_i_max_comment_line_length}\
      '"'\
```

```
      "${LFBFL_split_score_command1}"\
      ${LFBFL_i_split_score_command_properties1}
    LFBFL_new_file_name2="${repeated_split_last_line_result}"
    # -----
    LFBFL_new_file_directory2="${LFBFL_comment_prefix2}"
    LFBFL_new_file_directory2+="${LFBFL_new_file_directory}"
    # shellcheck disable=SC2248
    repeated_split_last_line "${LFBFL_new_file_directory2}"\
      "${LFBFL_comment_prefix2}"\
      ${LFBFL_i_max_comment_line_length}\
      '"'\
      "${LFBFL_split_score_command2}"\
      ${LFBFL_i_split_score_command_properties2}
    LFBFL_new_file_directory2="${repeated_split_last_line_result}"
    # Create new comments strings. ----------------------------------
    LFBFL_new_comment=""
    if [[ "${LFBFL_new_file_name}" != "${LFBFL_old_file_name}" ]];
    then
      LFBFL_new_comment="${LFBFL_comment_prefix}"
      LFBFL_new_comment+="${LFBFL_timestamp}"
      LFBFL_new_comment+=": This file was renamed from"
      LFBFL_new_comment+=$'\n'
      LFBFL_new_comment+="${LFBFL_old_file_name2}"'"'
      LFBFL_new_comment+=$'\n'
      LFBFL_new_comment+="${LFBFL_comment_prefix}to"
      LFBFL_new_comment+=$'\n'
      LFBFL_new_comment+="${LFBFL_new_file_name2}"'".'
    fi
    if [[ LFBFL_i_log_directory_change -eq 1
      && "${LFBFL_new_file_directory}" != "${LFBFL_old_file_directory}"
      ]];
    then
      if [[ "${LFBFL_new_file_name}" != "${LFBFL_old_file_name}" ]];
      then
        LFBFL_new_comment+=$'\n'
      fi
      LFBFL_new_comment+="${LFBFL_comment_prefix}"
      LFBFL_new_comment+="${LFBFL_timestamp}"
      LFBFL_new_comment+=": This file was moved from"
      LFBFL_new_comment+=$'\n'
      LFBFL_new_comment+="${LFBFL_old_file_directory2}"'"'
      LFBFL_new_comment+=$'\n'
      LFBFL_new_comment+="${LFBFL_comment_prefix}to"
      LFBFL_new_comment+=$'\n'
      LFBFL_new_comment+="${LFBFL_new_file_directory2}"'".'
    fi
```

```
    if [[ LFBFL_i_verbose -eq 1 ]]; then
      printf "LFBFL_new_comment: %s\n" "${LFBFL_new_comment}"
    fi
    # Find line with ©Copyright. ------------------------------------
    LFBFL_i_copyright_line_number=$(
      grep -n '©Copyright' "${LFBFL_new_file_path}"\
      | head --lines=1\
      | cut -d ':' -f 1
    )
    if [[ LFBFL_i_verbose -eq 1 ]]; then
      printf "LFBFL_i_copyright_line_number: %s\n"\
        "${LFBFL_i_copyright_line_number}"
    fi
    if [[ LFBFL_i_copyright_line_number -eq 0 ]]; then
      notify-send "${LFBFL_send_summary_2}"\
        "${LFBFL_send_summary_2} for: ${LFBFL_new_comment}"
      if [[ LFBFL_i_verbose -eq 1 ]]; then
        printf "No copyright line. Skipping file.\n"
      fi
      continue
    fi
    # Get total number of lines. ------------------------------------
    LFBFL_i_line_count=$(
      wc -l\
      < "${LFBFL_new_file_path}"
    )
    if [[ LFBFL_i_verbose -eq 1 ]]; then
      printf "LFBFL_i_line_count: %s\n" "${LFBFL_i_line_count}"
    fi
    # Compute number of lines after. --------------------------------
    LFBFL_i_lines_after=$((
      LFBFL_i_line_count - LFBFL_i_copyright_line_number
    ))
    if [[ LFBFL_i_verbose -eq 1 ]]; then
      printf "LFBFL_i_lines_after: %s\n" "${LFBFL_i_lines_after}"
    fi
    # Update file. --------------------------------------------------
    LFBFL_temp_file_path="${LFBFL_new_file_path}.LFBFL.temp"
    {
      head --lines="${LFBFL_i_copyright_line_number}"\
        "${LFBFL_new_file_path}"
      printf "%s\n" "${LFBFL_new_comment}"
      tail --lines="${LFBFL_i_lines_after}" "${LFBFL_new_file_path}"
    } >> "${LFBFL_temp_file_path}"
    mv "${LFBFL_temp_file_path}" "${LFBFL_new_file_path}"
    # Add file to stage. --------------------------------------------
```

```
    git add "${LFBFL_new_file_path}"
    LFBFL_i_renaming_happened=1
  done
  readonly LFBFL_i_renaming_happened
  if [[ LFBFL_i_renaming_happened -eq 1 ]]; then
    # Commit -------------------------------------------------------
    local LFBFL_message="commit_a_file_renamed_comment()"
    LFBFL_message+=" post-commit hook, ${LFBFL_timestamp}."
    readonly LFBFL_message
    printf "Commit message:\n%s\n" "${LFBFL_message}"
    git commit --no-verify --message="${LFBFL_message}"
  fi
}
```

3.57 git_helpers/pre-commit

```
#!/usr/bin/env bash
# This file is part of DevOrSysAdminScripts library.
#
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# you can redistribute it and/or modify it under the terms
# of the GNU General Public License
# as published by the Free Software Foundation,
# either version 3 of the License,
# or (at your option) any later version.
#
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# but WITHOUT ANY WARRANTY;
# without even the implied warranty of
# MERCHANTABILITY or FITNESS FOR A PARTICULAR PURPOSE.
# See the GNU General Public License for more details.
#
# You should have received a copy of
# the GNU General Public License
# along with DevOrSysAdminScripts.
# If not, see <https://www.gnu.org/licenses/>.
#
# ©Copyright 2023-2024 Laurent Frédéric Bernard François Lyaudet
# 2024-05-31T01:57+02:00: This file was renamed from
# "pre-commit"
# to
# "pre-commit".
# 2024-05-31T01:57+02:00: This file was moved from
# "."
# to
```

```
# "./git_helpers".

LFBFL_some_directory="../DevOrSysAdminScripts"
declare -i LFBFL_i_pushd_result3
pushd "${LFBFL_some_directory}" || {
  LFBFL_i_pushd_result3=$?
  printf "pre-commit no such directory\n"
  # shellcheck disable=SC2248
  exit ${LFBFL_i_pushd_result3}
}
# shellcheck source=git_checks.libr.sh
source "./git_helpers/git_checks.libr.sh"
declare -i LFBFL_i_popd_result3
popd || {
  LFBFL_i_popd_result3=$?
  printf "pre-commit no popd\n"
  # shellcheck disable=SC2248
  exit ${LFBFL_i_popd_result3}
}

declare -gi LFBFL_i_git_hook_error=0
if ! check_files; then
  LFBFL_i_git_hook_error=1
fi
if ! check_no_abusive_trailing_comma; then
  LFBFL_i_git_hook_error=1
fi
if ! check_black_code_formatting; then
  LFBFL_i_git_hook_error=1
fi
readonly LFBFL_i_git_hook_error
# shellcheck disable=SC2248
exit ${LFBFL_i_git_hook_error}
```

3.58 IMAP_transfert.py

```
"""
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or (at your option) any later version.

```

```
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Pour transférer d'un compte IMAP à un autre
Penser à autoriser les apps moins sûres sur Gmail si nécessaire
"""

import argparse
import email
import imaplib


class IMAPTransfertBadArgumentError(Exception):
    """
    A class to avoid a linter warning,
    that needs an Error suffix to avoid another linter warning,
    that needs a docstring to avoid another linter warning.
    If you don't see its intent, try badminton.
    """


class SurpriseError(Exception):
    """
    A class to avoid a linter warning,
    that needs an Error suffix to avoid another linter warning,
    that needs a docstring to avoid another linter warning.
    If you don't see its intent after seeing its use, try volleyball.
    But "Cheer up!", now you don't need "pass" XD.
    """


def IMAP_transfert(
    from_login: str,
    from_password: str,
    from_host: str,
```

```
    from_port: int,
    to_login: str,
    to_password: str,
    to_host: str,
    to_port: int,
    *,
    to_new_folder: str = "",
    verbose: bool = False,
) -> None:
    """
    Transfert all emails from some IMAP account
    to another IMAP account.
    """
    empty = ""
    if from_login == "":
        raise IMAPTransfertBadArgumentError("From login is empty.")
    # if from_password == "":
    if from_password == empty:
        # https://github.com/PyCQA/bandit/issues/714
        # Dumb
        raise IMAPTransfertBadArgumentError("From password is empty.")
    if from_host == "":
        raise IMAPTransfertBadArgumentError("From host is empty.")
    if from_port <= 0:
        raise IMAPTransfertBadArgumentError(
            "From port is not positive.",
        )
    if to_login == "":
        raise IMAPTransfertBadArgumentError("To login is empty.")
    if to_password == empty:
        # And Dumber The infinite collection XD
        raise IMAPTransfertBadArgumentError("To password is empty.")
    if to_host == "":
        raise IMAPTransfertBadArgumentError("To host is empty.")
    if to_port <= 0:
        raise IMAPTransfertBadArgumentError(
            "To port is not positive.",
        )
    # pylint: disable=pointless-string-statement
    """
    Maybe one day Python will ship something like that in its standard
    library:
    https://www.php.net/manual/fr/function.filter-var.php
    But now urlparse doesn't do any validation.
    You may want to check that the hosts or other fields are not
    "anything".
```

```
    """

    # pylint: disable=unreachable
    raise SurpriseError(
        "I haven't tested this version of my script... Oooooh!\n"
        "But I linted it without this exception ;). Aaaaaaaah!\n"
        "XD XD XD\n"
        "Untested Scream Horror Show (Soon on BrainfuckedFlix)",
    )
    # """
    # I haven't tested any version with an empty to_new_folder...

    if verbose:
        print("Connecting to from account.")
    from_account = imaplib.IMAP4_SSL(host=from_host, port=from_port)
    from_account.login(from_login, from_password)

    if verbose:
        print('Connecting to "to account".')
    to_account = imaplib.IMAP4_SSL(host=to_host, port=to_port)
    to_account.login(to_login, to_password)

    if to_new_folder != "":
        if verbose:
            print("Creating new folder in to account.")
        to_account.create(to_new_folder)

    if verbose:
        print("Fetching all emails ids in from account.")
    from_account.select()
    result, ids = from_account.uid("search", "all")
    if verbose:
        print(f'from_account.uid("search") result:{result} ids:{ids}')
    ids = ids[0].decode().split()
    if verbose:
        print("Fetching and transfering each email in from account.")
    for my_id in ids:
        result, raw_message = from_account.uid(
            "fetch",
            my_id,
            "(RFC822)",
        )
        if verbose:
            print(f'from_account.uid("fetch", {my_id}) result:{result}')
        message = raw_message[0][1]
        my_email = email.message_from_string(message.decode("utf-8"))
```

```
        flags = imaplib.ParseFlags(raw_message[0][0])
        # flags = " ".join(flags) mypy is unhappy but IMAP flags are
        # ASCII strings, we give mypy what it wants.
        flags_as_strs = [flag.decode("ascii") for flag in flags]
        flags_str = " ".join(flags_as_strs)
        # date = imaplib.Internaldate2tuple(raw_message[0][0])
        date = '"' + my_email["date"] + '"'
        if verbose:
            print(
                f'from_account.uid("fetch", {my_id})'
                f" flags:{flags} date:{date}",
            )
        to_account.append(to_new_folder, flags_str, date, message)


def main() -> None:
    """
    Parse command-line arguments and
    transfert all emails from some IMAP account
    to another IMAP account.
    """
    parser = argparse.ArgumentParser(
        description=(
            "Transfert all emails from some IMAP account"
            " to another IMAP account"
        ),
    )
    parser.add_argument(
        "from_login",
        type=str,
        help="Email account login of user one.",
    )
    parser.add_argument(
        "from_password",
        type=str,
        help="Email account password of user one.",
    )
    parser.add_argument(
        "to_login",
        type=str,
        help="Email account login of user two.",
    )
    parser.add_argument(
        "to_password",
        type=str,
        help="Email account password of user two.",
```

```
    )
    parser.add_argument(
        "--to_new_folder",
        type=str,
        default="",
        help="Transfert emails into a new folder of this name.",
    )
    parser.add_argument(
        "--from_host",
        type=str,
        default="imap.gmail.com",
        help="Host of first email account",
    )
    parser.add_argument(
        "--from_port",
        type=int,
        default=993,
        help="Port on host of first email account",
    )
    parser.add_argument(
        "--to_host",
        type=str,
        default="imap.gmail.com",
        help="Host of second email account",
    )
    parser.add_argument(
        "--to_port",
        type=int,
        default=993,
        help="Port on host of second email account",
    )
    parser.add_argument(
        "-v",
        "--verbose",
        action="store_true",
        help="Print details of operations",
    )

    args = parser.parse_args()

    if args.verbose:
        print(
            "-----------------------------------------------------\n"
            "IMAP transfert:\n"
            "-----------------------------------------------------\n"
            "From account:\n"
```

```
            f" - Login: {args.from_login}\n"
            f" - Password: {args.from_password}\n"
            f" - Host: {args.from_host}\n"
            f" - Port: {args.from_port}\n"
            "To account:\n"
            f" - Login: {args.to_login}\n"
            f" - Password: {args.to_password}\n"
            f" - Host: {args.to_host}\n"
            f" - Port: {args.to_port}\n"
            "-----------------------------------------------------\n",
        )
        if args.to_new_folder is not None:
            print(
                f"Emails will go in folder {args.to_new_folder}"
                " after it is created.",
            )
    try:
        IMAP_transfert(
            args.from_login,
            args.from_password,
            args.from_host,
            args.from_port,
            args.to_login,
            args.to_password,
            args.to_host,
            args.to_port,
            to_new_folder=args.to_new_folder,
            verbose=args.verbose,
        )
    except (IMAPTransfertBadArgumentError, SurpriseError) as e:
        print(e)


if __name__ == "__main__":
    main()
```

3.59 LiensUtiles.txt

```
https://support.google.com/a/answer/9476255?hl=fr#zippy=%2C%C3%A9tape\
-configurez-le-service-de-migration-de-donn%C3%A9es
---------------------------------------------------------------------
https://stackoverflow.com/a/78413334/5796086
https://stackoverflow.com/questions/21612980/\
why-is-usr-bin-env-bash-superior-to-bin-bash/78413334
https://web.archive.org/web/20240501105801/https://stackoverflow.com\
/questions/21612980/why-is-usr-bin-env-bash-superior-to-bin-bash/\
```

```
78413334#78413334
----------------------------------------------------------------------
https://news.ycombinator.com/item?id=22214690
https://github.com/koalaman/shellcheck/issues/2005
https://opensource.stackexchange.com/questions/6271/\
calling-gpld-programs-from-a-shell-script
----------------------------------------------------------------------
```

3.60 miscellaneous/funny.sh

```
#!/usr/bin/env bash
# This file is part of DevOrSysAdminScripts library.
#
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#
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unprotected_funny(){
  # Works in a script but not in shell if job control is active.
  shopt -s lastpipe
  declare -i LFBFL_i_result=0
  local LFBFL_line
  printf "hahaha\n" | while read -r LFBFL_line; do
    if [[ "${LFBFL_line}" == "hahaha" ]]; then
      LFBFL_i_result=1
    fi
  done
  [[ LFBFL_i_result -eq 1 ]] && printf "I had a good laugh.\n"
}
```

```

protected_funny(){
  # Always works regarding the echoes.
  printf "hahaha\n" | while read -r LFBFL_line; do
    unprotected_funny
  done
}
```

3.61 miscellaneous/spampty_data.txt

```



a
 a
empty.txt:
one_space.txt:
all_space.txt:
normal.txt:a
normal_again.txt: a
```

3.62 miscellaneous/string_escaping.html

```
<!DOCTYPE html>
<html lang="en">
<head>
  <meta charset="utf-8">
  <title>Truc</title>
</head>
<body>
<header id="title-block-header">
<h1 class="title">Truc</h1>
</header>
&lt;abc> &amp; &lt;de>
<p title="<&>'
 abcd`\ machin">
This is some text
</p>

<pre>
"&lt;&amp;>'&quot;
abcd`\ machin"
This is some text
</pre>

<script>
```

```
let a = 'abcd & < > \\ \n  \"\'';
alert(a);
let b = "abcd & < > \\ \n  \"\'";
alert(b);
</script>

</body>
</html>
```

3.63 miscellaneous/string_escaping.xhtml

```
<!DOCTYPE html PUBLIC "-//W3C//DTD XHTML 1.0 Transitional//EN"
          "http://www.w3.org/TR/xhtml1/DTD/xhtml1-transitional.dtd">
<html xmlns="http://www.w3.org/1999/xhtml" xml:lang="fr" lang="fr">
<head>
<meta content="text/html; charset=UTF-8" http-equiv="Content-Type" />
  <title>Truc</title>
</head>
<body>
<h1 class="title">Truc</h1>
&lt;abc> &amp; &lt;de>

<p title="&lt;&amp;>'&#10;
 abcd`\ machin">
This is some text
</p>

<pre>
"&lt;&amp;>'&quot;
abcd`\ machin"
This is some text
</pre>

<script type="text/javascript">
let a = 'abcd &amp; &lt; > \\ \n  \"\'';
alert(a);
let b = "abcd &amp; &lt; > \\ \n  \"\'";
alert(b);
</script>

</body>
</html>
```

3.64 README.md.tpl

```
# DevOrSysAdminScripts

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[![Code-style:black:badge]][Black-GitHub.com]
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[![Typecheck:mypy:badge]][Typecheck-mypy-lang.org]
[![Linting:pylint:badge]][Pylint-GitHub.com]
[![Linting:ruff:badge]][Ruff-GitHub.com]

|                **Some "small" useful scripts**               |
|:-------------------------------------------------------------:|
| **for enhancing development or system-administrators tasks** |

Currently, the biggest part of this repository is in
a suite of shell scripts about build and checks that
I use in some of my other repositories.

I tried to follow some conventions for my bash code:

- I tried to encapsulate most of the code in functions.
- Unless used as return values, variables in functions are defined
  as local variables (`local` keyword or `declare`)
  with prefix "LFBFL_".
- When used as return values, variables in functions are defined
  as global variables (`declare -g`) with prefix
  "@function_name@_result_".
- Whenever a variable is no more modified after some point,
  add the keyword `readonly` or use `declare -r`.
- Whenever a variable will only contain integer values (or boolean
  values as 0 or 1), use `declare -i`.
- I named bash files with suffixes ".exec.sh" when the script can
  be runned
  (executable, abbreviation stops before a vowel).
- I named bash files with suffixes ".libr.sh" when the script contains
  only functions definitions
  (library, abbreviation stops before a vowel).
- But the files "pre-commit" and "post-commit" were not renamed,
  since it is not possible to give them other names to use them as
  pre-commit/post-commit hooks in `git`.
- When some ".exec.sh" code is encapsulated into functions,
```

```
  either these functions can be reused and go in some ".libr.sh" file,
  either they are truly specific to this script and they are kept
  in ".exec.sh" file.
- When using `source` builtin, although any path/filename with a slash
  will only be searched for from the current directory (tested),
  we add the prefix "./", to make it explicit.
- We use mapfile instead of read -r most of the time, but with the
  following exceptions: real use of read -r to read the command line of the
  shell, comments (of course), and a file funny.sh where it is kind of
  required to use read -r.
- We replaced all echo calls by printf calls to follow
  "The Open Group Base Specifications" since at least 2 decades:
  "New applications are encouraged to use printf instead of echo.",
  <https://pubs.opengroup.org/onlinepubs/000095399/>.
- We used `_?i_` or `_?j_`, etc. in integer variables names in Bash
  scripts.

[CodeFactor-badge]: https://www.codefactor.io/repository/github/\
llyaudet/DevOrSysAdminScripts/badge

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github/llyaudet/DevOrSysAdminScripts

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3dd6ef85e8db4460991d1ebc404b68dd

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DevOrSysAdminScripts/dashboard?utm_source=gh&utm_medium=referral\
&utm_content=&utm_campaign=Badge_grade

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languages/top/llyaudet/DevOrSysAdminScripts

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languages/code-size/llyaudet/DevOrSysAdminScripts

[Security:bandit:badge]: https://img.shields.io/badge/\
security-bandit-yellow.svg

[Bandit-GitHub.com]: https://github.com/PyCQA/bandit

[Code-style:black:badge]: https://img.shields.io/badge/\
code%20style-black-000000.svg

[Black-GitHub.com]: https://github.com/psf/black

```

```
[Imports:isort:badge]: https://img.shields.io/badge/\
%20imports-isort-%231674b1?style=flat&labelColor=ef8336

[Isort-GitHub.io]: https://pycqa.github.io/isort/

[Typecheck:mypy:badge]: https://www.mypy-lang.org/static/\
mypy_badge.svg

[Typecheck-mypy-lang.org]: https://mypy-lang.org/

[Linting:pylint:badge]: https://img.shields.io/badge/\
linting-pylint-yellowgreen

[Pylint-GitHub.com]: https://github.com/pylint-dev/pylint

[Linting:ruff:badge]: https://img.shields.io/endpoint?url=\
https://raw.githubusercontent.com/astral-sh/ruff/main/assets/\
badge/v2.json

[Ruff-GitHub.com]: https://github.com/astral-sh/ruff
```

3.65 string_escaping.libr.php

```
<?php
/*
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```

```
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*/

namespace string_escaping\HTML {
  function escape_text(string $s_string) : string {
    return str_replace(
      array(
        '&',
        '<',
      ),
      array(
        '&amp;',
        '&lt;',
      ),
      $s_string,
    );
  }



  function escape_attribute(string $s_string) : string {
    # in double quotes
    return str_replace(
      array('"'),
      array('&quot;'),
      $s_string,
    );
  }



  function escape_pre(string $s_string) : string {
    return str_replace(
      array(
        '&',
        '<',
      ),
      array(
        '&amp;',
        '&lt;',
      ),
      $s_string,
    );
  }
}

```

```
namespace string_escaping\XHTML {
  function escape_text(string $s_string) : string {
    return str_replace(
      array(
        '&',
        '<',
      ),
      array(
        '&amp;',
        '&lt;',
      ),
      $s_string,
    );
  }



  function escape_attribute(string $s_string) : string {
    # in double quotes
    return str_replace(
      array(
        '&',
        '<',
        '"',
        "\n",
      ),
      array(
        '&amp;',
        '&lt;',
        '&quot;',
        '&#10;'
      ),
      $s_string,
    );
  }



  function escape_pre(string $s_string) : string {
    return str_replace(
      array(
        '&',
        '<',
```

```
      ),
      array(
        '&amp;',
        '&lt;',
      ),
      $s_string,
    );
  }
}



namespace string_escaping\JS {
  use function string_escaping\XHTML\escape_text as XHTML_escape_text;

  function escape(string $s_string) : string {
    return str_replace(
      array(
        '\\',
        "\n",
        "'",
        '"',
      ),
      array(
        '\\\\',
        '\n',
        '\\\'',
        '\\"',
      ),
      $s_string,
    );
  }



  function escape_for_string_definition_inside_XHTML(string $s_string)
  : string {
    return XHTML_escape_text(escape($s_string));
  }
}

/*
$ php
<?php
require_once('string_escaping.libr.php');
```

```
var_dump(string_escaping\HTML\escape_text('<&"\'truc\\'));
var_dump(string_escaping\HTML\escape_attribute("<&\"\n'truc\\"));
var_dump(string_escaping\HTML\escape_pre("<&\"\n'truc\\"));
var_dump(string_escaping\XHTML\escape_text('<&"\'truc\\'));
var_dump(string_escaping\XHTML\escape_attribute("<&\"\n'truc\\"));
var_dump(string_escaping\XHTML\escape_pre("<&\"\n'truc\\"));
var_dump(string_escaping\JS\escape("abcd & < > \\ \n  \"'"));
var_dump(string_escaping\JS\escape_for_string_definition_inside_XHTML(
  "abcd & < > \\ \n  \"'"
));
*/
```

3.66 unicode.libr.php

```
<?php
/*
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This file was renamed from "unicode.php" to "unicode.libr.php".
*/

function decimal_code_point_to_UTF8(
  int $i_code_point_in_decimal_notation
) : string {
  // var_dump($i_code_point_in_decimal_notation);
  // 0xxxxxxx ASCII
```

```
33    if($i_code_point_in_decimal_notation < 128){
34      return chr($i_code_point_in_decimal_notation);
35    }
36    $i_continuation_base_value = 128;
37    // 110xxxxx 10xxxxxx
38    if($i_code_point_in_decimal_notation < 2048){  // 2**11
39      $i_first_byte_base_value = 192;
40      $i_last_byte_significant_bits = (
41        $i_code_point_in_decimal_notation % 64  // 2**6
42      );
43      $i_first_byte_significant_bits = intdiv(
44        (
45          $i_code_point_in_decimal_notation
46          - $i_last_byte_significant_bits
47        ),
48        64,  // 2**6
49      );
50      $s_result = chr(
51        $i_first_byte_base_value
52        + $i_first_byte_significant_bits
53      );
54      $s_result .= chr(
55        $i_continuation_base_value
56        + $i_last_byte_significant_bits
57      );
58      return $s_result;
59    }
60    // 1110xxxx 10xxxxxx 10xxxxxx
61    if($i_code_point_in_decimal_notation < 65536){  // 2**16
62      $i_first_byte_base_value = 224;
63      $arr_arr_data_per_byte_reverse = [
64        ["i_bits" => 6, "i_base_value" => $i_continuation_base_value],
65        ["i_bits" => 6, "i_base_value" => $i_continuation_base_value],
66        ["i_bits" => 4, "i_base_value" => $i_first_byte_base_value],
67      ];
68      $i_rest = $i_code_point_in_decimal_notation;
69      $s_result = "";
70      foreach($arr_arr_data_per_byte_reverse as $arr_data){
71        // _Pragma unroll
72        // J'aurais pu "roller" les 4 cas des ifs.
73        // Mais pour les perfs, il faudrait tout unroller vu le nombre
74        // d'itérations. Et j'aimerais bien comparer du code unrollé
75        // aux 2 niveaux à la main avec celui unrollé par le compilateur
76        // sur les 2 niveaux. Je préjuge/présuppose qu'il unroll
77        // l'intérieur d'abord (pour ne pas refaire le même boulot)
78        // puis l'extérieur et qu'il oublie de réoptimiser l'intérieur
```

```
      // en fonction de l'extérieur unrollé. Est-ce que j'ai raison ?
      $i_significant_bits = $i_rest % (2**$arr_data["i_bits"]);
      $i_rest = intdiv(
        $i_rest - $i_significant_bits,
        2**$arr_data["i_bits"],
      );
      $s_result = (  // .= or =.
        chr($i_significant_bits + $arr_data["i_base_value"])
        .$s_result
      );
    }
    return $s_result;
  }
  // 11110xxx 10xxxxxx 10xxxxxx 10xxxxxx
  if($i_code_point_in_decimal_notation < 2097152){  // 2**21
    $i_first_byte_base_value = 240;
    $arr_arr_data_per_byte_reverse = [
      ["i_bits" => 6, "i_base_value" => $i_continuation_base_value],
      ["i_bits" => 6, "i_base_value" => $i_continuation_base_value],
      ["i_bits" => 6, "i_base_value" => $i_continuation_base_value],
      ["i_bits" => 3, "i_base_value" => $i_first_byte_base_value],
    ];
    $i_rest = $i_code_point_in_decimal_notation;
    $s_result = "";
    foreach($arr_arr_data_per_byte_reverse as $arr_data){
      // _Pragma unroll
      $i_significant_bits = $i_rest % (2**$arr_data["i_bits"]);
      $i_rest = intdiv(
        $i_rest - $i_significant_bits,
        2**$arr_data["i_bits"],
      );
      $s_result = (  // .= or =.
        chr($i_significant_bits + $arr_data["i_base_value"])
        .$s_result
      );
    }
    return $s_result;
  }
  throw new Exception(
    "UTF-8 avec jusqu'à 6 octets a été abandonné il y a longtemps."
    ." Autre anecdote \"débile\" : MySQL a de base UTF-8 avec jusqu'à"
    ." 3 octets au lieu de 4 :face-palm:."
    ." Du coup, ça peut donner quelques \"surprises\"."
    ." Sinon, y'a aussi ASCII dans MySQL."
    ." Et comme ça joue sur la mémoire et les performances,"
    ." pourquoi ils n'ont pas fait ASCII, UTF-8-2, UTF-8-3, UTF-8-4 ?"
```

```
    ." :)"
  );
}



function hexa_code_point_to_UTF8(
  string $s_code_point_in_hexadecimal_notation
) : string {
  $s_code_point_in_hexadecimal_notation = strtoupper(
    $s_code_point_in_hexadecimal_notation
  );
  $i_code_point_in_decimal_notation = 0;
  $i_ord_0 = ord('0');
  $i_ord_A = ord('A');
  $i_digit_offset = -$i_ord_0;
  $i_letter_offset = 10 - $i_ord_A;
  for(
    $i = 0, $i_max = strlen($s_code_point_in_hexadecimal_notation);
    $i < $i_max;
    ++$i
  ){
    $s_hexa_digit = $s_code_point_in_hexadecimal_notation[$i];
    $i_code_point_in_decimal_notation *= 16;
    if(ctype_digit($s_hexa_digit)){
      $i_code_point_in_decimal_notation += (
        ord($s_hexa_digit) + $i_digit_offset
      );
      continue;
    }
    if(ctype_xdigit($s_hexa_digit)){
      $i_code_point_in_decimal_notation += (
        ord($s_hexa_digit) + $i_letter_offset
      );
      continue;
    }
    throw new Exception("Not an hexadecimal digit.");
  }
  return decimal_code_point_to_UTF8(
    $i_code_point_in_decimal_notation
  );
}



function check_string_is_valid_ASCII(string $s_string) : bool {
```

```
171   $i_offset_in_octets_from_string_start = 0;
172   $i_current_line_number = 1;
173   $i_offset_in_octets_from_line_start = -1;
174   for($i = 0, $i_max = strlen($s_string); $i < $i_max; ++$i){
175     $i_current_octet = ord($s_string[$i]);
176     $i_offset_in_octets_from_string_start = $i;
177     $i_offset_in_octets_from_line_start += 1;
178     if($i_current_octet > 127){
179       throw new Exception(
180         "Non-ASCII character found on line "
181         .$i_current_line_number
182         ."; the octet/character "
183         .($i_offset_in_octets_from_line_start + 1)
184         ." has value "
185         .$i_current_octet
186         ." which is not ASCII."
187         ." (Sequential position without line splitting:"
188         ." This is at octet/character "
189         .($i_offset_in_octets_from_string_start + 1)
190         ".)",
191       );
192     }
193     // if($s_string[$i] === "\n"){
194     if($i_current_octet === 10){
195       $i_current_line_number += 1;
196       $i_offset_in_octets_from_line_start = -1;
197     }
198   }
199   return true;
200 }
201
202
203
204 function check_file_is_valid_ASCII(string $s_file_name) : bool {
205   $s_string = file_get_contents($s_file_name);
206   if($s_string === false){
207     throw new Exception("File ".$s_file_name." not found.");
208   }
209   return check_string_is_valid_ASCII($s_string);
210 }
211
212
213
214 function check_string_is_valid_UTF8(string $s_string) : bool {
215   $i_continuation_octet_needed = 0;
216   $i_offset_in_octets_from_string_start = 0;
```

```
217     $i_offset_in_characters_from_string_start = -1;
218     $i_character_start_position_from_string_start = 0;
219     $i_current_line_number = 1;
220     $i_offset_in_octets_from_line_start = -1;
221     $i_offset_in_characters_from_line_start = -1;
222     $i_character_start_position_from_line_start = -1;
223     for($i = 0, $i_max = strlen($s_string); $i < $i_max; ++$i){
224       $i_current_octet = ord($s_string[$i]);
225       $i_offset_in_octets_from_string_start = $i;
226       $i_offset_in_octets_from_line_start += 1;
227       if($i_continuation_octet_needed > 0){
228         if($i_current_octet < 128 || $i_current_octet >= 192){
229           throw new Exception(
230             "Non-UTF8 character found on line "
231             .$i_current_line_number
232             ."; the octet "
233             .($i_offset_in_octets_from_line_start + 1)
234             ", part of the character "
235             .($i_offset_in_characters_from_line_start + 1)
236             ", has value "
237             .$i_current_octet
238             ." which is not a continuation octet."
239             ." This character starts at octet "
240             .($i_character_start_position_from_line_start + 1)
241             ." of the current line."
242             ." (Sequential positions without line splitting:"
243             ." This is at character "
244             .($i_offset_in_characters_from_string_start + 1)
245             ." and octet "
246             .($i_offset_in_octets_from_string_start + 1)
247             "."
248             ." This character starts at octet "
249             .($i_character_start_position_from_string_start + 1)
250             ".)",
251           );
252         }
253         --$i_continuation_octet_needed;
254         continue;
255       }
256
257       $i_offset_in_characters_from_string_start += 1;
258       $i_offset_in_characters_from_line_start += 1;
259       $i_character_start_position_from_string_start = (
260         $i_offset_in_octets_from_string_start
261       );
262       $i_character_start_position_from_line_start = (
```

```
263         $i_offset_in_octets_from_line_start
264       );
265       if($i_current_octet < 128){ // 0xxxxxxx ASCII
266         // if($s_string[$i] === "\n"){
267         if($i_current_octet === 10){
268           $i_current_line_number += 1;
269           $i_offset_in_octets_from_line_start = -1;
270           $i_offset_in_characters_from_line_start = -1;
271         }
272         continue;
273       }
274       if($i_current_octet >= 128 && $i_current_octet < 192){
275         throw new Exception(
276           "Non-UTF8 character found on line "
277           .$i_current_line_number
278           ."; the octet "
279           .($i_offset_in_octets_from_line_start + 1)
280           ."; part of the character "
281           .($i_offset_in_characters_from_line_start + 1)
282           ."; has value "
283           .$i_current_octet
284           ." which is a continuation octet."
285           ." This character starts at octet "
286           .($i_character_start_position_from_line_start + 1)
287           ." of the current line."
288           ." (Sequential positions without line splitting:"
289           ." This is at character "
290           .($i_offset_in_characters_from_string_start + 1)
291           ." and octet "
292           .($i_offset_in_octets_from_string_start + 1)
293           ."."
294           ." This character starts at octet "
295           .($i_character_start_position_from_string_start + 1)
296           ".)",
297         );
298       }
299       if($i_current_octet >= 192 && $i_current_octet < 224){
300         // 110xxxxx 10xxxxxx
301         $i_continuation_octet_needed = 1;
302         continue;
303       }
304       if($i_current_octet >= 224 && $i_current_octet < 240){
305         // 1110xxxx 10xxxxxx 10xxxxxx
306         $i_continuation_octet_needed = 2;
307         continue;
308       }
```

```
309     if($i_current_octet >= 240 && $i_current_octet < 248){
310       // 11110xxx 10xxxxxx 10xxxxxx 10xxxxxx
311       $i_continuation_octet_needed = 3;
312       continue;
313     }
314     throw new Exception(
315       "Non-UTF8 character found on line "
316       .$i_current_line_number
317       ."; the octet "
318       .($i_offset_in_octets_from_line_start + 1)
319       ., part of the character "
320       .($i_offset_in_characters_from_line_start + 1)
321       ., has value "
322       .$i_current_octet
323       ." which is invalid."
324       ." This character starts at octet "
325       .($i_character_start_position_from_line_start + 1)
326       ." of the current line."
327       ." (Sequential positions without line splitting:"
328       ." This is at character "
329       .($i_offset_in_characters_from_string_start + 1)
330       ." and octet "
331       .($i_offset_in_octets_from_string_start + 1)
332       ."."
333       ." This character starts at octet "
334       .($i_character_start_position_from_string_start + 1)
335       .".)",
336     );
337   }
338   return true;
339 }
340
341
342
343 function check_file_is_valid_UTF8(string $s_file_name) : bool {
344   $s_string = file_get_contents($s_file_name);
345   if($s_string === false){
346     throw new Exception("File ".$s_file_name." not found.");
347   }
348   return check_string_is_valid_UTF8($s_string);
349 }
350
351
352
353 /*
354 ?>
```

```
<?php
require_once("./unicode.libr.php");
var_dump(hexa_code_point_to_UTF8("002B"));
var_dump(hexa_code_point_to_UTF8("00E6"));
var_dump(hexa_code_point_to_UTF8("1400"));
var_dump(hexa_code_point_to_UTF8("10111"));
// https://en.wikibooks.org/wiki/Unicode/Character_reference/2000-2FFF
// Some invisible unicode characters that are a problem:
// 2000-200F 2028-202F 205F 2060-206F
$arr = [
  '2000', '2001', '2002', '2003', '2004', '2005', '2006', '2007',
  '2008', '2009', '200A', '200B', '200C', '200D', '200E', '200F',
  '2028', '2029', '202A', '202B', '202C', '202D', '202E', '202F',
  '205F',
  '2060', '2061', '2062', '2063', '2064', '2065', '2066', '2067',
  '2068', '2069', '206A', '206B', '206C', '206D', '206E', '206F',
];
$arr_s = [];
$arr_s2 = [];
$arr_s3 = [];
foreach($arr as $s_hexa){
  $s = hexa_code_point_to_UTF8($s_hexa);
  $arr_s []= $s;
  if(strlen($s) === 2){
    $s2 = '\x'.bin2hex($s[0]).'\x'.bin2hex($s[1]);
  }
  if(strlen($s) === 3){
    $s2 = '\x'.bin2hex($s[0]).'\x'.bin2hex($s[1])
      .'\x'.bin2hex($s[2]);
  }
  $arr_s2 []= $s2;
  $s3 = "  | sed -e 's/".$s2."/ /g'\\";
  $arr_s3 []= $s3;
  echo($s3."\n");
}
//*/

```

3.67 upgrade_build_and_checks.exec.sh

```
#!/usr/bin/env bash
# This file is part of DevOrSysAdminScripts library.
#
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# you can redistribute it and/or modify it under the terms
# of the GNU General Public License
```

```
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# If not, see <https://www.gnu.org/licenses/>.
#
# ©Copyright 2023-2026 Laurent Frédéric Bernard François Lyaudet
# This file was renamed from "upgrade_build_and_checks.sh"
# to "upgrade_build_and_checks.exec.sh".

upgrade_build_and_checks(){
  local LFBFL_verbose=""
  if [[ "$*" == *--verbose* ]]; then
    printf "%s %s\n" "$0" "$*"
    LFBFL_verbose="--verbose"
  fi
  local LFBFL_fixed_point_build=""
  if [[ "$*" == *--fixed-point-build* ]]; then
    LFBFL_fixed_point_build="--fixed-point-build"
  fi
  readonly LFBFL_verbose
  (\
    cd build_and_checks_dependencies/\
    && ./update_common_build_and_checks.exec.sh "${LFBFL_verbose}"\
  )
  ./build_and_checks.exec.sh "." "${LFBFL_verbose}"\
    "${LFBFL_fixed_point_build}"
}

upgrade_build_and_checks "$@"
```

3.68 wget_sha512.libr.sh

```
#!/usr/bin/env bash
# This file is part of DevOrSysAdminScripts library.
#
```

```
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# This file was renamed from "wget_sha512.sh"
# to "wget_sha512.libr.sh".

wget_sha512(){
  # $1=file_path
  # $2=download_URL
  # $3=correct_sha512
  # $4=optional --verbose
  local LFBFL_verbose=""
  if [[ "$*" == *--verbose* ]]; then
    printf "%s wget_sha512 %s\n" "$0" "$*"
    LFBFL_verbose="--verbose"
  fi
  readonly LFBFL_verbose
  if [[ ! -f "$1" ]]; then
    wget ${LFBFL_verbose:+"${LFBFL_verbose}"} -O "$1" "$2"
  fi
  declare -r LFBFL_present_sha512=$(
    sha512sum "$1"\
    | cut --fields=1 --delimiter=' '
  )
  if [[ "${LFBFL_present_sha512}" != "$3" ]]; then
    printf "%s does not have correct sha512:\n - wanted %s\n - found %s\n"\
      "$1" "$3" "${LFBFL_present_sha512}"
    return 1
  fi
```

```
}
```

Merci Dieu ! Merci Père ! Merci Seigneur ! Merci Saint Esprit !